
Replacing the Seal (A11) In the Chuck Housing

1. Remove the circlip (A12) and spacing ring (A13) from the chuck housing.
2. Remove the old seal (A11) from the chuck housing.
3. Check the condition of the bearing.
4. Check for damage to the sealing surfaces in the housing.
5. Wipe the sealing side of the seal with a light coating.
6. Use a hydraulic press and an adapter to place the seal in the cover cap without damaging it.



Figure 14.47 - 9 Using a Hydraulic Press to Locate a Vacuum Seal into a Housing

7. Ensure that the seal is in contact with the bearing and it is parallel.
8. Refit the spacing ring (A13) and circlip (A12).

Replacing the Inner Bearing (A14)

The shaft (A6) has to be removed as described previously.

1. Remove the seal as described above first.
2. Use a drift to remove the bearing (A14) from the chuck housing.
3. Check for damage to the mating surface and repair if necessary.
4. Lubricate the new bearing with GV 010 oil and press into the chuck housing by hand. Excess pressure should not be used. A slip fit has been tolerated.
5. Ensure that the bearing is fully seated.



Figure 14.47 - 10 Photograph of Bearing Located In Chuck Housing

Replacing the Outer Bearing (A15)

1. In most cases, the shaft (A6) does not have to be removed.
2. Remove the housing cover cap (A5) as previously described.
3. Remove the bearing (A15) from the chuck housing.
4. Check for damage to the mating surface and repair if necessary.
5. Lubricate the new bearing with GV 010 oil and press into the chuck housing by hand. Excess pressure should not be used. A slip fit has been tolerated.
6. Ensure that the bearing (A15) is fully seated.
7. Grease and replace the O-ring (A16).
8. Check the condition of the seal.
9. Refit the cover (A5), tab washer (A2) and locking nut (A3).

Building up the Chuck Assembly

1. Initially place all the components on a clean surface to ensure that all the parts are available.



Figure 14.47 - 11 Photograph of Chuck Components

2. Place screws similar to the screws (A4) in the end of the chuck housing to protect the vacuum-mating surface.

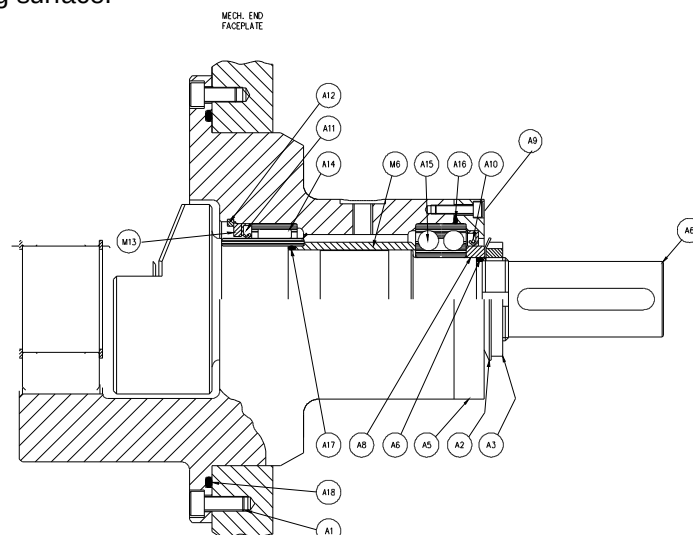


Figure 14.47 - 12 Cross Section Drawing Of Driven Chuck

3. Place the seal (A10) into the cover (A5) as previously described.
4. Place the wear sleeve (A17) onto the shaft (A6).
5. Place the bearing (A14) front seal (A11), spacer (A13) and circlip (A12) into the chuck housing as previously described.



Figure 14.47 - 13 Photograph Showing Mechanic Inserting Circlip into Chuck Housing

6. Place the bearing (A15) into the chuck housing as previously described.
7. Place the Chuck assembly in a clamping chuck.
8. Offer the shaft (A6) through the chuck housing ensuring that no damage occurs to the threads for the locking nut.



Figure 14.47 - 14 Photograph Showing Mechanic Assembling Driven Chuck

9. Use a rubber mallet to force the shaft fully into the housing of the chuck. The bearing (A16) may move when the shaft is being pushed through the housing.



Figure 14.47 - 15 Photograph Showing Mechanic Driving Shaft through the Chuck Housing

10. Rotate the shaft and place a piece of steel across the rear of the shaft, as shown.

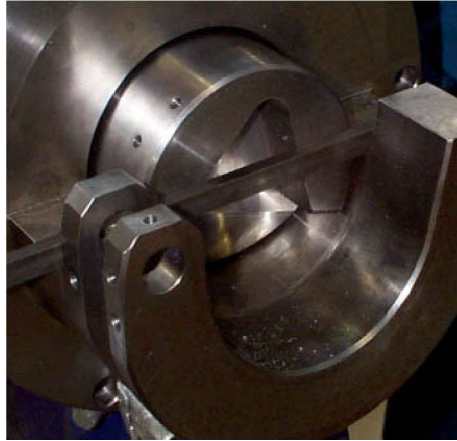


Figure 14.47 - 16 Photograph Showing a Steel Bar Holding Shaft in Chuck Housing

11. Use a drift to relocate the bearing (A16) back into the housing.



Figure 14.47 - 17 Photograph Showing a Mechanic Using a Drift to Relocate the Bearing

12. Install the cover as previously described.



Figure 14.47 - 18 Photograph Showing the Chuck Housing Cover Being Installed

14.48 Adhering Structures to Machine

Any modifications to the machine should be cleared with General Vacuum Equipment, before carrying out the work. This is especially relevant when attaching structures to the machine.

	Caution ! There should be no compromise of the vibration isolation of the machine by structures being added to it.
	Caution ! Do not carry out any electric welding upon the machine without firstly removing the main incoming earth cable to the machines electrical supply.
	Caution ! Do not drill into vacuum pipe work or the chamber, as air leaks can be formed.

14.49 Sight-Glass Windows

There are spacers provided to allow the location of the windows onto the chamber. These are installed as shown below and are used to stop damage to the window.

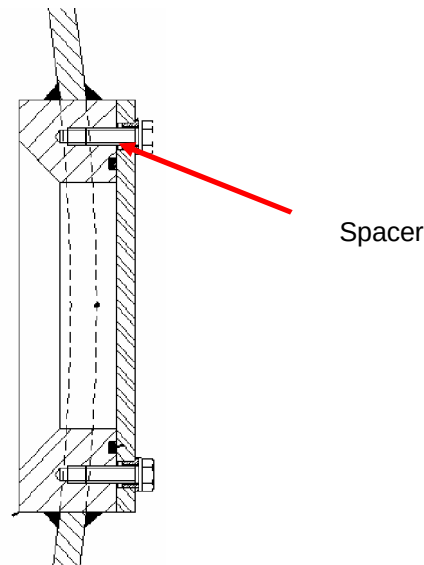


Figure 14.49 - 1 Cross Section View of Window

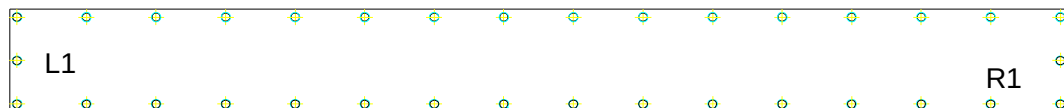
	Caution !
	Ensure Spacers are fitted between each location bolt. The bolts should be tightened using fingers only. When a vacuum is pulled on the machine, they should be then be finger tightened again.

Rectangular Windows

The bolts should be tightened in the following sequence (this is just an example);

T1, B16, T16, B1, T8, B9, T2, B15, T15, B2, L1, L2, T9, B8, T4, B12, B5, T11, T5, T3, B14, T11, B3, T14, B10, T6, T13, B11, B7, B4, B13, B6, T12, T7, T10,

T1 T2 T3 T4 T5 T6 T7 T8 T9 T10 T11 T12 T13 T14 T15 T16



B1 B2 B3 B4 B5 B6 B7 B8 B9 B10 B11 B12 B13 B14 B15 B16

Figure 14.49 - 2 Tightening Bolts Pattern - Rectangular Window

Round Windows

The bolts should be tightened in the following sequence;

1, 4, 6, 3, 5, 2

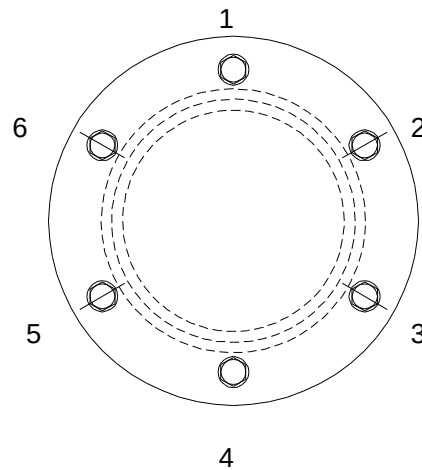


Figure 14.49 - 3 Tightening Bolts Pattern - Round Window

Do not over tighten the holding bolts on the window. Undue pressure will cause the Polycarbonate to fail and shatter. If unsure about replacing the windows, seek advice from General Vacuum Equipment.

	Warning !
	Do not use any type of solvents to clean the window. As there will be chemical erosion, that could lead to failure of the window.

14.50 AC Vector Drive Motors (Winding Mechanism)

AC motors have been supplied with the machine for the following reasons;

- Low maintenance
- Low replacement cost
- Wider range of operation

All AC motors and their relevant feedback device require periodic inspection.

The bearings are usually sealed for life and should be given the same attention as the other sealed bearings.

Regularly check that the motors are running smoothly (no vibration) and the accessible terminal screws are tight.

Check the cooling fan inlets for contamination and remove any debris found.

Check that the Encoder to motor coupling is fully engaged on a monthly basis.

Reference should be made to the manufacturer's manual that has been supplied for further details.



Figure 14.50 - 1 Photograph of AC Vector Motors on Winding Mechanism

14.51 Siemens MICROMASTER 440 – Booster Pumps

The plc control circuit powers up the drive when a request is made to create a vacuum environment in the chamber.

The controllers are set up to limit the rotational frequency, as the density of the gases at the inlet of the pump decreases, the torque required to turn the pump decreases and the frequency is increased.

The initial movement is slow as the motor ramps up to its operating speed.

The units have individual status displays and if any fault codes are shown, reference should be made to the vendor manual supplied.



Figure 14.51 - 1 Photograph of Siemens Micromaster 440 drive.

Parameter Settings

A CDfloppy disk is supplied containing “back ups” of the all the drives.

This includes the Micromaster 440 parameters for the booster pumps and the winding system drive configurations.

14.52 Siemens Combimaster Drive - Cart Traverse

This type of drive unit combines the inverter controller and the motor in one package. It is located on the traverse motor and does not require any major maintenance tasks or setting up.

The control circuit powers up the drive when a request is made to move the cart into the chamber.

The motor has two LEDs on the outside of the case. These are used to indicate the status of the motor.

Green LED	Yellow LED	Meaning
On	On	Mains power is on, Motor in Standby
On	Off	Motor running as per control signals
Flashing	Flashing	Current Limit Warning
Flashing	On	Over temperature of inverter
On	Flashing	Over temperature of motor
Off	On	Other Fault - i.e. trip - see Siemens Combimaster manual
Off	Flashing	Main power under voltage
Off	Off	Main power off

Table 14.52 - 1 Combimaster LED Troubleshooting Look up Table



Figure 14.52 - 1 Siemens Combimaster Unit on a motor

14.53 Coating Shield - Kinertrrol Unit

The coating shield that is normally located close to the winding mechanism, above the source can be rotated downwards for cleaning and set-up purposes.

A pivot shaft is provided which is driven by a compact Kinertrrol pneumatic actuator.

The actuator has been supplied to provide a reliable operating action.

There are mechanical stops built into the actuator that have been set up in the factory. These should not be adjusted.

The unit also contains a limit switch to allow monitoring of its position by the computer.



Figure 14.53 - 1 Photograph of Coating Shield Kinertrrol Actuator

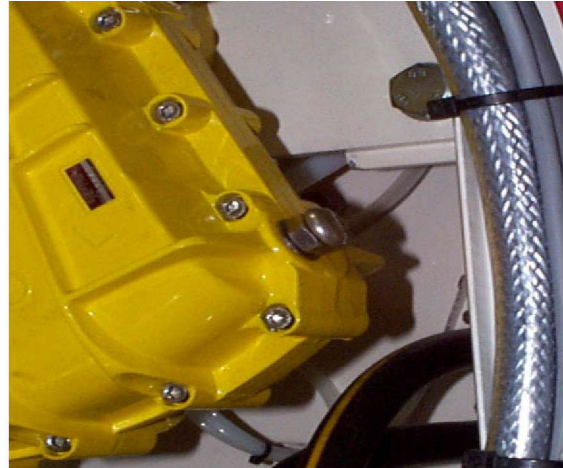


Figure 14.53 - 2 Photograph of Mechanical Stop - Kinertrrol Actuator.

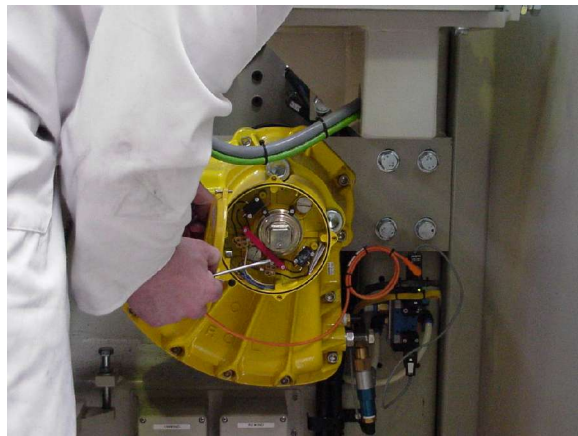


Figure 14.53 - 3 Photograph Showing User Adjusting the Unit's limit switches

14.54 Restoring Software

A back up copy of the contents of the computer's hard drive is provided to customers in the case of the loss or corruption of the software in the computer.

A second hard drive is also supplied. This is a copy of the primary hard drive on the machine.

If any modifications are made to the software on the primary drive. A new back up CD should be made and the back up hard drive be updated using the back up software provided.

The commercial software used to make the back ups is called Drive image.

A boot disk is provided within the software folder to be used to copy the contents of the back up CD supplied.

	Caution !
	If unsure, of the procedure, consult a local IT specialist

This procedure is to make a spare HDD from a back up CD

1. On the new HDD, set the jumper to Enable Cable Select. Ensure the computer is powered down.
2. Connect the IDE cable to the new HDD (remembering to remove it from the existing HDD and that the master is always connected to the last plug).
3. Place Disk 1(bootable) into the floppy drive.
4. Place the copied and tested CD into its drive.
5. Turn the computer on
6. Accept the 'New system config' when prompted, by pressing F1
7. From the main menu select 'Restore Image' (the software will properly prompt you to do this).
8. Click Browse on the Name Image File window
9. At the Drives selection, scroll down and select \\.\Pqcd0. This is the Read/Writer selection.
10. Make sure File name is in the File name box and Ok that screen. This is done by opening the *job name file*, opening files until the **PQI** file is found.
11. Next that screen. Pay attention to the Image file comments to make sure its selected ok
12. Select Destination Partition - Disk 1: with Partitions existing.
13. For the Resize Option, select 'leave remaining unused space'.
14. For the Safe Mode Option, select 'check for bad sectors & verify disks writes.
15. Double check that Source (CD-ROM) & Destination (HDD) are correct
16. Check copying from CD to new HDD.
17. Select to review results.
18. Remove the floppy & CD_ROM.
19. Reboot the computer

14.55 Gas Ballast Valve Maintenance – Leybold Sogevac Pump

The gas ballast valve should be checked for correct operation on a regular basis. Indication of correct operation will be a change in the tone of the pump. Another indication is that the oil level in the pump's sight-glass will rise.

Checks must be made on the filter inlet to ensure that it is not contaminated. If the filter becomes dirty and blocked it must be removed and cleaned.



Figure 14.55 - 1 Photograph Showing Gas Ballast Valve

14.56 Air Admittance Valve

The valves supplied on this machine have been selected for their high level of reliability. However the valves should be checked for correct sealing on a regular basis.

If the valve is removed for any reason, ensure that it is re-installed in the same location with the arrow on the body pointing towards the pipe-work.

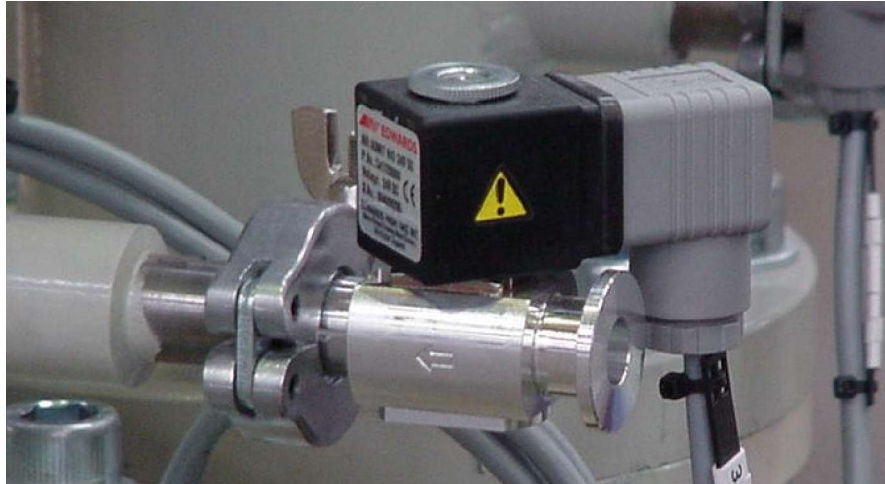


Figure 14.56 - 1 Air Admittance Valve

14.57 Cable Coiler Maintenance

The winding cart cable coiler should be checked for mechanical damage and be repaired/replaced as necessary.

	Caution !
	Check that the area under the track is kept clear, so no damage can occur. The coiler has not been designed to support weight being placed upon it. Do not stand on it.

The plastic protective cover has been supplied to protect the internal cables and hoses.

A section from the vendor's catalogue has been included in the vendor manuals folder.



Figure 14.57 - 1 Photograph of Cart Cable Coiler.

14.58 Rotary Union Removal

Depending upon the normal direction of travel, either a left-hand or right-hand threaded unit will have been supplied. The direction can be determined by the following:-

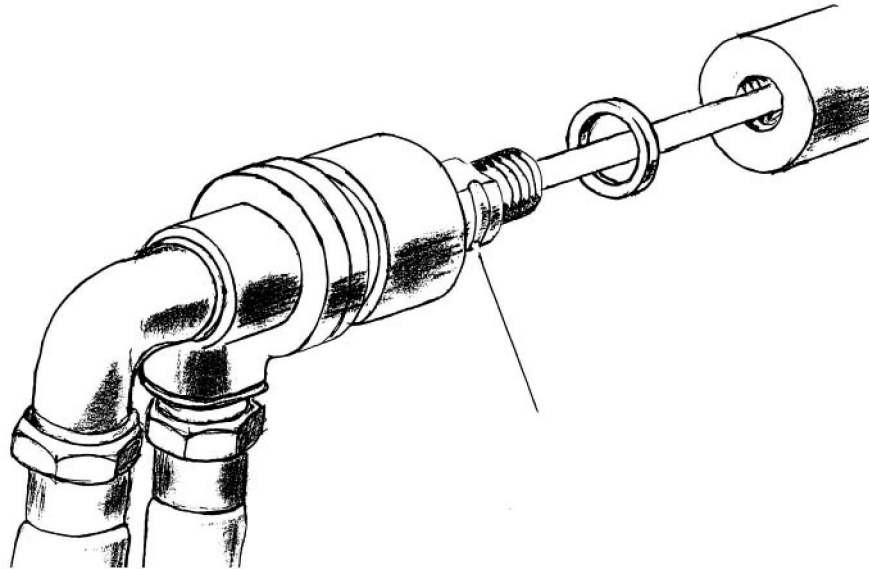


Figure 14.58 - 1 Determining If Rotary Union Is Right Or Left-Hand Threaded

	Caution !
	Do not attempt to carryout any of the following instructions unless you have referred to the assembly drawing supplied as cross references may have changed.

If the rotary union requires removing, the following procedure should be used.

1. Isolate the machine electrically.
2. Remove all the guarding from around the drive motor and drive belt to the leadthrough being worked upon.
3. Isolate the coolant supply by closing off the supply valve in the chiller. Refer to the chiller schematic for identification.
4. Close the return valve.
5. Drain the pipework and drum/roller of coolant. Note, the amount draining will be limited until the rotary union joint is broken.
6. Rotate the roller/drum by hand to ensure that all the liquid inside it is removed.

Note: If the cooling fluid is slow on draining from the pipework, turn the flexible hose fitting one turn to release internal pressure and allow air to be introduced into the cavity.

7. When all the coolant has been drained, remove the feed and return flexible pipes.
8. Remove the rotary union stay arm (if fitted).
9. Removing the rotary union will require a special tool making, as shown in figure A3-308572. This will have to be made to suit the size of the union.
10. Place the flat sections of the tool across the flat sections of the rotary union and unscrew the union. This will break the watertight seal between the union and the roller journal.
11. Note: If the rotary union will not unscrew, it maybe necessary to modify the tool as shown in figure A3-308573.

12. Fasten the two “ears” together over the union to form a clamp. Use a “Hide” hammer to hit the tool evenly to loosen the union. Heat the journal gently with a flame, but this should only be used as a last resort.

	Caution !
	The use of heat, should only used as a last resort.

13. Remove the union including the internal feed pipework.

Note: If there is difficulties encountered in removing the feed pipe, it can be unscrewed from the union the left inside the roller.

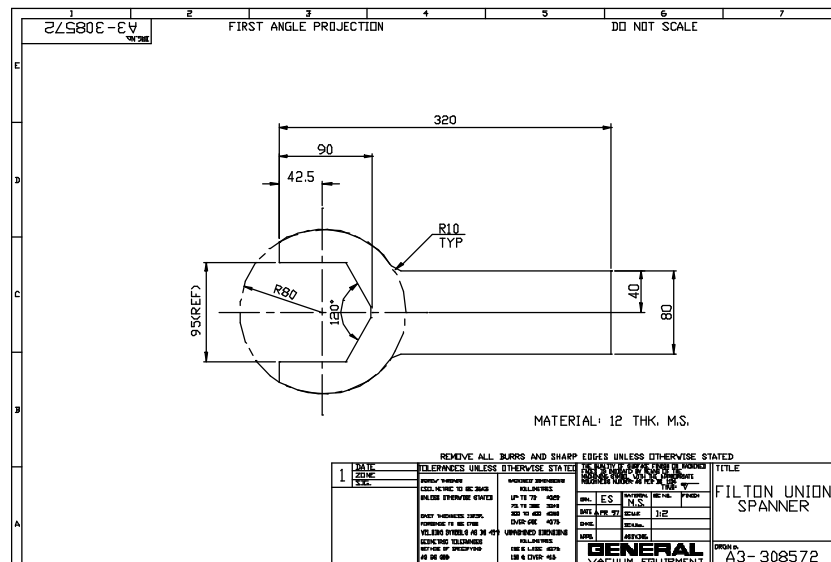


Figure 14.58 - 2 Filton Spanner

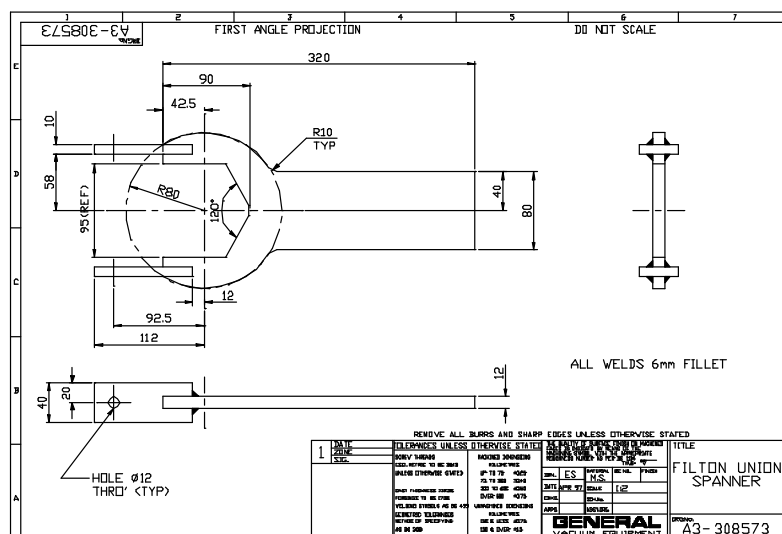


Figure 14.58 - 3 Filton Spanner

14.59 Replacing Drum Lip Seals

Initially remove the rotary Union as previously instructed.



Caution !

Do not attempt to carryout any of the following instructions unless you have referred to the assembly drawing supplied as cross references may have changed.

1. Release the tension on the drive belt by releasing the screws on the drive motor support bracket.
2. Slip the drive belt over the end of the roller journal. Remove the taper lock pulley and remove the pulley, taper-lock and key.
3. Drain the oil fed lubricator by unscrewing the flexible pipe on the outer seal plate (A11). This oil should be collected in a container for refilling the seal cavity later.
4. Remove the locking ring (A14), tab washer (A15) and spacer (A6).
5. Unscrew the outer seal plate screws (A18) and place two of the screws into the jacking holes in the plate.
6. Screw the screws in evenly to remove the plate (A1).

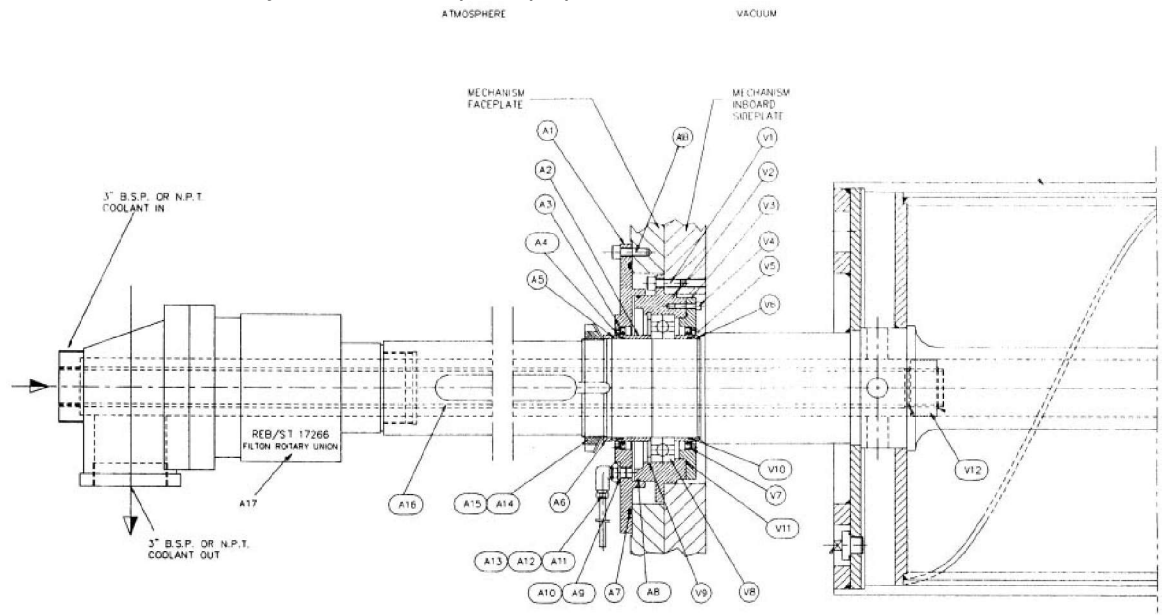


Figure 14.59 - 1 Cross Sectional Drawing Of Drum

Supporting Roller/Drum

Support the water-cooled drum roller by placing a hydraulic jack under it on close to the end that the seal is being worked upon. Use a timber post and a wooden Vee block under the drum.



Caution !

The drum must not be able to move, rotationally or sideways when further steps are taken. Place timber support blocks between the end of the drum and the adjacent steelwork. This will prevent the drum from being moved.

Removing The Housing

1. Once the drum is secure, proceed to remove the inner bearing housing (V3) and seal plate (A2).
2. Unscrew the remaining retaining screws (A18) from the housing.
3. Place two M10 x 120 caphead screws into the jacking points on the housing and use them to evenly jack the housing off.

	Note! Note: If the outer seal rings will not slide off the journal, continue to jack off the bearing housing (A1) off the journal for about 20 mm (3/4") and then release the jacking screws. Push the housing back up the journal. This will form a gap between the seal rings and the bearing housing, therefore enabling the release of the o-ring (A4) from the seal ring.
---	--

4. Remove the seal rings (A3) and o-ring (V7) from the journal.
5. With the housing off, remove the o-ring (V5) and seal rings (V6).
6. Remove the inner seal plate (V3) from the bearing housing (A1) and remove the o-ring (V5).
7. Remove the old lip seal (V7) from the housing (V3).

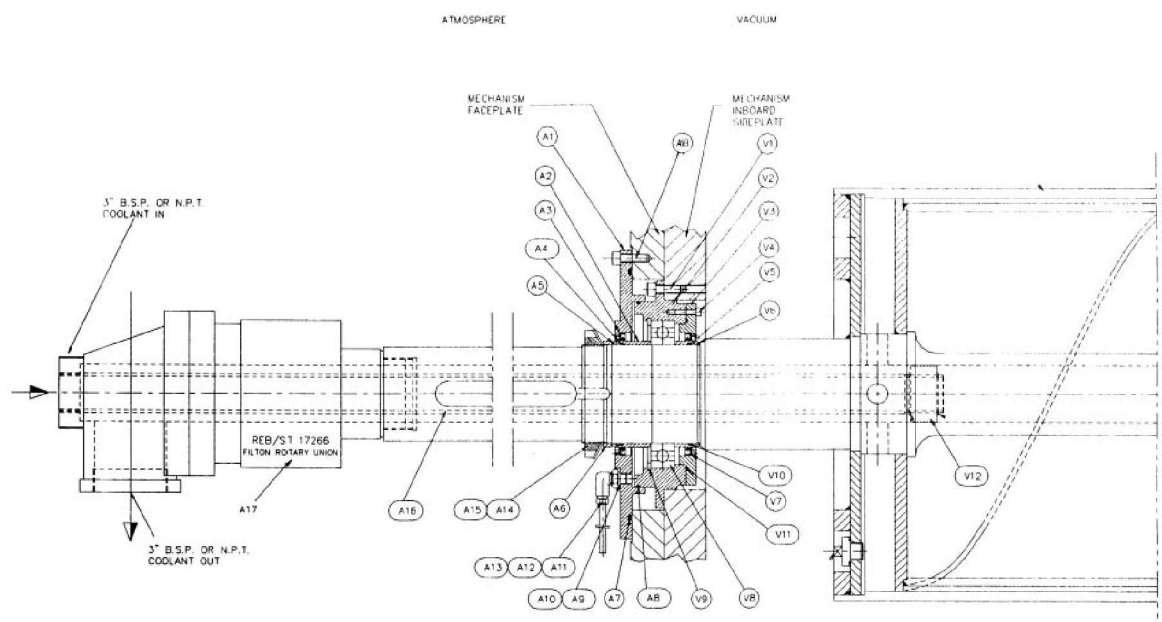


Figure 14.59 - 2 Cross Sectional Drawing Of Drum

Installing A New Lip Seal

1. Thoroughly clean the housing cover and remove any old sealing compound which may be left in the seal area. Remove any sharp edges or burrs.
2. Obtain or manufacture a steel spigot ring to allow even pressure to be applied the new seal when it is being installed.
3. Use a sealing compound to provide a vacuum seal behind the new lip seal as it is placed in the housing. Suggested compound is Kinneyseal. Alternatives are Blue Hylomar or Hermatite. Ensure that any residue is wiped away and does not come into contact with the sealing face of the seal.

	Note!
	Extreme care must be taken when inserting a new lip seal. The inner lip of the seal must not be damaged.

4. Place a new o-ring (A7) in the groove in the housing using vacuum grease.
5. Cover the housing to keep it clean and place it on one side.

Replacing A Lip Seal From The Housing Cover

1. Remove the old lip seal (A3) from the outer seal plate (A1).
2. Install the new lip seal as previously explained above.
3. Use a sealing compound to provide a vacuum seal behind the new lip seal as it is placed in the housing. Cover the housing to keep it clean and place it on one side.

Preparing To Re-Install The Bearing Housing

1. Cover the thread and key-way on the end of the shaft with adhesive tape to stop any possible damage to it.
2. Place the inner seal plate (V6) and o-ring (V10) back onto the journal and onto the inner seal ring (V5).
3. Use a small amount of oil to lubricate the lip seal to lubricate it as it passes over the inner ring.

	Note!
	If there is a problem when replacing a new o-ring (V10) between the seal rings, the o-ring needs to be placed inside the seal ring. This will allow it to fit correctly with the inner clamp ring (V5).

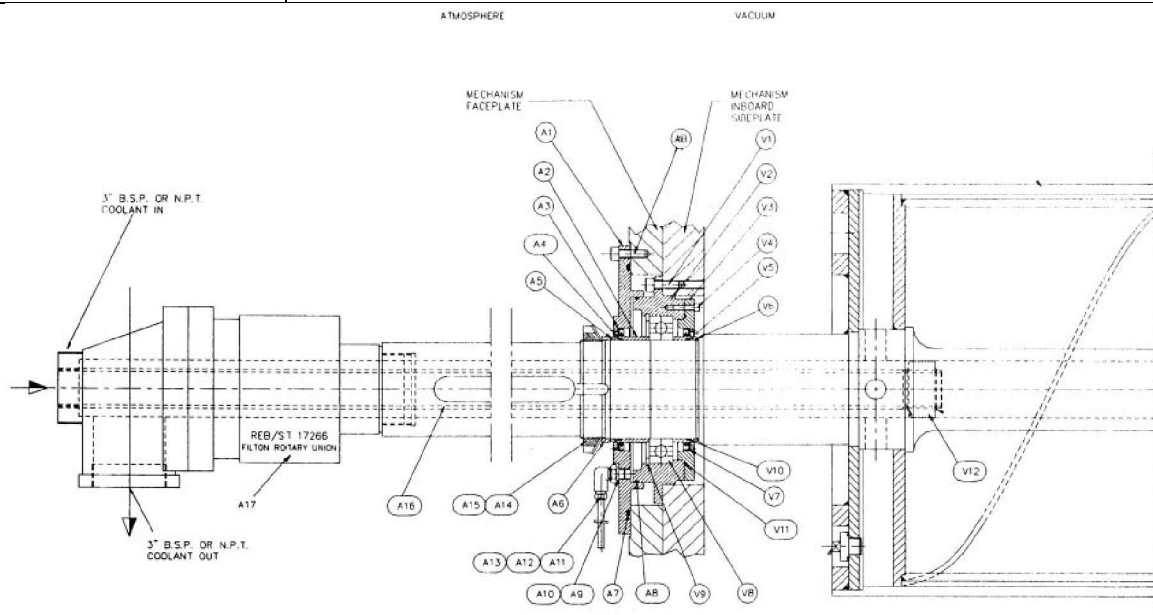


Figure 14.59 - 3 Cross Sectional Drawing Of Drum

Replacing The Bearing Housing

1. Remove the protective tape from the threaded section of the roller.
2. Remove the previously used jacking bolts from the bearing housing (V3).
3. Place wooden packers at opposite end
4. Place wooden packers at opposite end of drum to stop the drum being pushed towards the chamber.
5. Replace the outer seal rings (A2) (A5) and o-ring (A4).
6. Place a new o-ring (A8) into the bearing housing (V3).
7. Place the bearing housing (V3) back onto the journal and insert locating screws (V1)
8. Install the outer seal plate (A1) onto the journal with a new o-ring (A7) in place. Care must be taken not to tear the o-ring. Once again, use a small amount of oil to lubricate the seal as the housing is placed in position over the seal ring.
9. Place the locating screws (A18) back into the housing to locate it against the faceplate.

Installing The Lubrication System

1. Replace the oil lubricator flexible pipework.
2. Fill the oil bottle with the oil previously removed.

	Note!
	This procedure may take a period of time as the air from within the cavity has to escape. The oil bottle should be monitored and filled as required.

Removing Supports

1. Remove all the timber supports previously fitted.

	Warning !
	Ensure that it is safe to carry out these instructions.

2. Rotate the drum by hand and check for obvious leakage from the seals.
3. Leave for one hour and then repeat step 2.

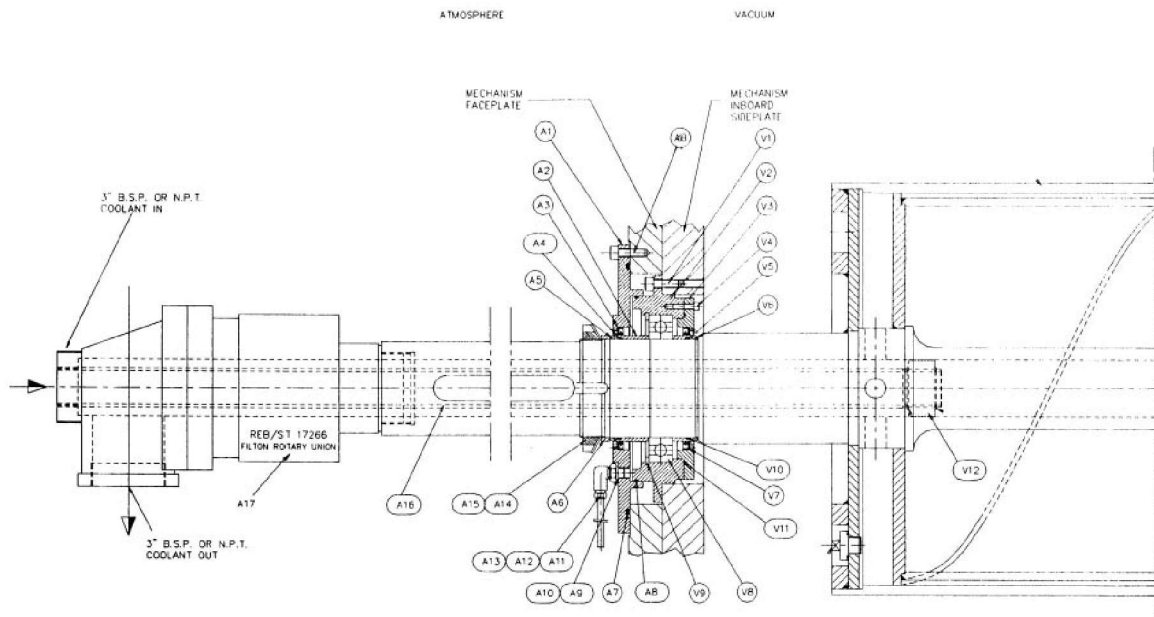


Figure 14.59 - 4 Cross Sectional Drawing Of Drum

Installing Drive Pulley

1. Replace the spacer (A6), tab washer (A14) and lock Nut (A15)
2. Replace the steel key (A16) and pulley. Use an anti seize compound on all surfaces to ensure that they can be removed again at a later date.
3. Align the pulley with the pulley on the motor.
4. Place the timing belt back between the two pulleys, and tighten the motor support bracket to tension the belt correctly.

Install the Rotary Union

1. Clean all previous sealing compound of the rotary union.
2. Clean the aluminium-sealing ring, and then cover both sides with a watertight sealing compound.
3. Place PTFE sealing tape on the rotary union thread.
4. Cover the thread tape with the compound.
5. Locate the union onto the journal using the special tool.
6. Screw the union into the shaft and tighten it up.

Installing The Cooling System Hoses Etc.

1. Replace the flexible hoses and tighten them.
2. Replace the stay arm (if fitted).
3. Close the drain valves.
4. Open the valves between the unions and the chiller.
5. Place the drained coolant into the "Top Up" tank on the chiller.
6. Energise the machine by switching the power back on.
7. Start the chiller in the heat mode and circulate the coolant mixture through the pipework.
8. Check for leaks and repair any that are found.
9. Replace all guards that were previously removed.



Caution !

Insulate all metal parts that insulation materials had been removed from.

14.60 Bellows Installation

The rubber bellows installed in the pipelines can be removed to allow access to remove sections of the pipework.

The hose clamps should be loosened. Then the bellows pulled away from the pipe-work that is to be removed.

Before re-installing check the pipework for any rough edges and clean them up.

There should be a maximum gap of 12mm between the two edges of the pipe.

Add vacuum grease to the inner surface of the seal and gently push the bellows back over the joint. Then re-tighten the hose clamps.



Figure 14.60 - 1 Gap Between Pipes

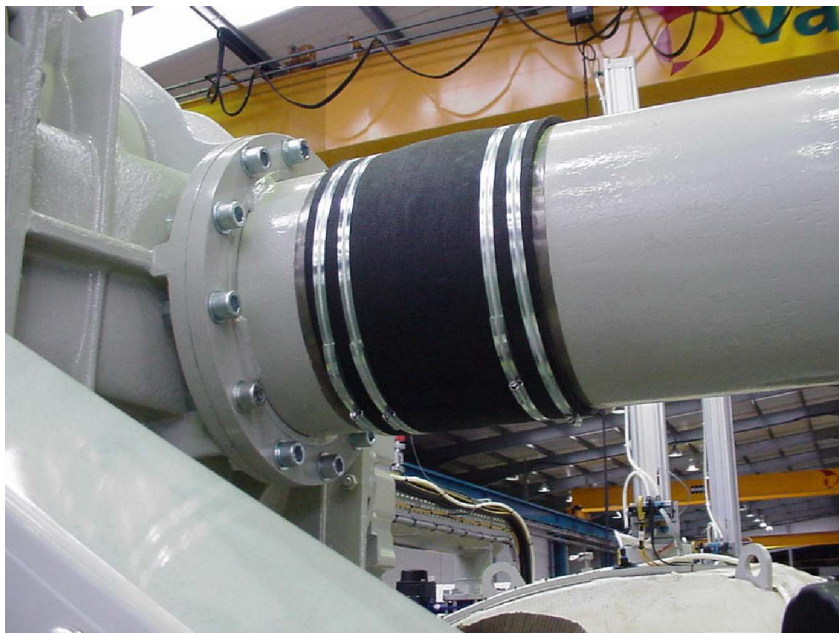


Figure 14.60 - 2 Bellows On Pipes

14.61 Evaporation Source Insulators and Springs

Insulators:

There are a number of insulators that are used to isolate the evaporator electrical supply from the surrounding steelwork. These should be inspected for damage caused either mechanically or by electrical shorts. If they are blackened, wipe down the surface with acetone, to clean them up.

Where appropriate, they should be replaced.

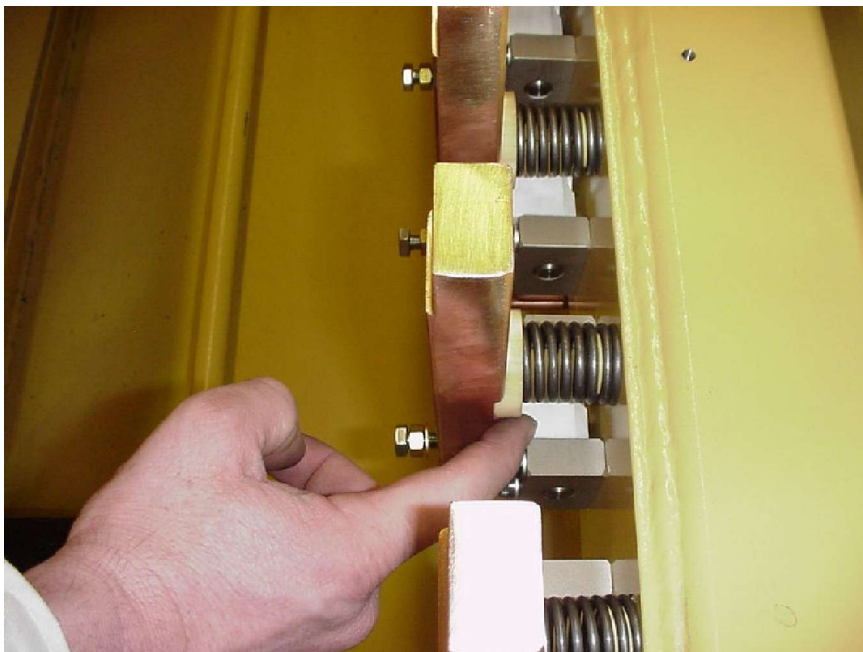


Figure 14.61 - 1 Photograph of Source Insulators

Springs:

Over a period of operation, the springs supplied will lose some of the force that they apply. It is normal that these are replaced every 12 months.

Caution !	
	Incorrect aluminium debris cleaning methods, may lead to shorter operating life of these components.

14.62 Calibration And Care Of The Operator Touch Screen

A vendor manual has been supplied with the unit. Its power supply is located in the desk panel and this should be initially checked if the panel does not display any of the screens.



Figure 14.62 - 1 Photograph of Operator Panel

Touch Segment

The touch screen can be calibrated from a onboard program. If calibration is required, please contact General Vacuum Equipment for advise on the procedure.

Replacement

If the screen fails, a standard VGA monitor screen can be used and a mouse can be plugged into the computer for operation. The relevant driver for the mouse will be required

Procedure For Replacing a Failed Touch screen.

1. Turn off computer.
2. Plug a serial mouse into com 1 on the computer, or plug a PS2 mouse into the PS2 mouse socket.
3. Turn computer back on.

Cleaning

The touch screen should be cleaned with a lint free, non-abrasive cloth. Avoid using any cleaning solution, glass cleaner or tissue paper.

14.63 Process Gas Wedge

Principle Of Operation

To allow the transfer of heat from the substrate to the cooling drum, air is fed through a specially manufactured gas wedge. It has been found that by passing air into the area between the substrate and the drum, major benefits have been recorded. This is more relevant in substrates that have a rough surface, such as BOPP.

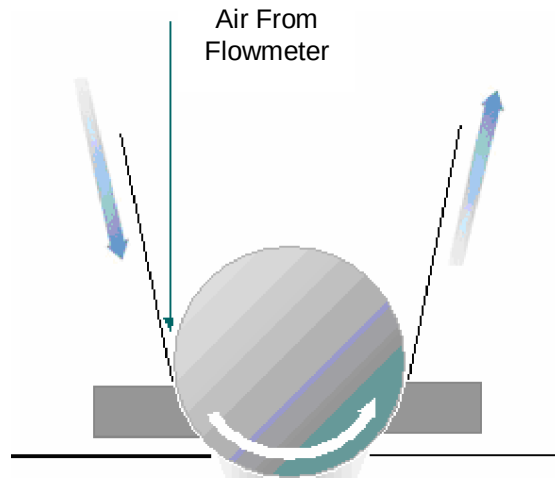


Figure 14.63 - 1 Principle Of Gas Wedge

Hardware

The system is very simple and consists of the following equipment:-

- Flow Control Regulating Valve
- Isolation Solenoid
- Gas Wedge
- Gas Supply Lines

Maintenance

The maintenance tasks are to check that the isolation solenoid valve operates as required from the control panel and that the nozzles in the gas wedge are kept clear.

To clean the nozzles, remove the gas wedge from the location in the machine. Attach a airline and blow through the assembly, checking for airflow through each nozzle.

If any are blocked, use a small drill to clear the blockage.

	Note!
	It has been found where machine users do not follow the recommended method of cleaning machines and allow a build up of dust within the winding mechanism, there is a higher possibility of blockages taking place.

Potential Problems

High Gas Flow

	Caution !
	High gas flow will lead to vacuum problems. Recommended levels up to 4 litres/sec maximum

Blocked Nozzles

Blocked nozzles will lead to uneven performance of the system. These can be cleaned by using a small drill to clean out debris.

Faulty Solenoid Gas Valve

Check that the valve is opening as per your requirements. Once again failure to open will lead to non-operation of this process equipment.

Normal indication of operation is a clicking as the valve opens and the flow meter indicator moving.

Low Gas Flow

Low gas flow will lead to non-operation of the process.

Poor Vacuum Levels

This potential problem may occur if the air supply to the gas wedge is too high and the vacuum in the machine "collapses". The regulating valve should be adjusted to overcome this problem.

Web Being Scratched

There is a hole in the side of the winding mechanism plate that allows the user to view the position of the gas wedge. Once installed, there should be no reason to move the wedge, unless for cleaning of the nozzles.

14.64 Substrate Threading Device – (Option)

The threading mechanism should be checked for chain tightness. If it is too slack, the chain is likely to “jump” the drive sprocket and cause problems. If it is too tight, it will place stress on the drive system. There should be 12mm of depression on the chain for normal operation.

Adjustments can be made on each side of the threader independently, if two are fitted, for example on paper machines

The chain should not be lubricated, but brushed using a steel brush if it becomes contaminated.

The sprockets supplied for supporting the system should be checked for bearing wear and be replaced when appropriate.

14.65 Source Rotary Window

There is an internal rotary cylinder provided to stop aluminium coating the source viewing window. It is operated at a preset speed, as developed by General Vacuum Equipment.

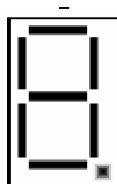
Electrical Information:

This is driven from a stepper motor within the chamber which is controlled by remote drive unit in the main electrical panel. There is an operation limit relating to maximum operating temperature for this unit. Although the temperature is not measured, from experimentation, the maximum running time has been determined.

The motor stepper drive unit is not powered up unless the chamber is closed.

Strogra Drive Units

Operational status and error indication is via 7-segment display, which indicates the actual status of the SERS.



Error and status indications are indicated by constant illuminating characters. Warnings are indicated by blinking characters.

Following table shows all possible indications:

Indication	Description	Comment
-	Initialisation phase	Is indicated 1 second after a power-on-reset
4	ready for switching phase current ON	
5	Phase current of motor is ON positioning jobs can be executed	
7	Check sum error of dates in E2Prom	Overwrite EPROM-parameters with backed up parameters (with SERS-software) or select standard parameters (with P1004=3 - see P1004)
8	Over temperature - from 85 °C (10%) on the power amplifier	Error check cooling of SERS – forced draft (fan and 24 VDC in ELK) o.k.?
9	Error under voltage - see 2.1.4	check power supply
8 (blinking)	Warning over temperature from 75 °C (10%) on the power amplifier	
9 (blinking)	warning under voltage - see 2.1.4	
A (blinking)	Warning position overflow – positioning job is reaching position limit	Parameter W (P47) to big-check positioning mode (for endless positioning P1014=1 see P1014)
c	Limit switch is open	check limit switches at machine and limit switch inputs of SERS
E	Short circuit in motor or at power amplifier board when installing the motor check phase connections	
F	Error step angle control - the motor could not follow the position command value (only with option step angle control) – see P1029 in manual	check encoder connections-acceleration to high- external load to big (not enough motor torque)

Indication	Description	Comment
F (blinking)	Warning step angle control – description as F – see P1029 in manual	See comment for F
H (blinking)	Warning program error – executable program stopped because of an error in the program	check parameters and labels in program(use SERS-software for debugging)
L (blinking)	Warning drive reached software limit switch.	See parameters P1039 and P1040 in manual

Table 14.65 - 1 Status And Error Displays On The Strogra Drive Units

The controller communicates through the Profibus Network and has all its operating parameters and control signals set up by the main computer.

Mechanical Information:

There are five components to the assembly

- Aluminium Cover
- Strobe Tube
- Motor
- Coupling
- Bearing Blocks

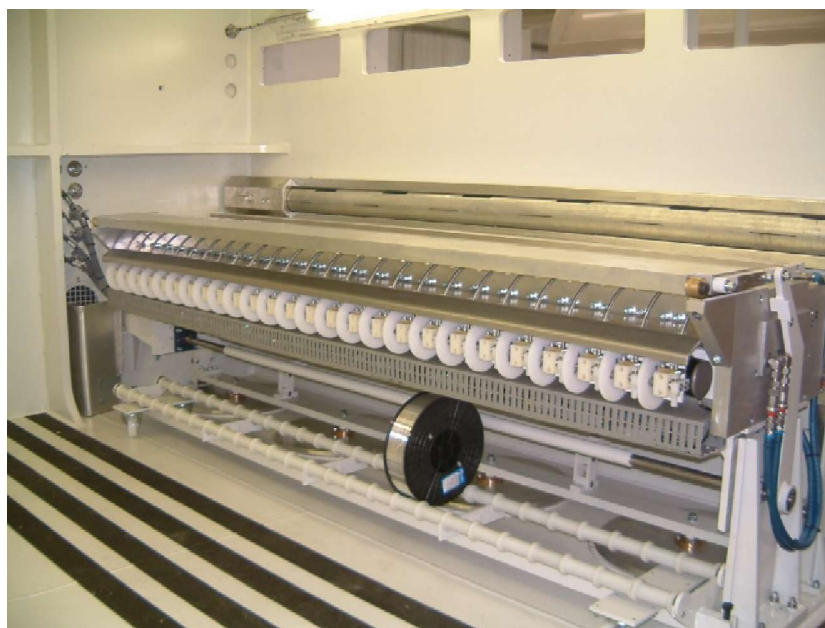


Figure 14.65 - 1 Photograph of Source Rotary Window in the Machine

Aluminium Cover

Manufactured from lightweight aluminium to allow easier removal. Should be painted with release paint to allow easier removal of aluminium coating. Located in place by 'wing nuts'. After refitting, tube should be rotated to ensure clearance by hand.

Strobe Tube

Manufactured from Aluminium and balanced to allow operation at the required speed. Should be painted with release paint to allow easier removal of aluminium coating.

Motor

Previously highlighted under electrical section. Non repairable parts by end user.

Coupling

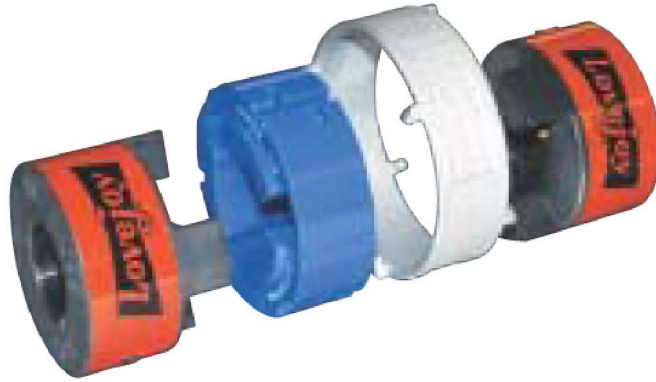


Figure 14.65 - 2 Photograph of a LoveJoy Coupling

The Jaw In-Shear coupling is used for applications requiring quick element replacement without moving either the driver or driven hub.

Installation & Alignment Instructions

All parts must be clean and free of any foreign materials before attempting assembly.

1. Slide one coupling hub onto each shaft, using keys when required. Make sure to place the 6 Pin ring over the hub on the left. Ensure that the keys should fit closely.
2. Position the hubs on the shafts. For best results, an equal length of shaft should be placed in each hub.
3. Tighten both hubs into its final position using the setscrew.
4. Check parallel alignment by placing a straight edge across the two coupling hubs, and measure the maximum offset at various points around the periphery of the coupling without rotating the coupling.
5. Examine the In-shear spider. One side of the spider contains raised writing and small bumps or nubs, while the other side is a machined flat surface. Install the spider with the writing side facing to the right.
6. Line up three of the pins to the corresponding channels that run through the element. Slide the pins of the ring all the way through the element until they are completely through to the other side (from left to right).
7. Twist the ring so that the pins move past the nubs, locking the element into place

Bearing Blocks

This require greasing weekly and opening have been made with slide covers to allow greasing



Figure 14.65 - 3 Photograph of the Drive Motor and Coupling

Removing the Cylinder

To remove the cylinder for maintenance.

1. Lockout the drive
2. The internal guard has to be removed.
3. Remove the coupling
4. Remove the bearing clamping assemblies.

14.66 Installing or Removing Optical Monitor Beams

The monitor beams should not normally require removal, unless a probe section is damaged for example.

Before any attempts to remove the beams are made, the power to the system should be removed. It is suggested that the main isolator to the machine be switched off to ensure that the whole system is isolated.

At the ends of the beams, there are cable connections, which should be unplugged, and the cable withdrawn to a safe position, where it will not become damaged.

The beams are located on support brackets. There are two cap-head screws in each end of the beam that should be loosened and removed, the individual beam will not fall as they are removed. After removal of the screws, the beam can be lifted out of the machine.



Figure 14.66 - 1 Photograph of Monitor Beam System

The beam should be checked to have an even gap between the inside surfaces of the two beams, which will allow the insertion of the calibration tool.

14.67 Optical Monitor Probe Calibration

Equipment Required

A calibration tool/jig is provided to allow the calibration of the optical monitor system

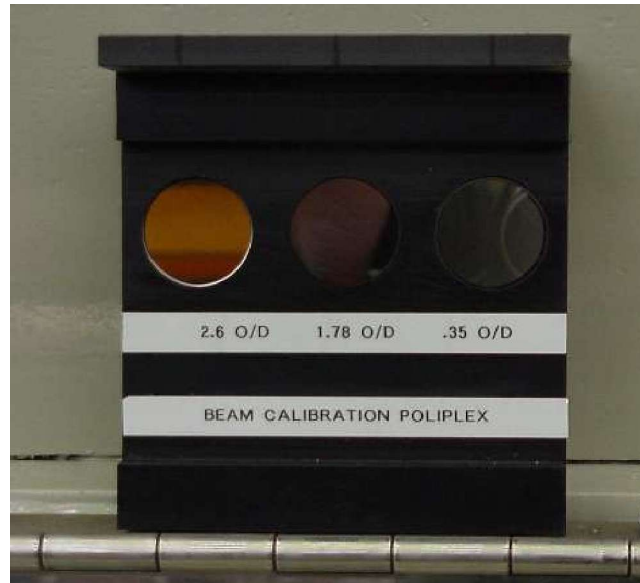


Figure 14.67 - 1 Photograph of Calibration Tool

Procedure

1. Switch the system on and allow it to proceed through its self-test procedure.
2. Allow the system to warm up over 30 minutes.
3. Within the ROLL SETUP screen, select the maximum width of the machine and enter the calibration unit OD value. Load the set-up information.

Select the supervisory level of operation by selecting the LOGIN screen, enter the password,

Select the CALIBRATION screen

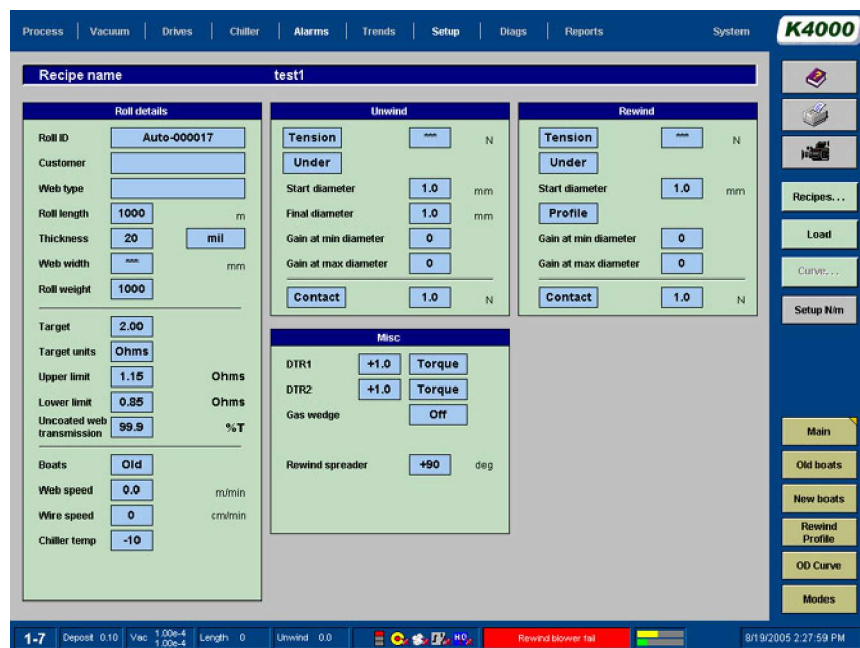


Figure 14.67 - 2 SETUP Screen

4. Log onto the machine at the supervisory level of operation to allow access to the CALIBRATION screen.

5. Select the CALIBRATION screen

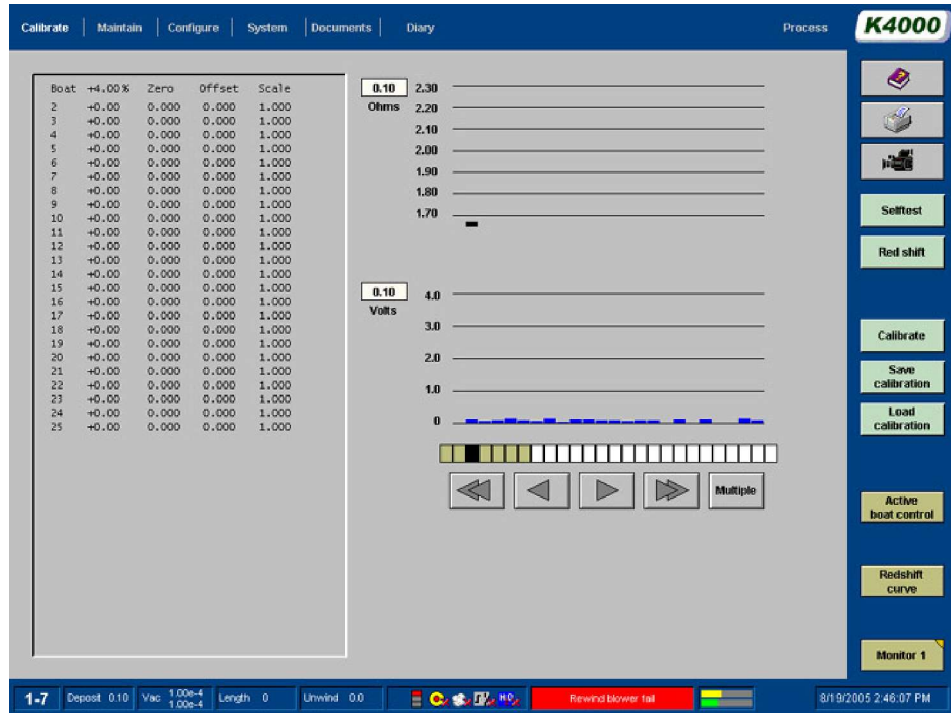


Figure 14.67 - 3 CALIBRATION Screen

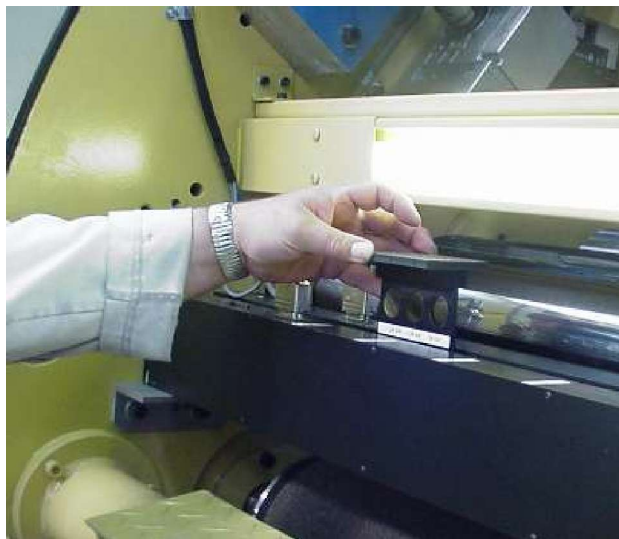


Figure 14.67 - 4 Inserting the Calibration Jig



Figure 14.67 - 5 Aligning The Jig

6. Place the jig in front of all the probes, as shown, so that the middle line on the jig can be aligned with the mark on the beam.
7. It is important that the jig is placed in the same position in front of every probe, with alignment to the middle calibration lens.
8. The cursor on the calibration screen should be moved to relevant identified probe position using the left and right direction cursors. The reading given on the left hand side is the deposition rate measured by the probe.
9. The value should be adjusted by pressing the calibration function button.
10. Repeat the above steps with the jig located in front of each of the probes as described above.
11. After all the probes have been checked/calibrated, press the exit button to return to the MAP screen

A further check can be made using the light OD coating lens and the heavy coated lens.

Reasons for Calibration Differences

It has been found that the difference in calibration between regular checks is usually negligible. However, sometimes under the following circumstances, calibration is definitely required;

- Removal of Monitor Beams
- Replacement of faulty Probe

It has been found that the following can cause the measurements to differ from that required;

1. Placing the system in a "Self Test" when light depositions are present on film in front of the probes, i.e.; 15 ohms and above.
2. Build up of coating medium dust around the probes.
3. Accidental changes in trim file

14.68 Plasma Treater

The principles, operation and cleaning of the plasma treater system is described within other sections of this manual.

In this section of the manual, the set up, maintenance and troubleshooting is described.

Plasma Treater System Components

There are three main components to the General Vacuum Equipment's plasma treater System, namely:

- The Generator, or Power Source Section (Contained in one unit with the heat exchanger)
- The Plasma Treater Assembly
- Process Gas Equipment



Figure 14.68 - 1 Photograph showing a power supply panel

Plasma Treater Assembly

The plasma treater assembly is installed within the vacuum chamber and is positioned to allow the film to pass through it for treatment.



Figure 14.68 - 2 Photograph showing a plasma treater

Plasma Treater Power Supply Generator

The plasma treater power supply generator is used to supply a high voltage medium frequency supply to the plasma treater assembly.

Power Supply Unit

The control unit is located in front of the control electronics; the control electronics are in a removable 19" module. The mains supply is three phase. A mains transformer for potential isolation is located in the panel.

For details of the control unit's operation refer to the vendor manual supplied.

Gas Connection and Distribution

In order to generate a plasma glow both power and a sufficient gas environment (or pressure regime) is required. As the plasma is located in a vacuum chamber where the pressure regime is below a satisfactory level for plasma generation, we need to feed gases into the associated local atmosphere to ignite and maintain plasma. It has been found various gases or mixtures can enhance certain physical properties of a passing substrate surface.

	Warning !
	THE FOLLOWING SAFETY PRECAUTIONS SHOULD BE ADHERED TO AT ALL TIMES. This equipment is designed to run on a process gas containing less than 20% oxygen. Use of process gases containing more than 20% require specialised pumping systems for safe operation. Therefore always use the correct gas mixture

In order to feed the necessary gas mixtures in specific ratios, it is necessary to incorporate a suitable gas system into the machine.

Type of Mass Flow Controller Used

A Type 2179A General Purpose Mass Flow Controller (With Shutoff Valve) is supplied

The Type 2179 is a general purpose, elastomer-sealed mass flow controller with an integral shut-off valve, designed to measure and control the flow of gases in a wide variety of applications. It can also be used as a pressure controller when connected to a suitable pressure transducer.

The normally closed shut-off valve provides positive shut-off to 4×10^{-9} sccm/sec He. The 2179 is available in both analogue and digital models with accuracies of 1% of Full Scale and 1% of Reading, respectively.

Power required for the 2179 is minimal: the nominal ± 24 VDC unit consumes only 100 mA during operation (200 mA at initial turn-on).

The 2179 type flow controller measures the mass flow rate of a gas and controls the flow rate according to a given set point. The control range is from 2 to 100% of full scale (F.S) with an accuracy of $\pm 1\%$ of F.S.

Although the gauge is located on the outside of the chamber, an internal flexible tube runs between the gauge connection and the treater assembly.

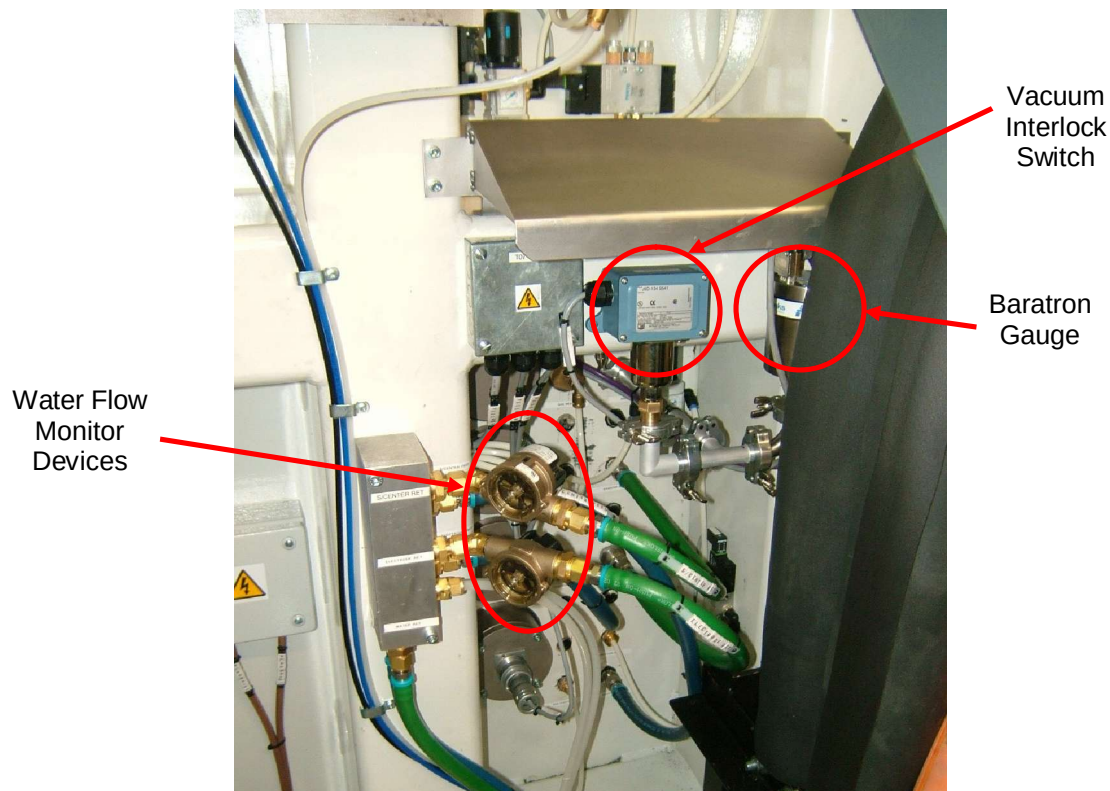


Figure 14.68 - 3 Photograph of Location of Baratron Gauge

Baratron Pressure Gauge

In order to measure the vacuum level of inside the plasma treater zone accurately when using a number of different process gases, a cold cathode baratron gauge is utilised.

The type 626 is the more economical of the process Baratron capacitance manometers. This transducer uses a dual-voltage power supply (± 24 VDC) and is available in Full Scale ranges from 1 to 1000 mmHg (Torr). This unit operates at ambient temperature. The measured level is communicated to the rest of the machine via the profibus network.

A complete pressure transducer system requires three components to convert pressure to a linear DC voltage output: a sensor, signal conditioner, and power supply. An analogue or digital meter is required to display the DC output in pressure units.

Plasma System Installation

The installation of all the equipment for the machine will have be completed prior to start up.

This includes all electrical wiring, water lines and mechanical installation of all the components.

Mechanical Drawings

There are three main mechanical drawings, which may prove useful in understanding the construction/ assembly of the plasma treater assembly, these are: -

- i) The Electrode Enclosure
- ii) The arrangement of The Plasma Treater
- iii) The Installation Assembly

These drawings are included in the drawing package.

Basic Safety Interlocks

Due to the high voltages and currents associated with the operation of the plasma treater, it is necessary to incorporate a number of interlocks into our system to prevent the operator coming into contact with any live components. Guarding and shielding has been provided over any exposed live parts together with a number of software (PLC) interlocks, preventing the operator from turning the unit on in an unsafe condition.

A hard-wired safety circuit has also been installed which is wired into the generator.

The circuit includes a number of devices wired in series, which prevent operation of the treater in an unsafe condition. These include: -

- A pressure switch (to ensure a partial vacuum is created before the switch is activated)
- A limit switch to prevent the unit being switched on while the chamber is open.
- The machines emergency stop button.
- Winding cart isolator in off position
- Flow switches

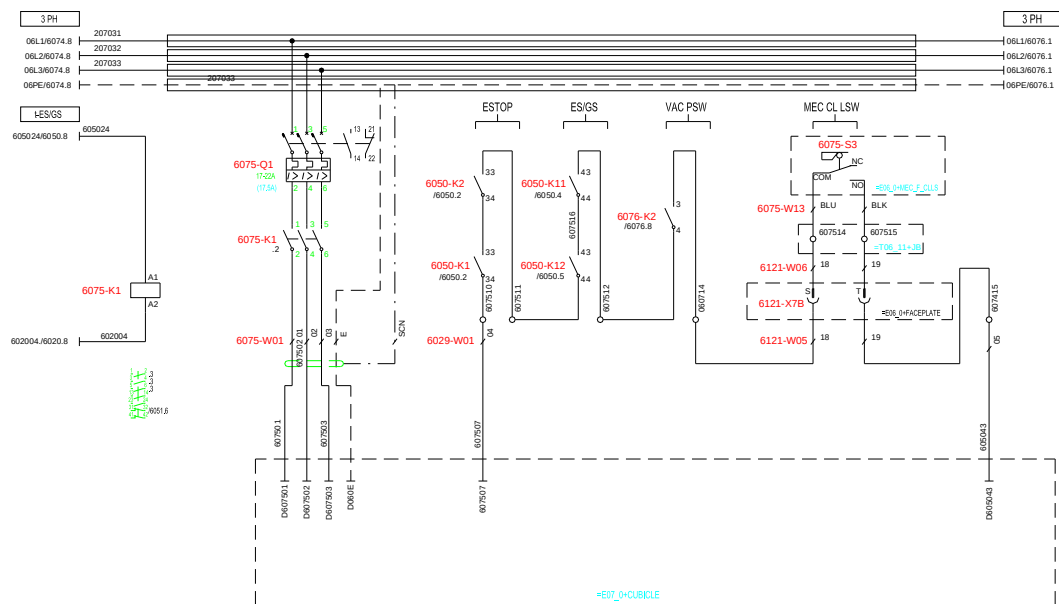


Figure 14.68 - 4 A typical schematic showing the hard-wired safety circuit, wired into the power supply.

The status of all interlocks can be viewed on the interlock diagnostics screen. Green means that they are in the correct mode to allow the treater to be used.



Warning !

These and all other interlocks which have been built into the machine, should under no circumstances be interfered with, or not used.

Installation Completion Checks

Before the plasma treater system is operated, it is required to perform a number of basic pre-start checks to prevent/avoid damage to the plasma treater and its associated components.

These include: -

- Check all water lines at piped up to the correct location as per the installation/services drawing, and are visually water tight
- Check all power connections are connected up to the correct location and are tight. Visually inspect all cables for damage.
- Check all gas lines are piped up correctly as per the installation drawing, and are secure

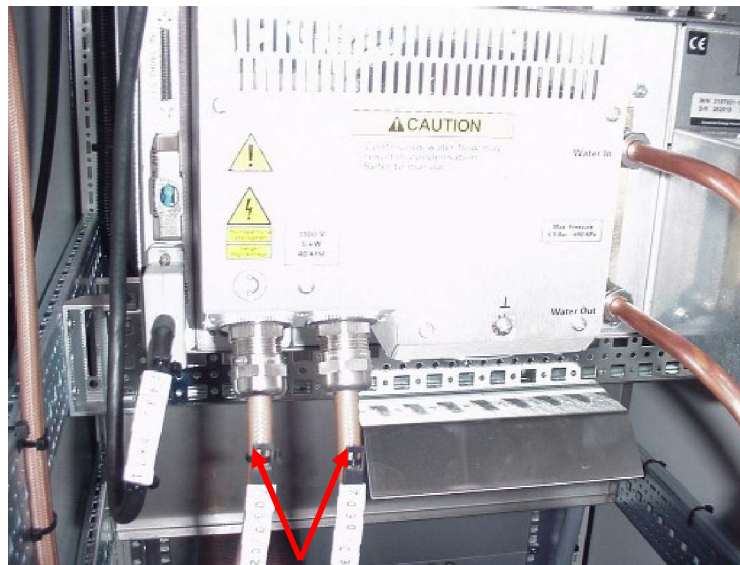


Figure 14.68 - 5 Photograph showing the MF/RF output triax cables leaving a power supply.

Removing the rear panel of the generator can access the connections for both the incoming three-phase cable and the output cable. Note: The location of the generator is found on the mechanism-moving end of the machine, and the cable is routed under the floor plate towards the faceplate.



Figure 14.68 - 6 Photograph showing the atmospheric RF/MF triax cables lead through terminal box, located on the faceplate of the machine.



Figure 14.68 - 7 Photograph showing the vacuum side insulating covers protecting the leadthroughs on the faceplate of the machine.

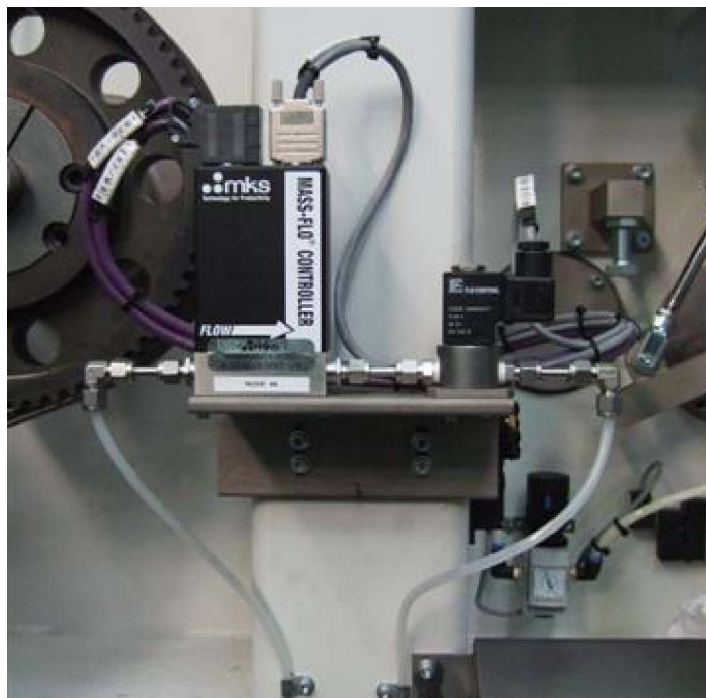


Figure 14.68 - 8 Photograph showing the MFC gas controller on the face plate

Setting Up of Mass Flow Controllers (MFC's)

As standard, a plasma treater system generally comes with a mass flow controller, which accurately control the volumetric flow of to process gases into the plasma zone.

Setting the Profibus Address

Address set-up may be performed via the PROFIBUS protocol or via a push button on the side of the device.

If the button is pushed on power up or it is pushed for a long time, the device will enter the address setting mode and it will display this by a 3 time fast blinking between red and green. In this mode the device start to count up the address with green flashes of the LED. The pushing must be interrupted for a short time only, to continue with counting in steps of ten. If the button is released for the second time, device will display a success address setting with the 3 time fast blinking between red and green.

To check the address setting it will be displayed again; e.g. if the device blinks three times green and after that for two times red, this means an address of 23. After displaying the address the device goes to the normal operation mode. Entering the address setting mode is possible at power up or if no communication is performed.

At power up or any time when the button is pushed the actual address setting will be displayed.

Baratron Gauge

In order to measure the vacuum level inside the plasma process, this gauge is provided.

Set-Up

Mount the transducer with the inlet port pointing (vertically) downward. Although the unit can be mounted in any orientation, mounting it as suggested allows any foreign matter entering the pressure port to fall away from the diaphragm.

Isolate the unit from vibration as much as possible. When not subject to gas damping at low pressure, the diaphragm may become susceptible to resonance. Te 0.05, 0.1 and 1 Torr F.S. units are most sensitive and you should isolate these units from any vibration that exists.

Remember to isolate the vibration through the cable as well as through the port.

Maintenance

Electrode Top Plate Inspection/ Replacement

Unfortunately, it is very difficult to estimate the approximate life span of an electrode plate as there are several factors which can significantly effect the life span, these include; power density, running hours, pressure regime, substrates, arcing events run etc.

A rough estimation would be between 6-24 months

Operational Checks

When the plasma treater is in operation, it is advised to perform regular checks of the electrode plates to monitor basic condition and more importantly erosion of the plates, as part of the normal preventative maintenance schedule.



Figure 14.68 - 9 Photograph showing erosion of the electrode plates prior to cleaning.

Pressure Checks

Assuming that the end user using the advised closed loop pressure control at a pressure regime of 6×10^{-2} mbar, sputtering should be minimised.

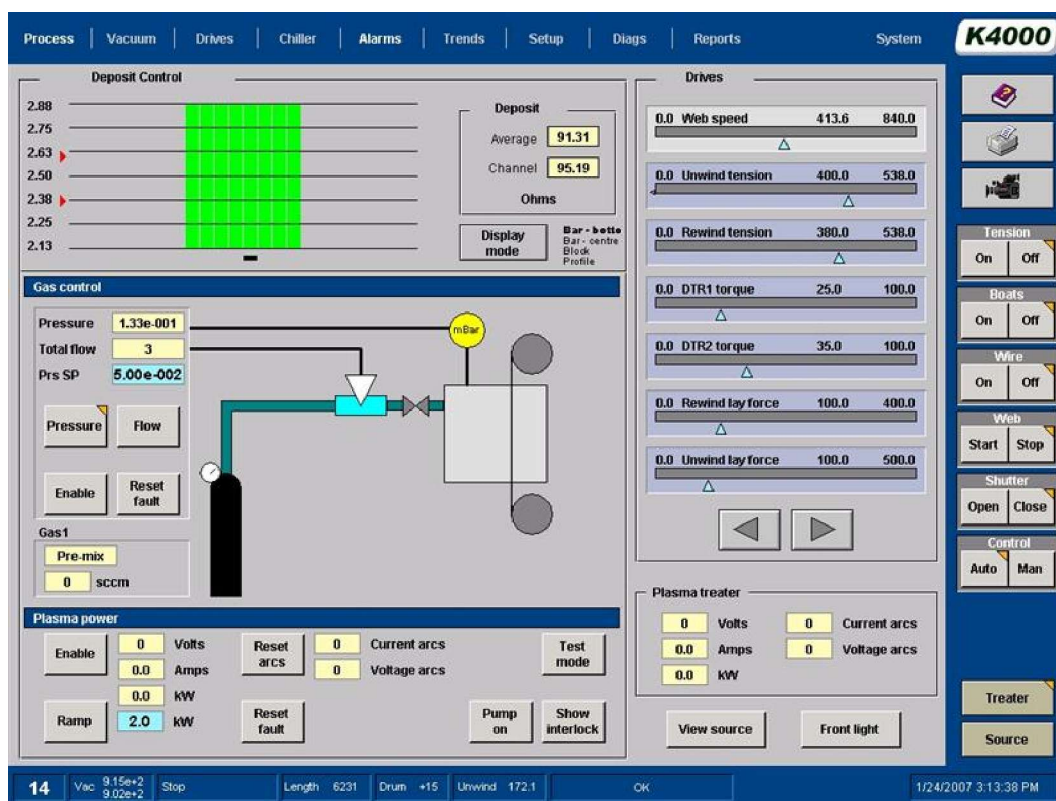


Figure 14.68 - 10 Plasma screen showing the vacuum level reading from the cold cathode gauge

Note: The zero of the gauge may drift over time, and provide a false vacuum level.

14.69 Mass Flow Controllers (MFC's)

How to Zero the Flow Controller

Ensure that no gas flow is entering the flow controller.

1. Apply gas, at a regulated pressure, to the flow controller.
2. Close the positive shutoff valve.
3. For an MFC: Command the control valve fully open by sending a full scale set point (5 VDC) signal, or:
 - 15 pin Type "D" connector: connect pin 4 (valve open) to pin 11 or 12 (signal ground).
 - 9 pin Type "D" connector: Supply +5 Volts to pin 1 (to open the valve).
 - Edge Card Connector: Connect pin D (valve open) to pins 10, B or C (signal ground).
4. Send a maximum set point through the PLC to the MFC.
5. A positive flow may occur momentarily while the gas pressure equalises across the flow controller.

Note: A set point command signal greater than 50mV (1% of full scale) is required for the flow controller to generate an output.

Adjust the Zero Pot

1. For an MFC: once flow through the controller has stopped (reached zero flow), remove the set point or valve open command.
2. Turn the zero pot (located on the inlet side of the flow controller) until the readout displays zero.
3. Open the positive shutoff valve.

An MFC indicate a small, positive flow (<1.0% F.S.) due to a leak through its control valve. However, do not 'zero out' this flow since it represents an actual flow measurement inherent in the system.

How to Adjust the Controller Gain (MFC only)

Adjust the controller again if the flow signal oscillates. Reducing the controller gain will reduce the signal oscillation. The controller gain adjustment pot is located on the upstream side of the controller.

- To decrease flow signal oscillation: Turn the controller gain counter-clockwise to decrease the controller gain setting.

Note: Lowering the supply pressure to the MFC will have the same effect as decreasing the gain since it will reduce the overflow/ underflow effect of the valve.

If the MFC responds too slowly to a change in set point, you may need to increase the controller gain slightly. To increase the controller gain, turn the controller gain pot clockwise.

14.70 Baratron Pressure Gauge

Checking the Transducer Zero

Check the transducer zero prior to the initial operation and then periodically as required. The zero can be set (or re-set) by adjusting the zero potentiometer located on the top cover of the transducer, or at the front panel of any MKS power supply/ readout, which is being used. The unit must be fully stabilized before you change the zero adjustment. Stabilization is accomplished by allowing the unit to warm up for approximately:-

- 15 minutes for Types 622 and 626 instruments.
- 2 and 4 hours for Types 627,628 and 629 instruments (greater than or equal to 1 Torr)
- 8 hours for the 0.05 Torr (627 and 629) and 0.1 Torr (627, 628, 629)

Pump the unit, with the power on, down to a pressure less than the transducer's resolution (0.01% of Full Scale).

Note: When the unit is properly warmed up and stabilized, the zero adjustment must be made at a pressure less than the transducer's resolution (0.01% of F.S.). Low range transducers (0.1 and 0.05 Torr) should be pumped for at least one hour after exposure to air to remove any moisture and to allow the pressure the pressure to stabilize.

Zeroing a transducer above its stated minimum resolution creates a zero offset relative (or unique) to the system in which the transducer is located. All subsequent readings are then linear and accurate relative to the offset value.

Note: If available pressure is not sufficiently low to set the transducer zero, you may use a vacuum leak detector with sufficient vacuum pumping (to achieve proper zeroing pressure). In this case, mount the transducer on the leak detector in the same plane or orientation, as it will be during actual use.

To correctly zero a transducer, follow the procedure below: -

1. Install the transducer
2. Power the transducer and allow it to warm up. Warm up time is:
 - a. 15 minutes for Types 622 and 626 transducer
 - b. 2-4 hours for types 627,628 and 629 transducer (greater and equal to 1 Torr)
 - c. 8 hours for the 0.05 and 0.1 Torr range units.

Pump the unit down to a pressure below its resolution
3. Using a small screwdriver, adjusts the ZERO pot until the readout displays zero (0000).

Highest Pressure Suggested for Proper Zero Adjustment	
Full Scale Range (Torr)	Highest Pressure for Zero Adjustment (Torr)
0.05*	$<2.5 \times 10^{-6}$
0.1**	$<5 \times 10^{-6}$
1	$<5 \times 10^{-5}$
2	$<1 \times 10^{-4}$
10	$<5 \times 10^{-4}$
100	$<5 \times 10^{-3}$
1K	$<5 \times 10^{-2}$
*Only available with the type 627 and 629 transducers	
** Only available with types 627,628 and 629 transducers	

Table 14.70 - 1 Recommended Pressure Levels

Coarse Zero Adjustments

All 600 Series transducers provide additional zero range capabilities through the coarse zero adjustment feature. The zero potentiometer provides ample control under normal conditions.

Note: Use the coarse zero adjustment only if the zero potentiometer fails to provide sufficient adjustment.

The coarse zero control is accessible through the side of the instrument.

Follow the steps below to adjust the coarse zero:

1. Install the transducer in a system with a power supply/ readout.
2. Turn the transducer on
3. Be sure the transducer is properly warmed up and pumped down below its resolution (for best results pump the transducer while it is warming up.)
4. Centre the ZERO pot located at the top of the transducer (adjust the screw to leave an equal amount of adjustment both clockwise and counter clockwise).
5. Turn the COARSE ZERO's multi-position switch (located on the side of the unit) to a position that produces the output signal closest to Zero Volts.
6. Re-Install the plastic ring
7. Adjust the ZERO pot (located on the top of the unit) to bring the output to exactly zero Volts

14.71 Plasma System Basic Safety Interlocks Troubleshooting

Water Cooling

The generator is water cooled and divided into different cooling circuits, which in part are monitored with flow control instruments.

Cooling water connections with the following technical specification are needed for cooling the MF/RF generators.

	Caution !
	Do not use de-ionized water for cooling AE power supplies You must stop coolant flow when the PE II is not operating or severe damage may occur from condensation. Use of water outside the specified range is not allowed. If the water supply is at a temperature below the local dew point, condensation can occur which will cause damage to internal electronic components,
Temperature	+5°C to +30°C inlet temperature
Flow Rate	7.57 litres (2gallons) per minute minimum
Pressure	100 psi (6.9 bars) maximum inlet pressure
Pressure Drop	6 psi (0.41 bars) @ 2gpm, inlet to outlet (new)
Contaminates	The following specifications are recommended for water used to cool the PEII 5kW: • pH between 7 and 9 • total chlorine < 20 ppm • total nitrate < 10 ppm • total sulphate < 100 ppm • total dissolved solids < 250 ppm • total hardness expressed as calcium carbonate equivalent less than 250 ppm • specific resistivity of 2500 Ω/cm or higher at 25°C • total dissolved solids (TDS) as estimated by the following TDS ≤ 640,000
	Specific resistivity (in Ω/cm)

Table 14.71 - 1 Cooling Water Specification

14.72 Motorised Spreader Roller Set-Up

1. Look for the apex indicator on the spreader roller.

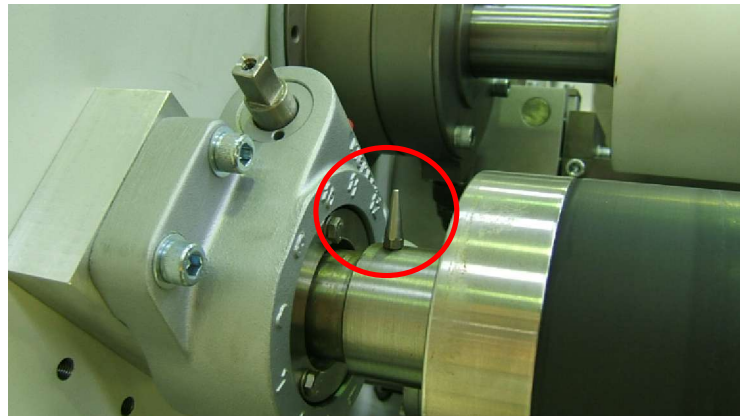


Figure 14.72 - 1 Photograph Showing Kickert Spreader Roller Apex Indicator

2. Remove the power source (fuses/circuit breaker) feeding the circuit/motor.
3. Loosen the coupling and locating bracket at the end of the spreader.



Figure 14.72 - 2 Photograph of Potentiometer and Coupling

4. Rotate the coupling attached to the potentiometer and monitor for full range of indication on the drives screen. The indicator shown below will move.
5. Check the position of the apex and determine at what angle it is in reference to the indicator on the DRIVES screen. Note fully vertical is +90 degrees. If necessary, remove the motor fan cover and move the apex to the +90 degree position.

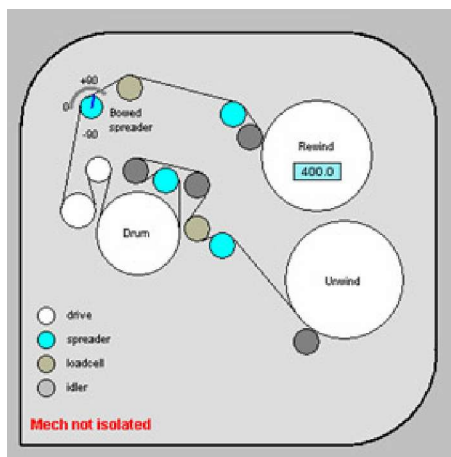


Figure 14.72 - 3 Drive Screen Indicating Spreader Roller feedback.

6. Move the coupling slowly until the feedback is giving the actual angle of the spreader roller.

7. Tighten the coupling onto the shaft.

	Caution !
	Take care when adjusting the coupling between the roller and the potentiometer, as it is delicate.

8. Using the angle control, change the set angle and monitor for the response.

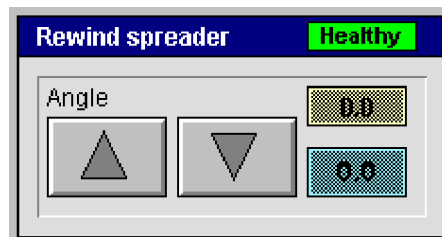


Figure 14.72 - 4 Rewind Spreader Section of Drives Screen

14.73 Layarm Spreader Roller Apex Position Adjustment

On some machines, a bow spreader has be supplied on the rewind layarm.

The apex position of the bow spreader roller can be moved on this type of roller. The location of the apex is by a red indicator dot on the roller end closest to the faceplate.

On wider machines there is a pointer built into the roller.



Figure 14.73 - 1 Photograph Showing Layarm Spreader Roller Apex Indication Pointer



Figure 14.73 - 2 Photograph Showing Layarm Spreader Roller Retaining Bolts Being Loosened

1. Lower the layarm towards the chucks using the pushbuttons.

	Warning !
	Switch off and isolate the movement isolator and lock it off using a lock.

2. Loosen the locating clamp on either end the roller by releasing the 10mm hex head bolts and few turns.
3. Use a 17mm Allen key or socket to rotate the apex of the roller to the position that is required.



Figure 14.73 - 3 Photograph of Apex Position Being Changed on Layarm Spreader Roller

4. Once at the required location, tighten the clamping plates/screws.

SECTION 15

MACHINE DOCUMENTATION

Section 15.1 Machine Specification
Section 15.2 Electrical and Mechanical Drawing Lists
Section 15.3 Bill of Materials
Section 15.4 Drive System Configurations
Section 15.5 Software
Section 15.6 Proprietary Manuals Listing
Section 15.7 Documentation On CD

15.1 Specifications

This machine has been built to perform to a technical specification. The main relevant aspects of this specification are listed below;

GENERAL VACUUM EQUIPMENT - K4000 2450 - Vacuum Metallizer

Substrate

Substrates must be suitable for vacuum metallizing:-

- BOPP 10 to 50µm
- PET 9 to 50µm
- CPP 20 to 40µm
- LDPE 25 to 50µm
- Nylon 12 to 50µm
- UPVC 20 to 100µm

Substrate Widths:

Maximum Width	2450mm
Minimum Width	1250mm
Maximum Roll Diameter:	1000mm
Core Size:	152 mm

Vacuum

Evaporation Zone Operating Pressure:	5.0 x 10 ⁻⁴ mbar
Winding Zone Operating Pressure:	2 x 10 ⁻² mbar - 2 x 10 ⁻¹ mbar
Pumpdown Time, Clean Dry And Empty:	5.0x10 ⁻⁴ mbar in 5 minutes
Ultimate pressure: after overnight pumping:	< 2x10 ⁻⁵ mbar
Pressure Rise Test:	<2.0 x 10 ⁻² mbar in first hour

Winding Mechanism

Maximum Line Speed:	840 metres/min.
Inch Speed	6 metres/min
Maximum Winding Tension:	538N
Minimum Winding Tension:	49N
Maximum Drum Tension:	1291N

Evaporation Sources

Evaporator Source Type:	Resistance heated intermetallic
Number:	26
Evaporator Size:	80 x 40 x 10mm
Aluminium Wire Size and purity:	2.0mm - 99.8% pure
Aluminium Wire Spool Size:	6.5Kg

Metallizing Data

Maximum Deposition Rate:	2.2 OD at 630 meters/min.
Deposition Uniformity:	+/- 5% Dependent upon material and running speed.
Thickness Range:	0.5 to 10 ohms/sq.
Evaporator Life:	

15.2 Electrical and Mechanical Drawing Lists

A copy of the machines 'BOM', bill of materials is provided as an appendix in this manual. This includes a list of drawings

The documentation CD also provides this listing with links to the drawings.

Printed A3 size drawings are supplied for the user.

Any request for further copies of the drawings, etc. must be made to General Vacuum Equipment.

Unless prior arrangements have been made, these drawings and the information on them belong to General Vacuum Equipment and should not be copied.

15.3 Bill of Materials

A computer generated and print out of the bill of materials will be supplied. This listing will be referenced in this manual.

15.4 Drive System Configurations

The drive system designer has supplied a set of wiring schematic drawings for their system.

Some systems also use a PLC or proprietary software to allow configuration of the set up parameters. A printed copy of any program will be provided for the user.

15.5 Software

Copies of some of the software program files used on the machine will be backed up onto various media.

A copy of the computer programs will be supplied on a CD-ROM.

Any other disks supplied with vendor-supplied equipment, will be passed onto the customer.

If General Vacuum Equipment Engineer's engineers make any modifications to the machine, the backup discs will be updated.

Software Supplied	Version	Number of Disks (CD ROM's)
Symantec pcAnywhere	12.0 Host	CD
Siemens Simatic PC Microbox/Box/Rack PC Documentation And Drivers		CD
Pinnacle Systems Instant CD/DVD	V 8SE	CD
Siemens Simatic WinCC Single License		Floppy Disk
Siemens Simatic WinCC Emergency License		Floppy Disk
Logitech MouseWare	9.79	CD
Siemens Simatic WinCC	V6.0 SP4	CD
Siemens Simatic SQL Server 2000 Service Pack 4 for WinCC V6.0		CD
Siemens Simatic WinCC V6.0 SP4 Documentation		CD
Siemens Simatic NET PC/Software	Edition 2005 + HF1	CD
Siemens MS Service Packs & Tools	2005.01	CD
Siemens Recovery W2KProSP4		CD
Siemens Restore W2KPiSP4	03/2004	2 x CD
ELO TouchTools	Rev N.2	CD
Cyberlink PowerDVD	XP 4.0	CD
Siemens Simatic WinCC/RT 8192 Single License		Floppy Disk
Siemens Simatic WinCC/RT 8192 Emergency License		Floppy Disk

Table 15.5 - 1 Software Supplied

15.6 Proprietary Manuals Listing

The following lists of manuals have been supplied in the proprietary manual binders.

ABB Load Cell System
Advanced Energy PE11 Power supply
Aerzen Booster Pump
B&R Stepper Unit
Berger Lahr Servo Motor
Bussman Fuses 170 M1566
Butterreit Cable Coiler
Edwards Active Inverted Magnetron Gauge (Penning)
Edwards Barometrically Compensated Dial Gauge
Edwards Speedivalves
ELO Touchscreen
ESP Witness CCTV monitor
ETP Coupling - Shutter Shaft
Eurotherm Profibus DP Interface
Eurotherm TU Comms
Eurotherm TU1451/71
Falk Couplings
FC Lane Electrical Leadthroughs
Festo Double Acting Cylinder 163439 DNC-80-160PPV-A
Festo Double Solenoid Valve 150980 JMFH-5/2-D-1C
Festo Exhaust Flow Control 10352 GRE-1/4
Festo Pressure Regulator 33322 LRS-1/4-S-B
Festo Proximity Switch, Electrical 150857 SME-8-S-LED-24
Festo Single Pilot Pneumatic Valve 14330 VLHE-3-1/4
Festo Solenoid Coil (F-coil) 34411 MSFG-24/42-50/60-OD
Festo Solenoid Coil (X-coil) 4527 MSFG-24/42-50/60
Festo Solenoid Valve 150981 MFH-5/2-D-1C
Festo Solenoid Valve 150983 MFH-5/3E-D-1C
Filton Rotary Union
Gates Timing/Polychain Belts
GEMS Sensors RFS Rotorflow Sensor – Plasma
HMS Anybus Unit
Honeywell Miniature Enclosed Switches (Track Switches)
IFM Asi DP Controller
IFM IND Asi Proximity Switch
Keb Combicom – Profibus DP Operator
Keb Combiline – E4/E5 - Built In Filter Size R
Keb Combiline – E4/E5 - Submounted Filter Size E
Keb Combivert – Installation Manual – Housing E
Keb Combivert – Safety Stop In Accordance With EN954-1
Keb Combivert 1 – Getting Started
Keb Combivert 2 – Instruction Manual - Power Circuit
Keb Combivert 3 – Instruction Manual - From V2.3
Keystone Isolation Valve
Keystone Isolation Valve – Installation
Keystone Pneumatic Actuator
Keystone Pneumatic Actuator – Installation
Kickert Spreader Roller
Kinetrol Actuator For Coating Shield
Kinetrol AP Positioner
Kinetrol Universal Limit Switch
Lexan Lexgard Laminate (Windows)
Leybold DIP20000 Diffusion Pumps
Leybold SV750B Rotary Vane Pumps
Lovejoy Coupling (Source window)

Lovejoy Coupling (Source window) – Installation Instructions
Manes Electrical Cabinet
Meanwell DRT 480-24 PSU
Meanwell DRT 960-24 PSU
Netgear Hub
OMNI Phase Line Master OSW Unit
Platon Flow meter
Platon Flow meter
Polycold PFC 1102
Rectus Quick Disconnect Couplings
Rhodes Flow Indicator
Ringfeder Locking Assembly
Rittal Cooling Unit SK3304.100
RS Audible Alarm (cart)
RS Xenon Beacons
Siemens 3KA553-1AE01 Isolator
Siemens AS-I Safety Monitor (3RK11051-BE040CA0)
Siemens Combimaster (GB5139-H1731-U400-D2)
Siemens CP 318-2 Processor module
Siemens CP341 (4 NES 720 6018-02 Release 1)
Siemens CP343-2 (C79000-G8976-C149-02)
Siemens ET200M Distributed I/O station 153.1AA0.0XB0 CP345, SM331, SM321, SM322 (05/99 C79000-G7076-C152 Edition 4)
Siemens ET200S (Edition 07/2003 EWA-4NEB78060240209)
Siemens FM350-2 Module (PLC) (c79000-G7076-C353-01)
Siemens Profibus bus connector (A5E0009555801)
Siemens Profibus CB15 (Issue C1 20/03/01)
Siemens RS 485 Repeater (EWA 3NEW 812 67 26)
Siemens Separately Ventilated Three-Phase Motor 1PH7 (610.43 429.21b)
Siemens SITOP Power supply (3RX9307-1AA00)
Siemens SM321 module
Siemens SM322 module
SKF Bearings
SSD AC890 CS Filter – Engineering Reference
SSD AC890 CS Filter – Quick Start Manual
Star Layarm Slides
Stogra Stepper Drive
Stogra Stepper Motor
Svecom Mechanical Shaft
Telemcanique ABL 6RT2420 PSU
Telemcanique Activar 71 drive and programming module – Programming Manual
Telemcanique Activar 71 drive and programming module – Installation Manual
Telemcanique Lexium 05 servo drive
Telemcanique Preventa Type XPS-ECP safety relay
Telemecanique Contactors/Lockout Switch LC1
TES Chiller Unit
Tyco Touchscreen
Unted Electric Pressure Switch
Valtec Ball Valve
Waltec Camera
XP PSU

Table 15.6 - 1 Proprietary Manuals Listing

15.7 Documentation on CD

A copy of all the machine documentation will also be supplied on CD. This CD contains the following, all in a PDF format:

- Operation And Instruction Manual
- Electrical Schematics
- Mechanical Assembly Drawings
- Bill Of Materials
- Vendor Manuals

There is also an interface for navigating the data on the CD.

To run the CD simply place the CD in a CD-ROM and the CD will autorun the navigation interface. Click on the buttons for the desired data. There are navigation buttons to take the user back one menu or back to 'Home' on most pages.

SECTION 16

MACHINE SPARE PARTS

Section 16.1 Identifying Spare Parts

Section 16.2 Suppliers

Section 16.3 How To Order From General Vacuum Equipment

This machine has been manufactured using common components available from multinational companies. Other components have been manufactured by General Vacuum Equipment to high engineering standard.

16.1 Identifying Spare Parts

During the production of your machine, we work to a master document called a “**Schedule**” consequently every part has a unique “**schedule number**”.

For ease of identification, all parts are grouped into sections relevant to the various processes & systems incorporated onto your machine. These sections are called “**sub-assemblies**”. There are approximately 15 sub-assemblies in a typical machine build. This number can vary slightly if additional equipment was included in the final specification.

Typical List of Machine Sub-Assembly's

- 01 – Vessel
- 02 – Mechanism
- 03 – Source
- 04 – Evaporation Source
- 05 – Source Power
- 06 – Web Transport
- 07 – Web Transport Drive & Controls
- 08 – Pumping Group
- 09 – Vacuum Controls & Instruments
- 10 – Services & Ancillaries
- 11 – Cryogenic Pumping
- 12 – Optional Equipment
- 13 – Operator Control Desk
- 14 – Machine Wiring
- 15 – Syscon Monitoring & Control

There are a number of ways to accurately identify parts used in the construction of your machine.

The sub-assemblies are then broken down into a list of smaller assemblies. This is where individual component parts can be identified by a sequence of identifying numbers, letters & descriptions.

These smaller assemblies are known as “**Sub-Sections**”. These are listed numerically in the same format as above, 01, 02, 03 Etc

Sub-sections are also issued with a “**General Assembly**” (GA) drawing No. This drawing shows all the parts included in the assembly & is a good point of reference when trying to identify a part you may not be familiar with.

Each sub-section is listed in 3 specific groups as follows:

- M – Manufactured Parts
- S – Stock Items
- B – Bought-In Items

(We refer to these numbers as “**Bubble Numbers**” – See fig 9.1 for examples)

Therefore each component part of an assembly can be listed by a unique “**Schedule Number**”.

For example Schedule Number: **06/14/M001** has 3 points of reference

1. Sub-Assembly 06 – Web Transport Sub-Assembly
2. Sub-Section 14 – Unwind/Rewind Chucks
3. Bubble Number M1 – Inboard Chuck

For a more descriptive reference you could also quote the Drawing or BoM's Number & the component or part description.

For example: **320G818 - Inboard Chuck** is also **06/14/M001**

This combination also gives a unique reference, ensuring you always receive the parts you require.

For information on other components within this Chuck assembly you would refer to General Assembly Drawing – 320G817

A complete list of all components & general assembly drawings is available on the CD supplied with your machine documentation.

A bill of material is provided of all the components used upon the machine. Some of the larger systems purchased by General Vacuum Equipment, such as the drive system will have a separate bill of materials provided by the manufacturer.

All General Vacuum Equipment's Assembly drawings reference a schedule number, for example;

A list of schedules and drawings can be found in section M -5

Below is a schedule for the vessel light within the chamber.

The drawing will identify components upon it by a letter/number combination within a bubble.

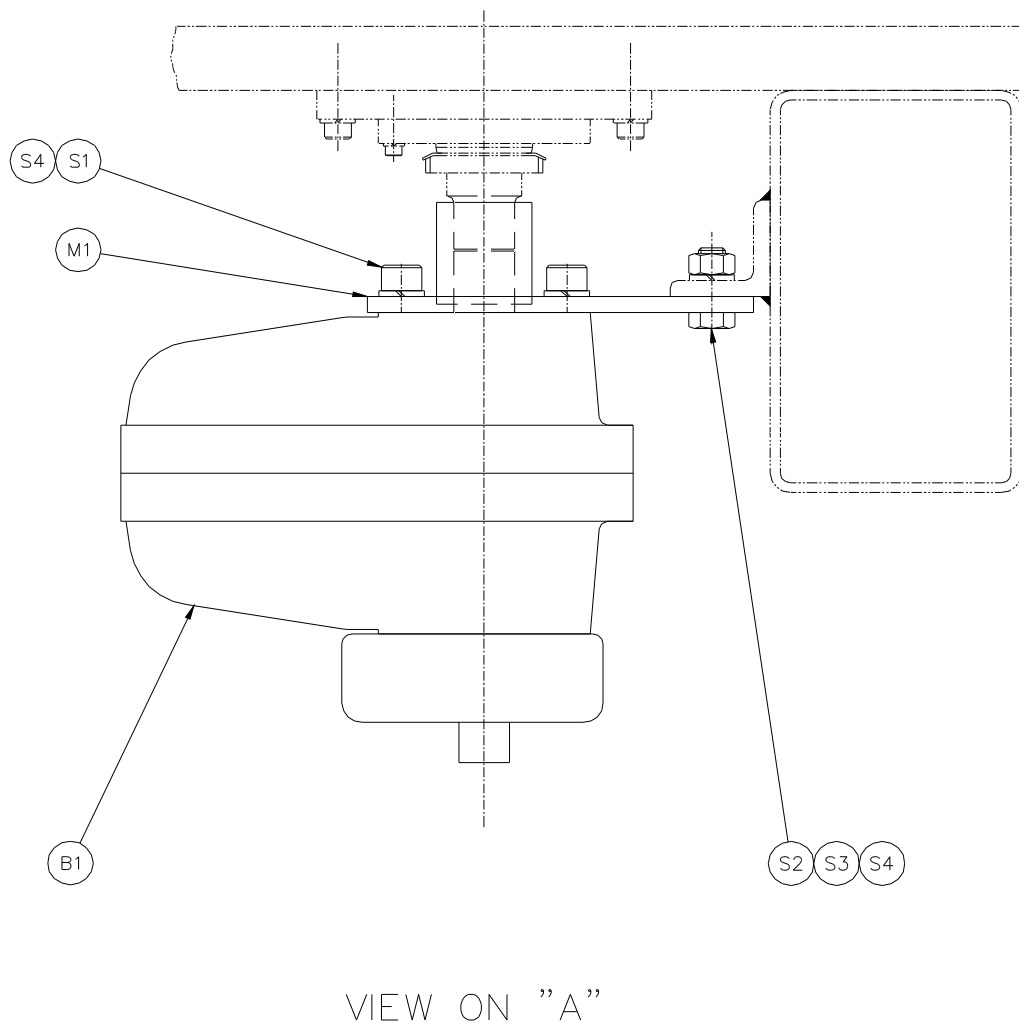


Figure 16.1 - 1 Part Identification On Drawing

Basically;

- B4 - indicates a bought item , for example a bearing.
- S2 - indicates a stock item, generally bolts and nuts
- M1 - indicates a manufactured item

On the schedule, the item will be described with the number used and if required a component drawing number. These component drawings are not supplied to end users.

16.2 Suppliers

As previously stated, General Vacuum Equipment purchase such items as;

- Vacuum Pumps
- Vacuum Gauges
- Drive Systems
- Pneumatic components
- Hydraulic components
- Control systems
- Bearings and seals

The manufacture's part numbers and details are to be found on the components to help the end user source the components locally.

General Vacuum Equipment will supply, if possible, local distribution/suppliers information as requested by the end users.

16.3 How to Order From General Vacuum Equipment

Enquires should be sent to General Vacuum Equipment's headquarters in the UK.

Customer Support

The machine is supported by a customer support system designed to offer the user, a fast, courteous service. If you need assistance beyond the scope of the manuals, spare parts or a site visit by a service engineer, please contact General vacuum equipment at the following headquarters address.

GENERAL VACUUM EQUIPMENT LTD,
Pennine Business Park,
Heywood,
Lancashire,
United Kingdom

Telephone: (44) 1706 622442
Facsimile(44) 1706 622772

See below for a full list of Contact Names & Numbers for all your Spares requirements, Technical Assistance & Customer Support.

Spares & Service Contacts:

For all your requirements concerning Warranty, Technical Support & Spares & Service please direct your correspondence to the personnel below.

Customer Support & Warranty:

Mr Graham Simpson – Senior Service Engineer
E-mail: graham.simpson@bobstgroup.com & bill.powell@bobstgroup.com

Technical Support & Service Enquiries:

Mr Bill Powell – Service Engineer
E-mail: bill.powell@bobstgroup.com

Spare Parts & Service Support:

Mr Paul Gardner – Spares Co-ordinator
E-mail: paul.gardner@bobstgroup.com

Mrs Judi Tobias – Spares Co-ordinator
E-mail: judi.tobias@bobstgroup.com

Shipping & Order Enquiries:

Miss Nicky Jones – Spares Administrator
E-mail: nicky.jones@bobstgroup.com

APPENDIX A

BILL OF MATERIALS

Bill Of Materials								
SECT	BUBB NO	DEPT	NUMBER	PARAM	VER	DESCRIPTION	ASSEMBLY DRAWING	QTY
	M007	M	318G910		2	BRG RETAINING CAP		2
	S001	M	421515		2	Screw Caphead M20x70		12
	S002	M	421127		2	Screw Caphead M16X55		8
	S003	M	421401		2	Screw Caphead M16X80		4
	S004	M	421125		2	Locknut M16		4
						Plain M20		12
						Plain M30		2
						M30		4
	S008	M	421122		2	Screw Caphead M10x30		8
06-14			902G355			Arrgt of Un/Rewind Lock Chucks	320G817	1
	B001	M	200585		2	Bearing Ball Angular Contact 3211		2
	B002	M	239245		2	Seal Simmerring BABSL70-90-7/72NBR		2
	B003	M	200015		2	BEARING 6008-2Z		2
	B004	M	422008		2	Toggle Clamp DE-STA-CO Model 341		4
	B005	M	200587		2	Bearing Needle Roller RNA4914		2
	B006	M	200586		2	Inner Ring IR70x80x54		2
	B007	M	239213		2	Seal O Ring RM 2393-57 Nitrile		2
	B008	M	239218		2	Seal O Ring RM0995-30Nitrile		2
	B009	M	239246		2	Seal O Ring RM0645-30Nitrile		2
	B010	M	239T027		2	'O' RING [RM 0495-30]		2
	B011	M	242011		2	LOCKNUT KM11		2
	B012	M	242012		2	LOCKWASHER MB11		2
	B013	M	239231		2	Seal Simmerring BABSL80-100-7/72NBR		2
	B014	M	234123		2	CIRCLIP EXTERNAL 1400-40		2
	B015	M	233138		2	CIRCLIP INTERNAL 1300-102		2
	B016	M	233139		2	CIRCLIP INTERNAL 1300-68		2
	M001	M	320G818		2	Inboard Chuck		2
	M002	M	320G819		2	Outboard Chuck		2

06 Denotes **Sub-Assembly 6**
(Web Transport)
-14 Denotes **Sub-Section 14**

Refers to the BoM's
(Bill of Material's)
Assembly No

Title of **Sub-Section**

This Number refers to
the **General Assembly**
drawing for Sub-
Section 14

Bill Of Materials								
SECT	BUBB NO	DEPT	NUMBER	PARAM	VER	DESCRIPTION	ASSEMBLY DRAWING	QTY
	M003	M	320G820		2	Locking Clamp		4
	M004	M	306G878					2
	M005	M	303G823					2
	M006	M	306G876					2
	M007	M	303G775			SPACER		2
	M009	M	303G797		2	STOP BLOCK		4
	M010	M	303G796		2	PLAIN BUSH		4
	M011	M	310G478		2	PIVOT PIN		4
	M012		320G823		-	6"Mech Expanding Reel Shaft Ref only 320G823		3
	M013	M	306G175		2	THRUST WASHER		2
	M014	M	306G875		2	STOP SHAFT		2
	M015	M	306G874	IT.A	2	WEAR PLATE		4
	M016	M	306G874	IT.B	2	WEAR PLATE		4
	M017	M	302G112		2	LATCH		4
	M018	M	320G821		2	Spacer		4
	M019	M	320G822	IT.1	2	Locking Chuck Protector		2
	M020	M	320G822	IT.2				2
	M024	M	320G894					4
	M025	M	320G895	IT.1				2
	M026	M	320G895	IT.2	2	Pillar		2
	M029	M	322G488		2	OUTBOARD SHIELD PLATE R/W		1
	M030	M	322G489		2			2
	M031	M	322G494		2			1
	S001	M	421111		2			16
	S002	M	421122		2	Screw Caphead M10x30		12
	S003	M	421149		2	Screw Caphead M10x20		20
	S004	M	421138		2	Screw Caphead M8x25		8
	S005	M	421091		2	Screw Caphead M6x25		8

BoM's Reference. This is a unique parts reference number. Quoting this number guarantee's you receive the correct part.

Quantity required for a full assembly.

Variant or Parameter of an individual Part.

Internal reference only, not important

Bill Of Materials								
SECT	BUBB NO	DEPT	NUMBER	PARAM	VER	DESCRIPTION	ASSEMBLY DRAWING	QTY
	S006	M	421226		2	Screw Grub M6x10		4
	S007	M	421258		2	DOWEL 6 dia x 40 mm LONG		4
	S008	M	421019		2	Screw Caphead M6x20		16
	S009	N	S Numbers refer to Stock items , Nuts, Bolts Washers Etc			Screw Caphead M8x25		16
	S010	N				Washer Plain M8		16
	S011	N				Washer Spring M10		12
	S012	N				Washer Spring M6		16
	S017	M	421249		2	Screw Countersunk Head Socket M8x20		12
06-15			902G356			Ebonite Covered Roller	320G653	1
	B001	M	200572		2	Carb BEARING C2210 TN9		1
	B002	M	200408		2	BEARING DOUBLE ROW SELF ALIGNING 1211E		1
	B003	M	239232		2	Seal Simmerring BABSL65-90-7NBR		2
	B004	M	242012		2	LOCKWASHER MB11		1
	B005	M	242011		2	LOCKNUT KM11		1
	B006	N	B Numbers refer to Bought-in items , Bearings, Seals Fittings Etc			Seal O Ring RM 2193-57 Nitrile		1
	B007	N				Seal O Ring RM1445-30 Nitrile		1
	B008	N				Seal O Ring RM1045-30 Nitrile		1
	B009	M				Seal O Ring RM0545-30Nitrile		2
	B010	M	233001		2	CIRCLIP INTERNAL 1300-100 (DIA 100 BORE)		1
	B011	M	423111		2	Elbow Adaptor Hobbs 8 O/D - 1/4" BSP		2
	B012	M	423038		2	Tubing Nut 8mm O/D		2
	B013	M	M Numbers refer to Manufactured Parts , these parts are unique to GVE			Tubing Sleeve 8mm O/D		2
	B014	M				Seal 1/4"bsp Enots 34-0156-02		2
	B015	M				Coned Locknut 1/4"BSP Enots 36-0540-02		2
	B016	M				LOCKNUT KM9		1
	B017	M			2	LOCKWASHER MB9		1
	M001	M	320G621	1650	2	Ebonite Roller		1

Uflex - V06G004 - Bill Of Materials



SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
00					Layouts & Schemes		
00-01			904G706		Layouts & Schemes - 2450 K4000 (Jay)		1
	01	M			General Arrangement - 2450 K4000	325G283	1
	02	M			Foundation Plan - 2450 K4000	325G284	1
	03	M			Mechanism Layout K4000	325G285	1
	04	M			Vacuum Schematic - 2450 K4000	325G286	1
	05	M			Drive Schematic - 2450 K4000	325G287	1
	06	M			Water Schematic - 2450 K4000	325G288	1
	07	M			Pneumatic Schematic - 2450 K4000	325G289	1
01					Vessel Sub Assy		
01-01			903G512		Vessel Assembly with roof tracks GRP 2450	403G952	1
	B1	M	239193		Seal O Ring RM0945-30 Nitrile		4
	M1	M	403G886		Vacuum Vessel 2450		1
	M2	M	404G107	2450	Runway Beam LH		1
	M3	M	404G108	2450	Runway Beam RH		1
	M4	M	306G558		RUNWAY STOP		2
	M5	M	302G901		RUNWAY PACKER		8
	M6	M	308G972	IT.28	Special Flange		4
	S1	M	421403		SOCKET CAP SCREW M12 x 50		48
	S2	M	421030		PLAIN WASHER M12		48
	S3	M	421249		M8 X 20MM SOCKET HEAD COUNTERSUNK SCREW		4
	S4	M	421156		SOCKET CAP SCREW M8x40		16
01-02			903G484		Rectangular Sightglass Assy (Mech Viewing)	403G961	4
	B001	M	204119		Bush Nylon Flanged 16diax12 i/dx21 o/d flange 19lg		16
	B002	M	239207		O Ring Cord 8.4mm c/s Nitrile		2
	M001	M	403G963	IT.1	Polycarbonate Window		1
	S001	M	421061		Screw HEXHD Setscrew M10x40		16
	S002	M	421029		PLAIN WASHER M10		16
01-04			903G578		Mesh Screens for Roughing Ports		1
	M001	M	403G003		Mesh Screen 12" Port		2
	M002	M	323G059		Mesh Screen 8" Port		1
	S001	M	421172		SOCKET CAP SCREW M5x12		12
	S002	M	421164		PLAIN WASHER M5		12
	S003	M	423458		Pipe Clip 6dia Enots 34-0218-04		12
01-05			902G307		Vessel Light Assy	320G232	3
	B001	M	421078		Terry Clip 1.1/2" 481150		2
	M001	M	320G230	2450	PVC Light Cover		2
	S001	M	421032		Screw Panhead M4x6		2
	S002	M	421033		SOCKET CAP SCREW M6x12		12
	S003	M	421034		Washer Plain M6		12

Uflex - V06G004 - Bill Of Materials



SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
01-08			902G308		Arrangement of 300 dia sightglass	307G478	2
	B001	M	204117		32774 nylon flanged bush		6
	B002	M	239190		Seal O-Ring RM2693-57 Nitrile		1
	M001	M	306G497		300DIA S/GLASS PAD (REF ONLY) 306G497		1
	M002	M	307G479		POLYCARB WINDOW		1
	S001	M	421114		SOCKET CAP SCREW M10x40		6
	S002	M	421029		PLAIN WASHER M10		6
01-09			903G572		Rectangular Sightglass Assy (Source Viewing)	404G220	4
	B001	M	204117		32774 nylon flanged bush		14
	B002	M	239207		O Ring Cord 8.4mm c/s Nitrile		2
	M001	M	404G221	IT.1	POLYCARBONATE WINDOW		1
	S001	M	421144		Screw HEXHD Setscrew M10x30		14
	S002	M	421029		PLAIN WASHER M10		14
01-11			903G615		ROTARY VIEWING WINDOW	404G378	1
	B001	M	202236		BEARING PILLOW BLOCK MP 25		2
	B002				STEPPER MOTOR (REF ONLY)		1
	B003	M	232312		LENZE HUB U4-14 710		1
	B004	M	232313		LENZE HUB U4-75 766		1
	B005	M	232311		SPIDER U4-26 752		1
	M001	M	403G981	2450	ROTOR		1
	M002	M	403G976		Rotor End Stub Shaft		1
	M003	M	403G977		Rotor End Stub Shaft		1
	M004	M	403G983	.IT.660	Glass Panel		4
	M005		404G382		PLATE		8
	M006		404G381	2450	SHROUD		1
	M007		404G379		BRACKET		1
	M008		404G380		BRACKET		1
	M009	M	403G998		SUPPORT		1
	M010	M	403G997		SUPPORT		1
	S001	M	421149		SOCKET CAP SCREW M10x20		4
	S002	M	421234		SOCKET CAP SCREW M12x30		4
	S003	M	421245		SOCKET CAP SCREW M5x10		16
	S004	M	421033		SOCKET CAP SCREW M6x12		3
	S005	M	421019		SOCKET CAP SCREW M6 x 20		4
	S006	M	421418		Screw Socket Countersunk Head M6 x 16		14
	S007	M	421029		PLAIN WASHER M10		4
	S008	M	421030		PLAIN WASHER M12		4
	S009	M	421034		Washer Plain M6		7
	S010	M	421164		PLAIN WASHER M5		16
	S011	M	421182		SOCKET CAP SCREW M8x16		3

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S012	M	421054		PLAIN WASHER M8		3
	S013	M	421052		Nut Wing M8 Steel		2
	S014	M	268025		Key Steel 5x5		1
	S015	M	421826		Screw Caphead M6x12 Fine Thread (0.75)		1
01-12			903G583		DIFFUSION PUMP LID ASSEMBLY	403G888	1
	B001	M	210910		Cylinder DNC-125-160-PPV-A		1
	B001	M	211373		REED SWITCH & CONNECTOR		2
	B001	M	211374		Cable Connector		1
	B001	M	211358		Swival Flange		1
	B002	M	427142		Rod Clevis with Pin & Locknut 14987		1
	B003	M	232309		COUPLING FENNER HRC180 WITH 'H' AND 'F' STYLE		1
	B004	M	205097		CAM FOLLOWER KR30		8
	B005	M	2005474		BEARING 61912		2
	B006	M	242111		LOCKNUT M12 x 1.5P		8
	B007	M	203079		BUSH DRY GLACIER MB2020 DU		16
	B008	M	203284		Bush Lenze ETP Express HYD		1
	B009	M	239356		Seal O Ring RM0895-30Nitrile		1
	B010	M	239357		Seal O Ring RM1095-30 Nitrile		1
	B011	M	202235		BEARING SL50		2
	B012	M	239360		Seal Simmerring BABSL60-75-8NBR		2
	B013	M	233005		CIRCLIP INTERNAL D1300-0750		1
	B014	M	234006		CIRCLIP EXTERNAL D1400-0600		2
	B015	M	203068		BUSH DRY GLACIER		2
	B016	M	239407		Seal O Ring 8.4c/sx540mm mean dia Nitrile		4
	M001		403G889	2450	DIFFUSION PUMP LID LAYSHAFT		1
	M002		403G901	2450	DIFFUSION PUMP LID SUPPORT BEAM		1
	M003	M	402G413		Cam Arm		1
	M004		403G904		DIFFUSION PUMP LID PIVOT BLOCK		4
	M005		403G905		DIFFUSION PUMP LID LINK PLATE		8
	M006		403G906		DIFFUSION PUMP LID LIFT BRACKET		2
	M007		403G907		DIFFUSION PUMP LID BEARING BLOCK		2
	M008		403G908		DIFFUSION PUMP LID GUIDE POST		2
	M009		403G971		DIFFUSION PUMP LID ACTUATION ARM		4
	M010	M	404G086		Pivot Bracket		1
	M011		404G264		PIVOT PIN		8
	M012	M	402G418		Bearing Spacer		1
	M013	M	402G419		Bearing Seal Housing		1
	M014	M	402G420		Outer Seal Cap		1
	M015	M	402G421		Spacer		1
	M016		403G940		DIFFUSION PUMP LID PLATE		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M017		403G942		DIFFUSION PUMP LID CONNECTION STUD		4
	M018		403G943		DIFFUSION PUMP LID RETAINING PLATE		4
	M019		404G026		SHAFT DIFFUSION PUMP		1
	M020	M	401G109		DIFF PUMPS BAFFLE RG		4
	S001	M	421114		SOCKET CAP SCREW M10x40		8
	S002	M	421161		SOCKET CAP SCREW M10x50		8
	S003	M	421210		SOCKET CAP SCREW M12x35		8
	S004	M	421403		SOCKET CAP SCREW M12 x 50		8
	S005	M	421289		GRUB SCREW M5x8		16
	S006	M	421038		Nut Full Hex M16		4
	S007	M	421030		PLAIN WASHER M12		4
	S008	M	421133		SOCKET CAP SCREW M5x16		4
	S009	M	421138		SOCKET CAP SCREW M8x25		16
	S010	M	421237		SOCKET CAP SCREW M8x35		6
	S011	M	421402		Screw Caphead M12x60		4
01-13			903G759		Panel Assembly Vessel Front	404G806	1
	M001	M	404G807	IT1	Panel		4
	M002	M	404G808	2450	Panel Support		1
	M003	M	404G097		BRACKET		2
	M004	M	404G095		BACK STRIP		3
	M005	M	404G498		SIDE PIECE		1
	M006	M	404G500		SIDE PIECE		1
	M007	M	404G809	2450	Panel Support		1
	M008	M	404G810	2450	Top Piece		1
	S001	M	421145		SOCKET CAP SCREW M6x30		16
	S002	M	421019		SOCKET CAP SCREW M6 x 20		13
	S003	M	421138		SOCKET CAP SCREW M8x25		4
	S004	M	421466		Screw Countersunk Head Socket M6x20		20
	S005	M	421057		Nut Full Hex M6		20
	S006	M	421034		Washer Plain M6		49
	S007	M	421054		PLAIN WASHER M8		4
01-15			903G648		VESSEL COVERS	404G536	1
	M001	M	404G098		ELECTRICAL LEADTHROUGH COVER		1
	M002	M	404G537		COVER PLATE		6
	M003	M	404G538		COVER PLATE		2
	S001	M	421033		SOCKET CAP SCREW M6x12		3
	S002	M	421134		Screw Socket Buttonhead M5x10		28
	S003	M	421034		Washer Plain M6		3
	S004	M	421164		PLAIN WASHER M5		28
01-16			903G848		Collector Plate Assy	405G322	1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M001	M	405G318	IT1	Collector Plate		3
	M002	M	405G319	IT1	Collector Plate		1
	M003	M	405G320		Collector Plate		1
	M004	M	405G321		Collector Plate		1
	M005	M	404G444		Spacer		13
	S001	M	421564		Screw Panhead M6x16		13
01-17			904G872		Diffusion Pump Lid Modification	408G155	1
	B001	M	235221		DIE SPRING ENTEX NO 418		4
	B002	M	239407		Seal O Ring 8.4c/sx540mm mean dia Nitrile		4
	B003	M	239240		Dx72 Draught Excluder		8
	M001	M	408G156		Spring Support Stud		4
	M002	M	408G157		Sealing Ring		4
	M003	M	408G158		Lid Cover		4
	M004	M	408G168		WEAR BUSH		4
	S001	M	421353		Screw Socket Buttonhead M6x16		32
	S002	M	421042		NUT FULL HEX M12		8
	S003	M	214025		Nipple Grease M6		4
02					Mech Drives Sub Assy		
02-01			904G597		GA BASEFRAME & FLOOR PANELS	407G565	1
	B101	M	421353		Screw Socket Buttonhead M6x16		31
	M001	M	321G971		MECH. BASEFRAME		1
	M002	M	325G057		Mech Baseframe Floor plate details		1
	M003	M	320G812		Baseframe Cable Tray Route		1
	M004	M	323G019		GUARD MECH DRIVES (TOP) (M3 M/C)		1
	M005	M	323G020		GUARD MECH DRIVES (R.H.) (M3 M/C)		1
	M006	M	323G021		GUARD MECH DRIVES (L.H.) (M3 M/C)		1
	M007	M	323G022		TIE PLATE MECH DRIVES GUARDS (M3 M/C)		2
	M008	M	323G024		SUPPORT BRACKET		1
	M009	M	407G290	IT6	Wire Mesh Guard Panel		1
	M010	M	407G290	IT4	Wire Mesh Guard Panel		1
	M011	M	407G562		Support Post		4
	M012	M	407G563		MOUNTING ANGLE		1
	S001	M	421234		SOCKET CAP SCREW M12x30		16
	S002	M	421030		PLAIN WASHER M12		12
	S003	M	421467		Screw Caphead M16x60		16
	S004	M	421059		PLAIN WASHER M16		16
	S005	M	421034		Washer Plain M6		31
	S006	M	421147		SOCKET CAP SCREW M4 x 12		16
	S007	M	421049		SOCKET CAP SCREW M8x20		16
	S008	M	421194		Washer Plain M4		16

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S009	M	421054		PLAIN WASHER M8		16
	S010	M	4211004		GRUB SCREW M8x16		8
02-02			904G244		Face Plate/Side Plate and Ties Assy	406G269	1
	B01	M	240470		Dowel Extractable 20 Dia x 40Lg		2
	B02	M	268121		3M Safety - Walk General Purpose 51 wide Black		1
	B03	M	239374		O Ring Cord 16.0mm c/s Nitrile		11
	M01	M	406G265		Mechanism Faceplate		1
	M02	M	406G266		Mechanism Sideplate		1
	M03	M	406G263	2450	Tie Bar		1
	M04	M	406G268	2450	Tie Bar		1
	M05	M	406G264	2450	Tie Bar		1
	M06	M	406G582	2450	Cable Guard		1
	S01	M	421115		SOCKET CAP SCREW M16x45		24
	S02	M	421114		SOCKET CAP SCREW M10x40		6
	S03	M	421116		Washer Spring M16		24
	S04	M	421021		Washer Spring M10		6
	S05	M	421134		Screw Socket Buttonhead M5x10		6
	S06	M	421164		PLAIN WASHER M5		6
02-03			904G679		G.A Mech Drives (In-line Winders & Ebonite)	407G060	1
	B001				AC MOTOR REF ONLY 1PH7167 Flange Mtg 13Kw (400rpm)		2
	B002	M	232354		ARCOFLEX Coupling size 75. Model 314.75.1.1		2
	B003	M	232355		ARCOFLEX Coupling size 65. Model 314.65.1.1		1
	B004	E	662086		Helical Gear Unit Flender D48-K4-(80)		1
	B005				AC MOTOR (BRKE F'TD) REF ONLY 664210 0.82Kw		2
	B006	M	232237		Coupling Renolds Crown Pin Flex PT57C/4P		1
	B007	M	224280		Bush Taperlock 1215 Bore 25 W/- Std Key TL1215/25		2
	B008	E	662087		Helical Gear Unit Flender D38-K4-(80)		1
	B009				AC MOTOR REF ONLY 664208 1PH7163 34Kw		1
	B010	M	225320		Sprocket Polychain 14M-34S-68/55		1
	B011	M	225294		Sprocket Polychain 14M-90S-90		1
	B012	M	224285		Bush Taperlock 4030 Bore 115 W/- Std Key TL4030/115		1
	B013	M	221765		Belt Polychain 14M-1750-68		1
	B014				AC MOTOR REF ONLY 1PH7133 Flange Mtg 13.5Kw (1150)		1
	B015	M	225296		Sprocket Polychain 14M-44S-20		2
	B016	M	221767		Belt Polychain 14M-1120-20		1
	B017	M	427008		Pneumatic Actuator dble acting Kinetrol 160-100		1
	B018				AC MOTOR REF ONLY 664209 1PH7103 6.25Kw		1
	B019	M	224290		Bush Taperlock 3020 Bore 38 W/- Std Key TL3020/38		1
	B020	M	224291		Bush Taperlock 3020 Bore 65 W/- Std Key TL3020/65		1
	B021	M	225297		Sprocket Polychain 8M-34S-36		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B022	M	224284		Bush Taperlock 1610 Bore 30 W/- Std Key TL1610/30		1
	B023	M	224292		Bush Taperlock 1610 Bore 25 W/- Std Key TL1610/25		1
	B024	M	221768		Belt Polychain 8M-1000-36		1
	B025	M	210858		Pneumatic Cylinder DNC-32-80-PPV		1
	B026	M	210859		Rod Clevis SG-M10 x 1.25		1
	B027	M	210860		Swivel Flange SNCS-32		1
	B028	M	232266		Locking Bush - ETP 25		1
	B029	M	234128		CIRCLIP EXTERNAL 1400-0100		1
	B032	M	423036		Adaptor Male 8mm-1/4"BSP 36-0530-11		5
	B033	M	423039		Tubing Sleeve 8 OD		15
	B034	M	423038		Tubing Nut 8 OD		15
	B035	M	423053		Straight Connector 8mm 36-0503-05		5
	B038	M	427295		'SMC' KQS08-02S Hex Soc HD male conn 8mm-1/4"BSP		4
	B039	M	423046		Tube Half Hard Copper 8mm O/D CS7008003		1
	B040	M	423049		Nylon 12 BS5409 Normal Duty 8mm O/D		10
	M001				MECHANISM FACEPLATE REF ONLY 406G265		1
	M002				MECH END DOUBLE CASTOR TRAVERSE REF ONLY 323G013		1
	M003				BASEFRAME REF ONLY 321G971		1
	M004				SINGLE CASTOR ASSY (VEE RAIL) REF ONLY 320G417		1
	M005				SINGLE CASTOR ASSY (PLAIN WHEEL) REF ONLY 320G418		1
	M006	M	407G099		Motor Mounting		2
	M007	M	320G959		W.C. Roller Support		1
	M008	M	407G101		Adaptor Plate		2
	M009	M	314G260		UNWIND L/ON DRVE SUP		1
	M010	M	312G382		REWIND L/ON DR"VSUPP		1
	M011	M	320G853		W.C.D. Drive Support		1
	M012	M	407G097		Motor Mounting		1
	M013	M	312G385		HINGED SHLD ACT SUPP		1
	M014	M	303G654		OIL BOTTLE BRACKET		1
	M015	M	316G292		OIL BOTTLE BRACKET		1
	M016	M	323G016		Tie Bar		1
	M017	M	320G899		Tie Bar Shim Set		2
	M018	M	313G592		PNEUMATIC ACTUATOR COUPLING		1
	M019	M	320G858		Motor Baseplate		1
	M020	M	323G017		Jacking Bracket (For Shipping)		2
	M021	M	321G955		Stop Block		1
	M022	M	321G956		Cylinder Mounting Spindle		1
	M023	M	321G957		Actuator Hub		1
	M024	M	323G209		Loadcell Control Unit Mgt PI,t		1
	S001	M	425003		Oil Reservoir 1/4"BSP (Ref OR 7730)		5

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S002	M	421172		SOCKET CAP SCREW M5x12		4
	S003	M	421035		SOCKET CAP SCREW M10x25		4
	S004	M	421161		SOCKET CAP SCREW M10x50		6
	S005	M	421234		SOCKET CAP SCREW M12x30		12
	S006	M	421210		SOCKET CAP SCREW M12x35		8
	S007	M	421403		SOCKET CAP SCREW M12 x 50		12
	S008	M	421470		Screw Caphead M12 x 70		4
	S009	M	421105		SOCKET CAP SCREW M16x35		8
	S010	M	421303		SOCKET CAP SCREW M16x40		8
	S011	M	421104		SOCKET CAP SCREW M16x50		16
	S012	M	421634		Screw Caphead M16x80		12
	S013	M	421309		SOCKET CAP SCREW M20x60		4
	S014	M	421533		SCREW, SOCKET HEAD CAP, M24x60		4
	S015	M	421138		SOCKET CAP SCREW M8x25		7
	S016	M	421524		SOCKET CAP SCREW M8x55		4
	S017	M	421285		Screw HEXHD SetscrewM10x60		4
	S018	M	421362		Screw HEXHD Setscrew M16x100		4
	S019	M	421246		Screw HEXHD Setscrew M16x70		5
	S020	M	421029		PLAIN WASHER M10		10
	S021	M	421030		PLAIN WASHER M12		28
	S022	M	421059		PLAIN WASHER M16		56
	S023	M	421070		PLAIN WASHER M20		4
	S024	M	421069		PLAIN WASHER M24		4
	S025	M	421054		PLAIN WASHER M8		15
	S026	M	421164		PLAIN WASHER M5		4
	S027	M	421027		NUT FULL HEX M10		4
	S028	M	421042		NUT FULL HEX M12		4
	S029	M	421038		Nut Full Hex M16		21
	S030	M	421036		Nut Full Hex M8		4
	S035	M	268064		Key Steel 32x16		1
	S036	M	268065		Key Steel 18x9		1
	S037	M	268027		Key Steel 14x9		3
	S038	M	268011		Key Steel 8x7		2
02-05			904G210		MECH END DOUBLE CASTOR TRAVERSE	324G421	1
	B001	E	662085		Siemens Geared Combi Mastercm		1
	B002	M	225279		Chain Wheel 3/4"P x 20T		1
	B003	M	225280		Chain Wheel 3/4"P x 25T		1
	B004	M	225292		Chain 3/4"P Duplex		1
	B005	M	225282		Connecting Link 3/4"P Duplex No 26		3
	B006	M	202016		BEARING MFC 50		8

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B007	M	225283		Chain Wheel 3/4"P x 20T		2
	B008	M	225293		Chain 3/4"P Duplex		1
	B009	M	225285		Connecting Link 3/4"P Duplex No 30		1
	B010	M	200605		Bearing Deep Ball Groove with 1 Seal 6310-RS1		4
	B101	M	421888		Screw Countersunk Head Socket M10x50		70
	M001	M	320G409		Driven Castor Shaft (Primary)		1
	M002	M	324G418	ITEM.8	Vee Track Mech End		1
	M003	M	320G412		Vee Castor		2
	M004	M	325G094		Double Castor housing (K4000)		2
	M005	M	321G529		STOP BLOCK		2
	M006	M	323G014		CASTOR PACKER		2
	M007	M	324G417	ITEM.8	TRACK INSERT		4
	M009	M	303G529		SPACER		8
	M010	M	312G785		SPACING BLOCK		1
	M011	M	312G786		SPACING BLOCK		1
	M012	M	320G410		Driven Castor Shaft (Secondary)		1
	M013	M	320G998		PLAIN CASTOR WHEEL		2
	M014	M	324G419	ITEM.8	MECH TRACK BASE		1
	M015	M	324G420	ITEM.8	MECH TRACK PLAIN		1
	S001	M	421527		Screw Caphead M16X110		4
	S002	M	421059		PLAIN WASHER M16		16
	S003	M	421251		Screw HEXHD Setscrew M16x50		8
	S004	M	421027		NUT FULL HEX M10		6
	S005	M	421038		Nut Full Hex M16		4
	S006	M	268017		Key Steel 12x8		1
	S007	M	268018		Key Steel 14x9		4
	S008	M	421114		SOCKET CAP SCREW M10x40		16
	S009	M	421029		PLAIN WASHER M10		16
	S010	M	421161		SOCKET CAP SCREW M10x50		4
	S011	M	421341		Screw HEXHD Setscrew M10x120		2
	S014	M	421173		GRUB SCREW M8x12		1
	S015	M	421583		GRUB SCREW M12x12		6
02-06			902G315		Single Castor Assembly (Plain Wheel)	320G418	1
	B001	M	202218		Bearing Flanged Cartridge MFC 50		2
	M001	M	303G909		CASTOR HOUSING		1
	M002	M	303G910		PLAIN CASTOR		1
	M003	M	303G934	IT.A	CASTOR PACKER		1
	M004	M	303G934	IT.B	CASTOR PACKER		1
	M005	M	303G934	IT.C	CASTOR PACKER		1
	M006	M	303G942		FREE RUNNING C/SHAFT		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M007	M	311G839		SPACER		2
	S001	M	421110		Screw HEXHD Setscrew M10 x 35		8
	S002	M	421251		Screw HEXHD Setscrew M16x50		4
	S003	M	421029		PLAIN WASHER M10		8
	S004	M	421059		PLAIN WASHER M16		4
	S005	M	268018		Key Steel 14x9		1
02-07			902G428		Single Castor Assembly (Vee Rail)	320G417	1
	B001	M	202218		Bearing Flanged Cartridge MFC 50		2
	M001	M	303G909		CASTOR HOUSING		1
	M002	M	320G412		Vee Castor		1
	M003	M	303G934	IT.A	CASTOR PACKER		1
	M004	M	303G934	IT.B	CASTOR PACKER		1
	M005	M	303G934	IT.C	CASTOR PACKER		1
	M006	M	303G942		FREE RUNNING C/SHAFT		1
	M007	M	311G839		SPACER		2
	S001	M	421110		Screw HEXHD Setscrew M10 x 35		8
	S002	M	421251		Screw HEXHD Setscrew M16x50		4
	S003	M	421029		PLAIN WASHER M10		8
	S004	M	421059		PLAIN WASHER M16		4
	S005	M	268018		Key Steel 14x9		4
	S006	M	214025		Nipple Grease M6		1
02-08			904G612		Filton Gland pipe work	325G047	1
	B001	M	4231164		Brewers Hose Assy 2" NB coned ends 1930mm lg		2
	B002	M	4231166		Brewers Hose Assy 1.1/2" NB coned ends 1800mm lg		1
	B003	M	4231165		Brewers Hose Assy 2" NB coned ends 1200mm lg		1
	B004	M	4231167		Brewers Hose Assy 1.1/2" NB coned ends 1780mm lg		3
	B005	M	4231168		Brewers Hose Assy 2" NB coned ends 1030mm lg		1
	B006	M	423011		2" Bspt male to 2" BSP Coned faced hex nipple		8
	B007	M	423012		1.1/2" BSPT Male to 1.1/2" BSP Coned faced Hex Nip		8
	B008	M	423013		Elbow Female 2" BSP M.I.		3
	B009	M	423485		Elbow Female 1.1/2" BSP M.I.		2
	B010	M	423015		Nipple Crane 2"BSP M.I. Hex 144/50		4
	B011	M	423498		Socket Crane M.I. 1.1/2" BSP Type 177/40		1
	B012	M	423022		Elbow Male/Female 1.1/2" BSP M.I.		1
	B013	M	423017		Red/Socket Crane 3"x2"BSP M.I. 179/80-50		1
	B014	M	423018		Red/G Bush Crane 3"-2"BSP M.I. 140/80-50		1
	B015	M	423086		Nipple Crane 1.1/2"BSP M.I. Hex 144/40		2
	B016	M	423028		Red/Tee on Run Crane 2"x1/2"x2"BSP 161/50-15-50		2
	B017	M	423098		Red on Run Tee TP161 Crane 161/40/15/40		2
	B018	M	423025		Nipple Crane 1/2"BSP M.I. Hex 144/15		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B019	M	423026		Hose Adaptor 1/2" BSP x 1/2" Hose		4
	B020	M	424017		Valve Gate 1/2" BSP Brass		4
	B021	M	423024		Elbow Male/Female 1/2" BSP M.I.		4
	B022	M	423016		Ball Valve 2"BSP Female c/w Lock Shield D171LS		1
	B023	M	2442060		U-Bolt UBS10		2
	B024	M	2442070		U-Bolt (4" NB)		1
	M001	M	325G659		Hose Support Bracket		1
	M002	M	325G695		Filton Hose Support Bracket		1
	S001	M	421149		SOCKET CAP SCREW M10x20		2
	S002	M	421029		PLAIN WASHER M10		8
	S003	M	421035		SOCKET CAP SCREW M10x25		3
	S004	M	421027		NUT FULL HEX M10		3
02-10			904G195		Hinged Shield Pivot Leadthrough (Adjustable)	406G318	1
	B01	M	200562		Bearing Deep Groove Ball 6013		2
	B02	M	239191		Seal Simmering BA72-90-10		2
	B03	M	242041		LOCKNUT KM13		1
	B04	M	242042		LOCKWASHER MB13		1
	B05	M	239248		Seal O-Ring RM0646-24 Nitrile		1
	B06	M	239193		Seal O Ring RM0945-30 Nitrile		1
	B07	M	239194		Seal O Ring RM1045-30 Nitrile		1
	B08	M	239195		Seal O Ring RM1443-57 Nitrile		1
	B09	M	240M275		M12 STUD WDS 405-204x90 LG.		6
	M01	M	406G319		Actuating Shaft		1
	M02	M	406G320		Bearing Housing		1
	M03	M	406G321		Spacer		1
	M04	M	406G322		Coupling		1
	M05	M	406G323		Key		1
	M06	M	312G673		BEARING HSNG END COV		1
	M07	M	312G674		BEARING HSNG END COV		1
	M08	M	314G081		INNER SHAFT SLEEVE		1
	M09	M	314G082		SHAFT SLEEVE CAP		1
	M10	M	406G353		Spacer Plate		1
	S01	M	421018		SOCKET CAP SCREW M10x35		6
	S02	M	421121		SOCKET CAP SCREW M6 x 16		3
	S03	M	421019		SOCKET CAP SCREW M6 x 20		3
	S04	M	421020		SCREW COUNTERSUNK HEAD SOCKET M5 x 20		3
	S05	M	421955		Bolt HEXHD M6x35		2
	S06	M	421042		NUT FULL HEX M12		6
	S07	M	421029		PLAIN WASHER M10		6
	S08	M	421030		PLAIN WASHER M12		6

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S09	M	421034		Washer Plain M6		3
02-11			904G788		Unwind / Rewind Servo Motor Layon Drive	407G972	1
	B1	M	225396		Sprocket Polychain 8M-34S-21		2
	B2	M	221822		Belt Timing Poly Chain 8M-1000-21		1
	B3	E	664393		AC MOTOR SERVO BERGER LAHR SER3910		2
	B4	M	224350		Bush Taperlock 1215 Bore 20 W/- Std Key TL1215/20		1
	B5	M	224016		TAPERLOCK BUSH 1610-20		1
	M1	M	407G974		REWIND LAYON DRIVE SUPPORT		1
	M2	M	407G975		UNWIND LAYON DRIVE SUPPORT		1
	M3	M	408G281		Drive Adaptor		2
	M4	M	408G283		Motor Mounting Flange		2
	S1	M	421104		SOCKET CAP SCREW M16x50		8
	S2	M	421470		Screw Caphead M12 x 70		4
	S3	M	421722		SOCKET CAP SCREW M4x20		8
	S4	M	421542		Screw Caphead M6X65		8
	S5	M	421042		NUT FULL HEX M12		4
	S6	M	421030		PLAIN WASHER M12		8
	S7	M	421059		PLAIN WASHER M16		8
	S8	M	421194		Washer Plain M4		8
	S9	M	421034		Washer Plain M6		8
03					Source End Sub Assy		
03-02			903G610		Arrg't of Busbars & Transformers	323G087	1
	M001	M	323G088	2450	SUPPORT BEAM		1
	M002	M	323G089		LIVE BUSBAR		26
	M003	M	323G090		NEUTRAL BUSBAR		5
	M004	M	323G091		NEUTRAL BUSBAR		5
	M005	M	323G092		NEUTRAL BUSBAR		4
	M006	M	323G093		NEUTRAL BUSBAR		1
	M007	M	323G094	ITEM1	LIVE BUSBAR CLAMP BLOCK		26
	M008	M	323G094	ITEM2	LIVE BUSBAR CLAMP BLOCK		26
	M009	M	323G095	ITEM1	NEUTRAL BUSBAR CLAMP BLOCK		5
	M010	M	323G095	ITEM2	NEUTRAL BUSBAR CLAMP BLOCK		5
	M011	M	323G096		LIVE BUSBAR SUPPORT BLOCK		26
	M012	M	323G097		NEUTRAL BUSBAR SUPPORT BLOCK		5
	M013	M	323G098		IN-HOUSE TEST NEUTRAL BUSBAR		5
	M014	M	323G334		SPACER		5
	M015	M	323G100		LIVE CONNECTING CABLE		52
	S001	M	421478		Screw HexHd Setscrew M12x40 Brass		52
	S002	M	421801		Screw HEXHD Setscrew M10x40 Brass		20
	S003	M	421201		Screw HEXHD Setscrew M10x35 Brass		102

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S004	M	421146		SOCKET CAP SCREW M8x30		4
	S005	M	421286		SOCKET CAP SCREW M6x75		62
	S006	M	421825		Screw Caphead M6X45		31
	S007	M	421145		SOCKET CAP SCREW M6x30		20
	S008	M	421019		SOCKET CAP SCREW M6 x 20		104
	S009	M	421205		Nut Full Hex M12 Brass		52
	S010	M	421202		Nut Brass Full M10		122
	S011	M	421057		Nut Full Hex M6		31
	S012	M	421206		Washer Brass M12		104
	S013	M	421203		Washer Brass M10		244
	S014	M	421054		PLAIN WASHER M8		4
	S015	M	421034		Washer Plain M6		62
	S016	M	421073		Washer Spring M12		52
	S017	M	421021		Washer Spring M10		122
04					Evaporation Souce Sub Assy		
04-01			904G265		K4000 Evaporation Source 2005	406G456	1
	B001	M	268015		Intermetallic Evaporation Boat 150x30x10		26
	B002	M	205097		CAM FOLLOWER KR30		14
	B003	M	239404		Seal, Viton Face Seal		8
	B004	M	239384		Seal, Viton Face Seal		52
	B005	M	235184		Compression Spring M60-75-12		26
	B006	M	203260		Ringfeeder 28x55		1
	B007	M	200634		Flange Bearing Unit		2
	B008	M	203212		BUSH GLACIER MB 3040 DU		1
	B009	M	203079		BUSH DRY GLACIER MB2020 DU		2
	B010	M	421437		Shoulder Screw Socket Head 16mm (M12) x 30 LG		3
	B011	M	240446		CHAIN 510-205X300LG		1
	B012	M	421652		Screw Shoulder Screw 8mm (M6) x 16lg		1
	B013	E	724007		TERRY CLIP 3/4 INCH		1
	B014	M	423671		Blanking Plug 3/4"BSPP		19
	B016	M	2441788		SELF-ALIGNING PAD, M12		4
	B017	M	268014		Coated Glass Fabric PTFE 0.5		1
	B018	M	427078		Tube 1/2" P.O. Multi purpose hose Blue		1
	B019	M	214020		Nipple Straight Grease 1/8"BSP GN1/8BSP		4
	M001		403G533		Source Water Cooled Back Plate		1
	M002	M	407G321		Source Water Cooled Front Plate		1
	M003	M	407G268	2450	Source Water Cooled Neutral Plate		1
	M004		403G529	2450	Source Water Cooled Live Side Plate		1
	M005		403G616	IT.2	Live End Block		26
	M006		403G534	2450	Source Water Cooled Trough		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M007	M	407G324		Trough Cover		1
	M008		403G590		Source Spacing Plate		6
	M009		403G624		Live Bracket Spacing Plate		14
	M010		403G757		Location Barrel		104
	M013		404G099	2450	Wire Feed Cover		1
	M014		404G102		Shutter Support Rail		1
	M015		404G065		Shutter Support Rail		1
	M016		404G104		Shutter Wear Plate		2
	M017		403G587	2450	Source Live End Mounting Beam		1
	M018		403G531		Live Extension Block		26
	M019		403G673		Pivoting Cold End		26
	M020	M	320G648		Pivot Block		26
	M021	M	311G194		INSULATION PAD		26
	M022	M	320G650		Pivot Shaft Support		52
	M023	M	320G863		Pivot Pin		26
	M024		403G737		Live End Adjusting Plate		26
	M025	M	311G193		SPRING LOCATOR		26
	M026	M	320G882		STOP		26
	M027		403G649	2450	Shutter Torque Shaft		1
	M028	M	406G468		Shutter Arm Front Right		1
	M029	M	406G469		Shutter Arm Rear Right		1
	M030	M	406G467		Shutter Lifting Lever		1
	M031		403G559	2450	Water Cooled Shutter		1
	M032	M	406G470		Shutter Yolk		1
	M033		403G627		Shutter Pivot Bush		3
	M034		403G628		Shutter Wheel		2
	M035		403G633	2450	Water Cooled Viewing Flap		1
	M036	M	406G475		Viewing Flap Pivot Bracket		1
	M037	M	406G476		Viewing Flap Drive Bracket		1
	M038		404G106		Viewing Flap Pivot Pin		1
	M039		403G947		Source Mounting Bracket Front		1
	M040		403G948		Source Mounting Bracket Rear		1
	M041		403G950		Source Lift Bracket Front		1
	M042		403G951		Key		4
	M043		403G953		Insulation Plate		2
	M044		403G954		Insulation Bush		8
	M045		403G955		Clamp Washer		16
	M046		403G959		Source Lift Bracket Rear		1
	M047		403G960		Source Support Bracket Rear		1
	M048	M	406G472		Locking Pin		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M049		404G978		CLAMP WASHER		2
	M050		405G001		TROUGH PLATE LATCH		1
	M051	M	406G477		SHUTTER LOCKING ARM		1
	M052		405G041		SHUTTER HANDLE		1
	M053		405G158		Hose Guide		1
	M054	M	405G257		Lifting Screw		4
	M055	M	405G353		Adjusting Block		2
	M056	M	405G354		Adjusting Block		2
	M057	M	408G321	2450	Stepper Drive Support		1
	M058	M	406G154	IT1	Clamp Bar		5
	M059	M	406G154	IT2	Clamp Bar		1
	M060	M	406G154	IT3	Clamp Bar		1
	M061	M	406G165		Edge Retainer		6
	M062	M	407G166	2450	Debris cover		1
	M063	M	407G262	IT.2	END BLOCK - WATER COOLED		26
	M064	M	406G166		Kant Seal 16 x 2 Viton		52
	M065	M	407G291		Stop		1
	M066	M	408G176		latch		1
	S001	M	421368		SOCKET CAP SCREW M5x25		8
	S002	M	421121		SOCKET CAP SCREW M6 x 16		131
	S003	M	421219		SOCKET CAP SCREW M6 x 40		104
	S004	M	421122		SOCKET CAP SCREW M10x30		36
	S005	M	421045		SOCKET CAP SCREW M10x45		34
	S006	M	421931		Screw Caphead M10 x 55 S.S.		52
	S007	M	421165		SOCKET CAP SCREW M10x75		8
	S008	M	421452		SOCKET CAP SCREW M12 x 25		53
	S009	M	421762		Screw Socket M12x30 Countersunk Head		2
	S010	M	421195		SOCKET CAP SCREW M12x40		28
	S011	M	421310		SOCKET CAP SCREW M12x65		37
	S012	M	421932		Screw Caphead M10 x 110 S.S.		52
	S013	M	421418		Screw Socket Countersunk Head M6 x 16		6
	S014	M	421226		GRUB SCREW M6x10		52
	S015	M	421259		SOCKET CAP SCREW M6x70		12
	S016	M	421245		SOCKET CAP SCREW M5x10		2
	S017	M	421058		Screw HEXHD Setscrew M10x45		4
	S018	M	421057		Nut Full Hex M6		24
	S019	M	421175		Washer Spring M5		12
	S020	M	421034		Washer Plain M6		181
	S022	M	421029		PLAIN WASHER M10		40
	S023	M	421030		PLAIN WASHER M12		111

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S024	M	421036		Nut Full Hex M8		26
	S025	M	421027		NUT FULL HEX M10		4
	S027	M	421653		SOCKET CAP SCREW M10x100		28
	S028	M	421526		Screw Caphead M12x45		3
	S029	M	421470		Screw Caphead M12 x 70		2
	S030	M	421402		Screw Caphead M12x60		5
	S033	M	422042		STUD M8 x 50 Lg		26
	S034	M	559205		KEYSTEEL 8X7		1
	S035	M	421387		Screw HEXHD Setscrew M6x25		27
	S036	M	421091		Screw Socket Cap Head M6 x 25		12
	S037	M	421403		SOCKET CAP SCREW M12 x 50		2
	S040	M	421073		Washer Spring M12		1
	S041	M	421042		NUT FULL HEX M12		1
04-03			904G398		2mm Wirefeed unit (non lubricated)	324G850	26
	B001				Stepper Motor SM87-1-18-ML-E-3-V-Z20		1
	B002	M	251095		BOSTON GEAR 28 TEETH H3228		2
	B003	M	203297		Iglidur Bush		6
	B004	M	203311		Spacer 6.5 x 10 x 25		4
	B005	M	204135		Insulating Sleeve		2
	M001	M	324G849		Drive Body for Non-Lubricated Wire Feed		1
	M002	M	405G328		Outer Side Plate (Non-Lubricated Wire Feed)		1
	M003	M	320G342		DRIVE ROLLER - 2MM WIRE		1
	M004	M	320G341		DRIVEN ROLLER - 2MM WIRE		1
	M005	M	324G767		Lead In Bush For 2mm Dia Wire Feed		1
	M007	M	407G073		Mounting Bracket for Non-Lubricated Wire Feed		1
	M008	M	303G866		SPUR GEAR 86T		1
	M009	M	304G350		SPUR GEAR 20T		1
	M010	M	303G865		BOSS		1
	M013	M	407G074		Wire Feed Insulator Plate		1
	S001	M	421091		Screw Socket Cap Head M6 x 25		4
	S002	M	421287		PIN SELOK 2.5DIA X 20		2
	S003	M	234124		CIRCLIP EXTERNAL 1400-0120		2
	S004	M	421288		GRUB SCREW M4x10		2
	S005	M	421290		Washer BRASSM8		2
	S006	M	421121		SOCKET CAP SCREW M6 x 16		4
	S007	M	421023		Screw Countersunk Head Socket M5x16		4
	S008	M	421034		Washer Plain M6		4
	S009	M	421576		GRUB SCREW M4x8		1
	S011	M	421259		SOCKET CAP SCREW M6x70		2
04-04			903G838		Wire Feed Spout Assembly	323G482	26

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M001	M	322G955		Fixed Spout		1
	M002	M	323G481		Removable Spout		1
	M003	M	325G508		Inner Shield Wire Feed Unit		1
	S001	M	421108		Nut Full Hex St.St. M10		1
	S002	M	421133		SOCKET CAP SCREW M5x16		2
04-09			903G563		Source Mechanical Lead Throughs	404G210	1
	B001	M	210909		Cylinder DNC-100-100-PPV-A		1
	B001	M	211373		REED SWITCH & CONNECTOR		2
	B001	M	211374		Cable Connector		1
	B001	M	211370		CYLINDER SWIVEL FLANGE SNCB-100		1
	B003	M	427004		Rod Clevis with Pin & Locknut 6147		1
	B004	M	200253		BEARING 61908		1
	B005	M	200077		BEARING NO SEALS 6008		1
	B006	M	234005		CIRCLIP EXTERNAL 1400-0400		2
	B007	M	233014		INTERNAL CIRCLIP D1300-0520		1
	B008	M	239347		Seal Simmerring BA40-52-7-72NBR		2
	B009	M	239198		Seal O Ring RM0745-30 Nitrile		1
	B010	M	239356		Seal O Ring RM0895-30Nitrile		3
	B011	M	423111		Elbow Adaptor Hobbs 8 O/D - 1/4" BSP		2
	B012	M	423038		Tubing Nut 8 OD		2
	B013	M	423039		Tubing Sleeve 8 OD		2
	B014	M	423113		Coned Locknut 1/4"BSP Enots 36-0540-02		2
	B015	M	423112		Seal 1/4"bsp Enots 34-0156-02		2
	B016	M	427113		Pneumatic Actuator Kinetrol 070-104B002000		1
	B017	M	234006		CIRCLIP EXTERNAL D1400-0600		2
	B018	M	2005474		BEARING 61912		2
	B019	M	239360		Seal Simmerring BABSL60-75-8NBR		2
	B020	M	233005		CIRCLIP INTERNAL D1300-0750		2
	B022	M	239357		Seal O Ring RM1095-30 Nitrile		2
	B023	M	203284		Bush Lenze ETP Express HYD		1
	B024	M	203067		BUSH PLAIN BEARING MB2030DU		2
	B025	M	203079		BUSH DRY GLACIER MB2020 DU		1
	M001	M	404G052		Flap Telescopic Universal Joint		1
	M002		404G038		Shutter Telescopic Universal Joint		1
	M003		404G039		SHUTTER ACTUATION ARM		1
	M004		404G156		CYLINDER MTG PIVOT BRACKET		1
	M005		404G050		INSULATION PLATE		1
	M006		404G051		INSULATION BUSH		4
	M007	M	403G794		Bearing Housing		1
	M008	M	402G339		Bearing spacer		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M009		404G053		LEAD THROUGH SHAFT		1
	M010	M	402G338		Seal Retaining Cap		1
	M011	M	402G340		Seal Dust shield		1
	M012	M	403G792		Spacer Ring Insulator		1
	M013	M	403G793		Flanged Bush Insulator		6
	M014	M	402G805		Kinetrol Adaptor		1
	M015		404G034		SHUTTER LEAD THROUGH SHAFT		1
	M016	M	402G418		Bearing Spacer		1
	M017		404G046		BEARING HOUSING		1
	M018	M	402G420		Outer Seal Cap		1
	M019	M	402G421		Spacer		1
	M020	M	403G793		Flanged Bush Insulator		6
	M021		404G045		INSULATION PAD		1
	S001	M	421278		SOCKET CAP SCREW M8x45		6
	S002	M	421425		Screw Caphead M8x50		4
	S003	M	421164		PLAIN WASHER M5		4
	S004	M	421416		SOCKET CAP SCREW M5x30		4
	S005	M	421030		PLAIN WASHER M12		4
	S006	M	421133		SOCKET CAP SCREW M5x16		4
	S007	M	421402		Screw Caphead M12x60		4
04-10			904G216		Source Electrical & Water Lead thro	406G471	1
	B001	M	427078		Tube 1/2" P.O. Multi purpose hose Blue		2
	B002	M	427098		Tube 3/4" P.O. Multi purpose hose Blue		1
	B003	M	203261		Plain Clamping Ring GR10-19.05 Plain Bore 19.05mm		26
	B004	M	423693		Stauff Pipe Clamp 3190PP		26
	B005	M	423712		Stauff Pipe Clamp 3254PP		5
	B006	M	239243		Seal O Ring RM0265-30 Nitrile		10
	B007	M	239229		Seal O Ring RM0545-30Nitrile		1
	B008	M	239244		Seal O Ring BS210 Nitrile		52
	B010	M	427073		Swagelok Male Hose Adaptor		6
	B011	M	427060		Tube Adaptor 12 OD x 1/2" Hose		8
	B012	M	423806		12mm-3/8" Male Connector		2
	B013	M	427143		MALE CONNECTOR 3/4" TO 3/4" BSPP		3
	B014	M	427147		TUBE ADAPTOR 3/4" TUBE TO 3/4" HOSE		6
	B015	M	427144		MALE CONNECTOR 3/4" TO 1/2" BSPP		2
	B016	M	427148		MALE ELBOW 12mm TO 1/2" BSPP		1
	B017	M	423441		Q.C. Coupling Brass 1/2"BSPP		2
	B018	M	427083		SWAGELOK MALE HOSE ADAPTOR		34
	B019	M	423442		Q.C. Nipple 1/2"		2
	B020	M	427163		Swagelok Male Connector B-12MO-1-12RT		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B021	M	427082		SWAGELOK SEALING WASHER S-8-RS-2V		3
	B022	M	427164		SWAGELOK SEALING WASHER S-12-RS-2V		3
	B023	M	427215		Male Adaptor 12 OD x 1/2" BSPP		1
	B024	M	423588		12mm Union Elbow		1
	B025	M	427183		Gasket ISO Parallel 3/8"		2
	B026	M	427262		Goodridge Fire Proof Sleeve 3/4" ID		4
	B027	M	427282		Goodridge Fire Proof Sleeve 1" ID		4
	B028	M	423253		Jubilee Clip Number 00		3
	B029	M	423146		Jubilee Clip Number 1		7
	B030	M	427290		Positionable Male Elbow 3/4" OD x 3/4" BSPP		1
	M000	M	325G001		VESSEL WATER LEADTHROUGH PIPING SCHEME, Ref Only		1
	M001		404G005		Live Lead Through		26
	M002		404G198		Neutral Water Return Bush		1
	M003	M	320G800		Connection Block		26
	M004		404G055	2450	Live Bar Support		1
	M005	M	320G732		Lead Through Plate		26
	M006	M	320G733		Lead Through Bush		52
	M007		404G199		Neutral Connection Block		5
	M008		404G209	IT2	LIVE BRAIDED CABLE		36
	M008		404G209	IT1	LIVE BRAIDED CABLE		36
	M009		404G332		Neutral Lead Through		5
	S001	M	421035		SOCKET CAP SCREW M10x25		10
	S002	M	421219		SOCKET CAP SCREW M6 x 40		62
	S003	M	421138		SOCKET CAP SCREW M8x25		36
	S004	M	421404		Screw Caphead M8x45		4
	S005	M	421019		SOCKET CAP SCREW M6 x 20		52
	S006	M	421091		Screw Socket Cap Head M6 x 25		62
	S007	M	421290		Washer BRASSM8		118
	S008	M	421218		Screw HEXHD Setscrew M8x25 Brass		118
04-11			903G586		WIRE REEL SUPPORT	404G298	1
	B001	M	268087		WIRE REEL 10KG 2.0mm DIA		26
	B002	M	2441729		CASTOR FLEXELLO		4
	M001		403G567		WIRE REEL BOBBIN		52
	M002		403G568	2450	WIRE REEL SUPPORT FABRICATION		1
	M003		403G569		SPACER		54
	M004		403G570		WIRE REEL SUPPORT BRACKET		3
	M005		403G571	2450	WIRE REEL SUPPORT BAR		2
	M006		403G572		ENDPLATE		4
	M007		404G064		SHOULDER BOLT		2
	S001	M	421121		SOCKET CAP SCREW M6 x 16		6

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S002	M	421182		SOCKET CAP SCREW M8x16		16
	S003	M	421249		M8 X 20MM SOCKET HEAD COUNTERSUNK SCREW		4
	S004	M	421034		Washer Plain M6		6
04-13			904G878		Source Trial Assembly	325G710	1
	B001	M	427305		Union Elbow 1/2" OD		2
	B002	M	427075		Tube Adaptor 1/2" OD x 1/2" Hose		8
	B003	M	427368		Union Tee 1/2" OD		2
	M001	M	325G711		Shield Plate		1
	M002	M	325G712		Support Angle		2
	M003	M	325G713		Front Cranked Link		1
	M004	M	325G714		Rear Cranked Link		1
	M005	M	325G715	IT1	Link Plate		1
	M006	M	325G715	IT2	Link Plate		1
	M007	M	325G716	IT1	Shutter Pivot Boss		1
	M008	M	325G716	IT2	Shutter Pivot Boss		1
	M009	M	325G717		Shutter Arm Pivot Boss		2
	M010	M	325G718		Pivot Bush		4
	M011	M	325G719		Double Pivot Bush		2
	M012	M	325G720		Pivot Washer		8
	M013	M	325G721		Pivot Pin		2
	M014	M	325G722		Clamp Washer		12
	M015	M	325G723		Guide Shaft		2
	M016	M	325G724	IT1	Shutter Guide Plate		1
	M017	M	325G724	IT2	Shutter Guide Plate		1
	M018	M	325G725		Location Plate		1
	M019	M	325G726		Locking Pin		2
	M020	M	325G971		Shield Plate - Wavy Edge		1
	M021	M	325G972		Stop Block		3
	S001	M	421051		Screw Socket Countersunk Head M6 x 12		4
	S002	M	421418		Screw Socket Countersunk Head M6 x 16		8
	S003	M	421026		Screw Socket Countersunk Head M6 x 25		5
	S004	M	4211385		Screw Socket Countersunk Head M6 x 45		6
	S005	M	421057		Nut Full Hex M6		5
	S006	M	421034		Washer Plain M6		10
	S007	M	421022		Washer Spring M6		5
	S008	M	421157		Screw Socket Csk Head M8 x 25		2
	S009	M	4211218		Screw Socket Csk Head M12 x 30		2
	S010	M	421049		SOCKET CAP SCREW M8x20		6
	S011	M	421569		Screw Caphead M8x100		7
	S012	M	421036		Nut Full Hex M8		14

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S013	M	421054		PLAIN WASHER M8		20
05					Source Power Sub Assy		
05-01			903G528		Source Transformers	D1312-4	1
	001	E	709293		TRANSFORMER 380:13V @ 1000A/40A		26
	002	E	616858		HAN-MODULAR 2 WAY MODULE - FEMALE - 100A		26
	003	E	616859		HAN-MODULAR 6 WAY FRAME HOUSING -FEMALE (HINGED FR		9
	004	E	616719		HARTING HAN 24B HIGH HOOD M40 SIDE ENTRY		9
	005	E	616302		HAN-MODULAR DUMMY MODULE		1
05-02			903G540		Source & Vacuum Control Cubicle		1
	001	E	244G825		2450 STD SOURCE CUBICLE		1
	002	E	429G041		ROTARY PUMP 3 ELECTRICAL BUILD OPTION		1
	003	E	429G042		POLYCOLD 2 ELECTRICAL BUILD OPTION		0
	004	E	429G043		ISOLATOR ELECTRICAL BUILD OPTION		0
	005	E	429G044		BOOSTER PUMP 4 ELECTRICAL BUILD OPTION		0
	006	E	429G022		ZINC SOURCE ELECTRICAL BUILD OPTION		0
	010				ISSUE 2: ECO 06G00343 WIRE FEED STEPPERS ADDED		0
	011	E	650762		B&R POWER PANEL PP210 (WIRE FEEDER SOFTWARE)		1
	012	E	606090		B&R X2X LINK SUPPLY X67PS1300 24VDC		1
	013	E	606091		B&R X67SM2436 STEPPER MTR MODULE, 2 x 3AMP		13
	014	E	606093		B&R X67CA0X11.0002, X2X LINK CABLE		6
	015	E	606094		B&R X67CA0P30.0050, POWER CABLE 5m		9
	016	E	606095		B&R X67CA0P10.0002, POWER CABLE 0.2m		6
	017	E	606097		B&R X67CA0X31.0050, X2X CABLE 5m		2
	018	E	606098		B&R X67CA0X51.0020, X2X CABLE 2m		1
	019	E	606096		B&R X67AC2A00 M12 MALE MOTOR PLUG, PG7		0
	020	E	606092		B&R X67SM4320 STEPPER MTR MODULE, 4 x 1AMP		1
	021				ISSUE 3 ECO 06G00431 : B&R ADDITIONS		0
	022	E	606105		B&R X67CA0X01.0010, X2X LINK CABLE		8
	023	E	606106		B&R X67AC0X01 M12 5 PIN MALE X2X PLUG		8
	024	E	606107		B&R X67AC0X21 M12 5 PIN FEMALE X2X PLUG		8
	025				Issue 4: ECO-06G0440 B&R Motor plugs changed		0
	026	E	606108		HIRSCHMANN ELST5012PG9, M12 MALE, 5 PIN, PG9		30
05-03			903G733		Source Misc Items		1
	001	E	603072		FLUORESCENT TUBE 8FT (125W)		3
	002	E	603073		FLUORESCENT LIGHT FITTING T8/T12		6
	003	M	421078		Terry Spring Clip 1.1/2"		6
	004	E	616817		HAN HARTING PG11 CABLE SEAL 6.5/12		26
	005	E	616644		HAN HARTING HA4 SOCKET (FEMALE) (10 of)		26
	006	E	616645		HAN HARTING HA3 HOOD TOP ENTRY		26
	007	E	606033		STEPPER MOTOR SM87-1-18-ML-E-3-V-Z267		26

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	009	E	677204		ZB4-BZ101 1 N/O CONTACT BLOCK ASSY		1
	010	E	677265		P/B OPERATOR GREEN (I) (FLUSH)		1
	011	E	652103		ASI 2 INPUT 2 OUTPUT MODULE 8 WIRE XAL-SZ1E		1
	012	E	652134		ASI 4AI [ij] MODULE CURRENT (4...20mA)		2
	013	E	652063		ASI EMS CONNECTION BASE		2
	014	E	616895		M12 PLUG 0/M12 SOCKET 90, 4WIRE 1M CABLE		7
	015	E	606065		STEPPER MOTOR SM87-3-18-M-V-6		1
	020				ISSUE 2: ECO 07G00074 STEPPER DRIVES INTO VESSEL		0
	021	E	616817		HAN HARTING PG11 CABLE SEAL 6.5/12		22
	022	E	616644		HAN HARTING HA4 SOCKET (FEMALE) (10 of)		22
	023	E	616645		HAN HARTING HA3 HOOD TOP ENTRY		22
	024	E	606105		B&R X67CA0X01.0010, X2X LINK CABLE		13
	025	E	606110		B&R X67CA0P20.0050, POWER CABLE 5m		23
	026	E	606111		B&R X67CA0A41.0020, STEPPER CABLE 2m		23
	027	E	616598		FC LANE 25 PIN SOCKET CODE 0		1
	028	E	616599		FC LANE 25 PIN PLUG CODE 0		1
	029	E	616603		FC LANE 25 PIN SOCKET CODE 1		1
	030	E	616604		FC LANE 25 PIN PLUG CODE 1		1
	031	E	616601		FC LANE CABLE OUTLET		6
	032	E	616602		FC LANE OUTLET GASKET		6
	033	E	616606		FC LANE 25 PIN SOCKET CODE 2		1
	034	E	616607		FC LANE 25 PIN PLUG CODE 2		1
05-04			903G562		Source Leadthrough Details		1
	001	E	616598		FC LANE 25 PIN SOCKET CODE 0		2
	002	E	616599		FC LANE 25 PIN PLUG CODE 0		2
	003	E	616600		FC LANE 25 PIN BULKHEAD CODE 0		2
	004	E	616601		FC LANE CABLE OUTLET		4
	005	E	616602		FC LANE OUTLET GASKET		4
	006				[-SKT5A, -SKT5G]		
	011	E	616603		FC LANE 25 PIN SOCKET CODE 1		1
	012	E	616604		FC LANE 25 PIN PLUG CODE 1		1
	013	E	616605		FC LANE 25 PIN BULKHEAD CODE 1		1
	014	E	616601		FC LANE CABLE OUTLET		2
	015	E	616602		FC LANE OUTLET GASKET		2
	016				[-SKT5B]		
	021	E	616606		FC LANE 25 PIN SOCKET CODE 2		1
	022	E	616607		FC LANE 25 PIN PLUG CODE 2		1
	023	E	616608		FC LANE 25 PIN BULKHEAD CODE 2		1
	024	E	616601		FC LANE CABLE OUTLET		2
	025	E	616602		FC LANE OUTLET GASKET		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	026				[-SKT5C]		
	031	E	616609		FC LANE 25 PIN SOCKET CODE 3		1
	032	E	616610		FC LANE 25 PIN PLUG CODE 3		1
	033	E	616611		FC LANE 25 PIN BULKHEAD CODE 3		1
	034	E	616601		FC LANE CABLE OUTLET		2
	035	E	616602		FC LANE OUTLET GASKET		2
	036				[-SKT5D]		
	041	E	616612		FC LANE 25 PIN SOCKET CODE 4		1
	042	E	616613		FC LANE 25 PIN PLUG CODE 4		1
	043	E	616614		FC LANE 25 PIN BULKHEAD CODE 4		1
	044	E	616601		FC LANE CABLE OUTLET		2
	045	E	616602		FC LANE OUTLET GASKET		2
	046				[-SKT5E]		
	051	E	616615		FC LANE 25 PIN SOCKET CODE 5		1
	052	E	616642		FC LANE 25 PIN PLUG CODE 5		1
	053	E	616617		FC LANE 25 PIN BULKHEAD CODE 5		1
	054	E	616601		FC LANE CABLE OUTLET		2
	055	E	616602		FC LANE OUTLET GASKET		2
	056				[-SKT5F]		
	061	E	616609		FC LANE 25 PIN SOCKET CODE 3		1
	062	E	616610		FC LANE 25 PIN PLUG CODE 3		1
	063	E	616611		FC LANE 25 PIN BULKHEAD CODE 3		1
	064	E	616601		FC LANE CABLE OUTLET		2
	065	E	616602		FC LANE OUTLET GASKET		2
	066	M	303G349	IT.B	LEADTHRO" FLANGE		1
	067	M	239250		Seal O Ring RM0793-57 Nitrile		1
	068	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	069	M	421034		Washer Plain M6		4
	070				[-SKT11A : CAMERA]		
	071	E	616612		FC LANE 25 PIN SOCKET CODE 4		1
	072	E	616613		FC LANE 25 PIN PLUG CODE 4		1
	073	E	616614		FC LANE 25 PIN BULKHEAD CODE 4		1
	074	E	616601		FC LANE CABLE OUTLET		2
	075	E	616602		FC LANE OUTLET GASKET		2
	076	M	303G349	IT.B	LEADTHRO" FLANGE		1
	077	M	239250		Seal O Ring RM0793-57 Nitrile		1
	078	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	079	M	421034		Washer Plain M6		4
	080				[-SKT11B : CAMERA]		0
	081	E	616615		FC LANE 25 PIN SOCKET CODE 5		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	082	E	616642		FC LANE 25 PIN PLUG CODE 5		1
	083	E	616617		FC LANE 25 PIN BULKHEAD CODE 5		1
	084	E	616601		FC LANE CABLE OUTLET		2
	085	E	616602		FC LANE OUTLET GASKET		2
	086	M	303G349	IT.B	LEADTHRO" FLANGE		1
	087	M	239250		Seal O Ring RM0793-57 Nitrile		1
	088	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	089	M	421034		Washer Plain M6		4
	090				[-SKT11C : CAMERA]		0
	151	E	616619		FC LANE 6 PIN SOCKET CODE 0		1
	152	E	616620		FC LANE 6 PIN PLUG CODE 0		1
	153	E	616623		FC LANE 6 PIN BULKHEAD CODE 0		1
	154	E	616601		FC LANE CABLE OUTLET		2
	155	E	616602		FC LANE OUTLET GASKET		2
	156	M	303G349	IT.B	LEADTHRO" FLANGE		1
	157	M	239250		Seal O Ring RM0793-57 Nitrile		1
	158	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	159	M	421034		Washer Plain M6		4
	160				[-SKT4A]		
	161	E	616624		FC LANE 6 PIN SOCKET CODE 1		1
	162	E	616625		FC LANE 6 PIN PLUG CODE 1		1
	163	E	616626		FC LANE 6 PIN BULKHEAD CODE 1		1
	164	E	616601		FC LANE CABLE OUTLET		2
	165	E	616602		FC LANE OUTLET GASKET		2
	166	M	303G349	IT.B	LEADTHRO" FLANGE		1
	167	M	239250		Seal O Ring RM0793-57 Nitrile		1
	168	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	169	M	421034		Washer Plain M6		4
	170				[-SKT4B]		
	171	E	616627		FC LANE 6 PIN SOCKET CODE 2		1
	172	E	616628		FC LANE 6 PIN PLUG CODE 2		1
	173	E	616629		FC LANE 6 PIN BULKHEAD CODE 2		1
	174	E	616601		FC LANE CABLE OUTLET		2
	175	E	616602		FC LANE OUTLET GASKET		2
	176	M	303G349	IT.B	LEADTHRO" FLANGE		1
	177	M	239250		Seal O Ring RM0793-57 Nitrile		1
	178	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	179	M	421034		Washer Plain M6		4
	180				[-SKT4C]		0
	181	E	616630		FC LANE 6 PIN SOCKET CODE 3		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	182	E	616631		FC LANE 6 PIN PLUG CODE 3		1
	183	E	616632		FC LANE 6 PIN BULKHEAD CODE 3		1
	184	E	616601		FC LANE CABLE OUTLET		2
	185	E	616602		FC LANE OUTLET GASKET		2
	186		404G415		ELECTRICAL LEAD THROUGH		1
	187	M	239250		Seal O Ring RM0793-57 Nitrile		1
	188				[-SKT12 : STROBE]		0
	196	M	303G349	IT.C	LEADTHRO" FLANGE		3
	197	M	239250		Seal O Ring RM0793-57 Nitrile		3
	198	M	4211314		M6 X 25 SKT CAP HEAD SCREW		12
	199	M	421034		Washer Plain M6		12
	200				[Blank - 10 Flanges to cover less wire feeds]		
06					Web Transport Sub Assy		
06-04			902G336		150MM/6"DIA Mech Expanding Shaft	320G823	3
	B001	M	318G018	2450	MECH EXPANDING SHAFT		1
	B002	M	200583		Spherical Roller Bearing 22214CC		2
	B003	M	234122		CIRCLIP EXTERNAL 1400-0700		4
	M001	M	303G810		BEARING SHIELD		4
06-05			904G702		Bowed Roller Driven Apex	404G635	1
	B001	E	620066		Pot Mounting Kit Farnell		1
	B002	E	620065		FARNELL POTENTIOMETER		1
	B003	M	232235		Universal Joint Telescoping Bore 1/8"		1
	M001	M	325G282	2450	Kickert Bowed Roller Driven Apex		1
	M002	M	404G639		Packer		1
	M003	M	318G392		POS"N IND"TR JOURNAL		1
	M004	M	318G393		POTENTIOMETER COVER		1
	M005	M	318G394		POTENTIOMETER MT PLT		1
	S001	M	421161		SOCKET CAP SCREW M10x50		4
	S002	M	421160		SOCKET CAP SCREW M10x90		4
	S003	M	421703		SOCKET CAP SCREW M4x16		2
	S004	M	421172		SOCKET CAP SCREW M5x12		8
	S005	M	421029		PLAIN WASHER M10		8
	S006	M	421194		Washer Plain M4		2
	S007	M	421164		PLAIN WASHER M5		8
06-06			902G342		156 DIA Idler Roller Tendency Driven No6	320G684	1
	B001	M	200571		BEARING SELF ALIGNING BALL 2210E		1
	B002	M	200572		Carb BEARING C2210 TN9		1
	B003	M	200573		Bearing Self Aligning Ball 1216		1
	B004	M	200574		Carb BEARING C2216		1
	B005	M	242001		LOCKNUT KM9		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B006	M	242105		LOCKWASHER MB9		1
	M001	M	320G685	2450	Idler Roller Body		1
	M002	M	320G686	2450	Idler Roller Shaft		1
	M003	M	320G687		Collar		2
	M004	M	320G688		Retaining Ring		2
	M005	M	320G569		Inboard & Outboard Bearing Housing		2
	M006	M	320G570		Bearing Retaining Cap		2
	M007	M	320G572		Bearing Retaining Spacer		1
	M008	M	320G571		Bearing Retaining Washer		1
	S001	M	421035		SOCKET CAP SCREW M10x25		6
	S002	M	421146		SOCKET CAP SCREW M8x30		2
	S003	M	421033		SOCKET CAP SCREW M6x12		10
	S004	M	421147		SOCKET CAP SCREW M4 x 12		12
	S005	M	421148		GRUB SCREW M8x30		4
	S006	M	214024		Nipple 45Deg Grease M6		2
	S007	M	214025		Nipple Grease M6		2
	S008	M	421021		Washer Spring M10		6
	S009	M	421103		Washer Spring M8		2
	S010	M	421227		Washer Spring M4		12
06-07			902G343		6"O/D SP'L Spreader Roll No2	321G126	1
	B001	M	200502		BEARING NO SEALS 2206E		1
	B002	M	200608		CARB BEARING C2206		1
	M001	M	321G125	2450	6" O/D Special Spreader Roller		1
	M002	M	321G083		Inboard & Outboard Bearing Housing		2
	M003	M	321G081		Bearing Retaining Cap		2
	M004	M	321G082		Bearing Retainer Washer		2
	S001	M	421035		SOCKET CAP SCREW M10x25		6
	S002	M	421145		SOCKET CAP SCREW M6x30		4
	S003	M	421033		SOCKET CAP SCREW M6x12		6
	S004	M	214024		Nipple 45Deg Grease M6		2
	S005	M	421021		Washer Spring M10		6
	S006	M	421022		Washer Spring M6		4
06-08			904G684		Layon Assy Rewind (No.13 & 14)	407G158	1
	B001	M	325G018	2450	KICKERT BOWED ROLLER MANUAL APEX RUBBER COVER		1
	B002	M	200502		BEARING NO SEALS 2206E		1
	B003	M	200608		CARB BEARING C2206		1
	B004	M	2061067		Runner Block - Igus DryLin WJ200UM-01-20		8
	B005	M	2061066		Guide Rail - Igus DryLin WS-20-80 x 800 Lg		2
	B006	E	698208		LIMIT SWITCH PARALLEL ROLLER 1NO/1 NC (14CE SERS)		1
	B007	M	214025		Nipple Grease M6		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M001	M	321G091	2450	148 DIA ALUMINIUM ROLLER		1
	M002	M	321G090		BEARING BUSH		2
	M003	M	309G125		DRIVE BELT CLAMP		2
	M004	M	325G241		Internal Drive Layon Arm (Rewind) - REF ONLY		1
	M005	M	407G214		Bearing Housing		2
	M006	M	407G215		Cap		2
	M007	M	407G308		Stop		4
	M008	M	407G422	2450	Tie Beam		1
	M009	M	407G423		Guide Bracket		2
	M010	M	407G424	IT.1	Belt Clamp		1
	M011	M	407G424	IT.2	Belt Clamp		1
	M012	M	407G425		Packer		2
	M013	M	407G426		Striker		1
	S001	M	421035		SOCKET CAP SCREW M10x25		12
	S002	M	421526		Screw Caphead M12x45		4
	S003	M	421033		SOCKET CAP SCREW M6x12		2
	S004	M	421019		SOCKET CAP SCREW M6 x 20		2
	S005	M	421221		SOCKET CAP SCREW M6x35		8
	S006	M	421049		SOCKET CAP SCREW M8x20		28
	S007	M	421138		SOCKET CAP SCREW M8x25		24
	S008	M	421146		SOCKET CAP SCREW M8x30		8
	S009	M	421027		NUT FULL HEX M10		2
	S010	M	421042		NUT FULL HEX M12		4
	S011	M	421057		Nut Full Hex M6		4
	S012	M	421029		PLAIN WASHER M10		12
	S013	M	421030		PLAIN WASHER M12		8
	S014	M	421034		Washer Plain M6		10
	S015	M	421054		PLAIN WASHER M8		32
	S016	M	4211324		Screw HEXHD Setscrew M10x80		2
	S017	M	4211323		Screw HEXHD Setscrew M6x50		4
06-09			904G685		Internal Drive Layon Arm (Rewind)	325G241	1
	B001	M	200329		Bearing Deep Groove Ball 6206		1
	B002	M	242009		LOCKNUT KM6		1
	B003	M	242010		LOCKWASHER MB6		1
	B004	M	200578		Bearing Flanged Cartridge MFC40		1
	B005	M	239247		Seal Simmerring BA40-55-7		2
	B006	M	239239		Seal O Ring RM0993-57 Nitrile		1
	B007	M	239246		Seal O Ring RM0645-30Nitrile		1
	B008	M	239248		Seal O-Ring RM0646-24 Nitrile		1
	B009	M	239249		Seal O-Ring RM0295-30 Nitrile		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B010	M	233136		CIRCLIP INTERNAL 1300-62		1
	B011	M	225286		Sprocket Polychain 8M-50S-21		2
	B012	M	224276		Bush T/Lock Bore 40 with STD Keyway TL2012		2
	B013	M	221764		Belt Polychain 8M-1792-21		4
	B014	M	200588		Bearing Deep Ball Groove with Shield 6205-Z		2
	B015	M	234118		CIRCLIP EXTERNAL 1400-0250		2
	B017	M	233127		CIRCLIP INTERNAL 1300-52		2
	B018	M	421744		Screw Caphead M8x1mm (Fine Pitch)x10		1
	M001	M	325G240	2450WEB	Drive Shaft		1
	M002	M	401G338		INBOARD BEARING HSG		1
	M003	M	309G160		IN/BD BEARING HOUS		1
	M004	M	309G161		IN/BD BEARING HOUS		1
	M005	M	315G903	IT.1	INNER SHAFT SLEEVE		2
	M006	M	315G904		SHAFT SLEEVE CAP		2
	M007	M	315G905		HOLDING COLLAR		1
	M009	M	309G197		IDLER PULLEY		2
	M010	M	309G168	IT.2	STUB SHAFT		2
	S001	M	421018		SOCKET CAP SCREW M10x35		7
	S002	M	421122		SOCKET CAP SCREW M10x30		4
	S003	M	421019		SOCKET CAP SCREW M6 x 20		6
	S004	M	268017		Key Steel 12x8		2
	S005	M	268021		Key Steel 5x5		1
	S006	M	268011		Key Steel 8x7		1
06-12			902G346		135Dia Rub Cover Nip Roll No 5	320G775	1
	B001	M	200577		BEARING DEEP GROOVE 61815		4
	B002	M	200578		Bearing Flanged Cartridge MFC40		1
	B003	M	200329		Bearing Deep Groove Ball 6206		1
	B004	M	233136		CIRCLIP INTERNAL 1300-62		1
	B005	M	239236		Seal Simmerring BABSL40-55-6/72NBR		2
	B006	M	239237		Seal O Ring RM0296-24 Nitrile		2
	B007	M	239238		Seal O Ring RM0616-24 Nitrile		1
	B008	M	239239		Seal O Ring RM0993-57 Nitrile		1
	B009	M	242009		LOCKNUT KM6		1
	B010	M	242010		LOCKWASHER MB6		1
	B011	M	214025		Nipple Grease M6		2
	B012	M	421744		Screw Caphead M8x1mm (Fine Pitch)x10		1
	M001	M	309G669		Bearing Housing		1
	M002	M	320G783		Bearing Collar Inboard		1
	M003	M	320G811	2450	Rubber Covered Nip Roller Body		1
	M004	M	320G780	2450	Nip Roller Shaft		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M005	M	320G782		Bearing End Cap		1
	M006	M	320G784		Bearing Collar Outboard		1
	M007	M	309G726		SHAFT SLEEVE CAP		2
	M008	M	309G727		INNER SHAFT SLEEVE		2
	M009	M	309G698		SEAL HOUSING		1
	M010	M	309G730		SPACER		1
	S001	M	421122		SOCKET CAP SCREW M10x30		3
	S002	M	421018		SOCKET CAP SCREW M10x35		4
	S003	M	421172		SOCKET CAP SCREW M5x12		3
	S004	M	421173		GRUB SCREW M8x12		4
	S005	M	421121		SOCKET CAP SCREW M6 x 16		3
	S007	M	421021		Washer Spring M10		7
	S008	M	421022		Washer Spring M6		3
	S009	M	421175		Washer Spring M5		3
06-13			902G347		WCD 500D Single Web TH'D 3" Filton-15c	320G573	1
	B001	M	239234		Seal Filton Rotary Union RH TH'D 17266/GV/6-RH		1
	B002	M	239220		Seal Simmerring BABSL130-150-7.5NBR		2
	B003	M	200575		Bearing Self Aligning Ball 1224		1
	B004	M	200572		Carb BEARING C2210 TN9		1
	B005	M	234072		CIRCLIP EXTERNAL 1400-0470		2
	B006	M	233132		CIRCLIP INTERNAL 1300-220		1
	B007	M	239190		Seal O-Ring RM2693-57 Nitrile		1
	B008	M	239221		Seal O-ring RM2295-30nitrile		1
	B009	M	239222		Seal O-ring RM1795-30 nitrile		1
	B010	M	239223		Seal O-ring RM1195-30 nitrile		2
	B011	M	423111		Elbow Adaptor Hobbs 8 O/D - 1/4" BSP		2
	B012	M	423038		Tubing Nut 8 OD		2
	B013	M	423039		Tubing Sleeve 8 OD		2
	B014	M	423112		Seal 1/4"bsp Enots 34-0156-02		2
	B015	M	239216		Seal O Ring RM 0246-24 Nitrile		2
	B016	M	423113		Coned Locknut 1/4"BSP Enots 36-0540-02		2
	B017	M	242078		LOCK WASHER MB24		1
	B018	M	242077		LOCKNUT KM24		1
	B019	M	242001		LOCKNUT KM9		1
	B020	M	242105		LOCKWASHER MB9		1
	M001	M	320G575		Outboard Housing		1
	M002	M	308G939		INBOARD BRG HOUSING		1
	M003	M	307G961		BRG HOUSING COVER		1
	M004	M	320G867		Inboard Bearing Hsg (Outside Cover)		1
	M005	M	304G239		INBOARD SHAFT SLEEVE		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M006	M	307G438		SHAFT SLEEVE CAP		2
	M007	M	303G502		HOLDING COLLAR		1
	M008	M	303G503		BUSH		1
	M009	M	309G116	IT.A	Centre Tubing 1.1/2"BSP		1
	M010	M	320G574	2450	Water Cooled Drum Wreb Th'd -15C		1
	M011	M	320G830		Plug 1/2"BSP (Thin)		2
	M012	M	320G569		Inboard & Outboard Bearing Housing		1
	M013	M	320G570		Bearing Retaining Cap		1
	M014	M	320G572		Bearing Retaining Spacer		1
	M015	M	320G831		Drum End Cover Plate		2
	S001	M	421122		SOCKET CAP SCREW M10x30		12
	S002	M	421149		SOCKET CAP SCREW M10x20		6
	S003	M	421033		SOCKET CAP SCREW M6x12		5
	S004	M	214024		Nipple 45Deg Grease M6		1
	S005	M	421021		Washer Spring M10		18
	S006	M	421418		Screw Socket Countersunk Head M6 x 16		24
06-14			902G348		170DIA L/Cell Roll No3 & 12	320G791	2
	B001	E	631044		Pressductor Radial Tensiometer Load Cell		2
	B002	E	631045		Connecting Cable c/w 90deg Plug Contact 20m lg		2
	B003	E	631071		2 Channel Control Unit PFEA111-65		1
	B004	M	200576		Carb BEARING C2208		1
	B005	M	233135		CIRCLIP INTERNAL 1300-0800		4
	B006	M	200500		BEARING NO SEALS 2208E		1
	M001	M	320G792	2450	170dia Carbon Fibre L/Cell Roll		1
	M002	M	319G830		INBOARD MTNG BOSS		1
	M003	M	319G831		OUTBOARD MTNG BOSS		1
	M004	M	314G782		BEARING SPACER		2
	M005	M	312G210		CAP WASHER		2
	S001	M	421089		SOCKET CAP SCREW M10x70		8
	S002	M	421035		SOCKET CAP SCREW M10x25		8
	S003	M	421021		Washer Spring M10		15
	S004	M	214024		Nipple 45Deg Grease M6		1
	S005	M	214025		Nipple Grease M6		1
06-16			904G260		Tendency Drive Assy	324G561	1
	B001	M	220641		PULLEY 90-8M-50		1
	B002	M	224277		Bush Taperlock 3020 Bore 40 W/- Std Key TL3020/40		1
	B003	M	220642		PULLEY 30-8M-50		1
	B004	M	224278		Bush Taperlock 1610 Bore 28 W/- Std Key TL1610/28		3
	B005	M	200591		BEARING 6005-Z		4
	B006	M	200153		BEARING 6205-2Z		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B007	M	220094		PULLEY 36H100		1
	B008	M	234118		CIRCLIP EXTERNAL 1400-0250		4
	B009	M	220645		PULLEY 27H100		1
	B010	M	220643		Twin Power Double Sided Timing Belt TP-2400-8M-50		1
	B011	M	220644		Twin Power Double Side Timing Belt		1
	M001	M	318G341		BOSS		1
	M002	M	318G306		TENSIONING SHAFT		1
	M003	M	318G126		30-8M-50 PULLEY		2
	M004	M	320G790		Boss		1
	M005	M	318G336		BOSS		1
	M006	M	318G337		TENSIONING SHAFT		1
	M007	M	306G005		18HING PULLEY		2
	M008	M	305G770	IT.V	BOSS DETAIL		1
	M009	M	406G610		Tendency Drive Guard		1
	S001	M	421035		SOCKET CAP SCREW M10x25		15
	S002	M	421021		Washer Spring M10		15
	S003	M	268024		Key Steel 12x8		1
	S004	M	268011		Key Steel 8x7		3
	S005	M	421029		PLAIN WASHER M10		2
	S006	M	421046		Screw HEXHD Setscrew M10x25		2
	S007	M	421033		SOCKET CAP SCREW M6x12		4
	S008	M	421034		Washer Plain M6		4
06-17			903G176		Light & Cover For Mech Assy (1650 - 2450 Machine)	320G737	1
	B001	E	714288		Cable Trunking Swift R33-130		1
	B002	M	421078		Terry Spring Clip 1.1/2"		2
	M001	M	309G347		SUPPORT BRACKET		2
	S001	M	421172		SOCKET CAP SCREW M5x12		4
	S002	M	421214		SOCKET CAP SCREW M3 x 8		2
	S003	E	714201		Swiftrack SC200 standard 3m length		2
	S004	M	421164		PLAIN WASHER M5		4
	S005	M	421034		Washer Plain M6		2
	S006	M	421117		Washer Plain M3		2
	S008	M	421022		Washer Spring M6		2
	S009	M	421221		SOCKET CAP SCREW M6x35		2
	S010	M	427100		6mm Single Sided Tubing Clip (1 Tube)		9
	S011	M	427101		6mm Single Sided Tubing Clip (2 Tubes)		1
	S012	M	421712		SCREW PANHEAD M4x6		10
	S013	E	603073		FLUORESCENT LIGHT FITTING T8/T12		2
06-18			904G217		Partition Assy	406G375	1
	B01	M	2441912		Inflatable vacuum seal (2450)		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B02	M	4211077		GRUB SCREW M4x6 NYLON		44
	M01	M	406G484	2450	Horizontal Partition		1
	M02	M	406G485	2450	Vertical Partition		1
	M03	M	406G486		Out Board Partition		1
	M04	M	406G487	IT1	Seal Channel		2
	M05	M	406G487	IT2	Seal Channel		1
	M06	M	406G487	IT3	Seal Channel		1
	M07	M	406G488		Seal Channel Corner		1
	M08	M	406G489		Seal Channel Corner		1
	M09	M	406G490		Seal Channel Corner		1
	M10	M	406G491		Partition Fill-in Plate		1
	M11	M	406G492		Partition Fill-in Plate		1
	M12	M	406G493	IT1	Shim		1
	M13	M	406G493	IT2	Shim		2
	M14	M	406G494	IT1	Shim		1
	M15	M	406G494	IT2	Shim		2
	M16	M	406G495	IT1	Shim		1
	M17	M	406G495	IT2	Shim		1
	M18	M	406G496		Shim		1
	M19	M	406G524		Bridge Piece		6
	S01	M	421035		SOCKET CAP SCREW M10x25		23
	S02	M	421020		SCREW COUNTERSUNK HEAD SOCKET M5 x 20		40
	S03	M	421170		Nut Full Hex M5		40
	S04	M	421164		PLAIN WASHER M5		40
	S05	M	421029		PLAIN WASHER M10		23
06-19			904G220		Drum Shield Latch Assy	406G497	1
	B01	M	421212		Pin Split M3x25		2
	B02	M	236012		SPRING ENTEX NO 536		1
	B03	M	4211088		PIN SELOK 5.0DIA X 30		2
	M01	M	406G498		Latch Support Housing		1
	M02	M	406G534		Latch Pin		1
	M03	M	406G535		Latch		1
	M04	M	406G536		Pull Rod		1
	M05	M	406G537		Spacer		2
	S01	M	421035		SOCKET CAP SCREW M10x25		4
	S02	M	421029		PLAIN WASHER M10		4
	S03	M	421030		PLAIN WASHER M12		2
06-20			904G399		Hinged Shield Assy (Extruded Seals)	406G596	1
	B01	M	202022		BEARING MFC 30		1
	B02	M	320G465	IT.4	3/8" NB Conv' PTFE ST.ST Braided Hose		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B03	M	239240		Dx72 Draught Excluder		7
	B04	M	239199		Seal O Ring RM0251-16 Nitrile		4
	B05	M	421791		Nuts Square M6 with position fixing No. 21.1330/2		13
	B06	M	423051		Jubilee Clip Number 0		40
	B07	M	423135		3/8"BSP Cone Seat M/F 90° Elbow 1037-6-6		4
	B08	M	423225		Adaptor Male 3/8"BSP - 9/16 UNF C/W "O" RING		4
	B09	M	423126		Bulkhead Adaptor 1/2" O/D Tube		4
	B10	M	423121		Stem Tailpiece Adaptor 1/2" O/D x 1/2" Hose		8
	B11	M	423128		Elbow Stem 1/2"		6
	B12	M	423055		Pipe Clip 12dia Enots 34-0218-07		4
	B13	M	423127		Bulkhead Locknut		4
	B14	M	423132		Tubing Sleeve 1/2" O/D		14
	B15	M	423131		Tubing Nut 1/2" O/D		14
	B16	M	423133		3/8"BSP Cone Seat BulkHD E79/43-6		2
	B17	M	423129		1/2 Hose Adaptor 60° Cone		2
	B18	M	423134		3/8"BSP Locknut TD190-6		2
	B19	M	423136		Union Nut 1/2" BSP		2
	B20	M	423130		1/2"x3/8" BSP M/F Adaptor XA8000-8-6		2
	B21	M	423026		Hose Adaptor 1/2" BSP x 1/2" Hose		4
	B22	M	423052		Rubber Water Hose 1/2" Bore (13MM)		10
	B23	M	268059		Graphite Foil roll 0.5mm thk x 40mm wide x 10m lg		1
	B24	M	240427		FIXED GRIP WDS 8191-204		1
	M01	M	406G318		Hinged Shield Pivot Leadthrough Assy - REF ONLY		1
	M02	M	406G497		Drum Shield Latch Assy - REF ONLY		1
	M03	M	406G598	2450	Drum Seal Machining		2
	M04	M	406G599		Inboard Arm		1
	M05	M	406G600		Outboard Arm		1
	M06	M	406G601	2450	V Rail		4
	M07	M	406G603	2450	Shielding		1
	M08	M	406G604	2450	Shielding (Pivot Side)		1
	M09	M	406G606		Connection Pipe		1
	M10	M	406G607		Connection Pipe		1
	M11	M	406G608		Connection Pipe		1
	M12	M	406G624		Inboard End Cover		1
	M13	M	406G625		Outboard End Cover		1
	M14	M	406G635	2450	Fill in Piece		2
	M15	M	406G873		Support Angle		2
	M16	M	320G275	2450	Shield Shaft		1
	M17	M	320G446		Clamp Block		1
	M18	M	320G449		Short End Shield		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M19	M	320G450		Drum End Seal		2
	M20	M	320G451		Guide Tube		4
	M21	M	320G457		Seal Mounting (END)		2
	M22	M	320G460	2450	Seal Mounting (Side)		2
	M23	M	320G487		Shield		12
	M24	M	320G508	2450	Connecting Pipe		1
	M25	M	320G824		Pipe Steady		2
	M26	M	320G832		Stop Angle		1
	M27	M	320G833		Stop Block		1
	M28	M	322G522	ITEM3	PARKING BLOCK		4
	M29	M	322G528		HOSE RETAINER (PIVOT SIDE)		4
	M30	M	322G529		HOSE RETAINER		6
	M31	M	322G858		Wavy Edge Shield (2450-M1)		2
	M32	M	322G862		EDGE SHIELD (GRP 2450 M1)		2
	M33	M	322G527		SCRAPER BLOCK		1
	S01	M	421195		SOCKET CAP SCREW M12x40		6
	S02	M	421122		SOCKET CAP SCREW M10x30		4
	S03	M	421114		SOCKET CAP SCREW M10x40		4
	S04	M	421161		SOCKET CAP SCREW M10x50		12
	S05	M	421181		SOCKET CAP SCREW M10x65		2
	S06	M	421033		SOCKET CAP SCREW M6x12		4
	S07	M	421121		SOCKET CAP SCREW M6 x 16		42
	S08	M	421019		SOCKET CAP SCREW M6 x 20		28
	S09	M	421145		SOCKET CAP SCREW M6x30		20
	S10	M	421147		SOCKET CAP SCREW M4 x 12		4
	S11	M	421051		Screw Socket Countersunk Head M6 x 12		14
	S12	M	421418		Screw Socket Countersunk Head M6 x 16		67
	S13	M	421466		Screw Countersunk Head Socket M6x20		3
	S14	M	421739		Screw Machine SLCSKHD M6x25		8
	S15	M	421190		Screw Countersunk Head Socket M6x30		24
	S16	M	421249		M8 X 20MM SOCKET HEAD COUNTERSUNK SCREW		4
	S17	M	421189		Screw HEXHD Setscrew M10x50		2
	S18	M	421906		SCREW HEX M6 x 20 LONG		50
	S19	M	421387		Screw HEXHD Setscrew M6x25		12
	S20	M	421027		NUT FULL HEX M10		2
	S21	M	421057		Nut Full Hex M6		8
	S22	M	421034		Washer Plain M6		116
06-21			903G861		Arrg't of Additional Heat Shield on Hinged Shield	323G537	1
	B001	M	268059		Graphite Foil roll 0.5mm thk x 40mm wide x 10m lg		1
	M001	M	405G307	2450	Heat Shield		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S001	M	421894		Screw HEXHD Setscrew M6x12		15
06-22			902G353		166.6 DIA St Idler no4 (Chromed)	320G654	1
	B001	M	200571		BEARING SELF ALIGNING BALL 2210E		1
	B002	M	200572		Carb BEARING C2210 TN9		1
	B003	M	242001		LOCKNUT KM9		1
	B004	M	242105		LOCKWASHER MB9		1
	M001	M	320G655	2450	166.6dia St Idler Roll (Chromed)		1
	M002	M	320G569		Inboard & Outboard Bearing Housing		2
	M003	M	320G570		Bearing Retaining Cap		2
	M004	M	320G572		Bearing Retaining Spacer		1
	M005	M	320G571		Bearing Retaining Washer		1
	S001	M	421035		SOCKET CAP SCREW M10x25		6
	S002	M	421146		SOCKET CAP SCREW M8x30		2
	S003	M	421033		SOCKET CAP SCREW M6x12		10
	S004	M	214024		Nipple 45Deg Grease M6		2
	S005	M	421021		Washer Spring M10		6
	S006	M	421103		Washer Spring M8		2
06-25			903G530		Mechanism Carriage	323G011	1
	B001	M	200200		BEARING TAPER ROLLER 32010X		2
	B002	M	200569		BEARING TAPER ROLLER 32011X		2
	B003	M	214025		Nipple Grease M6		2
	M001	M	312G290	IT-A	Carriage Roller Bracket		1
	M002	M	312G290	IT.B	Carriage Roller Bracket		1
	M003		320G994		PLAIN WHEEL (RUNWAY BEAM)		2
	M004	M	307G606		SHAFT		2
	M005	M	306G864		LOCATION COLLAR		2
	M006	M	307G607		RUNWAY ADJUSTER		4
	M007	M	318G910		BRG RETAINING CAP		2
	S001	M	421065		Bolt HEXHD M20x70		12
	S002	M	421127		SOCKET CAP SCREW M16x55		8
	S003	M	421126		Screw HEXHD Setscrew M16X80		4
	S004	M	421125		Locknut M16		4
	S005	M	421070		PLAIN WASHER M20		12
	S006	M	421124		Washer Plain M30		2
	S007	M	421123		Locknut M30		4
	S008	M	421122		SOCKET CAP SCREW M10x30		8
06-26			904G605		Arrgt of Un/Rewind Lock Chucks (In-line drive)	325G039	1
	B001	M	200585		Bearing Ball Angular Contact 3211		2
	B002	M	239245		Seal Simmerring BABSL70-90-7/72NBR		2
	B003	M	200703		BEARING 6008-Z		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B004	M	422008		Toggle Clamp DE-STA-CO Model 341		4
	B005	M	200587		Bearing Needle Roller RNA4914		2
	B006	M	200586		Inner Ring IR70x80x54		2
	B007	M	239213		Seal O Ring RM 2393-57 Nitrile		2
	B008	M	239218		Seal O Ring RM0995-30Nitrile		2
	B009	M	239246		Seal O Ring RM0645-30Nitrile		2
	B010	M	239T027		'O' RING [RM 0495-30]		2
	B011	M	242011		LOCKNUT KM11		2
	B012	M	242012		LOCKWASHER MB11		2
	B013	M	239231		Seal Simmerring BABSL80-100-7/72NBR		2
	B014	M	234123		CIRCLIP EXTERNAL 1400-0400		2
	B015	M	233138		CIRCLIP INTERNAL 1300-102		2
	B016	M	233139		CIRCLIP INTERNAL 1300-68		2
	M001	M	325G038		Chuck - Inboard		2
	M002	M	320G819		Outboard Chuck		2
	M003	M	320G820		Locking Clamp		4
	M004	M	306G878		DRIVE SHAFT		2
	M005	M	303G823		END CAP		2
	M006	M	306G876		BEARING SPACER		2
	M007	M	303G775		SPACER		2
	M009	M	303G797		STOP BLOCK		4
	M010	M	303G796		PLAIN BUSH		4
	M011	M	310G478		PIVOT PIN		4
	M012				6"Mech Expanding ReelShaft Ref only 320G823		3
	M013	M	306G175		THRUST WASHER		2
	M014	M	306G875		STOP SHAFT		2
	M015	M	306G874	IT.A	WEAR PLATE		4
	M016	M	306G874	IT.B	WEAR PLATE		4
	M017	M	302G112		LATCH		4
	M018	M	320G821		Spacer		4
	M019	M	320G822	IT.1	Locking Chuck Protector		2
	M020	M	320G822	IT.2	Locking Chuck Protector		2
	M024	M	320G894		Spacer		4
	M025	M	320G895	IT.1	Pillar		2
	M026	M	320G895	IT.2	Pillar		2
	M029	M	322G488		OUTBOARD SHIELD PLATE R/W		1
	M030	M	322G489		INBOARD SHIELD PLATE		2
	M031	M	322G494		OUTBOARD SHIELD PLATE U/W		1
	S001	M	421111		Screw Socket Csk Head M6 x 20		16
	S002	M	421122		SOCKET CAP SCREW M10x30		12

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S003	M	421149		SOCKET CAP SCREW M10x20		8
	S004	M	421138		SOCKET CAP SCREW M8x25		8
	S005	M	421091		Screw Socket Cap Head M6 x 25		8
	S006	M	421226		GRUB SCREW M6x10		4
	S007	M	421258		DOWEL 6 dia x 40 mm LONG		4
	S008	M	421019		SOCKET CAP SCREW M6 x 20		16
	S009	M	421040		Screw HEXHD Setscrew M8 x 25		16
	S010	M	421054		PLAIN WASHER M8		16
	S011	M	421021		Washer Spring M10		12
	S012	M	421022		Washer Spring M6		16
	S017	M	421249		M8 X 20MM SOCKET HEAD COUNTERSUNK SCREW		12
	S018	M	421452		SOCKET CAP SCREW M12 x 25		12
06-27			904G608		Ebonite covered roller (In-line Drive)	325G042	1
	B001	M	200572		Carb BEARING C2210 TN9		1
	B002	M	200408		BEARING DOUBLE ROW SELF ALIGNING 1211E		1
	B003	M	239232		Seal Simmerring BABSL65-90-7NBR		2
	B004	M	242012		LOCKWASHER MB11		1
	B005	M	242011		LOCKNUT KM11		1
	B006	M	239224		Seal O Ring RM 2193-57 Nitrile		1
	B007	M	239228		Seal O Ring RM1445-30 Nitrile		1
	B008	M	239194		Seal O Ring RM1045-30 Nitrile		1
	B009	M	239229		Seal O Ring RM0545-30Nitrile		2
	B010	M	233001		CIRCLIP INTERNAL 1300-100 (DIA 100 BORE)		1
	B011	M	423111		Elbow Adaptor Hobbs 8 O/D - 1/4" BSP		2
	B012	M	423038		Tubing Nut 8 OD		2
	B013	M	423039		Tubing Sleeve 8 OD		2
	B014	M	423112		Seal 1/4"bsp Enots 34-0156-02		2
	B015	M	423113		Coned Locknut 1/4"BSP Enots 36-0540-02		2
	B016	M	242001		LOCKNUT KM9		1
	B017	M	242105		LOCKWASHER MB9		1
	M001	M	320G621	2450	Ebonite Roller		1
	M002	M	320G569		Inboard & Outboard Bearing Housing		1
	M003	M	325G043		Inboard brg hsg for ebonite roller (In-line Drive)		1
	M004	M	308G954		INBOARD BRNG HOUSING		1
	M005	M	325G235		Inboard Brg Hsg End Cap		1
	M006	M	303G590		INNER SHAFT SLEEVE		2
	M007	M	303G591		SHAFT SLEEVE CAP		2
	M008	M	303G592		HOLDING COLLAR		1
	M009	M	320G705		Cork Retaining Plate		4
	M010	M	320G611		Outboard Housing		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M011	M	320G570		Bearing Retaining Cap		1
	M012	M	320G572		Bearing Retaining Spacer		1
	S001	M	421035		SOCKET CAP SCREW M10x25		6
	S002	M	421149		SOCKET CAP SCREW M10x20		6
	S003	M	421111		Screw Socket Csk Head M6 x 20		12
	S004	M	421033		SOCKET CAP SCREW M6x12		5
	S005	M	421122		SOCKET CAP SCREW M10x30		6
	S006	M	214024		Nipple 45Deg Grease M6		1
	S007	M	421021		Washer Spring M10		18
06-29			902G357		166.6Dia Post WCR.1-1/2" LH THD -15C	320G668	1
	B001	M	239233		Seal Filton Rotary Union LH TH'D 18105/GV/12-L		1
	B002	M	239231		Seal Simmerring BABSL80-100-7/72NBR		2
	B003	M	200137		BEARING SELF ALIGNING 1214		1
	B004	M	200572		Carb BEARING C2210 TN9		1
	B005	M	242026		LOCKWASHER MB14		1
	B006	M	242025		LOCKNUT KM14		1
	B007	M	233134		CIRCLIP INTERNAL 1300-125		1
	B008	M	239224		Seal O Ring RM 2193-57 Nitrile		1
	B009	M	239323		'O' RING RM1545-30 Nitrile		1
	B010	M	239226		Seal O Ring RM1345-30 Nitrile		1
	B011	M	239227		Seal O Ring RM0695-30 Nitrile		2
	B012	M	423111		Elbow Adaptor Hobbs 8 O/D - 1/4" BSP		2
	B013	M	423038		Tubing Nut 8 OD		2
	B014	M	423039		Tubing Sleeve 8 OD		2
	B015	M	423112		Seal 1/4"bsp Enots 34-0156-02		2
	B016	M	423113		Coned Locknut 1/4"BSP Enots 36-0540-02		2
	B017	M	234118		CIRCLIP EXTERNAL 1400-0250		2
	B018	M	239230		Seal O Ring RM 0151-16 Nitrile		1
	B019	M	242001		LOCKNUT KM9		1
	B020	M	242105		LOCKWASHER MB9		1
	M001	M	320G611		Outboard Housing		1
	M002	M	400G404		INBOARD BRNG HOUSING		1
	M003	M	320G869		Inboard Bearing Housing External Cap		1
	M004	M	400G406		INBOARD BRNG HOUSING (INTERNAL CAP)		1
	M005	M	400G413		SHAFT SLEEVE		2
	M006	M	400G414		SHAFT SLEEVE CAP		2
	M007	M	400G410		HOLDING COLLAR		1
	M008	M	400G412		CENTRE TUBING		1
	M009	M	320G669	2450	166.6 Dia W.C.Roller L.H. Th'd		1
	M010	M	400G411		BUSH		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M011	M	321G493		1/4" BSP PLUG		1
	M012	M	320G569		Inboard & Outboard Bearing Housing		1
	M013	M	320G570		Bearing Retaining Cap		1
	M014	M	320G572		Bearing Retaining Spacer		1
	S001	M	421122		SOCKET CAP SCREW M10x30		9
	S002	M	421149		SOCKET CAP SCREW M10x20		6
	S003	M	421021		Washer Spring M10		15
	S004	M	268011		Key Steel 8x7		1
	S005	M	421033		SOCKET CAP SCREW M6x12		5
	S006	M	214024		Nipple 45Deg Grease M6		1
06-30			902G358		6 1/2" O/D SP'L Spreader Roll No7	321G080	1
	B001	M	200502		BEARING NO SEALS 2206E		1
	B002	M	200608		CARB BEARING C2206		1
	M001	M	321G084	2450	6.5" o/d Special Spreader Roller		1
	M002	M	321G083		Inboard & Outboard Bearing Housing		2
	M003	M	321G081		Bearing Retaining Cap		2
	M004	M	321G082		Bearing Retainer Washer		2
	S001	M	421035		SOCKET CAP SCREW M10x25		6
	S002	M	421145		SOCKET CAP SCREW M6x30		4
	S003	M	421033		SOCKET CAP SCREW M6x12		6
	S004	M	214024		Nipple 45Deg Grease M6		2
	S005	M	421021		Washer Spring M10		6
	S006	M	421022		Washer Spring M6		4
06-32			904G616		Gas Wedge Assembly	325G078	1
	B001	M	424015		Needle Valve Port Size 1/4"		1
	B002	M	423057		Flowmeter NGIB211 GTF2AHDB		1
	B003	M	424014		Direct Acting Solenoid Valve		1
	B004	M	423036		Adaptor Male 8mm-1/4"BSP 36-0530-11		1
	B005	M	239217		Seal O Ring RM0245-30 Nitrile		1
	B006	M	239199		Seal O Ring RM0251-16 Nitrile		2
	B007	M	214023		Exhaust Filter 1/4"BSP M/1512		1
	B008	M	423054		Nipple Hex 1/4" BSPT		2
	B009	M	423090		Elbow Male Adaptor		1
	B010	M	424007		Tube 8mm O/D PUN-8		2
	B011	M	423038		Tubing Nut 8 OD		7
	B012	M	423039		Tubing Sleeve 8 OD		7
	B013	M	423058		Elbow Connector 8mm O/D		4
	B014	M	423046		Tube Half Hard Copper 8mm O/D CS7008003		1
	B015	M	423056		Pipe Clip 8dia Enots 34-0218-05		2
	M001	M	325G054	2450	Gas Wedge Manifold Beam		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M002	M	325G053		Mounting Brkt for Gas Wedge (Sideplate)		1
	M003	M	325G055		Mounting plate for Gas Wedge (Mech plate)		1
	M004	M	403G073		Gas Wedge End Cap		2
	M005	M	403G074		Gas Nozzle		24
	M006	M	325G077		Clamping Block (Gas Wedge)		2
	M007	M	311G885		Gas Leadthrough		1
	M008	M	312G411		Valve Mounting Bracket		1
	S001	M	421122		SOCKET CAP SCREW M10x30		4
	S002	M	421221		SOCKET CAP SCREW M6x35		2
	S003	M	421229		Screw Socket M8x20 CSKHD		4
	S004	M	421033		SOCKET CAP SCREW M6x12		2
	S005	M	4211088		PIN SELOK 5.0DIA X 30		4
	S006	M	421134		Screw Socket Buttonhead M5x10		2
06-33			904G686		Floating Unwind Layon Roller Assy	407G244	1
	B001	M	200357		BEARING 6012		1
	B002	M	200579		BEARING ROLLER NU1012		1
	B003	M	2061067		Runner Block - Igus DryLin WJ200UM-01-20		8
	B004	M	2061065		Guide Rail - Igus DryLin WS-20-80 x 700 Lg		2
	B005	M	203320		Bush Flanged Iglidur Z - 15 ID x 17 OD x 11 Lg		8
	B006	M	234006		CIRCLIP EXTERNAL D1400-0600		2
	B007	E	698208		LIMIT SWITCH PARALLEL ROLLER 1NO/1 NC (14CE SERS)		1
	B008	M	214025		Nipple Grease M6		4
	M001	M	407G506	2450	Dia 158 Neoprene/Alu Roller		1
	M002	M	407G507	2450	Roller Shaft		1
	M003	M	407G508		Bearing Shield		2
	M004	M	325G087		Internal Drive (Unwind Layon) - REF ONLY		1
	M005	M	407G333		Roller Mounting Arm		2
	M006	M	407G335		Support Plate - Inboard		1
	M007	M	407G336		Support Plate - Outboard		1
	M008	M	407G337		Link Arm		2
	M009	M	407G516		Belt Clamp		2
	M010	M	407G517		Pin		4
	M011	M	407G518		Pin Retainer		4
	M012	M	407G308		Stop		4
	M013	M	309G125		DRIVE BELT CLAMP		2
	S001	M	421122		SOCKET CAP SCREW M10x30		4
	S002	M	421138		SOCKET CAP SCREW M8x25		24
	S003	M	421049		SOCKET CAP SCREW M8x20		30
	S004	M	421171		SOCKET CAP SCREW M5x16		8
	S005	M	421245		SOCKET CAP SCREW M5x10		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S006	M	4211323		Screw HEXHD Setscrew M6x50		2
	S007	M	421057		Nut Full Hex M6		2
	S008	M	421029		PLAIN WASHER M10		4
	S009	M	421054		PLAIN WASHER M8		30
	S010	M	421034		Washer Plain M6		4
	S011	M	421164		PLAIN WASHER M5		8
06-34			904G687		Internal Drive Layon Arm (Unwind)	325G087	1
	B001	M	200592		BEARING 6306		2
	B002	M	242009		LOCKNUT KM6		1
	B003	M	242010		LOCKWASHER MB6		1
	B004	M	200578		Bearing Flanged Cartridge MFC40		1
	B005	M	239236		Seal Simmerring BABSL40-55-6/72NBR		2
	B006	M	239256		Seal O Ring RM1543-57Nitrile		1
	B007	M	239254		Seal O Ring RM0845-30Nitrile		1
	B008	M	239194		Seal O Ring RM1045-30 Nitrile		1
	B009	M	239249		Seal O-Ring RM0295-30 Nitrile		2
	B010	M	239255		Seal O Ring RM0795-30 Nitrile		1
	B011	M	225286		Sprocket Polychain 8M-50S-21		2
	B012	M	224276		Bush T/Lock Bore 40 with STD Keyway TL2012		2
	B013	M	225288		Timing Belt Polychain 8M-1600-21		4
	B014	M	200588		Bearing Deep Ball Groove with Shield 6205-Z		2
	B015	M	234118		CIRCLIP EXTERNAL 1400-0250		2
	B017	M	233127		CIRCLIP INTERNAL 1300-52		2
	B018	M	421744		Screw Caphead M8x1mm (Fine Pitch)x10		1
	M001	M	325G088	2450WEB	Layon Drive Shaft		1
	M002	M	315G900		INBOARD BRNG HSNG		1
	M003	M	315G901		SEAL HOUSING		1
	M004	M	315G902		INBOARD SEAL HOUSING		1
	M005	M	315G903	IT.1	INNER SHAFT SLEEVE		2
	M006	M	400G308		BEARING HSNG EXTENSN		1
	M007	M	315G904		SHAFT SLEEVE CAP		2
	M008	M	315G905		HOLDING COLLAR		1
	M009	M	309G168	IT.2	STUB SHAFT		2
	M010	M	309G197		IDLER PULLEY		2
	M012	M	400G309		BEARING SPACER		1
	S001	M	421018		SOCKET CAP SCREW M10x35		7
	S002	M	421045		SOCKET CAP SCREW M10x45		4
	S005	M	421029		PLAIN WASHER M10		4
	S006	M	268024		Key Steel 12x8		2
	S007	M	268025		Key Steel 5x5		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S008	M	421146		SOCKET CAP SCREW M8x30		4
	S009	M	421091		Screw Socket Cap Head M6 x 25		6
06-35			904G698		Layon Jig Plate Assy	407G564	1
	M01	M	407G519		Jig Plate - Rewind Layon		1
	M02	M	407G520		Jig Plate - Unwind Layon		1
	S01	M	421149		SOCKET CAP SCREW M10x20		9
	S02	M	421029		PLAIN WASHER M10		9
07					Web Transport Drive & Control Sub Assy		
07-02			903G514		Drive Control Cubicle		1
	001	E	244G707		2450 STD DRIVE CUBICLE		1
	002	E	429G011		PLASMA TREATER ELECTRICAL BUILD OPTION		0
	003	E	429G012		UNWIND LAYARM ELECTRICAL BUILD OPTION		1
	004	E	429G013		DRUM CLEANER ELECTRICAL BUILD OPTION		0
	005	E	429G014		MOTORISED BOW ADJUST ELECTRCIAL BUILD OPTION		1
	006	E	429G015		CHUCK LIMIT SWITCH ELECTRICAL BUILD OPTION		0
	007	E	429G016		WEB THREADER ELECTRICAL BUILD OPTION		0
	008	E	429G019	LFBEAM	MONITOR BEAM ELECTRICAL BUILD OPTION		1
	010				ECO 06G00423 : DRIVE'S PLC EQUIPMENT		0
	011	E	650733		SIMATIC S7-300 24V DC CPU 317-2 DP		1
	012	E	650715		512K MICRO MEMORY CARD FOR S7-300, ET200S & IM151		1
	013	E	650404		6ES7(300) MOUNTING RAIL 480mm		1
	014	E	650305		S7(300) 32 DIGITAL INPUT		1
	015	E	650588		FRONT CONNECTOR FOR S7-300, 40 SINGLE CORES, 2.5m		1
	016	E	650489		3S7(300) 16 DIGITAL OUTPUT 24V 0.5A		1
	017	E	650317		6ES7(300) 20 POLE FRONT CONNECTOR (SCREW)		1
	018	E	650773		S7-300 ANALOG INPUT 8AQ, 6ES7-331-7KF01-0AB0		1
	019	E	650317		6ES7(300) 20 POLE FRONT CONNECTOR (SCREW)		1
	020	E	650643		6ES7(300) SHIELD CONNECTING ELEMENT		1
	021	E	615533		6 AMP SINGLE POLE (M.C.B)		3
	022				Issue 4: ECO-06G044 6171-A3 DRIVE ESTOP ASI MODULE		0
	023	E	652063		ASI EMS CONNECTION BASE		1
	024	E	616911		M12/4WIRE/90DEG(M) 5M CABLE		1
	025	E	652130		ASI SAFE MONITOR 2 CHANNEL		1
07-03			903G551		Drive Misc Items		1
	001	E	603072		FLUORESCENT TUBE 8FT (125W)		1
	002	E	603073		FLUORESCENT LIGHT FITTING T8/T12		2
	003	M	421078		Terry Spring Clip 1.1/2"		2
	004	E	604110		AE ENCLOSURE (200 x 300 x 120) (PRIMER)		1
	005	E	652133		ASI 4 INPUT MODULE 2 WIRE INPUT		1
	006	E	652130		ASI SAFE MONITOR 2 CHANNEL		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	007	E	652063		ASI EMS CONNECTION BASE		1
	008	E	616647		M12/4WIRE/90DEG 2M CABLE		4
	010	E	637009		FOOTSWITCH FISUIUN FOOT SWITCH		0
	011				[Issue 3: ECO 04G486]		0
	012	E	613631		CABLE 3CORE 1MM PVC/PVC YY/NR		20
	013	E	619022		16MM SKINTOP COMPRESSION GLAND		1
	014	E	643078		BEACON AMBER FLASHING XENON 12/24Vdc		1
07-04			902G462		Drive Leadthrough Details		1
	001	E	616598		FC LANE 25 PIN SOCKET CODE 0		1
	002	E	616599		FC LANE 25 PIN PLUG CODE 0		1
	003	E	616600		FC LANE 25 PIN BULKHEAD CODE 0		1
	004	E	616601		FC LANE CABLE OUTLET		2
	005	E	616602		FC LANE OUTLET GASKET		2
	006	M	303G349	IT.B	LEADTHRO" FLANGE		1
	007	M	239250		Seal O Ring RM0793-57 Nitrile		1
	008	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	009	M	421034		Washer Plain M6		4
	010				[-SKT 1 : LOADCELL]		
	011	E	616603		FC LANE 25 PIN SOCKET CODE 1		1
	012	E	616604		FC LANE 25 PIN PLUG CODE 1		1
	013	E	616605		FC LANE 25 PIN BULKHEAD CODE 1		1
	014	E	616601		FC LANE CABLE OUTLET		2
	015	E	616602		FC LANE OUTLET GASKET		2
	016	M	303G349	IT.B	LEADTHRO" FLANGE		1
	017	M	239250		Seal O Ring RM0793-57 Nitrile		1
	018	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	019	M	421034		Washer Plain M6		4
	020				[-SKT 2 : BEAM]		
	021	E	616606		FC LANE 25 PIN SOCKET CODE 2		1
	022	E	616607		FC LANE 25 PIN PLUG CODE 2		1
	023	E	616608		FC LANE 25 PIN BULKHEAD CODE 2		1
	024	E	616601		FC LANE CABLE OUTLET		2
	025	E	616602		FC LANE OUTLET GASKET		2
	026	M	303G349	IT.B	LEADTHRO" FLANGE		1
	027	M	239250		Seal O Ring RM0793-57 Nitrile		1
	028	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	029	M	421034		Washer Plain M6		4
	030				[-SKT 7A : REWIND]		
	031	E	616609		FC LANE 25 PIN SOCKET CODE 3		1
	032	E	616610		FC LANE 25 PIN PLUG CODE 3		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	033	E	616611		FC LANE 25 PIN BULKHEAD CODE 3		1
	034	E	616601		FC LANE CABLE OUTLET		2
	035	E	616602		FC LANE OUTLET GASKET		2
	036	M	303G349	IT.B	LEADTHRO" FLANGE		1
	037	M	239250		Seal O Ring RM0793-57 Nitrile		1
	038	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	039	M	421034		Washer Plain M6		4
	040				[-SKT7B : UNWIND]		
	151	E	616619		FC LANE 6 PIN SOCKET CODE 0		1
	152	E	616620		FC LANE 6 PIN PLUG CODE 0		1
	153	E	616623		FC LANE 6 PIN BULKHEAD CODE 0		1
	154	E	616601		FC LANE CABLE OUTLET		2
	155	E	616602		FC LANE OUTLET GASKET		2
	156	M	303G349	IT.B	LEADTHRO" FLANGE		1
	157	M	239250		Seal O Ring RM0793-57 Nitrile		1
	158	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	159	M	421034		Washer Plain M6		4
	160				[-SKT 9 : LIGHT]		
	161	E	616624		FC LANE 6 PIN SOCKET CODE 1		0
	162	E	616625		FC LANE 6 PIN PLUG CODE 1		0
	163	E	616626		FC LANE 6 PIN BULKHEAD CODE 1		0
	164	E	616601		FC LANE CABLE OUTLET		0
	165	E	616602		FC LANE OUTLET GASKET		0
	166	M	303G349	IT.B	LEADTHRO" FLANGE		0
	167	M	239250		Seal O Ring RM0793-57 Nitrile		0
	168	M	4211314		M6 X 25 SKT CAP HEAD SCREW		0
	169	M	421034		Washer Plain M6		0
	170				[-SKT 3 : WEB THREADER]		0
	171	E	616627		FC LANE 6 PIN SOCKET CODE 2		1
	172	E	616628		FC LANE 6 PIN PLUG CODE 2		1
	173	E	616629		FC LANE 6 PIN BULKHEAD CODE 2		1
	174	E	616601		FC LANE CABLE OUTLET		2
	175	E	616602		FC LANE OUTLET GASKET		2
	176	M	303G349	IT.B	LEADTHRO" FLANGE		1
	177	M	239250		Seal O Ring RM0793-57 Nitrile		1
	178	M	4211314		M6 X 25 SKT CAP HEAD SCREW		4
	179	M	421034		Washer Plain M6		4
	180				[-SKT 8 : BOW SPDR ADJUST]		
	186	M	303G349	IT.C	LEADTHRO" FLANGE		5
	187	M	239250		Seal O Ring RM0793-57 Nitrile		5

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	188	M	4211314		M6 X 25 SKT CAP HEAD SCREW		20
	189	M	421034		Washer Plain M6		20
	190				[Blank - 10 Flanges to cover less wire feeds]		0
07-09			902G463		Drive Motors		1
	001	E	664377		VECTOR MOTOR 13kW 310NM 400RPM 37A IMB35		2
	002				[REWIND / UNWIND]		
	003	E	664208		VECTOR MOTOR 34KW 186NM 1750RPM 72A		1
	004				[DRUM]		0
	005	E	664209		VECTOR MOTOR 6.25KW 34NM 1750RPM 13 A		1
	006				[DTR1]		
	007	E	664210		AC MOTOR 0.82KW 2.6NM 3000RPM (BRAKE FITTED)		0
	007				Unwind layarm motor & gearbox (Berger Lahr)		1
	007				Rewind layarm motor & gearbox (Berger Lahr)		1
	008				[REWIND/UNWIND LAYARM]		
	009	E	664378		VECTOR MOTOR 13.5KW 112NM 1150RPM 29A IMB35		1
	010				[DTR2]		0
08					Pumping Group Sub Assy		
08-01			904G423		Pump Group Assy (SV750/Aerzen)	407G016	1
	B01	M	420137	400V50HZ	Aerzen High Vacuum Booster GL 8000 CM-Y		1
	B02	M	420136	400V50HZ	Aerzen High Vacuum Booster GM 2000 CM-C		3
	B03	M	420138	400V50HZ	Leybold SOGEVAC SV750BF Rotary Vane Pump		3
	B04	M	208119		Valve Pneumatic Butterfly 6" FIG320 T112		3
	B05	M	208152		Valve Pneumatic Butterfly 8" FIG320 T112		1
	B06	M	426110		Micro Pirani Transducer MKS 925C-11		1
	B07	M	424032		Centering Ring DN 16 ISO-KF		8
	B08	M	424002		Clamp Ring NW10/16KF		9
	B09	M	424031		Elbow 90 Deg DN 16 ISO-KF		1
	B10	M	424005		Blanking Flange NW16KF		6
	B11	M	424127		Centering Ring with Fine Filter DN 16 ISO-KF		1
	B12	M	424006		Vacuum Valve SP16k		1
	B13	M	306G490	IT.7	STND RUBBER SLVE		3
	B14	M	404G014		12" Bellow		1
	B15	M	423031		Jubilee Clip Number 7.1/2		12
	B16	M	423050		Jubilee Clip Number 12.1/2		4
	B17	M	239205		Seal O Ring RM1893-57 Nitrile		4
	B18	M	239215		Seal O Ring RM 2493-57 Nitrile		1
	B19	M	239206		Seal O Ring RM3193-57 Nitrile		1
	B20	M	239204		Seal O Ring RM3393-57 Nitrile		1
	B21	M	239386		Seal O Ring RM3593-57 Nitrile		2
	B22	M	239213		Seal O Ring RM 2393-57 Nitrile		5

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B23	M	420178		Leybold Dirt Trap NW150		3
	M01	M	308G972	IT.47	Special Flange		2
	M02		404G087		VACUUM PIPE		1
	M03	M	405G386		BLANKING PLATE		1
	M04	M	406G775		Vacuum Pipe		1
	M05	M	309G533	IT.216	PIPE SUPPORT ASSY		3
	M06	M	407G027		Vacuum Pipe		1
	M07	M	407G028	IT.1	Vacuum Pipe		3
	M08	M	407G029		Vacuum Pipe		3
	M09	M	407G030		Vacuum Pipe		3
	M10	M	407G190		Adaptor Flange (8"ANSI - GM8000)		1
	M11	M	407G055		Vacuum Pipe		1
	M12	M	407G056		Support Frame		1
	M13	M	407G059		Pump Foot Mounting		6
	M14	M	407G075		Adapter Flange (SV750 - GM2000)		3
	S01	M	421529		Screw Caphead M16X25		12
	S02	M	421104		SOCKET CAP SCREW M16x50		32
	S03	M	421518		Screw Caphead M20x130		24
	S04	M	421519		Screw Caphead M20x150		8
	S05	M	421522		Screw Caphead M20x80		44
	S06	M	421347		SOCKET CAP SCREW M20x55		8
	S07	M	4211299		Screw Caphead M20x90		44
	S08	M	421346		SOCKET CAP SCREW M24x80		4
	S09	M	421072		Nut Full Hex M20		144
	S10	M	421059		PLAIN WASHER M16		44
	S11	M	421070		PLAIN WASHER M20		272
	S12	M	421069		PLAIN WASHER M24		8
	S13	M	421811		Washer Spring M24		4
	S14	M	4211284		Screwed Rod M20x70		24
	S15	M	421195		SOCKET CAP SCREW M12x40		12
	S16	M	421073		Washer Spring M12		12
	S17	M	421101		Nut Full Hex M24		4
08-09			904G987		Diffusion Pump Piping and Guages	408G463	1
	B001	M	208153		Valve Pneumatic Butterfly 4" FIG320 T112		4
	B002	M	208155		Valve Pneumatic Butterfly 2" FIG320 T112		2
	B003	M	420175	400V50HZ	Oerlikon Leybold Diffusion Pump DIP20000		4
	B004	M	426110		Micro Pirani Transducer MKS 925C-11		6
	B005	M	426002		Gauge Head Penning AIM-S-NW25		1
	B006	M	424006		Vacuum Valve SP16k		6
	B007	M	424010		Clamp Ring NW20/25KF		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B008	M	424009		Centering Ring NW25KF		2
	B009	M	424008		Elbow 90deg NW16KF		7
	B011	M	424002		Clamp Ring NW10/16KF		20
	B012	M	424004		Centering Ring NW16KF		14
	B013	M	424068		Elbow 90deg NW25KF		1
	B014	M	424069		McMaster-carr vacuum silencer -1"NPT		1
	B015	M	424070		McMaster-carr vacuum silencer -2"NPT		1
	B018	M	424081		Centering Ring with Filter DN 16 ISO-KF		6
	B019	M	239214		Seal O Ring RM 1393-57 Nitrile		4
	B020	M	426001		GAUGE CAPSULE DIAL G-R4-1000		1
	B021	M	268050		Oil Silicone CVC4 (5kg can)		5
	B022	M	421884		Washer Nylon M16		8
	B023	M	239407		Seal O Ring 8.4c/sx540mm mean dia Nitrile		4
	B024	M	420176		Manifold as drawing DIP20000-DETAIL		4
	B025	M	424179		Claw Clamp DN 63-100 ISO-K		4
	M001	M	408G468		Inlet Adaptor Flange - Leybold DIP20000		4
	M003	M	402G884		Silencer Adaptor - Small		1
	M004	M	402G885		Silencer Adaptor - Large		1
	M005	M	404G015		Vacuum Pipe		4
	M006	M	323G390		CYLINDER GUARD		1
	M007	M	324G414		DIFF PUMP CABLE TRAY BRACKET		8
	S001	M	421634		Screw Caphead M16x80		32
	S002	M	421043		Screw HEXHD Setscrew M12x45		40
	S003	M	4211151		WASHER HEAVY DUTY M12		40
	S004	M	421517		Screw Caphead M16X120		40
	S005	M	421059		PLAIN WASHER M16		144
	S006	M	421030		PLAIN WASHER M12		80
	S007	M	421116		Washer Spring M16		68
	S008	M	421038		Nut Full Hex M16		68
	S009	M	421125		Locknut M16		8
	S010	M	421403		SOCKET CAP SCREW M12 x 50		80
09					Pumping Group Control & Inst'n Sub Assy		
09-03			903G550		Vacuum Misc Items		1
		E	616757		HARTING HAN K6/12 PLUG (MALE)		4
		E	616758		HARTING HAN K6/12 SOCKET (FEMALE)		4
		E	616863		HARTING HAN 16B HIGH HOOD M32 SIDE ENTRY		4
		E	616821		HAN K 4/2+E CHASSIS HOUSING SIZE 16		4
		E	616582		8/8 DATA PLUG		1
10					Services & Ancillaries Sub Assy		
10-01			902G375		M.E Water Fittings		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B001	M	423026		Hose Adaptor 1/2" BSP x 1/2" Hose		4
	B002	M	423052		Rubber Water Hose 1/2" Bore (13MM)		25
	S001	M	423051		Jubilee Clip Number 0		16
10-02			903G590		WATER PIPING (VESSEL END) 2450	404G214	1
	B001	M	423164		Adaptor Male 8mm - 1/2" BSP		1
	B002	M	423044		Elbow Stem 8 OD		1
	B003	M	423045		Adaptor Nippled 8mm-1/4"BSP Enots 36-0557-11		1
	B004	M	423038		Tubing Nut 8 OD		1
	B005	M	423039		Tubing Sleeve 8 OD		1
	B006	M	423025		Nipple Crane 1/2"BSP M.I. Hex 144/15		17
	B007	M	423479		Nipple Hex Reducing 1/2"-3/8" BSP SS		8
	B009	M	423435		3/4"BSP HEX NIPPLE M.I.		10
	B010	M	423704		Plug Solid 1 1/4"BSP Crane Type 148/32		1
	B011	M	423091		Elbow Female 1/2" BSP M.I.		18
	B012	M	423139		Elbow Female 3/4" BSP M.I.		10
	B013	M	423013		Elbow Female 2" BSP M.I.		8
	B014	M	423028		Red/Tee on Run Crane 2"x1/2"x2"BSP 161/50-15-50		2
	B015	M	423151		Socket Crane M.I. 2" BSP Type 177/50		6
	B016	M	423118		Union Male/Female 2" BSP M.I.		6
	B017	M	423026		Hose Adaptor 1/2" BSP x 1/2" Hose		5
	B018	M	423074		Adaptor Stem 2"BSP x 2" F352		10
	B019	M	427073		Swagelok Male Hose Adaptor		20
	B020	M	427072		Male Hose Adaptor		24
	B021	M	424017		Valve Gate 1/2" BSP Brass		9
	B022	M	424101		2" BSP BRONZE BALL VALVE		6
	B023	M	423690		IFM Efector 500 G1/4" - G 1/2" Adaptor E30000		6
	B024	M	423343		IFM Efector 300 M18x1.5/G 1/2" Adaptor E40096		1
	B025	M	423682		Strainer 3" Flange		1
	B026	M	423686		Gasket 3" PN16 Flange		5
	B027	M	208149		Valve Hand Butterfly 75 FIG320 Hand operated		2
	B028	M	423073		Jubilee Clip Number 3		16
	B029	M	423689		IFM Efector 500 Pressure Transmitter PA3024		6
	B030	M	424018		Gauge Pressure 65dia 6Bar1/4BSPT		1
	B031	M	423699		IFM Temperature Sensor TN2530		1
	B032	M	423020		'U' Bolt 2" Rolled Thread		2
	B033	M	423072		Rubber Water Hose 2" Bore		5
	B034	M	427078		Tube 1/2" P.O. Multi purpose hose Blue		8
	B035	M	427098		Tube 3/4" P.O. Multi purpose hose Blue		5
	B036	M	423604		Plug 1/2" BSPT Male HEX SKT Pressure plug		1
	B037	M	423691		'U' Bolt 3" Rolled Thread		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M001	M	403G185	IT.50.235	50 NB Mild Steel Tube BS1387 Heavy (2")		1
	M002	M	403G185	IT.50.350	50 NB Mild Steel Tube BS1387 Heavy (2")		1
	M003	M	403G185	IT.50.400	50 NB Mild Steel Tube BS1387 Heavy (2")		1
	M004	M	403G185	IT.50.550	50 NB Mild Steel Tube BS1387 Heavy (2")		1
	M005	M	403G185	IT.50.600	50 NB Mild Steel Tube BS1387 Heavy (2")		1
	M006	M	403G185	IT.50.700	50 NB Mild Steel Tube BS1387 Heavy (2")		1
	M007	M	404G215		INLET WATER MANIFOLD		1
	M008	M	404G216		MANIFOLD MOUNTING		2
	M009	M	404G217		OUTLET WATER MANIFOLD		1
	M010	M	404G218		SUPPORT BRACKET		1
	M011	M	404G224		SENSOR PIPE		6
	M014	M	324G302		MOUNTING PLATE (I/O MODULE)		1
	S001	M	421210		SOCKET CAP SCREW M12x35		8
	S002	M	421517		Screw Caphead M16X120		16
	S003	M	421634		Screw Caphead M16x80		8
	S004	M	421027		NUT FULL HEX M10		4
	S005	M	421038		Nut Full Hex M16		24
	S006	M	421029		PLAIN WASHER M10		4
	S007	M	421030		PLAIN WASHER M12		12
	S008	M	421059		PLAIN WASHER M16		48
	S011	M	421173		GRUB SCREW M8x12		2
	S012	M	421172		SOCKET CAP SCREW M5x12		4
	S013	M	421042		NUT FULL HEX M12		4
10-03			904G610		PUMPING MODULE WATER MANIFOLDS (25 DEG)	407G221	1
	B001	M	423705		Elbow Male/Female 1/4" BSP M.I.		1
	B002	M	423025		Nipple Crane 1/2"BSP M.I. Hex 144/15		16
	B003	M	423091		Elbow Female 1/2" BSP M.I.		16
	B004	M	423139		Elbow Female 3/4" BSP M.I.		2
	B005	M	423496		Plug Solid 1/4" BSP		1
	B006	M	423098		Red on Run Tee TP161 Crane 161/40/15/40		2
	B007	M	423021		Red/Socket Crane 2"x1.1/2"BSP M.I. 179/50-40		2
	B008	M	4231161		Branch Tee 1/2" X 1/2" X 1/4" SS		6
	B009	M	424017		Valve Gate 1/2" BSP Brass		2
	B010	M	423701		CAP M.I. 3/4"BSP		2
	B011	M	423104		Cap M.I. 1.1/2" BSP		2
	B012	M	424018		Gauge Pressure 65dia 6Bar1/4BSPT		1
	B013	M	423026		Hose Adaptor 1/2" BSP x 1/2" Hose		2
	B014	M	423081		Adaptor Stem 1.1/2"BSP x 1.1/2" F351		2
	B015	M	427072		Male Hose Adaptor		16
	B016	M	427073		Swagelok Male Hose Adaptor		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B017	M	423083		Jubilee Clip Number 2		8
	B018	M	423085		CAP M.I. 1/2"BSP		2
	B019	M	427285		Male Hose Adaptor 1/4" x 1/4" BSPT		12
	B020	M	4231162		Nipple Crane 1/4"BSP SS Hex 144/8		6
	B021	M	4231163		Elbow Female 1/4" BSP M.I.		6
	B022	M	423082		Rubber Water Hose 1.1/2" Bore		5
	B023	M	427078		Tube 1/2" P.O. Multi purpose hose Blue		20
	B024	M	427263		Tube 1/2" P.O. Multi purpose hose Green		20
	B025	M	427098		Tube 3/4" P.O. Multi purpose hose Blue		4
	B026	M	427265		Tube 3/4" P.O. Multi purpose hose Green		4
	B027	M	427169		Tube 1/4" P.O. Multi purpose hose Blue		7
	B028	M	427266		Tube 1/4" P.O. Multi purpose hose Green		7
	M001	M	407G224		INLET/OUTLET MANIFOLD		2
	S001	M	421122		SOCKET CAP SCREW M10x30		8
	S002	M	421029		PLAIN WASHER M10		16
	S003	M	421027		NUT FULL HEX M10		8
10-05			903G621		Pneumatic Items		1
	B001	M	214031		Air Service Unit		1
	B002	M	2121216		Source Flap		1
	B003	M	2121217		Diffusion Pump Lid Actuator		1
	B004	M	2121218		Moving Shield		1
	B005	M	2121219		Drum Shield Actuator		1
	B006	M	2121220		Inflatable Zone Seals		1
	B007	M	2121221		Nip Roller Actuator		1
	B008	M	2121222		Shut Off		1
	B009	M	427192		Festo 157601 AHB-MEB Rotation Key		2
	B010	M	2441707		Small padlock		2
	M001	M	324G294		MTG ADAPTOR PLATE (AIR SERVICE UNIT)		2
	S001	M	427016		'SMC' KQL12-04S ELBOW MALE PUSH IN 12mm 1/2"		7
	S002	M	427049		ELBOW MALE PUSH IN 8mm 1/4"		2
	S003	M	427050		ELBOW MALE PUSH IN 8mm 1/8"		9
	S004	M	427019		'SMC' KQT12-00 TEE EQUAL PUSH IN 12mm		19
	S006	M	427021		'SMC' KQR08-12 REDUCER PUSH IN 8mm - 12mm		20
	S008	M	427023		CONNECTOR MALE PUSH IN 12mm 1/2"		2
	S009	M	427024		UNION FEMALE PUSH IN 12mm 1/2"		1
	S010	M	423034		Tube 8mm O/D PUN-8		19
	S011	M	423035		Tube 12mm O/D 152589PUN-12		15
	S012	M	423145		Tube 4mm O/D 152587 PUN-4		1
	S014	M	427022		REDUCER PUSH IN 4mm - 8mm		2
10-06			903G780		TRAVERSE LIMIT SWITCH MECH END	323G366	1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B001	E	698208		LIMIT SWITCH PARALLEL ROLLER 1NO/1 NC (14CE SERS)		5
	M001	M	314G066		LIMIT SWITCH BRACKET		2
	M002	M	306G025		LIMIT SWITCH STRIKER		1
	M003	M	308G456		LIMIT SWITCH STRIKER		1
	M004	M	320G146		FIXING PLATE		4
	S001	M	421110		Screw HEXHD Setscrew M10 x 35		8
	S002	M	421029		PLAIN WASHER M10		8
	S003	M	421121		SOCKET CAP SCREW M6 x 16		4
	S004	M	421031		SOCKET CAP SCREW M5x20		6
	S005	M	421113		SOCKET CAP SCREW M5x35		2
10-08			903G560		CABLE COILER ASSY	404G160	1
	B001	E	619458		EKD PLASTIC ENERGY GUIDEWAY CHAIN TYPE PKK		1
	B002	M	423151		Socket Crane M.I. 2" BSP Type 177/50		2
	B003	M	423074		Adaptor Stem 2"BSP x 2" F352		8
	B004	E	619459		CABLE CLAMP KHF 24-38B		2
	B005	E	619460		CABLE CLAMP KHF16-25B		1
	B006	M	423072		Rubber Water Hose 2" Bore		11
	M001	M	404G165		L.H SUPPORT		1
	M002	M	404G166		R.H SUPPORT		1
	M003	M	404G167		COILER SUPPORT		1
	M004	M	404G168		DIVIDER SUPPORT		1
	M005	M	404G169		COVER		1
	M006		404G170		COVER		1
	M007	M	404G174	2450	WEAR PLATE		1
	M008	M	404G183		DIVIDER		1
	M009	M	404G187		COILER END COVER		1
	M010	M	404G194		PIPE MANIFOLD		1
	S001	M	421033		SOCKET CAP SCREW M6x12		4
	S002	M	421019		SOCKET CAP SCREW M6 x 20		27
	S003	M	421049		SOCKET CAP SCREW M8x20		2
	S004	M	423073		Jubilee Clip Number 3		12
	S005	M	421036		Nut Full Hex M8		5
	S006	M	421034		Washer Plain M6		31
	S007	M	421054		PLAIN WASHER M8		7
	S008	M	421819		Screw Countersunk Head Socket M8x120		6
10-09			902G384		Safety Lockouts		1
	B001	E	632128		Safety Lockout Brady 265375		1
	B002	E	632129		Padlocks Blue Brady 265374		3
10-10			904G681		Logo & I/D Plate (Jay)		1
	M01	M	325G231		Logo & ID Plate (Jay)		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
10-11			903G776		WARNING LABELS FOR K4000 MACHINE	323G361	1
	M001	M	323G382		WARNING LABEL (Warning/Winding System)		1
	M002	M	323G381		WARNING LABEL (Warning/Empty Chamber)		1
	M003	M	323G385		WARNING LABEL (Danger/Isolator)		3
	M004	M	314G491		WARNING LABEL (Attention/Air Supply Lockout)		1
	M005	M	323G386		WARNING LABEL (Lockout/Isolator)		11
	M006	M	323G387		WARNING LABEL (Caution/Windows)		6
	M007	M	323G536		WARNING LABEL (Warning Sign)		6
	M009	M	323G384		WARNING LABEL (Do not removed guards)		8
	M010	M	323G357		WARNING LABEL (Feed Open)		1
	M011	M	323G358		WARNING LABEL (Feed Closed)		1
	M012	M	323G359		WARNING LABEL (Return Open)		1
	M013	M	323G360		WARNING LABEL (Return Closed)		1
	M014	M	323G388		WARNING LABEL (Danger/Isolate)		5
	M015	M	325G682		WARNING LABEL (Guards/Large)		2
	M016	M	325G690		WARNING LABEL STRAP		8
	S001	M	421172		SOCKET CAP SCREW M5x12		16
	S002	M	421164		PLAIN WASHER M5		32
	S003	M	421170		Nut Full Hex M5		16
10-13			904G129		NAME PLATES	324G188	1
	M01	M	324G187		K4000 LOGO PLATE		1
	M02	M	323G863		GENERAL LOGO PLATE		1
	S01	M	421892		Screw Panhead M4x16		8
11					Cryo Pumping Sub Assy		
11-01			904G606		Cryo Connections Assy (25 Deg)	407G198	1
	B001	M	425050		Polycold Unit World Model PFC1100HC		1
	B002	M	423595		Elbow Male/Female 3/4" BSP M.I.		2
	B003	M	423718		Reducer 5/8"-3/8"		1
	B004	M	427073		Swagelok Male Hose Adaptor		2
	B005	M	423435		3/4"BSP HEX NIPPLE M.I.		1
	B006	M	423138		Socket Crane 316 M 3/4" BSP Type 177		1
	B007	M	425025		High Capacity Refrigerant Charge for PFC1100		1
	B008	M	423063		Copper Tube Fully Annealed 3/8" O/D		3
	B009	M	423064		Copper Tube Fully Annealed 5/8" O/D		3
	B010	M	423065		Armaflex Insulation AF 22mm x 25mm Wall		7
	B011	M	421363		BOLT-COLLARED EYE BS4278 M10		2
	B012	M	423061		Reducer 5/8"-1/2"		1
	B013	M	423062		Reducer 3/8"-1/2"		1
	M001	M	407G207		Polycold Surround		1
	M002	M	322G632	IT.1	Polycold Water Flow Restrictor		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S001	M	421033		SOCKET CAP SCREW M6x12		6
	S002	M	421034		Washer Plain M6		6
	S003	M	421027		NUT FULL HEX M10		2
	S004	M	421029		PLAIN WASHER M10		2
11-02			903G623		CRYO COIL EVAP ZONE	404G037	1
	B001	M	239218		Seal O Ring RM0995-30Nitrile		2
	M001	M	404G033	2450	CRYO COIL		2
	M002	M	404G035		CRYO COIL BRACKET		3
	M003	M	404G036		SUPPORT BLOCK		12
	M004	M	404G420	2450	CRYO COIL GUARD		1
	M005	M	308G933		Feed Through Flange		2
	M006	M	405G309	2450	TRAY		1
	M007	M	405G310		CRYO COIL CENTRE SUPPORT		2
	S001	M	421035		SOCKET CAP SCREW M10x25		6
	S002	M	421143		SOCKET CAP SCREW M6x50		12
	S003	M	421049		SOCKET CAP SCREW M8x20		8
	S004	M	421057		Nut Full Hex M6		12
	S005	M	421029		PLAIN WASHER M10		6
	S006	M	421034		Washer Plain M6		20
	S007	M	421054		PLAIN WASHER M8		8
	S008	M	421019		SOCKET CAP SCREW M6 x 20		8
12					Ancillary Equipment Sub Assy		
12-01			904G613		Chiller Unit (+20 to -15C) 2050/2450 400v 3ph 50Hz	325G050	1
	B001	M	325G049	OPTION2	Chiller Unit (+20 to -15C) 2050/2450 400v 3ph 50Hz		1
	S001	M	421705		Screw Caphead M10x35		12
	S002	M	421027		NUT FULL HEX M10		12
	S003	M	421029		PLAIN WASHER M10		24
12-02			903G658		Installation Kit K4000 Metalliser		1
	B001	M	421847		HSA M10x140-70/77 Anchor stud		62
	B002	M	421016		HSA M16x240-125/145 Anchor stud		22
	B003	M	421820		HSA M16x190-75/95 Anchor stud		6
	B004	M	267001		Hammer drill 110V 50hz SB2E20R A.E.G		1
	B005	M	267002		Hilti TE-CX 10/47 drill bit M10 x 400LG DRILL BIT		2
	B006	M	267003		Hilti TE-CX 16/47 drill bit M16x 400LG		1
	B007	M	267006		Spanner M20 ref CSM220		1
	B008	M	267007		Spanner M22 ref CSM222		1
	B009	M	267008		Spanner M24 ref CSM224		1
	B010	M	267009		Spanner M30 ref CSM230		1
	B011	M	267010		Bacho adjustable spanner 15" ref No 8074		1
	B012	M	267011		Allen key set 10M		1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B014	M	267013		Screw driver FL/Tip 6" ref No 66-175		1
	B015	M	267015		Barn tool box 19" ref No AB19		1
	B016	M	267016		Grease gun side lever light/W ref No 555		1
	B017	M	267017		Nylon flexible extension 9" (ref No RFN-9)		1
	B018	M	267018		Aluminium bar 2"x1"x3 metres long		1
	B019	M	268003		Ronacrete primer		2
	B020	M	268002		Ronacrete epoxy floor mortar		24
	B021	M	267005		TKC7M combination spanner set		1
	B022	M	267045		14mm A/F Allen key (Long Arm Type)		1
	B023	M	267046		17mm A/F Allen key (Long Arm Type)		1
	B024	M	267047		19mm A/F Allen key (Long Arm Type)		1
	B026	M	267064		TORX Tamperproof L-Shaped Driver Set		1
	B027	M	559191		STANLEY KNIFE RETRACTABLE		1
	B028	M	559058		STANLEY KNIFE - BLADES		1
	B029	M	267065		3 Metre Tape Measure		1
	B030	M	267066		8mm A.F Allen T Bar Key		1
	B031	M	267067		3/8" A.F Allen T Bar Key		1
	B032	M	267068		Side Wire Cutters		1
	B033	M	267069		3" Scraper		1
	B034	M	267070		1" Scraper		1
	B035	M	267071		Window Scraper		1
	B036	M	267072		10" Scaffold Sprit Level		1
	B037	M	267012		B/P Hammer 1 1/2LB ref No 5312RH		1
	B038	M	267079		5/8" A/F Allen key (Long Arm Type)		1
	M001	M	421837		Packer 250x200x1mm m/s		8
	M002	M	421838		Packer 250x200x2mm m/s		16
	M003	M	421839		Packer 250x200x3mm m/s		16
	M004	M	421840		Packer 250x200x10mm m/s		8
	M005	M	421007		Packer 200x50x1mm m/s		96
	M006	M	421008		Packer 200x50x2mm m/s		192
	M007	M	421009		Packer 200x50x3mm m/s		192
	M008	M	421010		Packer 200x50x10mm m/s		96
	M009	M	421841		Packer 120x120x1mm m/s		16
	M010	M	421842		Packer 120x120x2mm m/s		32
	M011	M	421843		Packer 120x120x3mm m/s		32
	M012	M	421844		Packer 120x120x10mm m/s		16
13					Operator Controls Sub Assy		
13-01			903G600		Vessel Desk	323G065	1
	B1	E	244G711		OPERATOR ARM CONSOLE [E05_0]		1
	M1	M	323G305		SPACER		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S1	M	421165		SOCKET CAP SCREW M10x75		4
14					Machine Cables Sub Assy		
14-02			903G894		MACHINE CABLES		1
		E	613482		STANDARD PROFIBUS CABLE		40
		E	613483		TRAILING PROFIBUS CABLE		20
		E	613620		CABLE 8 CORE 28AWG FCC68 FLAT DATA		15
		E	613621		CABLE 7CORE 1MM PVC/PVC YY/NR		150
		E	613858		CABLE 4CORE 6MM PVC/PVC CY/NR		10
		E	613624		CABLE 4CORE 50MM PCU/PVC/PVC GREY YY/NR		15
		E	613626		CABLE 4CORE 25MM PVC/PVC CY/NR SCREENED		10
		E	613627		CABLE 4CORE 16MM PVC/PVC YY/NR		30
		E	613629		CABLE 4CORE 1.5MM PVC/PVC CY/NR SCREENED		10
		E	613630		CABLE 3PAIR TWISTED + SCREEN 22AWG		60
		E	613631		CABLE 3CORE 1MM PVC/PVC YY/NR		120
		E	613633		CABLE 2PAIR 18AWG TWISTED + OVERALL SCREEN		440
		E	613635		CABLE 12CORE 1MM PVC/PVC YY/NR		70
		E	613639		CABLE 4CORE 10MM PVC/PVC CY/NR SCREENED		30
		E	613770		CABLE 4CORE 10MM PVC/PVC YY/NR		110
		E	613774		CABLE ASI PROFILE YELLOW		150
		E	6131037		CABLE 2PAIR TWISTED + SCREEN 20AWG		100
		E	613778		CABLE 4CORE 35MM PVC/PVC YY/NR		15
		E	613785		CABLE 4CORE 50MM PCU/PVC/PVC YY/NR		40
		E	613864		CABLE 4CORE 16MM PCU/PVC/PVC YY/NR		20
		E	613865		CABLE 7CORE 10MM PVC/PVC SY/NR		100
		E	613829		CABLE 5CORE 1.5MM PVC/PVC CY/NR SCREENED		20
		E	613775		CABLE ASI PROFILE BLACK		10
		E	613850		CABLE 4CORE 1MM PVC/PVC YY/NR		60
		E	613840		CABLE 4CORE 2.5MM PVC/PVC CY/NR SCREENED		20
		E	613639		CABLE 4CORE 10MM PVC/PVC CY/NR SCREENED		30
					ISSUE 2: ECO 06G00374 CABLES CHANGED		0
15					Monitor Sub Assy		
15-01			904G228		Eddy Current Monitor Beam	406G561	1
	001	E	244G517		BEAM WIRING		1
	B001	E	673139		LF Probe Assembly		24
	M001	M	320G580	2450	Probe Monitor Beam		1
	M002	M	309G541		BEAM BRACKET		2
	M003	M	406G562		Monitor Beam Clamp		1
	S001	M	421035		SOCKET CAP SCREW M10x25		4
	S002	M	421029		PLAIN WASHER M10		4
	S003	M	421138		SOCKET CAP SCREW M8x25		4

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	S004	M	421404		Screw Caphead M8x45		2
	S005	M	421511		Screw HEX HD Setscrew M8 x 40		1
	S006	M	421103		Washer Spring M8		10
	S007	M	421054		PLAIN WASHER M8		2
	S008	M	421036		Nut Full Hex M8		7
	S009	M	421273		Screw Countersunk Setscrew M6x25 Brass		97
	S010	E	616616		Connector DB15P		25
	S011	E	616618		Solder 15 Way D SKT		1
	S012	E	668031		Resistor 22 OHM		4
	S013	E	616621		Solder Tag (100 per pack) qty is per pack		1
	S014	E	616622		25WDS IDC		1
	S015	M	2441425		Right Angle Bracket (2 per pack) qty is per pack		4
	S016	M	421274		Spacer M3x25 (50 per pack) qty is per pack		4
	S017	M	421275		Screw Panhead M3x16		4
	S018	M	421276		Screw Countersunk Head Socket M3x20 SS		4
	S019	E	616639		Shell 15D		1
	S020	E	627047		RIBBON CABLE 14 WAY		15
	S021	M	235185		Probe Mounting Spring		96
15-07			902G768		Buffer Box Assembly		1
	001	E	673141		LF Buffer Box		1
	002	E	673135		Beam Interface (Buffer) Board Assembly		1
15-10			903G888		Video Camera		1
	001	E	430032		Camera Dome Top Colour Vandal Resistant		4
	002	E	618530		ESP COLOUR QUAD MONITOR DCQ3		1
	003	E	618263		6 PIN MINI DIN EXTENSION CABLE, 3M		2
	004	E	618435		6 PIN MINI DIN EXTENSION CABLE, 15M		2
	006	E	430043		WALL MOUNTED MONITOR BRACKET 360 x 240 AT 50KG		1
15-20			903G737		Desk Components, PC Hardware and Software		1
	002	E	618452		SIEMENS 40GB HD		1
	003	E	650712		WIN AC PRO LITE SLOT PLC (PCI SLOT) CPU416-2		0
	005	E	618340		PORTABLE MINI KEYBOARD		1
	006	E	618341		TRACKBALL MOUSE		1
	007	E	618435		6 PIN MINI DIN EXTENSION CABLE, 15M		2
	008	E	618425		SVGA MONITOR EXTENSION CABLE, MALE-FEMALE, 15.2M		1
	009	E	618576		COLOUR DESKJET PRINTER HP 6940		1
	011	E	618434		ZOOM 56K PCI INTERNAL MODEM		1
	012	E	860011		PCANYWHERE S/W V11.5 & NORTON GHOST V9.0		1
	025	E	618445		WinCC V6.0 SP4 RT8K E, G, SP, F, I		1
	026	E	618419		WinCC V6.2 SP3 User Archives Option for WinCC		1
	030	E	618468		NOT FOR SIEMENS!		0

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	030	E	618471		PC DISK DRIVE POWER LEAD TYPE 3		1
	031	E	618470		IDE 3 POINT HARNESS		1
	032	E	6131013		CAT5 CROSSOVER LEAD STP 2 METRE (BIEGE)		0
	033	E	618572		5M USB 2.0 CABLE PLUG B TO PLUG A		0
16					Plasma Treater Sub Assy		
16-01			904G248		STANDARD PLASMA TREATER MECHANICAL ASSEMBLY 2005	406G290	1
	B001	M	422054		Berger Hinge GN-129-16-D		4
	B002	M	424033		Clamp Ring DN 10-16 ISO-KF		1
	B003	M	424032		Centering Ring DN 16 ISO-KF		1
	B004	M	424021		Elbow 90 deg - Hose Adaptor DN 16 ISO-KF-20		1
	B005	M	424023		Screw - In Flange DN 16 ISO-KF x G1/2"		1
	B006	M	423989		Extended Swivel Elbow Connector 6 OD x 1/8" BSPP		1
	B008	M	427097		Swagelok PB8 Hose 10m		1
	B009	M	423706		Tube 12mm O/D x 1.0 WALL S.S. CUT TO LENGTH.		1
	B010	M	423980		Tube 10mm O/D x 1.0 WALL S.S. CUT TO LENGTH.		1
	B011	M	427183		Gasket ISO Parallel 3/8"		8
	B012	M	427234		Male Connector		2
	B013	M	423988		10mm-3/8" Male Elbow Connector		4
	B014	M	423586		10mm Union Elbow		2
	B015	M	423806		12mm-3/8" Male Connector		2
	B016	M	423987		Positionable Male Elbow 12 OD x 3/8" BSPP		2
	B017	M	423588		12mm Union Elbow		6
	B018	M	427245		Male Adaptor 12 OD x 3/8" BSPP		4
	B019	M	427087		3/8" Tube Adaptor		2
	B020	M	427060		Tube Adaptor 12 OD x 1/2" Hose		4
	B021	M	423693		Stauff Pipe Clamp 3190PP		1
	B023	M	240466		Handscrew		2
	B024	M	422053		Taper Knob M6 x 20 O.D. (Black)		1
	B025	M	422012		Taper Knob M10 x 40 O.D. (Black)		2
	M001	M	406G283	2450	Electrode Enclosure		2
	M002	M	406G284	2450	Magnet Shunt Plate		2
	M003	M	406G285		MAGNET RETAINER		4
	M004	M	406G286		END INSULATION		4
	M005	M	406G287		END PLATE		1
	M006	M	406G288	2450	Magnet retaining strip		4
	M007	M	406G289	2450	Insulation Plate		2
	M008	M	406G297	2450	Back plate		1
	M009	M	406G300	2450	Water cooled side plate		2
	M010	M	406G313	2450	Conductance seal 1		1
	M011	M	406G314	2450	Conductance seal 2		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M012	M	406G315		Conductance seal spacer		4
	M013	M	406G324		End Plate		1
	M014	M	406G327	2450	Cover plate LH		1
	M015	M	406G329		Back Treatment Bar		2
	M016	M	406G330		Location Block		2
	M017		406G332	2450	Impedance shield		10
	M018	M	406G340	2450	Strengthening channel		1
	M019	M	406G342		Mounting Bracket		4
	M020	M	406G343		Sight Glass Holder		2
	M021	M	406G348	2450	Cover plate RH		1
	M022	M	406G349	2450	Threading cover		1
	M023	M	406G351		Locking Block		1
	M024		406G352	2450	PTFE Strip		4
	M025	M	406G354		End Plate		2
	M026	M	406G355	2450	Ceramic shield clamp		2
	M027	M	406G356	2450	Ceramic shield clamps		1
	M028	M	406G358	2450	Conductance shield 3		1
	M029	M	406G359	2450	Gas Bar		1
	M030	M	406G372	2450	Water cooled centre plate		1
	M031	M	406G376		Power Stud		2
	M032	M	406G377		Insulation Sleeve		2
	M033	M	406G378		Insulation Block		1
	M034	M	406G379		Insulation Cover		1
	M035	M	406G384		Impedance block		2
	M036	M	406G389		Location Plate		2
	M037	M	406G395		Stop Block		2
	M038	M	406G396		Stop and Location block		1
	M039	M	406G397		Stop and Location Block		1
	M040	M	402G585		SINTERED N.I.B. MAGNET		4
	M041	M	402G590		ANISOTROPIC FERRITE MAGNET 150 X 25		30
	M042	M	402G647		ANISOTROPIC FERRITE MAGNET 136 X 25		2
	M043	M	402G650		ANISOTROPIC FERRITE MAGNET 150 X 12		64
	M044	M	402G652		ANISOTROPIC FERRITE MAGNET 90 X 12		4
	M045	M	402G654		ANISOTROPIC FERRITE MAGNET 82 X 12		4
	M046	M	402G656	IT0	CERAMIC STRIP (FIXED LENGTH)		84
	M047	M	402G656	IT1	CERAMIC STRIP (FIXED LENGTH)		4
	M048	M	402G656	IT2	CERAMIC STRIP (FIXED LENGTH)		8
	M049	M	402G809		SIGHT GLASS		2
	M050	M	406G566		Impedance block		2
	M051	M	402G656	IT3	CERAMIC STRIP (FIXED LENGTH)		12

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	M052	M	406G350		Locking Block		1
	S001	M	421149		SOCKET CAP SCREW M10x20		2
	S002	M	421339		Screw Caphead M5x10 SS		4
	S003	M	421121		SOCKET CAP SCREW M6 x 16		10
	S004	M	421019		SOCKET CAP SCREW M6 x 20		4
	S005	M	421091		Screw Socket Cap Head M6 x 25		1
	S006	M	421145		SOCKET CAP SCREW M6x30		18
	S007	M	421351		SOCKET CAP SCREW M5x50		4
	S008	M	421049		SOCKET CAP SCREW M8x20		24
	S009	M	421138		SOCKET CAP SCREW M8x25		28
	S010	M	421237		SOCKET CAP SCREW M8x35		7
	S011	M	421648		Screw Csk Head Slotted M5 x 8		56
	S012	M	421444		SOCKET C'SK SCREW M5x10		4
	S013	M	421752		SCREW C'SK HEAD SOCKET M5x12		48
	S014	M	421427		Screw Counter Sink M5 x 16 Lg. Nylon		24
	S015	M	421051		Screw Socket Countersunk Head M6 x 12		13
	S016	M	421418		Screw Socket Countersunk Head M6 x 16		4
	S017	M	421545		Screw Set Hex Head M8x20 Brass		2
	S018	M	421547		Shoulder Screw Socket Head 6mm (M5) x 30 LG		2
	S019	M	421652		Screw Shoulder Screw 8mm (M6) x 16lg		2
	S020	M	421034		Washer Plain M6		16
	S021	M	421328		Washer Plain M5 St.Stl.		4
	S022	M	421057		Nut Full Hex M6		16
	S023	M	421323		Screw Socket Countersunk Head M5x12 Nylon		24
16-02			904G249		PLASMA TREATER INSTALLATION 2005	406G363	1
	B001	M	239271		Seal O Ring RM0191-16V Viton		2
	B002	M	239199		Seal O Ring RM0251-16 Nitrile		2
	B003	M	239524		Seal O-Ring 18mm dia shaft 4mm section dia		2
	B004	M	4231001		Straight adaptor 6mm O/D to 1/4"BSPP male		1
	B005	M	4231000		Reducer 1/2"BSPT male to 1/4"BSPP female		2
	B006	M	427078		Tube 1/2" P.O. Multi purpose hose Blue		1
	B007	M	427088		Tube 3/8" P.O. Multi purpose hose Blue		1
	B008	M	424022		Hose PVC 20 Dia		1
	B009	M	423506		Nylon Tube Natural 3/8" O.D.		1
	B010	M	423265		Nylon Tube Natural 6mm O/D		1
	B011	M	427249		12MM-1/2" STRAIGHT ADAPTOR		8
	B012	M	427250		Swagelok Male Connector B-12MTA-1-8RS		2
	B013	M	427251		3/8"-3/8" STRAIGHT ADAPTOR		2
	B014	M	427244		Male Connector		2
	B015	M	423587		12mm-1/2" Male Connector		2

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B016	M	423645		12mm-1/2" Male Connector		2
	B017	M	423588		12mm Union Elbow		2
	B018	M	427087		3/8" Tube Adaptor		2
	B019	M	427060		Tube Adaptor 12 OD x 1/2" Hose		14
	B020	M	427183		Gasket ISO Parallel 3/8"		2
	B021	M	427082		SWAGELOK SEALING WASHER S-8-RS-2V		12
	B022	M	424033		Clamp Ring DN 10-16 ISO-KF		3
	B023	M	424032		Centering Ring DN 16 ISO-KF		3
	B024	M	424021		Elbow 90 deg - Hose Adaptor DN 16 ISO-KF-20		1
	B025	M	424023		Screw - In Flange DN 16 ISO-KF x G1/2"		2
	B026	M	427163		Swagelok Male Connector B-12MO-1-12RT		2
	B027	M	424005		Blanking Flange NW16KF		1
	B028	M	4231041		FLOW METER SWITCH AND VISUAL		2
	B029	M	423192		Male Connector 1/4" BSP - 1/4" O/D		1
	M001	M	317G047		Feedthrough Pin		2
	M002	M	317G048		Feedthrough Bush		2
	M003	M	317G049		Feedthrough Locking Ring		2
	M004	M	323G138		FeedThrough Enclosure Box		1
	M005	M	406G341		Plasma Mounting Bracket		2
	M006	M	406G383		Plasma Mounting Bracket		2
	M007	M	406G404		Water feed through nozzle		2
	M008	M	406G416		Live lead through block		1
	M009	M	406G417		Live lead through cover		1
	M010	M	406G558		Plasma water manifold		1
	M011	M	406G583		Size 16 aluminium extension nipple		1
	REF1	M	406G290		ELECTRODE ASSEMBLY		1
	REF2	M	406G387		Roller assembly		1
	REF3	M	400G430	1650	245 O/D TIE BAR		1
	REF4	M	321G429		Litz Cable Assembly		2
	S001	M	421031		SOCKET CAP SCREW M5x20		2
	S002	M	421890		SOCKET CAP SCREW M5x40		2
	S003	M	4211105		Screw Caphead M8x90		2
	S004	M	421035		SOCKET CAP SCREW M10x25		8
	S005	M	421215		Screw HEXHD Setscrew M8x20 Brass		2
	S006	M	421359		Nut Brass Full M16		2
	S007	M	421699		Bolt HEXHD M10x20		8
	S008	M	421296		Bolt HEXHD M10X40		4
	S009	M	421029		PLAIN WASHER M10		12
	S010	M	421054		PLAIN WASHER M8		8
16-03			904G250		PLASMA TREATER ROLLER K4000/M1 2005	406G387	1

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SECT	BUBB NO	DEPT	ITEM	PARAM	DESCRIPTION	ASSY	QTY
	B001	M	200502		BEARING NO SEALS 2206E		2
	B002	M	214025		Nipple Grease M6		2
	M001	M	321G091	2450	148 DIA ALUMINIUM ROLLER		1
	M002	M	321G090		BEARING BUSH		2
	M003	M	406G385		Inboard roller support bracket		1
	M004	M	406G386		Outboard roller support bracket		1
	M005	M	406G388		Bearing End Cap		2
	M006	M	406G567	A	Roller bracket shims		2
	M007	M	406G567	B	Roller bracket shims		4
	S001	M	421121		SOCKET CAP SCREW M6 x 16		2
	S002	M	421019		SOCKET CAP SCREW M6 x 20		6
16-04			904G278		PLASMA TREATER POWER SUPPLY UNIT		1
	B001	E	670117		PEII 10Kw AC power supply with active front panel		1
16-05			904G279		PLASMA TREATER MANUAL GAS SYSTEM	406G577	1
	B001	M	427129		1/4" BSPT Male - 1/4" Push in Fitting		2
	B002	M	4231058		1/4"-1/4" BSPT NIPPLE		2
	B003	M	426100		Flowmeter		1
	B004	M	424015		Needle Valve Port Size 1/4"		1
	B005	M	424014		Direct Acting Solenoid Valve		1
	B006	M	423212		Nylon Tube Natural 1/4" O/D		1
	B007	E	674002		Pressure Switch		1
	B008	M	424035		Tee DN 16 ISO-KF		1
	B009	M	424032		Centering Ring DN 16 ISO-KF		3
	B010	M	424033		Clamp Ring DN 10-16 ISO-KF		3
	M001	M	312G411		Valve Mounting Bracket		1
	M002	M	406G578		Plasma Treater Gas Information Plate		1
	S001	M	421121		SOCKET CAP SCREW M6 x 16		2
	S002	M	421057		Nut Full Hex M6		2
	S003	M	421034		Washer Plain M6		2

APPENDIX B

GLOSSARY OF TERMS

This appendix contains both explanations of terms and abbreviations. No item is described twice.

Words in capitals refer to other entries in the glossary

Word	Explanation
ADHESION	A term used to describe the bonding strength between the evaporated aluminium and the substrate.
ANGSTROM	A unit of length equal to 1 E-8 centimeters. Commonly used as a measure of thickness of vacuum deposited metals.
ANODE	The positive electrode in an electron device. This electrode may collect electrons and negative ions.
AUTO START	Operational mode used to allow the starting of the vacuum pumping group using a timer to pre-set the start up time and date.
BACK PRESSURE	The pressure produced at the exit side of a pump.
BACKGROUND SCRATCHES	Marks on a coated substrate, the source of which is in the metal deposition.
BARRIER COAT	A coating applied to a material to prevent migration of some substance into or through the material.
BIAXIAL ORIENTATION	The process of stretching a plastic film (usually at elevated temperature) in both the machine and transverse directions so as to achieve similar tensile, modulus, and elongation properties in both directions.
BLOCKING	Undesired adhesion between the plies of materials in roll or sheet form; can be due to adhesive cold flow, improper drying of inks, or improper curing of coatings.
BP	An abbreviation for Booster Pump
BOOSTER PUMP	An intermediate mechanical pump located between a diffusion pump (or other type of high vacuum pump) and its fore pump. It serves to increase the gas handling characteristics of the overall system or to act as a dynamic barrier to the penetration of specific vapours or gases into the high vacuum pump.
BOWED ROLL	A type of spreader roller used for the removal of MD wrinkles during web transport. The curvature in the roll causes the centre of the web to become tight and forces the wrinkles to the edges of the web. Sometimes referred to as a Banana Roll or a Mount Hope Roll.
BULKHEAD	A descriptive name sometimes given to assembly comprising of a large rectangular face plate assembly mounted on a wheeled carriage that can be retracted to facilitate operator access to the cantilever-mounted web transport system.
BUTT SPLICE	The joining of two webs without an over-lap; most often accomplished by the use of pressure-sensitive tapes, heat reactivated tapes, etc.
BUTTERFLY VALVE	A valve consisting of a rotating vane that blocks the supply when at right angles to the direction of flow (i.e. closed) and allows flow when in line with it.
CANTILEVERED	Equipment projecting from a wall or bulkhead primarily supported at one place only
CAST	A forming of a plastic film or sheet by pouring liquid resin onto a moving belt or by extruding a plastic material onto a highly polished stainless steel roll.

Word	Explanation
CATHODE	The negative electrode of an electron device. This electrode may emit electrons and collect positive ions.
CENTRE WINDING	A winding method whereby the driving force, supplying the tension, comes from the shaft through the centre of the roll.
CENTER/SURFACE WINDING	A winding method where tension is supplied both from the centre shaft and the rolls via a driven rider roller.
CHAMBER	The chamber is the fixed part of the machine, into which the winding cart is driven. It is designed to allow evacuation to the process vacuum levels required.
CHUCK	See CORE CHUCK
COATING ZONE	The zone in which the sputtering process takes place. The pressure in the coating zones is higher (i.e. lower vacuum) than in the winding zone when the process is running. This is because of the input of process gases into the coating zone.
COLD-CATHODE IONIZATION GAUGE	An ionization gauge in which electrons are released from the cathode by photon and ion bombardment at ambient temperature, and for which thermionic emission can generally be neglected.
CONDUCTANCE-VACUUM	The ratio of throughput, under steady rate conservation conditions, to the pressure differential between two specified cross-sections inside a pumping system.
CONSTANT TENSION	A winding or unwinding mode in which the web tension is kept constant with changing roll diameter.
COOLING ROLLER	The roller in the web path which serves the dual purpose of cooling the web and adjusting the tension on the web prior to the rewind.
CORE	A tube on which webs of material are wound.
CORE CHUCK	A mechanism for fixing a core onto a rotating shaft so that there is no slippage between the shaft and the core.
CORONA TREATMENT	A method for improving the surface adhesion of a substrate by altering its surface through an oxidation process initiated by an electrical discharge.
CPU (Central Processing Unit)	The "heart" of any computer or programmable logic system.
CREASES	Marks caused by fold overs usually due to wrinkles in a web material.
CRITICAL INLET PRESSURE	The inlet pressure of a vapour pump above which an abrupt decrease in pumping speed occurs.
CROSS-OVER PRESSURE	The highest pressure at the inlet of a high vacuum pump at which the pump may be operated. It refers to the high inlet pressure condition that exists following the initial vacuum system pump down by a roughing pump and crossing over to a high vacuum pumping system.
CRYOGENIC	Is the term used for the science of refrigeration.
CRYOPANEL	A refrigerated tube using liquid helium roller to collect any moisture present on the web face that will be treated.
CRYOPUMP	A vacuum pump that operates by the condensation of gas at surfaces maintained at temperatures sufficiently low for the vapour pressures of the condensed gases to be insignificant.
DEGASSING	The deliberate removal of gas from a material, often achieved by heating the material under vacuum (for example, in an ion gauge).
DIFFUSION PUMP	A vapour pump in which the pumped gas flows into a vapour stream under conditions in which molecular flow predominates. Momentum is transferred from the vapour to the gas carrying it along in the direction of the stream.

Word	Explanation
DIFFUSION PUMP (cont.)	Pump fluid is heated in vacuum to generate the vapour that is directed through a nozzle. It expands freely in the stream before it reaches a cool wall where it condenses and is returned to boiler to begin a new cycle.
DISHING	A term used to describe the movement on a roll of substrate.
DOWNTIME	Production time lost due to unscheduled maintenance on the machine.
DULL FINISH	Any finish that lacks luster but is somewhat smooth. Any smooth surface that lacks luster or brilliance.
DRAW CONTROL	Descriptive name given to operational mode of drive system when a roller drive motor is controlled to operate at a speed higher or lower speed than the master speed roller. Generally mode of operation applied to draw rollers(DTR)
DRUM	The component around which the web is wound in the process zone. The drum is double-skinned and the surface is cooled.
DRUM SHIELD	Name given to copper shield that is used to protect coating drum
DRYSTAR PUMP	This is a multi-stage positive displacement rotary pump that contains no sealing or lubricating fluids within the swept volume of the pump. Making the pump ideal for evacuating dangerous gases, or for use with a 'clean' process.
E-STOP	See Emergency Stop
EBONITE ROLLER	Type of roller manufactured from Ebonite. Ebonite is a hard plastic that is applied to the roller and then machined. Has no magnetic properties and is therefore used as backing roller for non contact resistance probes.
ELONGATION	The distance a material will stretch lengthwise before breaking, expressed as a percentage of its original length (ASTM D882).
ELECTRODE	An electrically conducting element of an electron device, usually connected to an external electric circuit. which emits. Collects. or influences the flow of current in the device.
EMERGENCY STOP	A mode of operation selected by pressing the red button (yellow background) on the operator panel. Activation will remove power from the drive, pumping and coating systems
EV (EQUALIZING VALVE)	A valve between each coating zone and the winding zone. The valve is opened when it is required to equalize the pressures between the zones.
EXHAUST FILTER	A baffle used to prevent loss of the pump fluid mixed with discharge gases from a vacuum pump. The trapped fluid is then returned to the pump.
FACE PLATE	The large rectangular plate on which the winding mechanism is mounted. The face plate is mounted on a wheeled cart unit via angle brackets and bracing ties.
FEATHERING	A term used to describe where the substrate moves slightly as it is unwound.
FEED THROUGH	A device for transmitting electrical current. fluids or mechanical motion through the wall of a vacuum system
FILM	A uniform, thin (0.00025 - 0.032 inch) substrate manufactured from synthetic polymers such as polyvinyl acetates, polycellulostics, polyester, polyethylene, polycarbonate, polypropylene, etc.
FILM CAST	Generally refers to film made by coating or casting a solution of a polymeric material on an endless belt or wheel, drying the solvents, stripping the film from the belt and winding after additional drying. Also applied to film thermally extruded onto a casting roll.

Word	Explanation
FLAG	A marker, usually a strip of colored paperboard, inserted in a roll of material and extended from an edge. This is to designate a deviation from standard, such as a splice, defect or specification change, or to mark a specific length.
FIXED SPREAD ROLLER	A rubber-covered roll, grooved with a narrow pitch from the centre of the face to each end, used to help maintain a flat web on thin film.
FLOW (GAS)	The motion or passage of a gas.
FLOW (MOLECULAR)	Mean free path is much greater than the largest dimension of a cross-section of the channel. At these pressures, the flow characteristics are determined by collisions of the gas molecules with the channel surface and flow effects from molecular collisions are insignificant
FLOW (Viscous)	The flow of a gas through a channel under conditions such that the mean free path is very small in comparison with the smallest dimension of a cross section of the channel. At these pressures, the flow characteristics are determined mainly by collisions between the gas molecules, i.e. the viscosity of the gas. The flow may be laminar or turbulent.
FORELINE	Descriptive name given to the exhaust section of the diffusion pump and the pipe work between the pump and the booster /rotary pumps
FORE PRESSURE	Sometimes referred to as backing pressure, and is the pressure measured downstream from the outlet or fore line of a vacuum pump.
FORE LINE VALVE	A valve in a fore line pipe work.
FUZZY EDGES	Edges of a slit substrate roll that exhibit protruding fibres due to poor slitting
GAS	A state of matter in which the molecules move freely to occupy the total volume of an enclosure.
GATE VALVE	An alternative type of valve used as an isolation valve between the high vacuum pumps and the vacuum chamber. Constructed as a plate that slides over the cross-section of pipeline. When in the closed position, total isolation can be achieved.
GAUGE (related to thickness)	Thickness of a material measured in hundred thousandths of inches, decimals, mils or microns. An instrument for exact measuring.
GAUGE BAND	A continuous lane of film web that has abnormal machine direction thickness compared to the adjacent film web.
HEAT LINES	Machine direction lines that have a crease-like appearance due to uneven shrinkage from excessive heat.
HELIUM DETECTOR	A leak detector in which a mass spectrometer is adjusted for maximum response to helium ions (4He^+).
HIGH EDGES	Raised or flared edges extending above the surface of the main body of a roll. Usually caused by dull or misaligned slitter knives causing slit edge flaring or thickening that stacks to cause pronounced high edges.
HIGH VACUUM VALVE	Reference to the valve placed between the high vacuum pumps and the vacuum chamber, and used to isolate the high vacuum pumps from the chamber during either roughing or venting the vacuum chamber.
HOLD MODE	Name given to operational mode relating to the pumping system. Placing the system into the HOLD mode stops all further evacuation of the chamber by the vacuum pumps.

Word	Explanation
INBOARD PLATE	Name given to plate that is some times attached to the face-plate and is used to support rollers.
INCH	Term used to describe slow winding operation of the web winding system. The function and operation of the inch mode are described in this manual.
INERT GAS	A gas that does not normally react chemically with other substances, i.e. an element that has completely filled electron shells such as helium, neon and argon.
INLET PRESSURE	The gas pressure at the entrance to a pump.
INTERLEAVE	A layer of material that is wound on to the rewind with the web to keep adjacent surface separate.
ION	An atom or molecule that has gained or lost one or more electrons and which thus has a negative or positive charge.
IONIZATION	A process that results in the formation of ions. Such a process can occur by adding (or removing) one or more electrons to (or from) an atom or molecule.
IONIZATION GAUGE	A vacuum gauge comprising a means of ionizing the gas and a means of correlating the ion current to the collector with the pressure of the gas.
JOG	See INCH
LEAK	Term used to describe void in an item. Generally used to describe leakage of water from cooled circuits or air into the chamber, vacuum leak.
LEAK (CAPILLARY)	A leak having a small cross-section and length many times it's cross-section.
LEAK (REFERENCE)	A calibrated leak that may be attached to a vacuum system be tested or directly to a leak detector. It is used to determine the value of an unknown leak. or the sensitivity of a leak detector.
LEAK (VIRTUAL)	An apparent leak because of the presence of contaminants which very slowly, or voids within the vacuum system.
LEAK DETECTOR	A device for locating leaks, and measuring the magnitude of the leak
LEAK TESTING	A procedure designed to locate leaks and measure their leakage rate
LEAK TIGHT	A description of an enclosure having a gas leakage rate less specified figure.
LEAKAGE RATE	The quantity of gas passing through a leak in a given time divided time.
LIGHT TRANSMISSION	Percentage of visible light transmitted through a material. Calculated from optical density measurements on an optical densitometer.
LOAD VAPOUR PUMP	The quantity of gas, not including pump fluid vapour. in mass flowing through the pump per unit time. It is also called capacity flow.
LOAD CELL	A transducer which in this machine converts tension into an e signal thus providing a feedback loop to control the tension accurately
LOCKOUT	Term used to describe the isolation of power source. Power sources such as electric, hydraulic, pneumatic.
MAGNETRON	Source of electrode used in the sputtering process, and can be positive (anode) or more commonly negative (cathode), using an it array to create a plasma.
MANDREL	In extrusion, the central finger of a pipe or tubing die. The support shaft for unwinding or rewinding rolls of paper, film, or foil.

Word	Explanation
MANOMETER	An instrument for measuring the pressure of gases and vapours.
MASS SPECTROMETER	An instrument that produces a beam of ions from a sample. The resulting mixture of ions according to the mass to charge ratio provides output signals that are measures of the relative abundance the ionic species present. A mass spectrometer usually consists of source, a mass analyzer and a detector.
mbar (millibar)	SI units of pressure. Nominal atmospheric pressure is 1015 mbar.
MEAN FREE PATH	The average distance a particle travels between successive collision other particles.
MECHANICAL PUMP	A device with moving parts such as rotating vanes, a piston or rotary members used for pumping gas or vapours.
MECHANISM	The roller assembly between the two side Plates.
MECHANISM BULKHEAD	see FACE PLATE
MICRON	One millionth of a metre or one thousandth of a millimeter. Twenty- five microns equal 0.001 inch.
MIL	A unit of measurement equal to 0.001 inch. Often used as a thickness measurement for films, foils, and coatings.
MIMIC PANEL	A schematic representation of the machine equipment displayed on a graphic screen for the convenience of the machine operator.
MOISTURE VAPOUR TRANSMISSION	The rate at which water vapour will permeate through a film or material.
MOUNT HOPE ROLLER	See BOW ROLLER
MONITOR PROBE	Name given to the non contact resistance measuring heads located close the web. Used to measure the thickness of the coating on the web.
OHM	A unit or electrical resistance equal to one ampere.
OHMS PER SQUARE	Measure of metal deposition by electrical resistance. When measuring resistance across a uniform square of metallized film, the value of ohms is identical whether the sample is one inch square, one foot square, or one yard square. This is because the width of the current path increases or decreases in exact proportion to the length of the current path. In actual practice, the resistance might be measured lengthwise on a ten inch length of film that is 0.5 inch wide. Such a sample has 20 squares (10x0.5) and the resistance in ohm is divided by 20 to arrive at ohms per square.
OIL MIST SEPARATOR	A housing mounted on the exhaust port of a mechanical pump, including baffle surfaces and absorbent material to trap oil vapours.
OIL SPOT	A surface contaminant that can occur randomly. Often difficult to detect in clear film, the contaminant usually becomes quite apparent after coating in the form of clear spots in transmitted light or dark spots in reflected light.
OUTGASSING	The evolution of gas from a liquid or solid under vacuum
OPP	Oriented (usually biaxially) polypropylene.
OPTICAL DENSITY	A measurement of opacity as 1/log (percent transmission).
OUTGASSING	A term used in the vacuum industry for the evaporation under vacuum of a volatile substance, which causes loss of vacuum and darkening and poor adhesion of metal coating.

Word	Explanation
OVERWIND	Term used to describe the direction that the web leaves the unwind roll or is wound onto the rewind roll.
PASCAL	The basic unit of pressure recorded for general use in vacuum technology. It is 1N/m ² . The recommended multiples and submultiples are 10E+3 and 10E-3 respectfully, of those which precede the basic unit.
PERMEABILITY	A measure of gas penetration through a solid barrier.
PICK OFF	Random or repeating areas in a web where the coating has been removed due to making contact with roller surface that is contaminated or moving at a different speed.
PIN HOLING	In coated film, very small holes through the thickness of the film that, normally, are not evident until the film is held against a strong light source.
PIN WINDOW	In coated film, a very small void in metal coating resulting from a contaminant being on the film surface during the metal deposition process.
PIRANI GAUGE	A thermal conductivity gauge containing a heated filament having a large temperature coefficient of resistance. Because heat dissipation from the filament is a function of the gas pressure in a certain pressure range. filament resistance or power required to maintain a constant filament resistance can be correlated with gas pressure.
PLC (Programmable Logic Controller)	The essence of the PLC control is that it will permit an operation to start only when certain specified conditions have been met.
POLYESTER FILM	A film produced from polyethylene (PET) which is usually biaxially oriented and typically has good resistance to moisture, solvents, oils, and many other chemicals.
POLYCOLD	Trade name used to describe a type of Cryo-generator unit.
PRESSURE (DIFFERENTIAL)	The difference in pressure between two regions in space that are at different absolute pressures.
PRESSURE (GAS)	The average normal force per unit area exerted by gas molecules impacting on a surface.
PRESSURE (GAUGE)	A term used in vacuum technology that refers to the pressure reading obtained from a vacuum gauge.
PRESSURE (PARTIAL)	The pressure of a designated component of a gaseous mixture. According to Dalton's Law the sum of the partial pressures of all the component species in a mixture is equal to the total pressure.
PRESSURE (ULTIMATE)	The limiting low pressure approached in a vacuum system after sufficient pumping time has elapsed to establish that further reduction in pressure will be negligible. Sometimes called the ultimate vacuum, or blank off pressure, or base pressure when referred to a pump under test.
PRESSURE (ULTIMATE)	That part of the ultimate pressure in a vacuum system caused by the partial pressure of a specific gas.
PARTIAL PRESSURE (VAPOUR)	The pressure of a vapour in a thermodynamic equilibrium with a condensed phase at a fixed temperature. The definition applies to single components as well as to multi component systems. In the latter case, it is necessary to distinguish between the total pressure over the condensed phase and the partial pressure of a given component
PRESSURE RISE	A term used to describe a reduction in vacuum level.

Word	Explanation
PROFILE	The thickness variation of a web. The statement "a good profile" describes a fairly "flat" web with minimal point-to-point thickness variation.
PUCKERING	The uneven, non flat condition in a web material.
PUMP	See DIFFUSION PUMP, ROTARY PUMP
PUMP FLUID	A liquid usually having a low vapour pressure, used as the working fluid in a mechanical or vapour pump
PUMP DOWN	Term used to describe the period of time it takes the pumping system to evacuate the chamber to the required process vacuum level.
PUMPING SPEED	A ratio of the throughput of a given gas to the partial pressure of that gas at a specified point near the inlet port of a pump.
RATE OF RISE	A leak detection technique in which a vacuum system is isolated from its vacuum pump by a valve and the rate of increase of pressure is observed. The pressure rise is a measure of the combined external and virtual leakage
RECOVERY TIME (LEAK)	The time, after probe gas is removed from a leak, required for the leak indicator of a leak detection system to return to 37% ($1/e$) of the maximum value reached during the period the probe gas was applied.
RECOVERY TIME (PUMPS)	The time required for a pump to return to a previously measured equilibrium pressure after being subjected to an air or other gas load for a specific time.
REEL	A term applied to the untrimmed roll of web material as removed from the producing machine master roll.
REEL SHAFT	This name given to the assembly that the roll of substrate is supported on.
RELATIVE HUMIDITY	The percentage of water vapour in the air compared to the maximum amount of water vapour that could be suspended in the air at the current temperature.
RESIDUAL GAS ANALYZER (RGA)	A device for measuring the amounts and species of various gases present in a vacuum system.
RESISTIVITY	The ability of a material to resist passage of electrical current, either through its bulk or on a surface. The unit of volume resistivity is the ohm-cm; of surface resistivity, the ohm.
RESPONSE TIME	The time required for a gauge reading to reach 63% of its total change when the pressure is changed instantaneously.
RESPONSE TIME (LEAK)	The time required after a probe gas is applied to a leak for the leak indication of a leak detection system to change from a zero or small leak rate indicator to a positive, or larger leak rate indication. This time is often considered equal to the time constant of the system.
REWINDING	The operation of winding the web material from the reel or master roll onto a core to produce rolls of desired width, diameter and tension.
RIGHT-ANGLED VALVE	A valve that operates within a right angle bend.
RHOMETER	A device used to measure the hardness of rolls of substrate.
ROOTS BLOWER	A mechanical vacuum pump that the pumping action is produced by rotating lobes. Effective at molecular flow regimes only.
ROTARY PISTON PUMP	A mechanical vacuum pump in that the pumping action is produced by a rotating piston or plunger.
ROTARY VANE PUMP	A mechanical vacuum pump in that the pumping action is produced by sliding vanes.

Word	Explanation
ROTATABLE	Negative electrode of a sputtering system that rotates around a cathode stationary magnet array.
ROUGHING VALVE	A valve positioned directly between the mechanical pumps (primary pumps) and the vacuum chamber. Its use is to control the initial pumping of the chamber from atmospheric pressure until crossover pressure.
RP (ROTARY PUMP)	A pump that consists of two figure eight elements rotating in opposite directions in a rigid but precisely shaped chamber and meshing together as they do so. They are known as "roughing pumps. a term which also describes the achieving of a rough vacuum that is. the main pumping down prior to the more sophisticated final stages. The coating and winding zones are connected together for roughing. Also known as
ROTARY PISTON.	ROTARY VANE or ROOTS BLOWER.
SIDE PLATES	The two side plates. inboard and outboard. earn the roller assembly between them The inboard side plate is attached to the face plate
SCRATCHES	The linear abrasions in the film surface which can vary in severity. They can be continuous replicating, or random. The usual cause is a process roller surface speed not matching web speed.
SHUTDOWN MODE	Name given to describe a mode of operation in controlling the pumping system.
SPEED (PUMP)	The volumetric rate of gas flow across a given cross-section at the pump inlet. It can be obtained from the ratio of the throughput of a gas to the partial pressure of that gas at a specified point near the inlet port of the pump. Often referred to as the pumping speed
SPLICE	A method of joining paper or plastic webs to produce an operational continuous web.
SPREADER	A special type of roller available in various configurations for the purpose of removing MD wrinkles during web transport via the application of a TD stress
STRETCH LANES	A permanent deformation in the web running in the longitudinal direction, usually caused by the build-up of transverse direction thickness non-uniformity's on a roll. The faults are usually generated by the same internal roll stresses that create wire and rope marks. Depending on the severity, stretch lanes can sometimes have transverse direction puckers (resembling "tank tracks" or a diamond pattern "diamond track" when the film is inspected unrestrained.
TD	The transverse or cross-web direction.
TENDENCY DRIVE	A drive where the roller is connected mechanically to its shaft by bearings. The shaft is driven at approximately the speed of the web. This reduces the force required to get the roller up to speed and to overcome any friction in the bearings.
TELESCOPING	A sideways movement of a web material within a roll, such that successive layers (wrapped) are offset, resulting in a conical configuration.
TENTER FILM	Name given to a film produced on a tenter process.
TENDENCY DRIVE	A roller drive system intended to remove any roller rotational and bearing frictional drag, (i.e. a "helper" drive system). Roller surface determined by web speed.
TENSION	The tensile stress in a web, usually expressed in force unit (lb.) or force width (lb./in) units.

Word	Explanation
TENSION ROLLERS	A web transport roller mounted on load cells or a strain gauge for the purpose of sensing web tension.
TENSIONING ROLLER	A roller that is generally covered with a cork material. It is used to pull the web tight over a preceding roller.
THERMAL CONDUCTIVITY GAUGE	A vacuum gauge containing two surfaces at different temperatures between which heat can be transported by gas molecules. Changes in the temperatures or in the heating power required to maintain constant temperature of one of the surfaces can be correlated with the gas pressure. Thermal conductivity gauges differ in the method of indicating the temperature change.
THERMO-COUPLE GAUGE	A thermal conductivity gauge which contains a heated filament and a thermocouple for the measurement of filament temperature as a function of gas pressure
THROUGHPUT	The quantity of gas in pressure-volume units, at a specified temperature, flowing per unit time across a specified open cross-section. Throughput may be referred to a specific constituent of a gas in which case the partial pressure of that constituent and the associated flow rate are the relevant quantities.
TORR	A unit of pressure defined as 1/760 of a standard atmosphere. It replaced the term millimeter of mercury (mm/Hg).
TRACKING	The ability of a web to follow a desired path through processing equipment.
TRAP	A device used in a vacuum pumping line to reduce vapour pressure in a vacuum system or prevents back streaming and migration of vacuum pump fluids.
TURBO-MOLECULAR PUMP	An axial flow turbine for operation in the molecular flow range consisting of a series of alternate circular rotor and stator disks. Both of which have inclined blades designed to impart momentum change to gas molecules.
UNWIND	The reel where the unprocessed web is loaded.
UNDERWIND	A term used to describe when the web is wound from the bottom of a roll.
ULTIMATE VACUUM	The designed minimum vacuum level achievable after evacuating the chamber for an extended time period.
UTILITIES	A term used to describe water, air and electric services
VACUUM	The condition of a gaseous environment in which the gas pressure is below atmospheric pressure
VACUUM (EXTREME) ULTRAHIGH	A vacuum in which the pressure is less than 7.5×10^{-13} torr.
VACUUM (HIGH)	A vacuum in which the pressure is less than or equal to 7.5×10^{-4} torr and greater than 7.5×10^{-7} torr.
VACUUM (LOW)	A vacuum in which the pressure is less than approximately 750 torr and greater than 25 torr.
VACUUM (MEDIUM)	A vacuum in which the pressure is less than or equal to 25 torr and greater than 7.5×10^{-4} torr.
VACUUM (ULTRAHIGH)	A vacuum in which the pressure is less than or equal to 7.5×10^{-10} torr and greater than 7.5×10^{-13} torr.
VACUUM (VERY HIGH)	A vacuum in which the pressure is less than or equal to 7.5×10^{-7} torr and greater than 7.5×10^{-10} torr.

Word	Explanation
VACUUM GAUGE	An instrument for measuring gas pressures below atmospheric pressure
VACUUM PUMP	A device for reducing and maintaining the gas pressure in a vessel below atmospheric pressure.
VACUUM SYSTEM	An enclosure within which a vacuum is maintained and the accessories for producing, controlling and measuring this vacuum.
VAPOUR	A gas at a temperature lower than its critical temperature. It is sometimes referred to as a condensable gas
VAPOUR PUMP	A fluid entrapment pump in which the transport of the gas is effected by one or more jets of vapour (see diffusion pump).
VVX ISOLATION VALVE	Description given to isolation valves in the vacuum pumping system.
WEB	The material being processed. May also be known as film, substrate or base material
WHEELED CARRIAGE	Any assembly that is mounted on wheels or rollers.
WEB BAG	The condition that exists when the centre section of the web droops or sags under tension while the sides are taut. The opposite of WEB SAG.
WEB BREAK	A term used to describe a break in the substrate being processed.
WEB SAG	The condition that exists when the sides of the web sag or droop under tension while the centre is taut. The opposite of WEB BAG.
WINDING MECHANISM	Description of the section of the machine that is built up of a number of rollers for transporting the substrate through the coating area.
WINDING CART	Name describing the section of the machine that is driven out of the chamber. The winding mechanism is generally located upon it.
WINDING ZONE	The vacuum chamber that houses the rollers not in the coating zones.
WRAP ANGLE	The angle or amount or arc a web forms when traveling around a roller.
WRINKLES	A very broad term for any typically longitudinal deviations in web flatness, which is usually not a direct result of thickness variations as are stretch lanes. The faults can range from corrugations to sharp fold overs. Causes include poor core starts, roller misalignments, winding tension transients, etc.

Table B - 1 Glossary Of Terms

APPENDIX C MEASUREMENT OF ALUMINIUM DEPOSIT THICKNESS

The measurement of the deposition thickness of aluminium can be defined in various units, these being:

- Resistance - ohms per square
- Light transmission - % Transmission
- Optical Density - O.D.
- Weight - gram/metre² (g/m²)
- Thickness - angstroms

The most commonly used units are resistance and light transmission/optical density. Below is brief description of each of these units and how they are measured.

Resistance

This type of measurement is stated in ohms per square (the resistance is the same irrespective of the size of the square). By using various types of monitoring equipment (i.e. VG non contact system), the resistance of a deposit can be measured. In the sketch, we can see that the deposit thickness can be calculated using the formula.

Advantages

- Clear and opaque substrates can be measured.
- Simple test methods can be used.
- Robust and little maintenance required.

Disadvantages

- Measurement changes when oxide layer is formed.
- Difficult to measure thin coatings accurately.

Optical Density/Light Transmission

These are measured in O.D. and % Transmission respectively. By passing the substrate between a light source and a light detector, the light passing through the substrate is measured.

NOTE: Light transmission 1/O.D.

Advantages

- Ideal for very light (thin) coatings.

Disadvantages

- Requires substrates to be uniform before coating.

- Unable to measure thick coating.
- Cannot be used on opaque substrates.
- Requires cleaning regularly.

There are a number of conversion tables and curves available, however they are only accurate for the substrate that they are based upon.

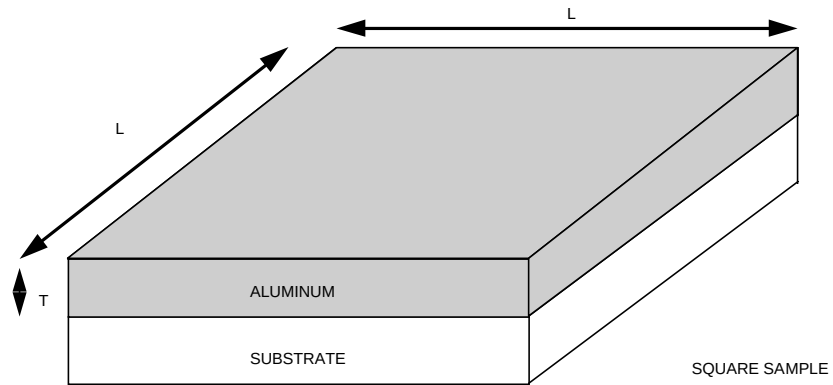
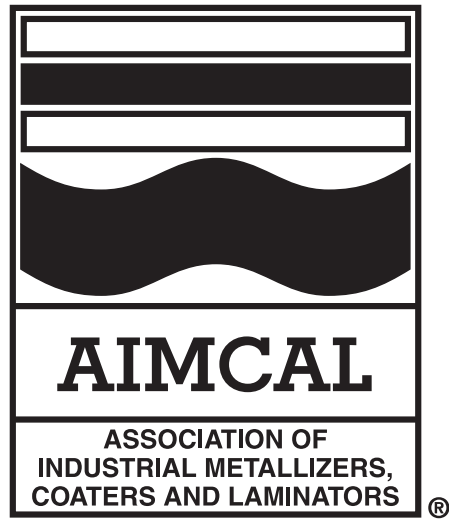


Figure C - 1 Relationship Of Ohms Per Square To Thickness

Measure of metal deposition by electrical resistance. When measuring resistance across a uniform square of metallized substrate, the value in ohms is identical whether the sample is one inch or one meter. This is because the width of the current path (l) increases or decreases in exact proportion to the length (l) of the current path.

APPENDIX D

AIMCAL – METALLIZING TECHNICAL REFERENCE



Metallizing Technical Reference

Third Edition

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AIMCAL

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Section 1: Test Methods and Procedures

1.1 AIMCAL Test Procedure for Determining Optical Density

TP-101-78

1) INTRODUCTION

The amount of metal deposited on the surface of a film will govern the oxygen (O₂), water vapor transmission (WVTR), and light barrier properties of that film. Hence a means to measure this coating is needed so that one can determine the amount of metal that will meet the particular requirement. A very rapid determination can be made with excellent reproducibility on a uniform sheet. USA Standard Number PH2-19-1959 provides the basis for the concept of diffuse optical density.

2) SCOPE

This procedure gives a means for determining the amount of metal as a function of the diffuse optical density.

3) APPARATUS

Densitometer: Several types of testing apparatus are available for use in this procedure. See Item 6 below for characteristics.

4) PROCEDURE

4.1 Refer to the operator instruction manual of any given instrument for calibration.

4.2 Mark the sample at 10 evenly spaced intervals across the width of the material with the metal side up.

4.3 Place the densitometer in the diffuse mode and set for visual light transmittance.

4.4 Test the sample with the metal side up. Special care should be given in handling the material to avoid scratches, pinholes, etc.

Note: This procedure may be used for virgin metallized film and also laminates providing the additional opacity of the second film is calculated into the reading.

5) CALCULATIONS

Average the measured values of each interval and record as the average diffuse optical density.

6) DENSITOMETERS

A densitometer is a device designed to measure the optical density of an exposed photographic film. Such an instrument can be utilized as a working part of the Processor Quality Assurance Program to determine the density of the individual steps that were obtained by using the densitometer.

There are several styles of densitometers available, with the greatest difference being the type of readout. The two types of readout are dial and meter with the latter being generally easier to use.

Note: Extreme caution should be exercised in the use of units that do not necessarily correlate to NIST measurements across the entire light spectrum. Equipment measuring light transmission using photo multiplier technology has severe limitations of tenacity due to saturation at the high intensity end and dark current noise at the low intensity level.

1) INTRODUCTION

This method is valuable when testing metallized films in which the amount of metal must be kept constant. It is recommended when measurement by the optical density method is not available. This method also saves testing time when compared to coating weight tests. In summary, a flat, uncreased strip of metallized film is placed on the electrode unit, either by holding it taut over the electrodes or placing it on a flat surface with the electrodes on the sample. The ohms per square reading is made directly from an ohmmeter. The thickness is related to ohms per square.

2) SCOPE

A method and test apparatus are described for measuring the relative ohms per square of a vacuum metallized film.

3) APPARATUS

3.1 Sensitive ohmmeter of the multi-scale type with leads ending with alligator clips.

3.2 A unit having electrodes precisely 10 in. apart mounted on a 2-in.-wide, suitably insulated rigid sheet (see Figure 1.2).

3.3 A cutting apparatus can be used to precisely cut a sample 1 in. wide by 12 in. long (the strip may also be cut by using two razor blades spaced 1 in. apart in a fixed width block).

3.4 One sheet of semi-hard rubber (50 durometer hardness).

4) PROCEDURE

4.1 Samples from a roll should be taken across the width of the metallized film for testing and should be at least 12 in. long. The sample must be free of dirt, grease, dust or fingerprints. The metallized surface also must be free of abrasions, scars or blemishes. The strips should be cut 1 in. wide and 12 in. long in the machine direction at 10% width intervals.

4.2 The ohmmeter leads should be connected to the twin electrodes. Short the circuit by placing a piece of copper wire between the electrodes. Turn on the ohmmeter and adjust it to zero on the 0-100 ohm scale. Remove the jumper. If the needle goes to the other side of the scale, the unit is ready for testing.

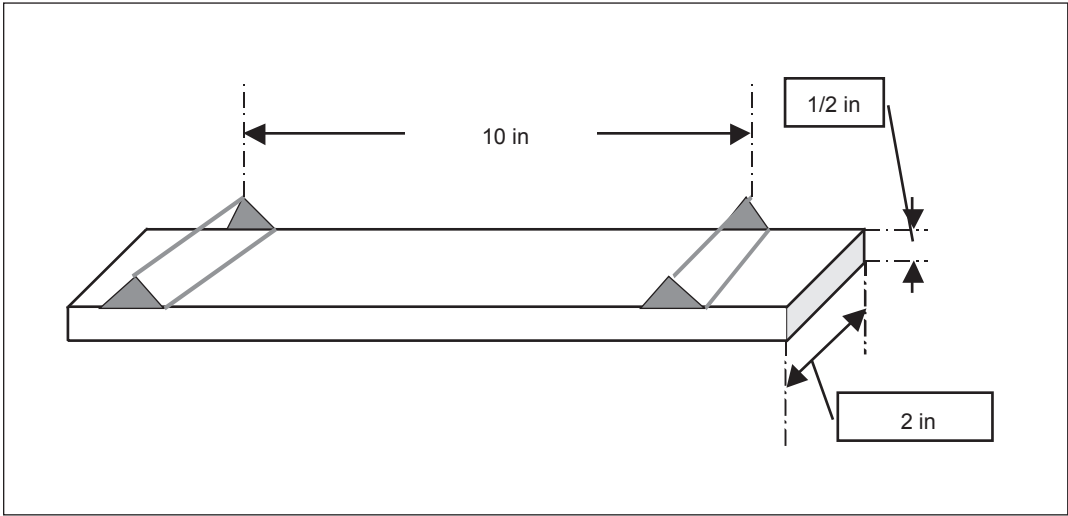
4.3 Either take the strip and press it with the thumbs onto the electrodes OR place the strip on the rubber backing and press the terminals firmly on the strip.

4.4 Read the resistance on the ohmmeter scale as ohms per square.

5) CALCULATIONS

Average the readings of the measurements on each sample.

Figure 1.2 Electrode Unit for Measuring Resistance



1.3 AIMCAL Standard Methods for Determining and Reporting Solar Control Film Parameters

TP-103-82

The Association of Industrial Metallizers, Coaters and Laminators suggests the following practices for determining and reporting solar control film performance criteria:

- 1) WINTER DESIGN CRITERIA (peak load conditions, night) shall be calculated at 70° F (21.1° C) indoors, 0° F (-17.8° C) outdoors, 15 mph wind, and indoor dew point below the glass temperature. Calculations shall be done in accordance with the Window 4.1 Window Thermal Performance software published by the Windows and Daylighting Group of Lawrence Berkeley Laboratories.
- 2) SUMMER DESIGN CRITERIA shall be calculated using 75° F (23.9° C) indoors, 89° F (31.7° C) outdoors, 7.5 mph wind.
- 3) SHADING COEFFICIENTS shall be calculated in accordance with ASHRAE, Chapter 26, 1977, using Item 2 conditions above. For simplicity and since calculations have shown no significant difference in results, shading coefficients shall be calculated using summer nighttime U value.
- 4) Other values can be used to calculate energy savings at specific conditions as long as those conditions are stated.
- 5) If reported, ultraviolet transmittance will be determined in accordance with ASTM E424-71 method A, with the following modifications:
 - 5.1 Solar UV will be defined as 300nm-380nm (agrees with ASTM solar group).
 - 5.2 Solar UV transmittance will be measured on double-strength clear glass or, if measured on film alone, weighted to adapt the results to include glass. The measurements will be made as described in ASTM method E424 or the ASTM techniques manual, and the results will be weighted using the following solar spectrum weighting factors:

WEIGHTING FACTORS

Wavelength	Measured on Glass	Measured on Film Alone*
380nm	.142	.12
370	.235	.205
360	.197	.17
350	.159	.13
340	.127	.10
330	.085	.05
320	.046	.01
310	.009	0
300	.000	0
	100%	78.5%

*Corrects to double-strength clear glass.

- 6) Total solar reflectance and transmittance will be determined in accordance with ASTM E424, by spectrophotometer, or pyranometer (sample and instrument at 20° angle).

1) INTRODUCTION

This test method quickly and easily identifies whether a vacuum-deposited thin metal layer has any substantial level of adhesion to a flexible substrate. It is intended to measure adhesion qualitatively, providing a check that the metallized substrate is acceptable or alternatively identifying a potential problem. Few precautions and minimal test equipment are required. If a more detailed qualitative measure of adhesion is required then the “AIMCAL Procedure for Qualitatively Determining the Metal Adhesion of Metallized Films” should be used (see Section 1.5).

2) SCOPE

A simple test method is described for measuring whether a metallized layer on a flexible substrate has any substantial adhesion to that substrate.

3) APPARATUS

Tape dispenser containing a reel of 1-in.-wide Scotch™ 610 pressure-sensitive tape.

4) PROCEDURE

- 4.1 Place a sample of the metallized substrate on a smooth, flat surface with the metallized layer facing up.
- 4.2 Apply approximately 6 in. of Scotch™ 610 adhesive tape to the surface and smooth with thumb and forefinger to ensure good contact. To facilitate separation, one end of the tape should be doubled over.
- 4.3 Hold the sample down with one hand, and with the other hand peel the tape back at approximately 180° in one smooth movement (see Figure 1.4a).

5) INTERPRETATION

- 5.1 Examine the test substrate for metal removal, and the used tape for metal transfer, using a strong backlight (see Figure 1.4b).
- 5.2 The significance of the visual assessment depends upon the application for which the metallized substrate is intended. If any metal is removed then either (a) visual comparisons must be made with specified visual standards or (b) the more quantitative AIMCAL procedure must be performed and the data obtained compared with numeric and/or visual standards (see Section 1.5).

Figure 1.4a

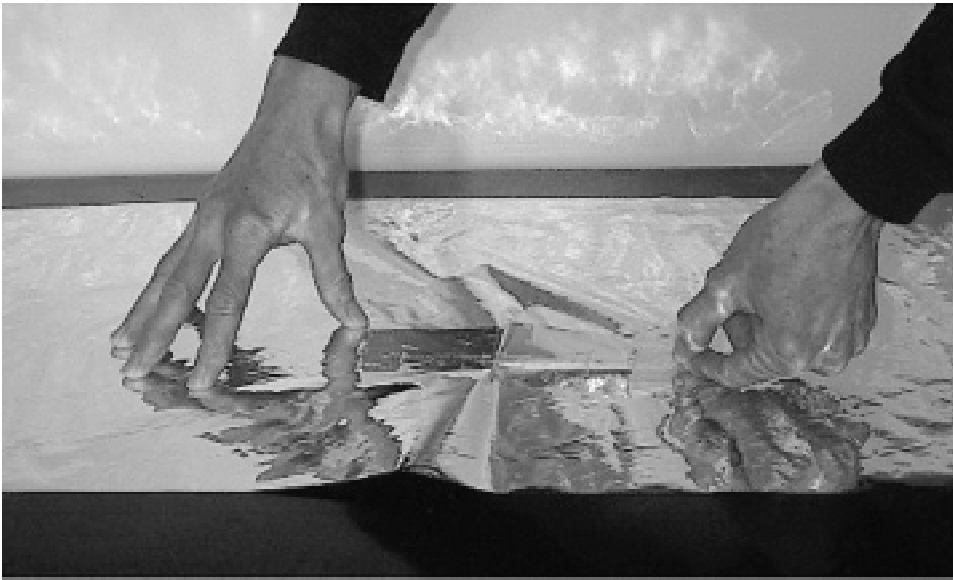
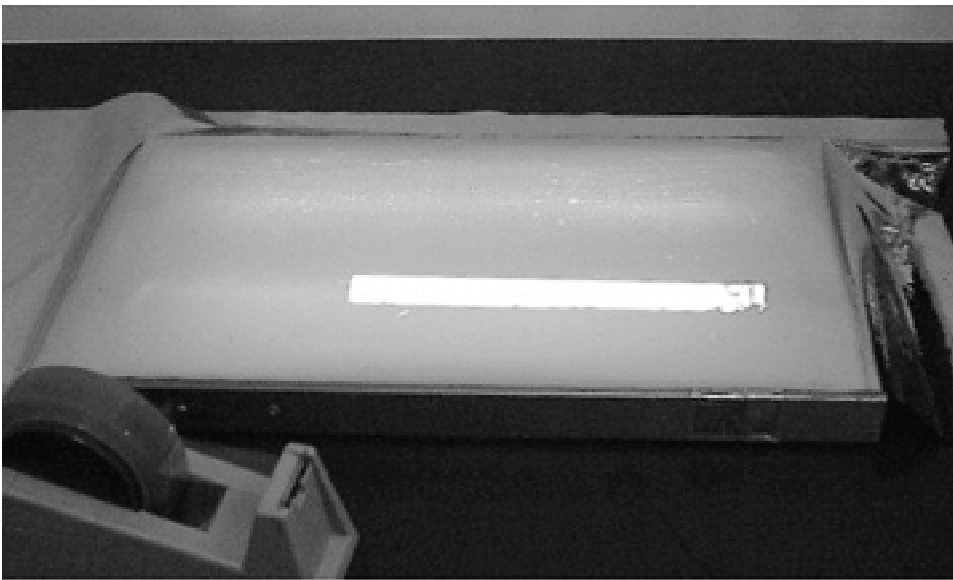


Figure 1.4b



1.5 AIMCAL Procedure for Qualitatively Determining the Metal Adhesion of Metallized Films

TP-105-92

1) SCOPE

This test is a means for comparing metal adhesion properties of metallized polyester films.

2) APPARATUS

- 2.1 A tensile tester capable of running at a crosshead speed of 12 in. per minute.
- 2.2 Precision cutters, templates, or other means to cut samples accurately.
- 2.3 A heat sealer (e.g., Sentinel) with controllable temperature, pressure, and dwell time and a 1-in.-wide jaw.
- 2.4 Silicone release-coated paper or wax paper, 50-lb. basis weight. Release coating on both sides.
- 2.5 Dow Chemical DAF 899*, 1-mil film of ethylene acrylic acid (EAA) resin.
- 2.6 A densitometer to measure the optical density (OD) of the metallized film.
- 2.7 A controlled environment; 50% RH, 72° F (22.2° C).
- 2.8 Scotch™ 610 adhesive tape, 2 in. wide to support the metallized film side of each test replicate.

*Available from Dielectric Polymers, Inc, 218 Race St., Holyoke, MA 01040; phone 413-532-3288

3) PROCEDURE

- 3.1 Condition the metallized film samples at 50% RH, 72° F (22.2° C) before testing. Test replicates must also be prepared and tested in this environment.
- 3.2 Measure and record the optical density range of the test films.
- 3.3 Cut a sample of metallized film approximately 12 in. (machine direction) by 4 in. (transverse direction). Place the sample on a smooth working surface with the metallized side down and secure to the work surface with tape in such a manner that no wrinkles are present in the film.
- 3.4 Attach a 2-in.-wide strip of Scotch™ 610 adhesive tape to the nonmetal side of the sample. This is done to prevent tearing the polyester film during the peeling procedure. Locate the tape at the approximate center of the film sample and apply for the full length of the sample. A hand roller is useful in providing a smooth, wrinkle-free adhesion. The tape is applied in the machine direction (MD) of the test sample (Diagram 1.5a)
- 3.5 Remove the tape/metallized film sample from the work surface.
- 3.6 Cut the tape/film sample into approximately 2.5-in. lengths (Diagram 1.5b).
- 3.7 Invert tape/film sample and place approximately 3.5-in.-long-by-3-in.-wide piece of DAF 899 film on top (metallized surface of film sample). Treated surface of 1-mil DAF 899 (1-mil EAA film) should be in contact with metallized surface of tape/film sample (Diagram 1.5c).
- 3.8 Heat sealer conditions for the test are 220° F (104.4° C), 15 sec., 15 psi.
- 3.9 Place tape/film/DAF 899 sandwich on silicone-release-coated paper and cover with second piece of silicone paper. Tape/film/DAF 899 sample should be located at top (leading) edge of slip sheets. This allows the entire layered sample to be inserted into the jaws of the heat sealer. (Diagrams 1.5d and 1.5e).

- 3.10 When the sealed sample is removed from the sealer, the slip-sheets are removed and a 1-in.-deep-by-3-in.-wide seal exists between the DAF 899 film and the metallized surface of the test film. The 1-in. dimension of the seal area results from the shorter dimension of the seal bar.
- 3.11 A center strip of the test specimen is now cut through its full length at a 1-in. width on a precision cutter.
- 3.12 For metal adhesion determination, the sample is mounted in the jaws of a tensile tester in such a manner as to effect a 180° peel of the DAF 899 from the metallized surface of the film. A pencil may be used to hold the tape/film/DAF 899 tail down to ensure the 180° peel, if necessary (see Figure 1.5f).
- 3.13 The tape/film tail of the sample is secured in the stationary jaw. The DAF 899 film is secured in the moving jaw and the rate of peel is set at 12 in./min.
- 3.14 Record the peel force for each of the five replicates.
- 3.15 An estimate of the percent metal removed in the 1-by-1-in. seal area can be made for each of the replicates.

Diagram 1.5a

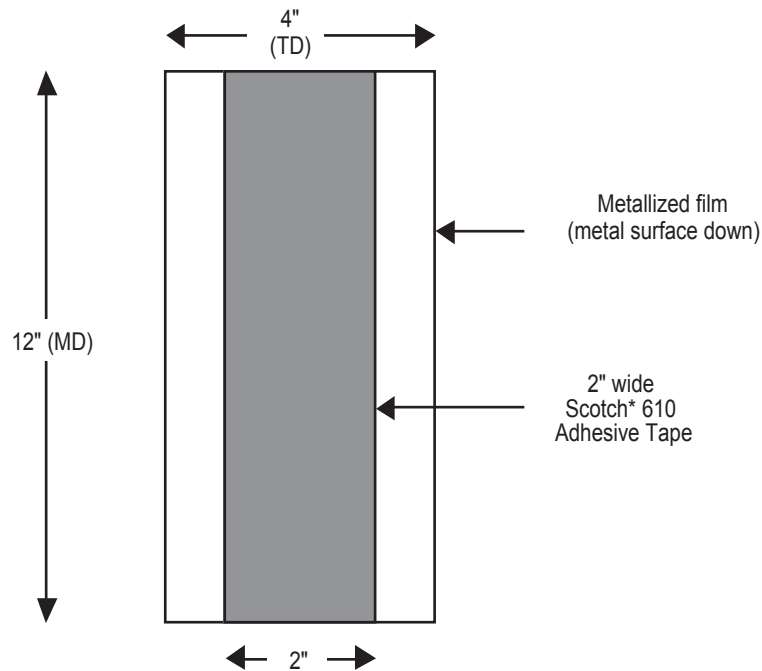


Diagram 1.5b

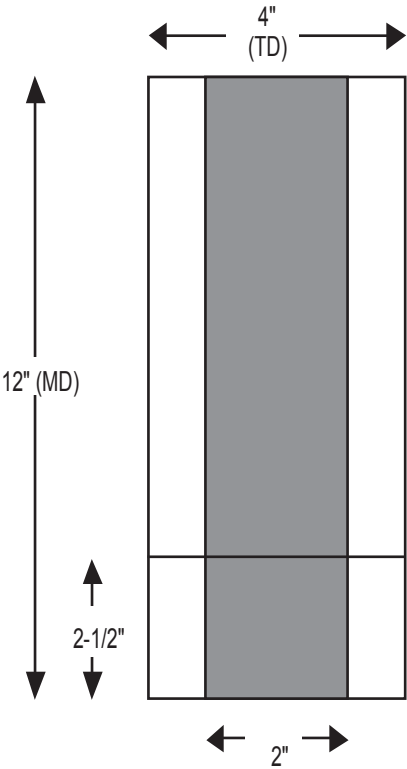
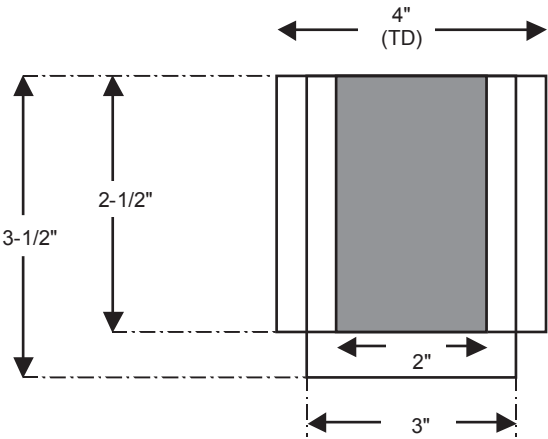


Diagram 1.5c



Inverted sample with metallized surface up and overlay of DAF 899 film (3-1/2" x 3")

Diagram 1.5d

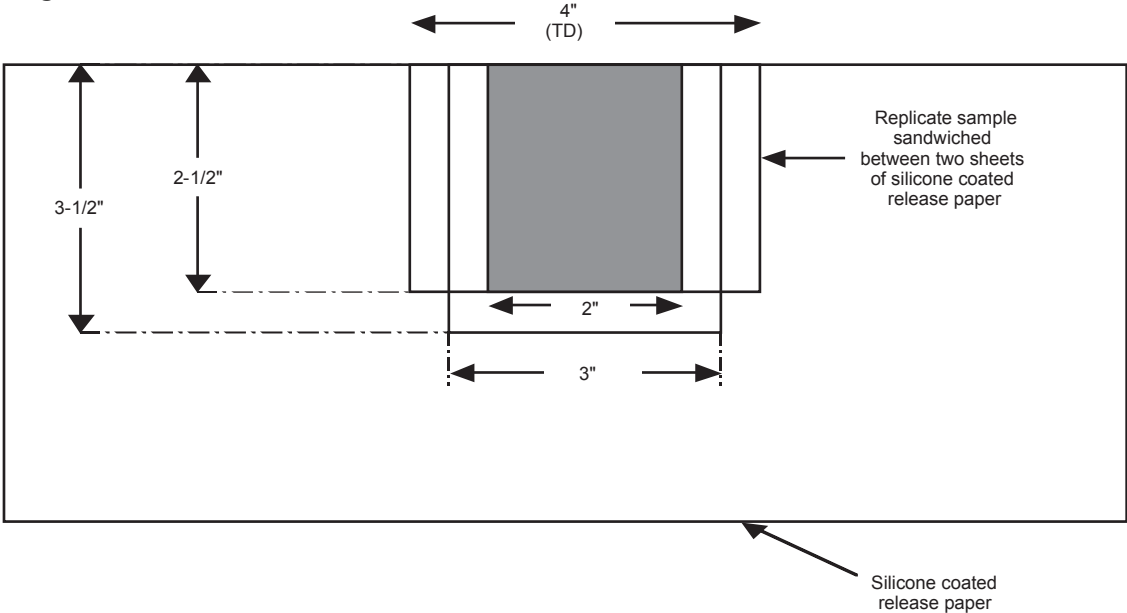


Diagram 1.5e

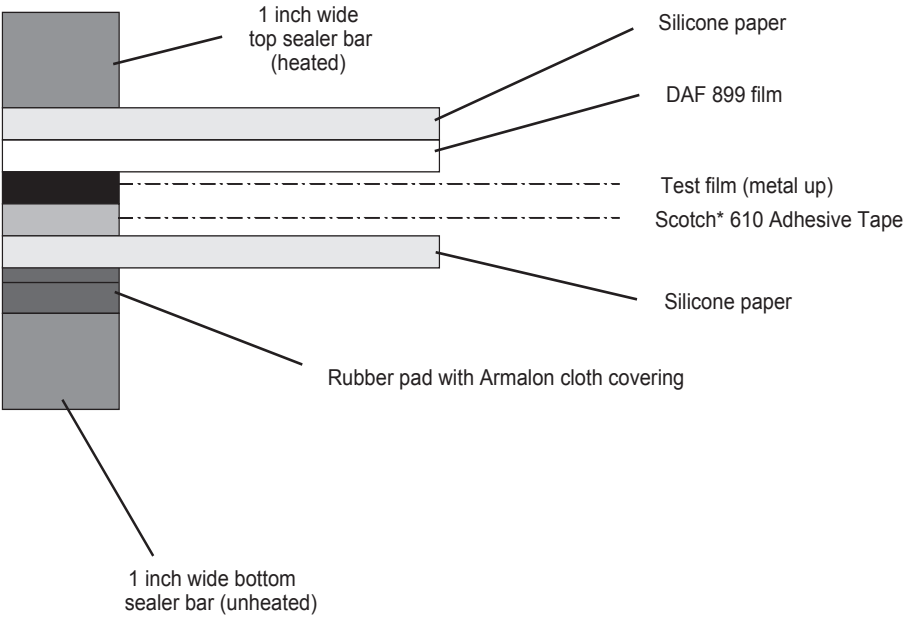
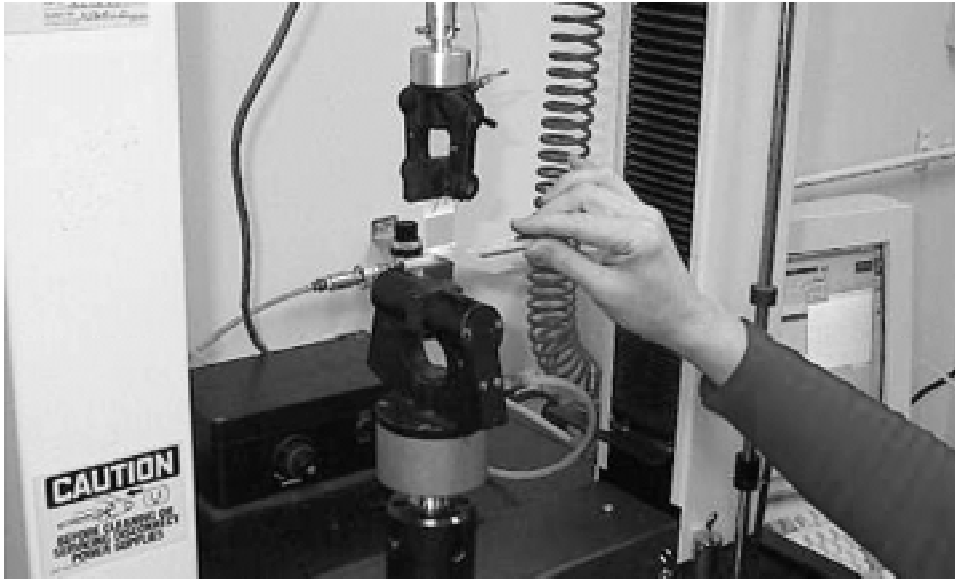


Photo 1.5



1.6 AIMCAL Typical Test Conditions for Determining the Barrier Properties of Metallized Films

1) WATER VAPOR TRANSMISSION RATE

Measurements should be made in accordance with ASTM 1249, which defines a procedure using an infrared diffusometer. Test conditions are normally 100° F (37.7° C) and 90% RH, or 73.4° F (23° C) and 50% RH. An alternative procedure is ASTM E96 (sometimes called the “dish” or “cup” method).

2) OXYGEN PERMEABILITY

Measurements should be made in accordance with ASTM D3985, which defines a procedure using a coulometric detector. Test conditions are normally 73.4° F (23° C) at 0 and 50% RH, respectively. An alternative procedure is ASTM D1434, which describes “volumetric” and “manometric” test methods.

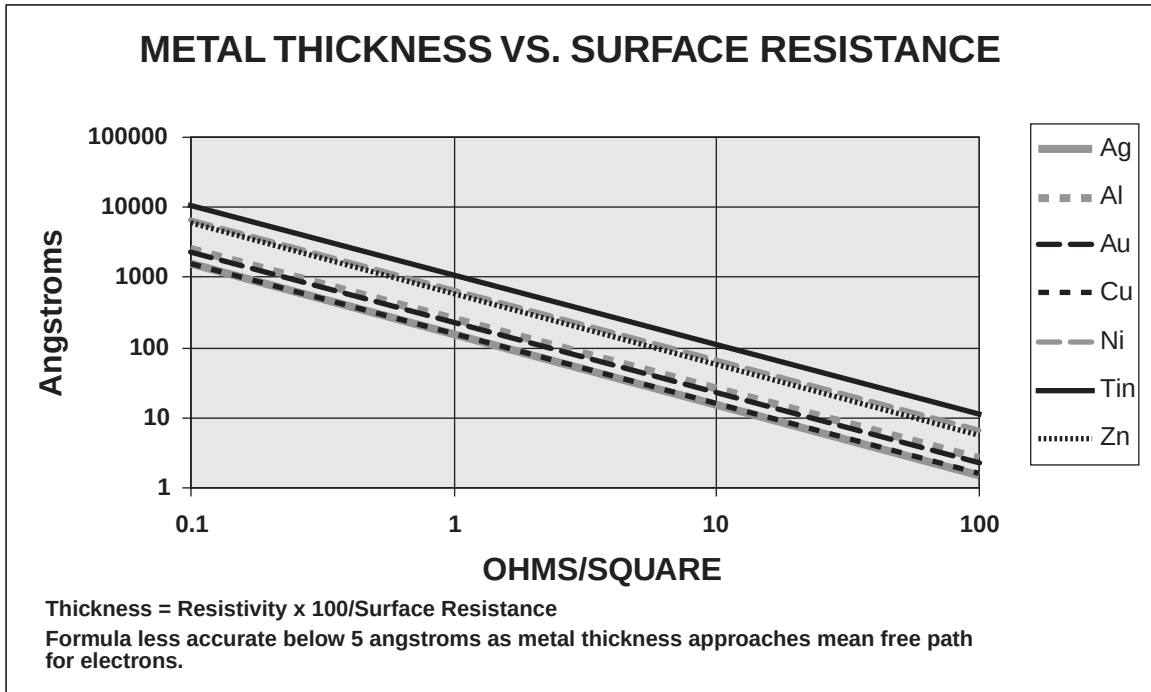
1.7 Index of ASTM Test Methods Applicable to Metallized Films and the Plain Substrates

Property	Test Method	Plain	Metallized
Clarity Transparency	ASTM D-1746	X	*
Distinctness of Image	Special	X	X
Gloss (Gardner)	ASTM D-2457	X	X
Haze	ASTM D-1003	X	*
Transmittance	ASTM D-1003	X	*
Solar Energy:	ASTM E-424	X	X
Transmittance/Reflectance			
Optical Density	ASTM D-1003	X	X
Surface Resistivity	ASTM D-257		X
Surface Roughness	Profilometer	X	X
	Interference Microscopy		X
Metal Adhesion	Special		X
Coating Adhesion	ASTM D-3359	X	X
Thickness (weight/area)	ASTM E-252	X	X
Coefficient of Friction	ASTM D-1894	X	X
Tensile Strength	ASTM D-882	X	X
Tear, Initial, Graves	ASTM D-1004	X	X
Running, Elmendorf	ASTM D-1922	X	X
Mullen Burst	ASTM D-774	X	X
Shrinkage - Oven	ASTM D-1204	X	X
- Dynamic	TMA	X	X
Density	ASTM D-1505	X	X
WVTR - Cup or Dish	ASTM E96-E	X	X
- Instrument	ASTM F-1249	X	X
Oxygen Permeability	ASTM D-3985	X	X
Surface Tension	ASTM D-2578	X	X
Melting Point	ASTM D-2117	X	X

*Suitable only for transparent samples

Section 2: Electrical, Optical, and Metal Thickness Relationships

Figure 2.1 Metal Thickness vs. Surface Resistance (Courtesy of Advanced Ceramics Corporation)



The lines are determined using resistivity values from Pollock¹ from Equation 2.1²

Equation 2.1

$$\text{Thickness} = \text{Resistivity} \times 100 / (\text{Surface Resistance})^{**}$$

The values of the bulk resistivities in micro ohm cm (moc) used in the calculation of Figure 2.1 are given below:

Ag	Al	Gold	Cu	Ni	Tin	Zn
1.59	2.65	2.35	1.67	6.84	11	5.91

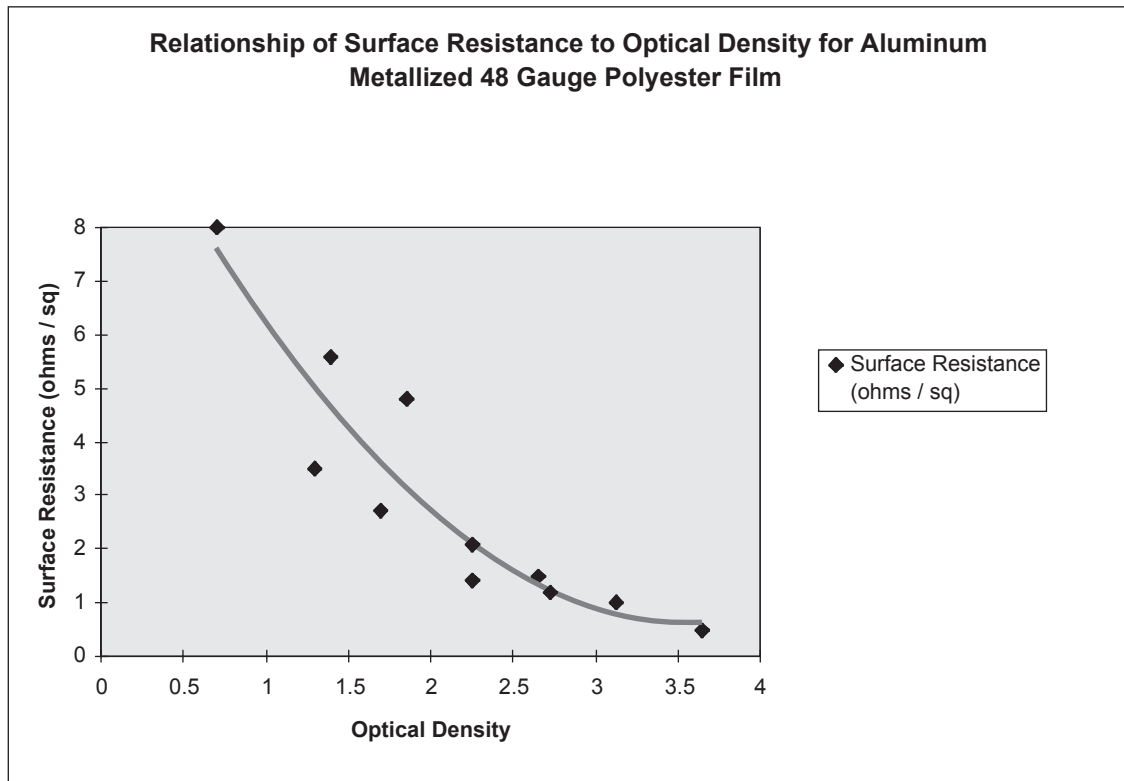
The surface resistance is measured in ohm/square.

******According to Powell and Rossnagel², the surface resistance R_s of a thin film of thickness, t , should be referred to as “sheet resistance” and not as is common practice as the sheet resistivity or sheet rho. Additional resistivity data³ for ideally pure elements are listed as a function of temperature.

Electrical Resistivity of Ideally Pure Elements ($\mu\Omega$ cm) Micro-ohm/cm (Adapted from Donnelly³)

T (K)	Ag	Al	Au	Cu	Ni
295	1.61	2.74	2.2	1.70	7.0
273	1.47	2.50	2.01	1.55	6.2
250	1.34	2.24	1.83	1.40	5.4

Figure 2.2 Relationship of Surface Resistance to Optical Density for Aluminum Metallized 48-gauge Polyester Film



Data Plotted from results of AIMCAL Round Robin conducted June 1978.

Regression of the data yields the polynomial curve:

Equation 2.2

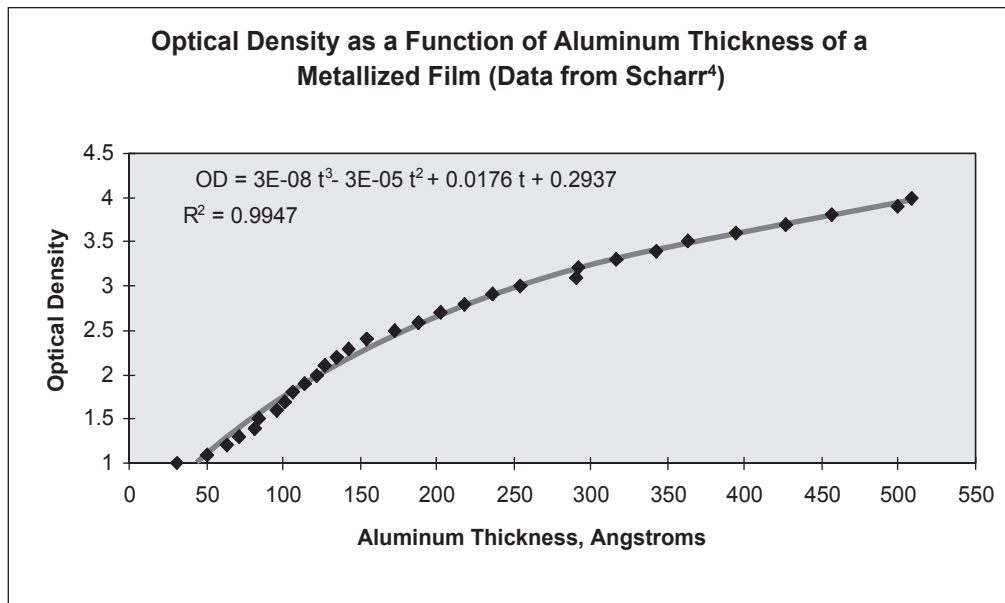
$$\text{Surface Resistance} = 0.8626 (\text{OD})^2 - 6.1352(\text{OD}) + 11.543 \quad R^2 = 0.8623$$

The optical density and surface resistance may be measured as described in the AIMCAL Test Procedures TP-101-78 and TP-102-78 respectively (see Sections 1.1 and 1.2).

Optical density is a measure of the percent light transmission of the film, Figure 2.4, and serves as a key measure of the light barrier of the product. However, as discussed in the Introduction in Section 3 for barrier properties, the optical density obtained for a given aluminum deposition is determined by the surface free energy of the polymer film surface. From this it would be expected that the aluminum thickness vs. optical density curve would be different for PET and oriented polypropylene or any other polymer film used for metallization. In addition, the optical density vs. resistivity curves should be determined experimentally for each film metallized if one measurement is to be used in place of another. Further correlations of resistivity or optical density with barrier properties and actual aluminum thickness should also be measured for each new base sheet to obtain the most accurate results.

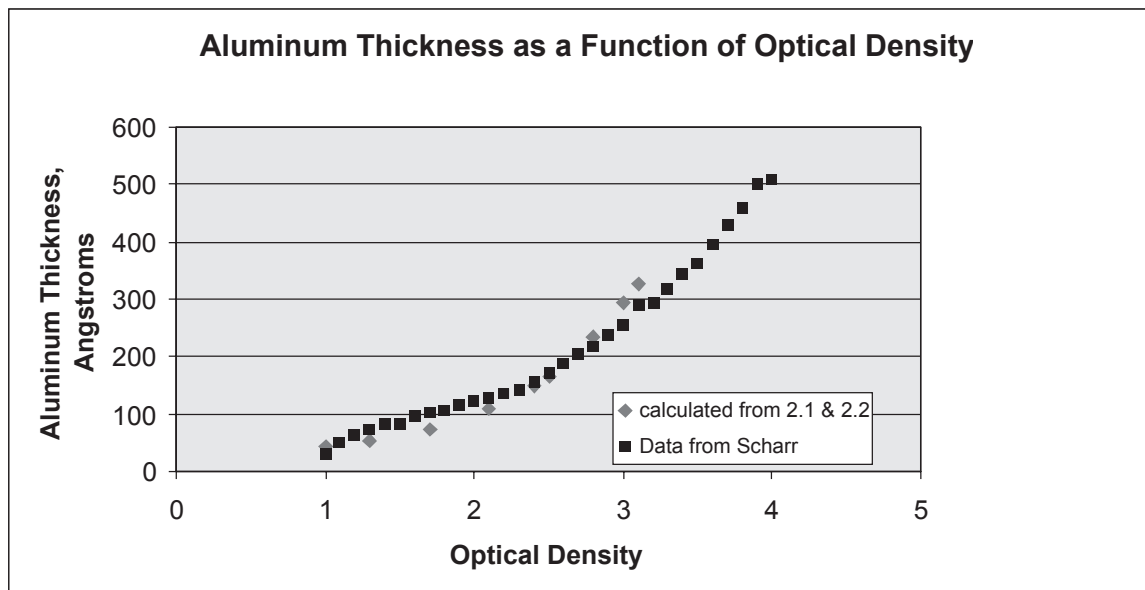
The results presented here represent one set of values obtained for the films and conditions listed and should only serve as a guide to the properties obtained from a particular set of films and metallizing chamber conditions.

Figure 2.3a Relationship of Aluminum Thickness to Optical Density⁴



An estimate of the aluminum layer thickness as a function of optical density may be made by substituting values from Figure 2.2 or from Equation 2.2, the curve fit data of surface resistance vs. optical density, into Equation 2.1 for the surface resistance to perform the thickness vs. surface resistance calculation. Substituting the second order curve fit into the thickness calculation of Figure 2.1 yields the calculated thickness as a function of optical density results in Figure 2.3b.

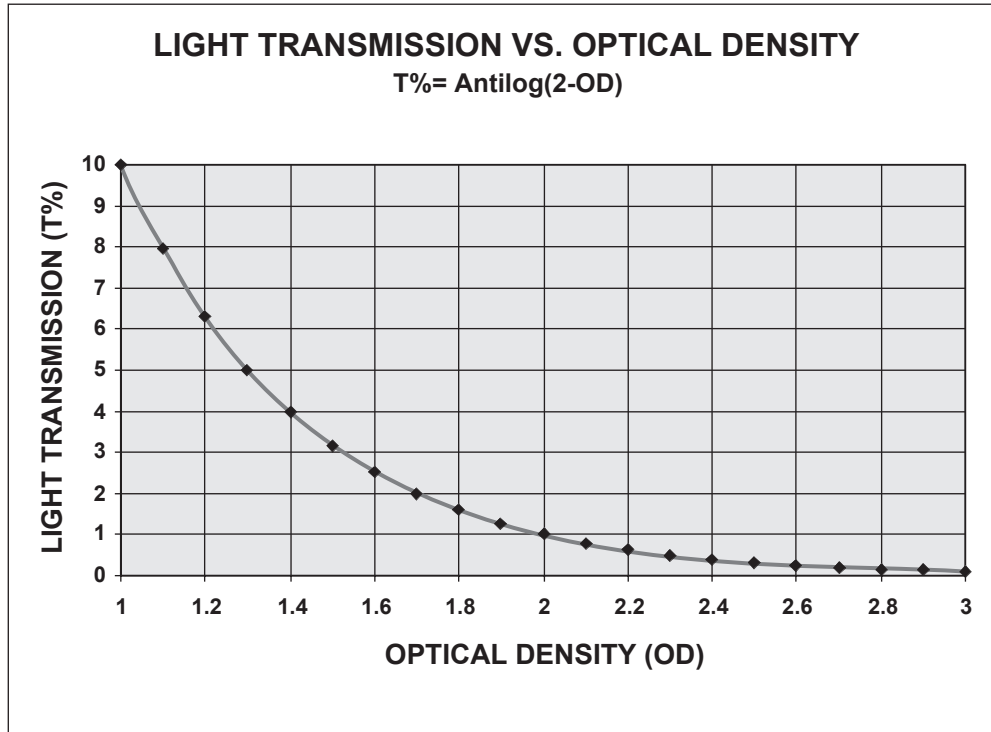
Figure 2.3b Comparison of Calculated Aluminum Thickness With Data of Scharr Industries



Comparison of the calculation results with the data used in Figure 2.3a shows a reasonable correspondence between the two curves up to an optical density of approximately 3.25. This would indicate that the Scharr data⁴ represents a reasonable set of values for the aluminum thickness as a function of optical density. The calculated data above an optical density of 3.25 showed large deviations which are most likely due to the numerical behavior of the second order curve fitting equation for surface resistance values less than 1 and

bears no relationship to actual results. Consequently the data from Scharr⁴ should be relied upon for relating aluminum layer thickness to optical density. While no reference is made to the base film used in the Scharr publication⁴ it is assumed to be PET film due to the approximate publication date.

Figure 2.4 Light Transmission vs. Optical Density (Courtesy of Advanced Ceramics Corporation)



The optical density is a term used to indicate the ability of a material such as a metallized film to block or absorb the radiant energy in an incident light beam. It is correctly termed the absorbance⁵ when used for measuring the attenuation of a light beam passing through a material.

Optical density is calculated as the $\text{Log}_{10}(1/T)$

Where: T is the transmittance defined as P/P_0
P is the unabsorbed energy remaining in the light beam passing through the sample
 P_0 is the energy of the incident light beam

The optical density is used to indicate the level of light barrier supplied by a metallized film. Thus an optical density (OD) of 2 represents a film with 1% light transmittance or 99% of the light is reflected and absorbed by the film.

2.5 Electrical, Optical and Barrier Properties of Various Metallized OPP Films (Courtesy of Quantum Performance Films)

Gauge	Optical Density	Light Transmission %	Resistivity (ohms/square)	WVTR*	OTR*
70	2.1	0.8	2.2	0.025	3.0-4.0
70	2.4	0.4	1.8	0.025	2.0
80	2.1	0.8	2.2	0.07	8.0-10.0

* flat laminated samples

WVTR = gm/100 in²/24 hrs. at 100 °F (37.7 °C), 90% RH

OTR = cm³/100 in²/24 hrs. at 73 °F (22.7 °C), 0% RH

Emmissivity of Polyester and Kapton H and HN Films (Courtesy of Metallized Products)

2.6: EMISSIVITY OF POLYESTER FILMS

	Ohms/Sq	Emissivity Range
25 Gauge	0.4	.03 - .04
	0.5	.03 - .045
	0.6	.025 - .04
	0.7	.025 - .04
	0.8	.025 - .04
	0.9	.025 - .04
	1.0	.025 - .045
	1.1	.03 - .04
	1.2	.035 - .05
	1.3	.03 - .04
	1.4	.025 - .06
	1.5	.03 - .04
	1.6	.04 - .045
	1.7	.035 - .04
	1.8	.03 - .05
	1.9	0.05
	2.1	0.05
	2.4	.04 - .06
	2.6	0.04
	3.0	0.05
	3.3	0.05
92 Gauge	0.6	0.03
	0.9	0.04
	1.0	.03 - .035
	1.1	.035 - .04
	1.2	0.04
	1.3	.025 - .04
	1.4	0.04
	1.5	0.04
	1.6	0.04
	1.7	0.025
2 mil	0.4	0.04
	0.5	.03 - .04
	0.6	.025 - .04
	0.7	.025 - .04
	0.8	.025 - .04
	0.9	.03 - .04
	1.0	.035 - .04
	1.1	.02 - .04
	1.2	.025 - .04
	1.3	.03 - .04
	1.4	.035 - .04
	1.7	0.04
	1.8	0.04
3 mil	0.6	0.03
	0.7	0.03
	0.8	.02 - .03
	1.0	0.025
	1.1	.025 - .03
	1.3	0.025
	1.4	.035 - .04
	1.6	0.03
	1.7	0.03
5 mil	0.4	0.03
	0.5	.025 - .035
	0.6	0.035
	0.7	0.04

2.7 EMISSIVITY OF KAPTON H AND HN FILMS

	Ohms/Sq	Emissivity Range
30H	0.3	0.025
	0.4	.02 - .035
	0.5	.025 - .04
	0.6	.025 - .04
	0.7	.03 - .04
	0.8	.025 - .04
	0.9	.025 - .04
	1.0	.03 - .04
	1.1	.03 - .04
	1.2	.03 - .04
	1.3	.04 - .045
	1.4	0.04
	1.5	0.045
	1.7	.045 - .05
	1.8	0.05
50H	0.6	.035 - .04
	0.7	.025 - .03
	0.8	.02 - .04
	0.9	.02 - .04
	1.0	0.03
	1.1	.03 - .04
	1.2	.03 - .04
	1.4	0.04
	1.7	.035 - .04
100HN	0.4	.03 - .04
	0.5	.03 - .035
	0.6	.03 - .04
	0.7	.03 - .04
	0.8	.03 - .04
	0.9	.025 - .045
	1.0	.03 - .04
	1.1	.03 - .04
	1.2	0.04
	1.3	.03 - .04
	1.4	.035 - .045
	1.5	0.04
200HN	0.4	0.03
	0.5	.02 - .04
	0.6	.025 - .04
	0.7	.015 - .04
	0.8	.025 - .04
	0.9	.02 - .045
	1.0	.025 - .04
	1.1	.03 - .04
	1.2	0.04
	1.3	.03 - .05
	1.4	.03 - .04
	1.5	0.04
	1.6	.03 - .035
300HN	0.6	.02 - .035
	0.7	.03 - .035
	0.8	.03 - .04
	0.9	.03 - .04
	1.0	.025 - .035
	1.1	0.035
	1.3	0.04
500HN	0.4	.02 - .035
	0.5	.02 - .04
	0.6	.02 - .04
	0.7	.02 - .04
	0.8	.02 - .04
	0.9	.025 - .04
	1.0	.03 - .035
	1.1	0.04

2.8 Section 2 References Cited

¹ Pollock, D., *Electrical Conduction In Solids*, American Society for Metals, Cahn's Publishing Company, Chicago, (1985), p123

² Powell, R.A., Rossmagel, S., "PVD for Microelectronics: Sputtered Deposition Applied To Semiconductor Manufacturing", *Thin Films*, V26, Academic Press, (1999), p 290

³ Donnelly, R.J., Table 7.06. "Cryogenic Design Information", Chap 7, "Cryogenics", *A Physicist's Desk Reference, Second Edition of Physics Vade Mecum*, Anderson, Herbert L., Editor, American Institute Of Physics, New York,(1989), p128

⁴ Scharr Industries Inc. Publication, "Conversion Table Metallized Film (Aluminum)", approximately 1985

⁵ Willard, H. H., Merritt Jr., L. L., Dean, J. A., *Instrumental Methods Of Analysis*, 4th edition, D. Van Nostrand Company, Inc., Princeton, 1968, p 76

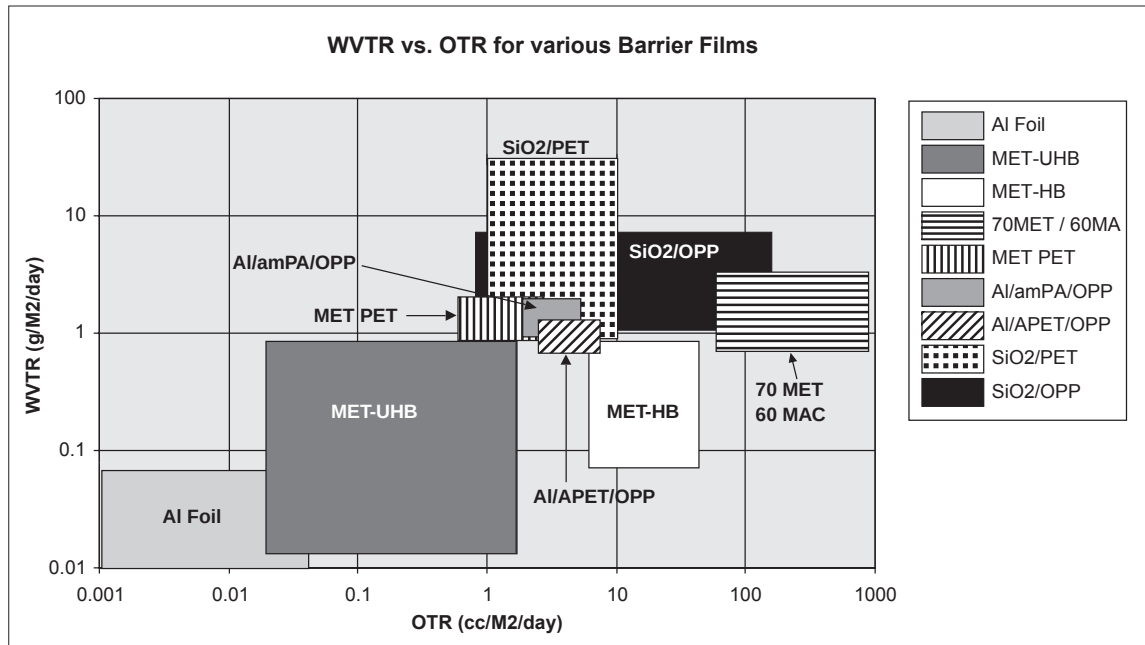
Section 3: Barrier Properties of Metallized Films

INTRODUCTION

The barrier properties of metallized films presented below represent a special set of values determined on particular base sheets and need to be understood in a larger context of barrier property development. During the last ten years the understanding of metallized film barrier development has improved and broad film barrier properties can be shown in terms of aroma, moisture and oxygen barrier for a wide range of metallized films¹. In addition, the chemical (aroma) barrier properties have been shown to be dependent upon the base sheet to a large degree². This is especially true for the barrier properties of oriented polypropylene films (OPP) where it has been shown that the barrier properties are dependent upon the film surface polymer³, the metallizing vacuum level⁴, an optimum optical density⁵ and the chemical composition of particular metallizing surfaces⁶. Consequently, the data presented here should serve as a guide and for applications requiring a particular barrier level the product should be checked by barrier testing the film produced from a particular chamber and the process capability of the chamber determined.

New technologies for the treatment of films in the metallizing chamber are also showing that the film optical density at a fixed aluminum deposition rate will depend upon the surface treatment level of the film surface⁷. Thus for a particular metallized film product the process used to make it will determine the ultimate properties of the aluminum layer deposited based upon the thickness and density of the aluminum and any other interfacial effects which might be important. From this it can be seen that it is unlikely that a unique relationship will exist for the optical density vs. barrier properties of a film, especially for OPP where the metallized surface may have many compositions. Polyester films should be more uniform in this regard as the surface polymer of various films are fairly uniform in composition from film to film. However, some PET films are surface modified to improve adhesion⁸ by coating and care should be taken to better understand the impact of surface polymer modifications in PET films on metallized film barriers. But in general the metallized PET films seem to be less likely to have surface polymer modifications by coextrusion or extrusion coating as found with metallized OPP films. Consequently, special care must be taken when specifying OPP films for applications requiring gas and moisture barrier versus metallized films for light barrier and decorative applications as described in reference 2. The first figure of this section shows a broad mapping of film properties on the WVTR-OTR space. It compares many film types versus aluminum foil. Many of the films shown are OPP films with high surface energy polymers such as, amorphous nylon⁹ (amPA), amorphous PET¹⁰ (APET), HDPE¹¹ (MET-HB), EVOH¹² (MET-UHB) and EP Copolymer¹³.

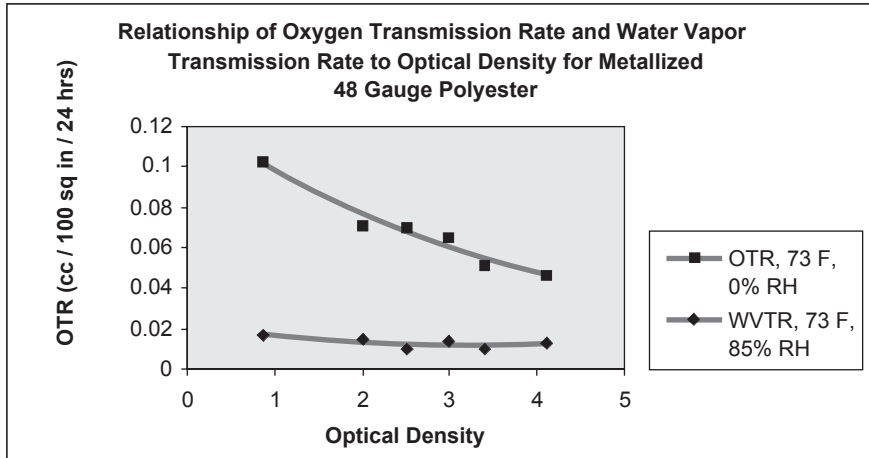
3.1 Plot of Moisture Barrier (WVTR) vs. Oxygen Barrier (OTR) for Various Barrier Films²



This graph demonstrates the wide range of barrier possible with a wide range of OPP based films and reinforces the comments above about the wide range of properties as a function of optical density, while, also showing the relative barrier property data range found for metallized PET and Silica coated films¹⁴.

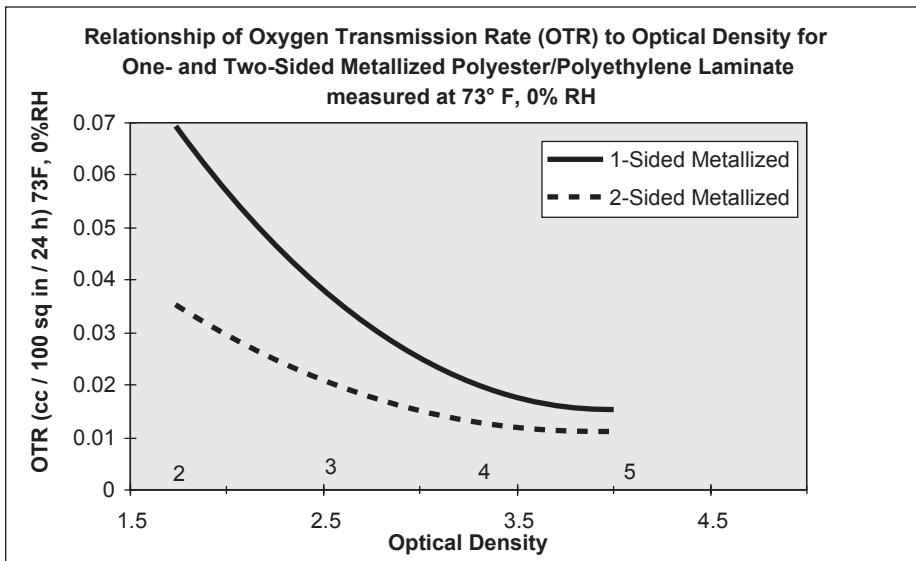
Care must be exercised in the handling of metallized films to insure that the barrier properties are not compromised during the many times the film is wound and unwound as it is processed into final products. All rolls must be clean and turning at film speed to minimize any scratching of the metal surface. Also, film tension must be at a level to prevent stretching of the films such that the metal layer is cracked. For example, film tension levels will need to be lower for OPP than for PET films to maintain the same range of film elongation due to differences in the films tensile strengths. Film suppliers should be able to specify appropriate tension ranges for the products they supply. This is also important in each of the subsequent lamination process to which the metallized film is subjected. Most films, used for their moisture and gas barrier properties, are laminated to place the metal layer in the middle of the structure to protect it from further scratching. While laminated films generally show some improvement over the flat film barrier properties, the metal layer is still subject to cracking if film tension levels and lamination conditions are not controlled.

3.2 Relationship of Oxygen and Water Vapor Transmission Rate to Optical Density for Metallized 48-gauge Polyester (Courtesy of Camvac International, Inc)



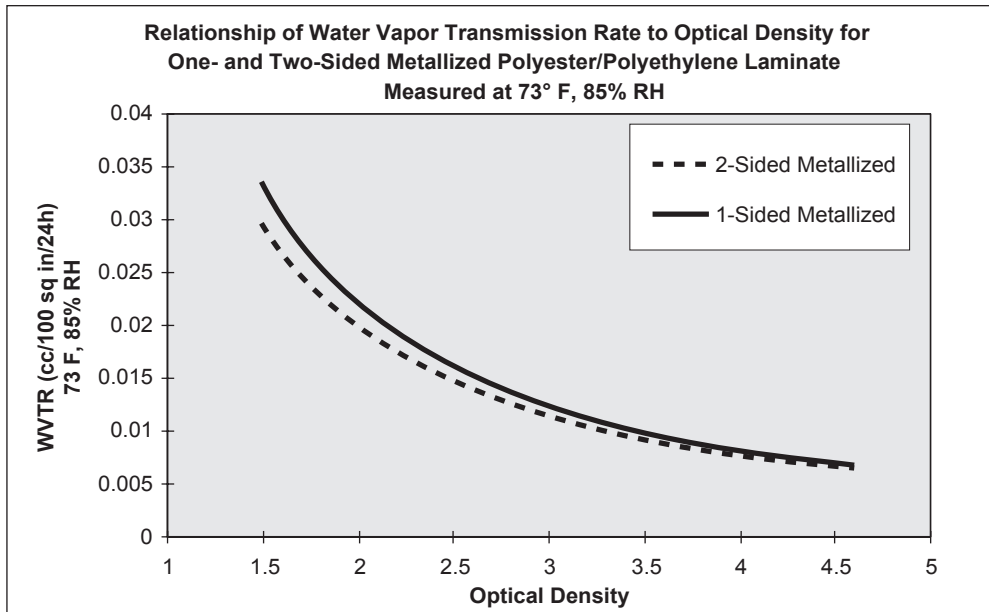
Note: Graph represents measured results

3.3 Relationship of Oxygen Transmission Rate (OTR) to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate (Courtesy of Camvac Ltd.)



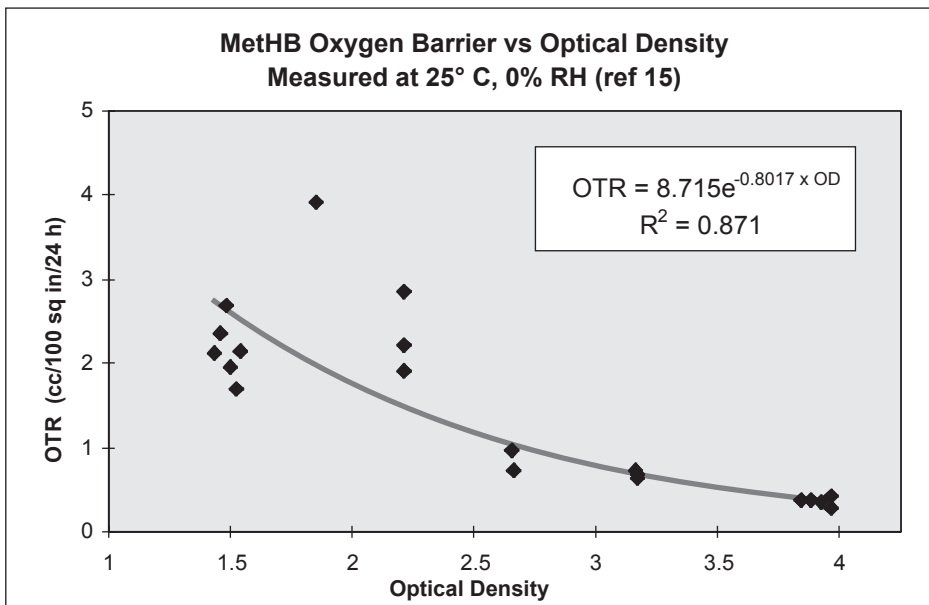
Note: Two-sided metallized film has an approximate equal thickness of aluminum on each side of the film.

3.4 Relationship of Water Vapor Transmission Rate to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate (Courtesy of Camvac Ltd.)

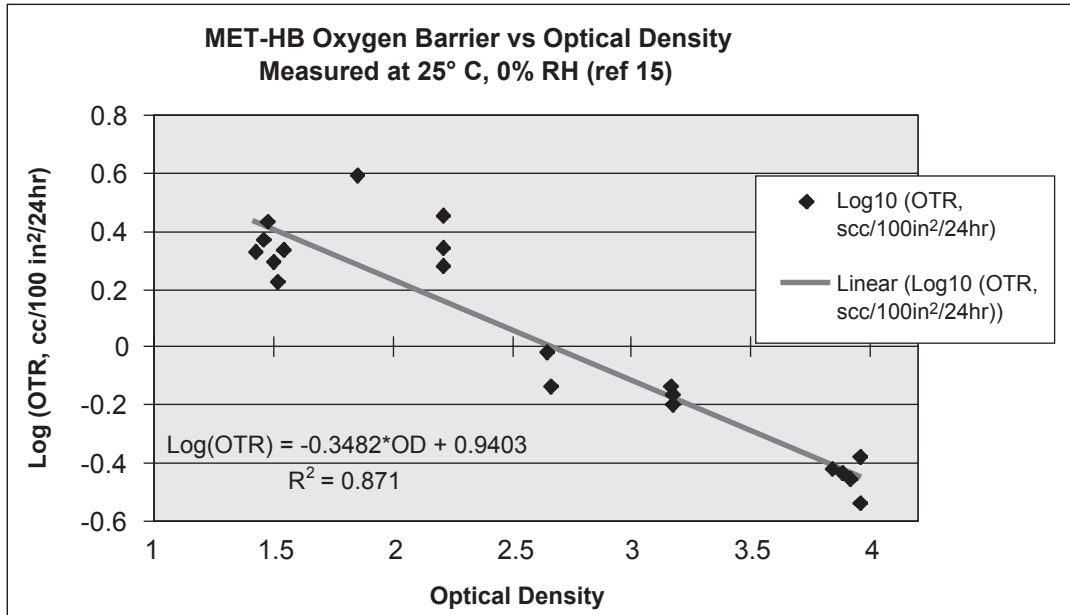


Note: Two-sided metallized film has an approximate equal thickness of aluminum on each side of the film.

3.5 Oxygen Barrier vs. Optical Density for Metallized OPP, MET-HB¹⁵



3.6 Log₁₀ (Oxygen Barrier) vs. Optical Density for Metallized OPP, MET-HB¹⁵



3.7 Average WVTR vs. OTR for Metallized Production, MET-HB¹⁵

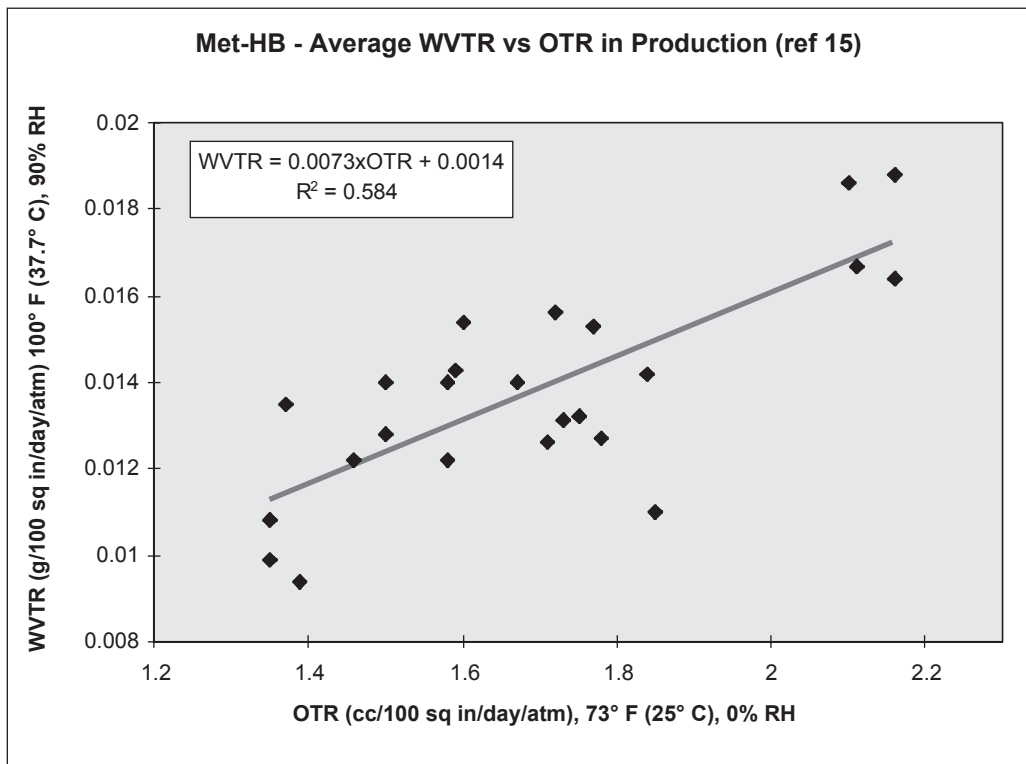
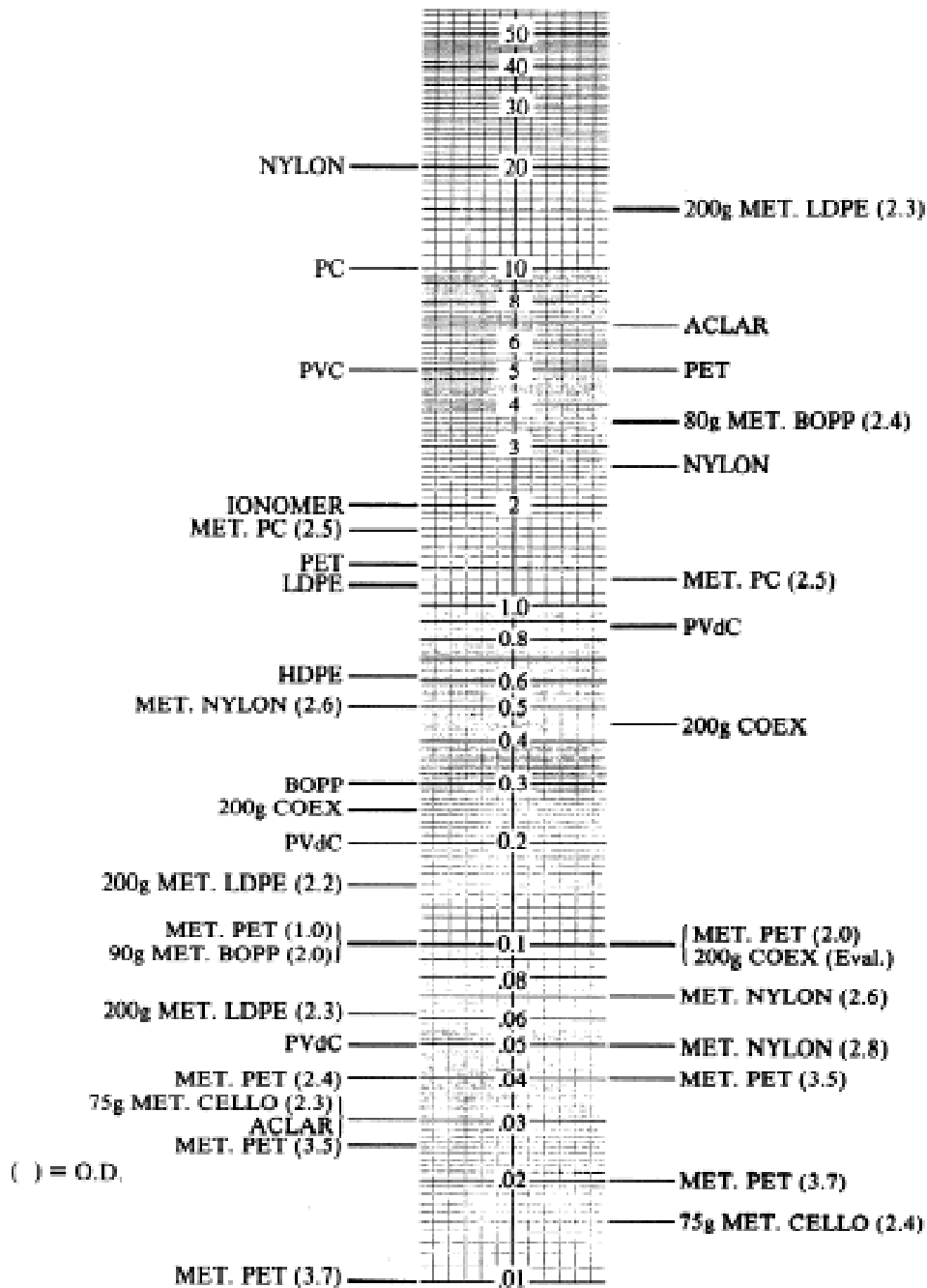


Figure 3.8 Barrier Properties for Metallized Films- 1 (Courtesy of ICI Americas Inc)¹⁶

Barrier Comparison

MVTR
(g/100 in²/24 hrs)

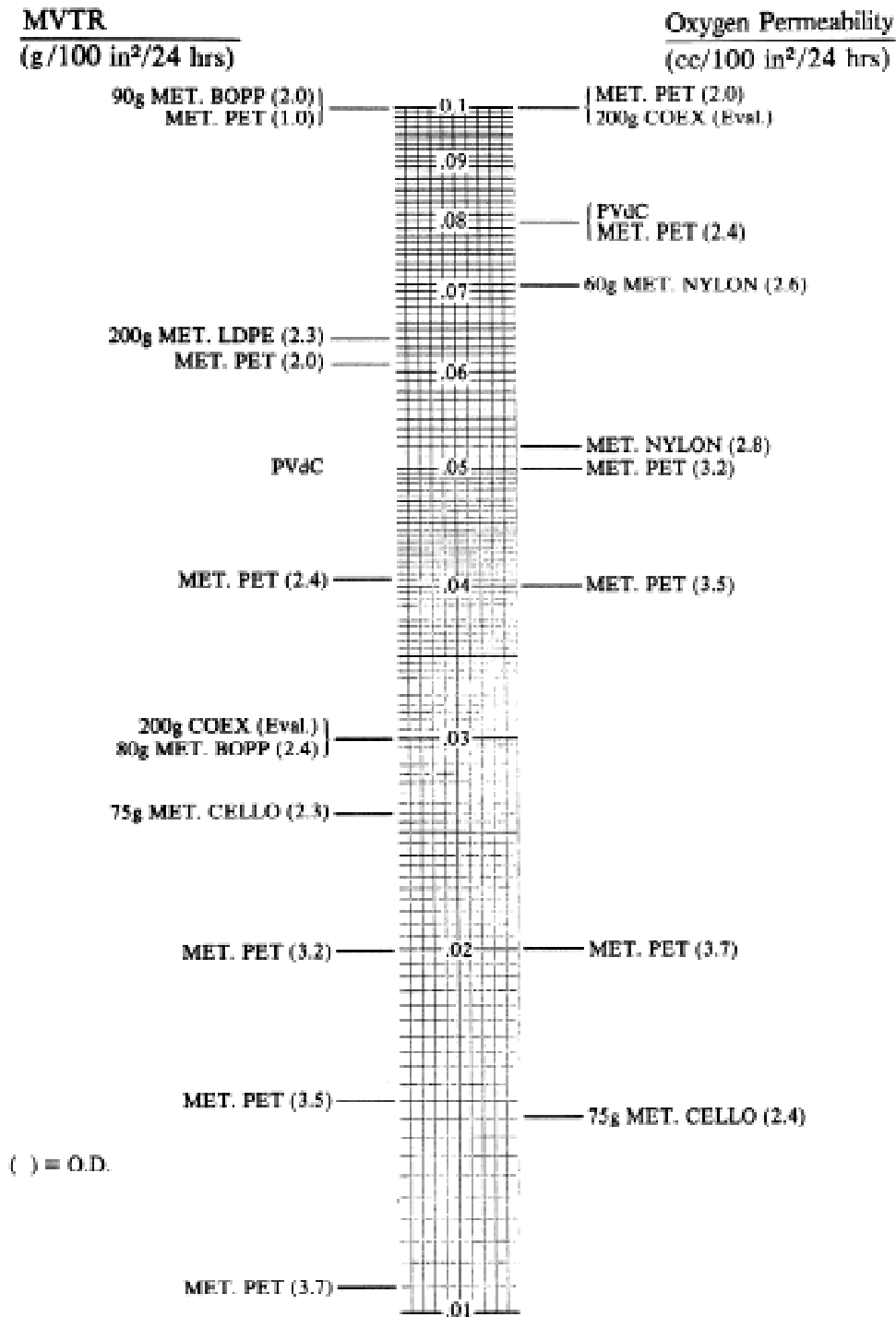
Oxygen Permeability
(cc/100 in²/24 hrs)



*Base substrates 1 mil/MET. substrates 48g unless noted otherwise

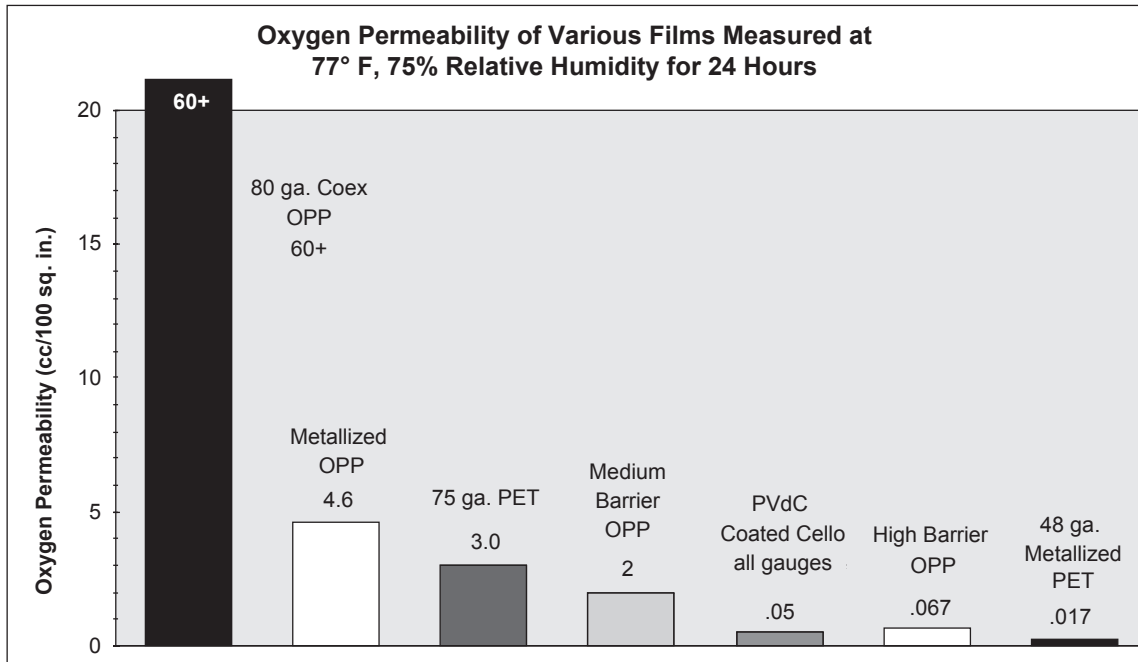
Figure 3.9 Barrier Properties for Metallized Films- 2 (Courtesy of ICI Americas Inc) ¹⁶

Barrier Comparison

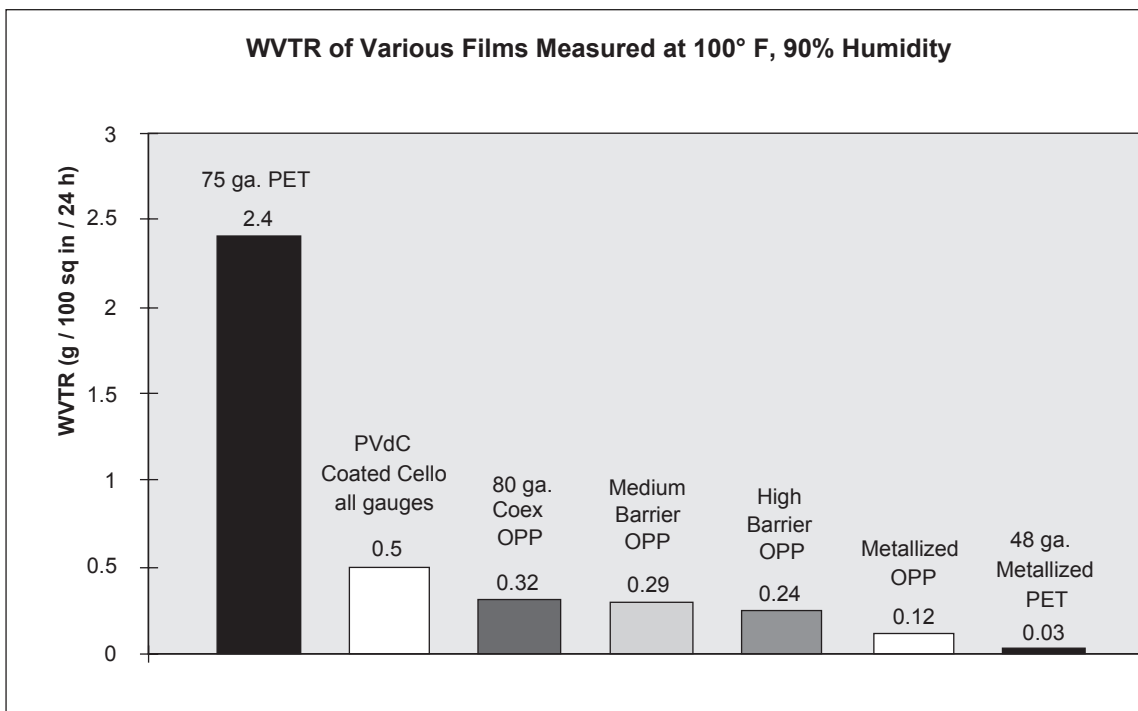


*All substrates 48g unless noted otherwise

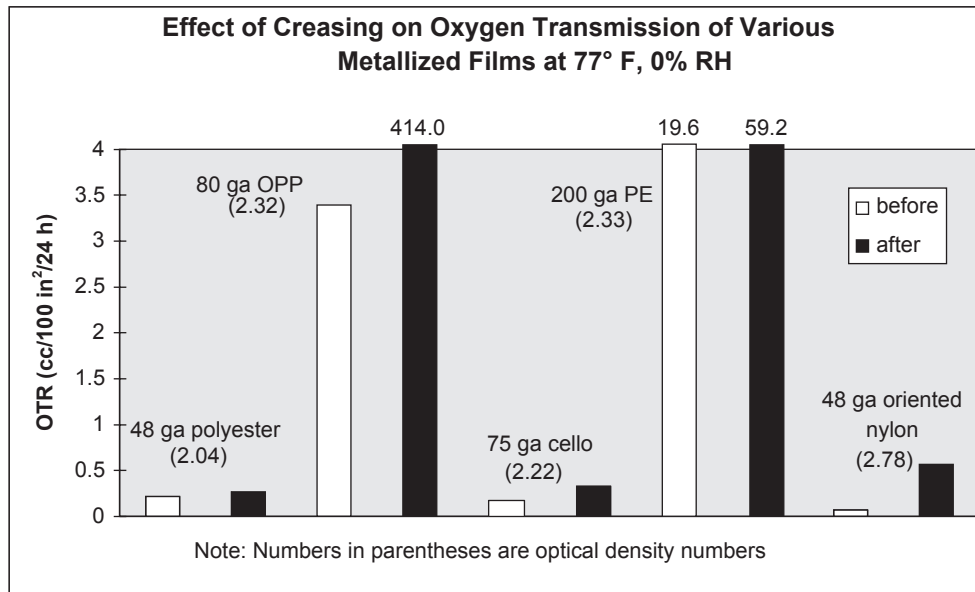
3.10 Oxygen Permeability of Various Films (Courtesy of ICI Americas Inc.)



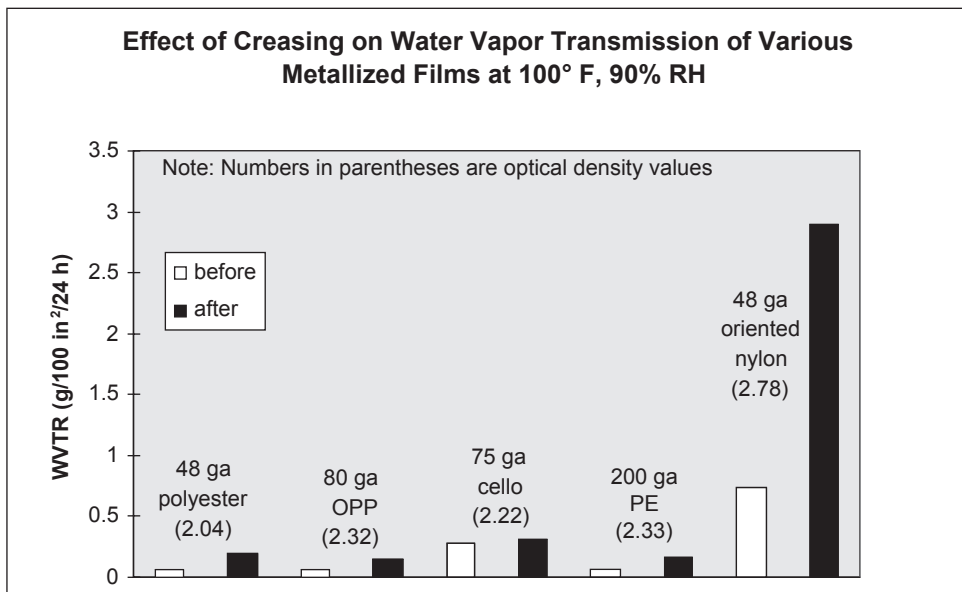
3.11 Water Vapor Transmission of Various Films (Courtesy of ICI Americas Inc.)



3.12 Effect of Creasing on Oxygen Transmission of Various Metallized Films (Courtesy of ICI Americas Inc.), values listed in Section 3.14



3.13 Effect of Creasing on Water Vapor Transmission of Various Metallized Films (Courtesy of ICI Americas Inc.), values listed in Section 3.14



3.14 Comparison of Barrier, Electrical and Optical Properties of Various Metallized Films (Courtesy of ICI Americas Inc.)

	OPTICAL DENSITY				RESISTIVITY (ohms/□)	WVTR (gms./100sq.in./24hrs.)		OXYGEN PERMEABILITY (cc./100sq.in./24 hrs.)	
	Metallized Film Type	Optical Density	%VLT	%UV		Flat	Creased	Flat	Creased
PET	48 ga.	2.04	0.91	1.71	1.8	0.061	0.190	0.105	0.258
	48 ga.	2.38	0.42	0.83	1.6	0.045	0.118	0.080	0.144
OPP	80 ga.	2.32	0.48	1.00	2.3	0.029	0.056	3.419	414.000
	90 ga.	1.91	1.22	2.52	2.5	0.103	0.120	16.850	37.300
Cello	75 ga.	2.22	0.61	0.99	1.6	0.266	0.271	0.143	0.304
PE	200 ga.	2.33	0.46	0.94	1.8	0.062	0.132	19.600	59.240
	125 ga.	2.01	0.98	1.92	2.6	0.279	0.429	16.720	152.200
Cast Nylon	100 ga.	2.22	0.60	1.08	1.9	8.335	8.335	0.129	0.387
Oriented Nylon	60 ga.	2.88	0.13	0.29	1.8	0.14	0.24	0.071	1.245
	48 ga.	2.78	0.16	0.38	1.2	0.15	0.61	0.052	0.561

WVTR conditions: 100° F, 90% RH

Oxygen permeability conditions: 77° F, 0% RH

3.15 Section 3 References Cited

- ¹ Mount III, E. M., "Aroma and Moisture Barrier Properties of Plain, Coated and Metallized OPP Films", *Technical Proceedings SME Barrier Technology for The Food Packaging Industry*, June 4-5, 1997
- ² Mount III, E. M., Wagner Jr., J. r., *TAPPI Technical Proceedings, Polymers, Laminations & Coatings Conference*, 1998, p875
- ³ Yializis, A., Mikheal, M. G., Ellwanger, R., Mount III, E. M., "Surface Functionalization of Polymer Films", *Proceedings of the 42nd Annual Technical Conference of the Society of Vacuum Coaters*, (1999), pp 469-474.
- ⁴ Szoke, R.L., "Vacuum Metallizing Plastic Films And Papers With Aluminum Control Parameters And Limitations A Converters Perspective", *Proceedings Of the First International Conference on Vacuum Web Coating*, Nov. 29 to Dec. 1, 1987, ed. R. Bakish, pp. 149-158
- ⁵ Yializis, A., Ellwanger, R., Harvey, J., "Barrier Degradation in Aluminum Metallized Polypropylene Films", *Proceedings of the 40th Annual Technical Conference of the Society of Vacuum Coaters*, (1997), pp 371-375
- ⁶ Mount III, E.M., Wagner, J.R., "Enhanced Barrier Vacuum Metallized Films," US Patent 5,981,079, Nov. 9, 1999
- ⁷ Decker, W., Yializis, A., "Surface Functionalization of Polymer Films and Webs Using Subatmospheric Plasma", *Proceedings of the 41st Annual Technical Conference of the Society Of Vacuum Coaters*, (1998), pp 355-357.
- ⁸ Rudd, D., Rice, S. W., "Polyester Film Coated With Metal Adhesion Promoting Coating And Having Superior Winding Performance ", U. S. Patent 4,921,764, Hoechst Celanese Corporation, issued May 1, 1990
- ⁹ Migliorini, RA, Mount III, EM, US Patent 5,591,520 Jan 7, 1997
- ¹⁰ Mount, E.M., Benedict, A.J.; European Patent 444340; "Metallisable Heat-Sealable, Oriented Polypropylene Film - Has Layer of Co-polyester To Improve Bonding To Metal"; (1991)
- ¹¹ Migliorini, R.A., Mount III, E.M., U.S. Patent 5194,318 Mar 16, 1993
- ¹² Migliorini, R.A. U.S. Patent 5,153,074, Oct. 6, 1992
- ¹³ All, F., Duncan, G., US patent 4,345,005 Aug. 17, 1982
- ¹⁴ Mapleston, P., "Barrier Film Coatings Add Performance Options", *Modern Plastics*, Aug 1992, pp 48-50.
- ¹⁵ Amon, M., "Relations Between Oxygen and Water Vapor Transmission Rates and Optical Density of Metallized Films", *AIMCAL Technical Proceedings*, Oct 4-6, 1999
- ¹⁶ Newton, J. R., "High Barrier Materials", *Future Trends In Vacuum Web Coating, AIMCAL Technical Proceedings*, Oct 20, 1983

Section 4: Useful Relationships for Metallized Films

4.1 Typical Area Yield for Various Films of Various Thickness (Courtesy of Camvac International Inc.)

GAUGE	TYPE	YIELD	GAUGE	TYPE	YIELD
		Sq. in/lb			Sq. in/lb
24	PP	128,800	120	PVC	16,985
28	PP	103,100	125	PE	24,000
32	PP	96,600	140	PVC	14,811
36	PP	85,700	150	PE	20,000
40	PP	77,300	170	PE	17,645
48	PP	64,400	175	PE	17,140
48	NYLON	51,000	200	PVC	10,560
55	PVC	40,000	200	PE	15,000
60	PVC	35,000	200	Tedlar	9,100
60	NYLON	40,800	280	PVC	7,400
70	ACRYLIC	33,000	500	PVC	4,100
70	PP	43,600	1000	Polystyrene	2,630
76	PP	40,700	40lb.	Paper	10,800
80	PP	38,200	43lb.	Paper	10,050
80	PVC	25,400	58lb.	Paper	7,450
90	PP	34,000	60lb.	Paper	7,200
100	PE	30,000	61lb.	Paper	7,080
100	PVC	23,625			
100	PP	31,000			
100	NYLON	24,500			
100	PP Cap	30,400			

4.2 Typical Area Yield for Polyester Film of Various Thickness (Courtesy of Hoechst Celanese Corporation)

THICKNESS		YIELD			
Gauge	Microns	Sq. in./lb.	Sq. ft./lb.	Sq. yd./lb.	Sq. m/kg
25	6.3	83,000	576.4	64.0	118.0
36	9.1	55,000	382.0	42.4	78.2
47	11.9	42,132	292.6	32.5	59.9
48	12.2	41,254	286.5	31.8	58.7
50	12.7	39,604	275.0	30.6	56.3
60	15.2	33,003	229.2	25.5	46.9
75	19.1	26,403	183.4	20.4	37.6
85	21.6	23,296	161.8	18.0	33.1
88	22.4	22,502	156.3	17.4	32.0
92	23.4	21,524	149.5	16.6	30.6
142	36.1	13,945	96.8	10.8	19.8
200	50.8	9,901	68.8	7.6	14.1
250	63.5	7,921	55.0	6.1	11.3
300	76.2	6,601	45.8	5.1	9.4
400	101.6	4,901	34.4	3.8	7.0
500	127.0	3,960	27.5	3.1	5.6
700	177.8	2,829	19.6	2.2	4.0
750	190.5	2,640	18.3	2.1	3.7
1000	254.0	1,980	13.7	1.5	2.8
1400	355.6	1,414	9.8	1.1	2.0

4.3 Formula for Calculating Length or Outer Diameter of a Roll of Polyester Film

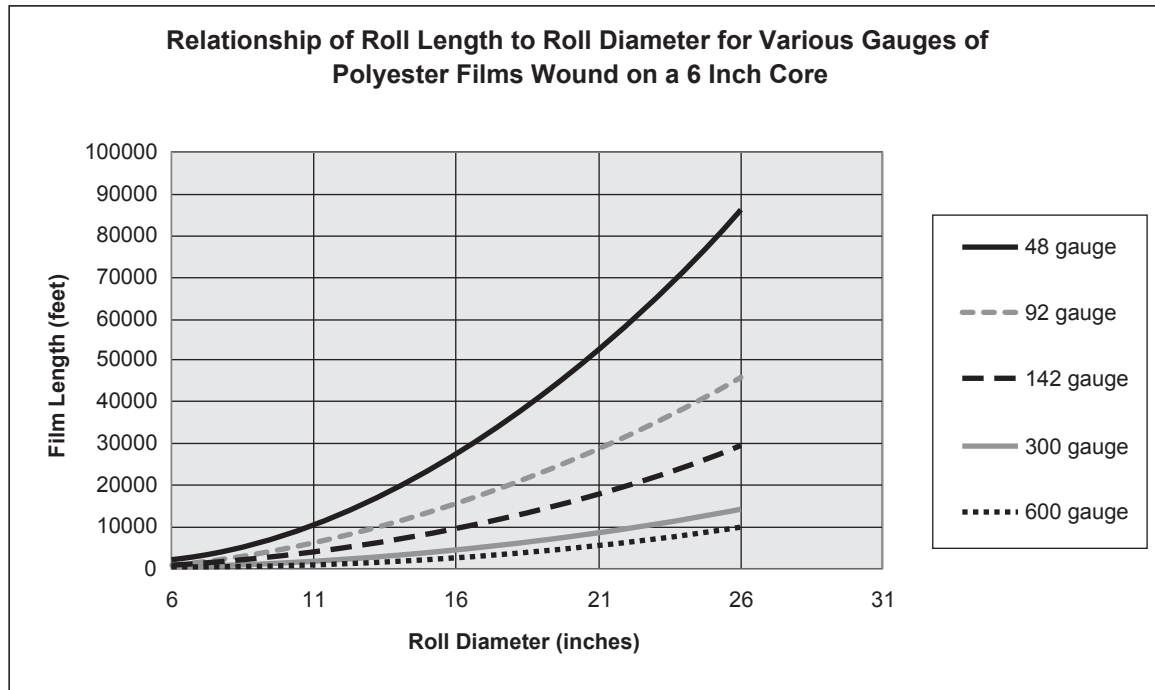
(Courtesy of ICI Americas Inc.)

$$Roll, OD(in) = \sqrt{\left[\frac{(thickness(ga) + 2.58) * length(ft)}{6545} + [coreOD^2(in)^2] \right]}$$

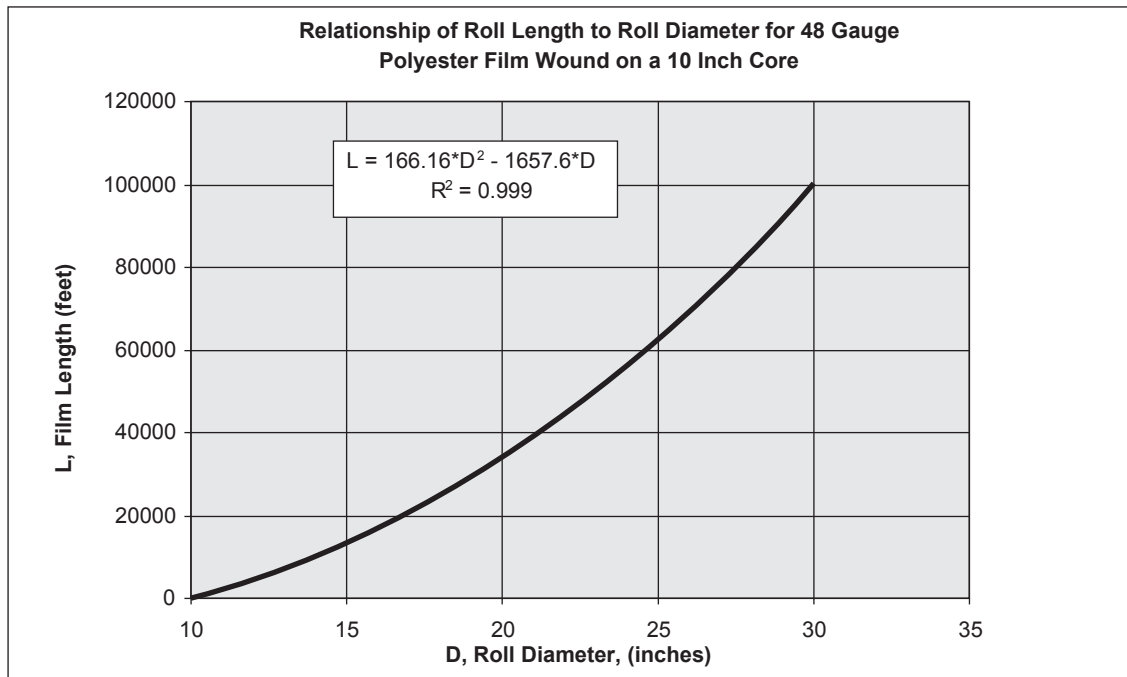
$$Length(ft) = \frac{[rollOD^2(in)^2 - coreOD^2(in)^2] * 6545}{thickness(ga) + 2.58}$$

If knurled film, substitute knurl height for 2.58. Use 2.58 for slit film without knurl.

4.4 Relationship of Roll Length to Roll Diameter for Various Thickness Polyester Films Wound on a Six Inch Core (Courtesy of ICI Americas, Inc)



4.5 Relationship of Roll Length to Roll Diameter for 48-Gauge Polyester Films Wound on a Ten Inch Core (Courtesy of Camvac International, Inc.)



4.6 Formula for Calculating Roll Weight of Polyester, Nylon and Propylene Film
(Courtesy of Hoechst Celanese Corporation)

$$\text{Roll Weight (pounds)} = \text{Th} \times \text{L} \times \text{W} \times \text{K}_1$$

Th = thickness in inches

L = length in feet

W = width in inches

K₁ = 0.606 for polyester

0.489 for nylon

0.394 for polypropylene

4.7 Formula for Calculating MSI (1000-square-inch) Cost of Polyester, Nylon and Propylene Film
(Courtesy of Hoechst Celanese Corporation)

$$\text{Cost (\$ / 1000 in}^2 \text{)} = \text{Price} \times \text{Th} \times \text{K}_2$$

Cost in \$/msi

Price in dollars per pound

Thickness in mils

K₂ = 0.0504 for polyester

0.0406 for nylon

0.0327 for polypropylene

4.8 Factors for Converting Various Basis Weights of Metallized Paper
(Courtesy of Camvac International, Inc.)

To change from one basis weight to another, use the following conversion factors:

<u>Convert from</u>	<u>Convert to</u>	<u>Do</u>
Grams per square meter (gsm)	lb./25x38	multiply by 0.675
lb./25x38	lb./24x36	Divide by 1.100
lb./25x38	msi/lb.	Divide 475 by lb./25x38
lb./24x36	msi/lb.	Divide 432 by lb./24x36

4.9 Conversion Table for Commonly Used English/Metric Units of Measure

(Courtesy of E. I. DuPont de Nemours & Company, Inc.)

<u>Abbreviation</u>	<u>START WITH:</u> <u>Description</u>	<u>MULTIPLY BY:</u>	<u>Abbreviation</u>	<u>TO OBTAIN:</u> <u>Description</u>
<u>AREA</u>				
in ²	square inch	6.452x10 ⁻⁴	m ²	meter ²
ft ²	square foot	9.290x10 ⁻²	m ²	meter ²
<u>ENERGY</u>				
Btu	Brit. Thermal Unit	1.056x10 ³	J	joule
cal	calorie	4.187	J	joule
kw-hr	kilowatt-hour	3.600x10 ⁶	J	joule
<u>FLOW</u>				
gal/min	gallons/minute	6.309x10 ⁻⁵	m ³ /s	meter ³ /second
<u>HEAT CAPACITY</u>				
c (heat capacity)	Btu/lb • °F	4.187x10 ³	J/kg•K	joule/kilogram-Kelvin
<u>LENGTH</u>				
Å	angstrom	1.000x10 ⁻¹⁰	m	meter
mil	thousandths of inch	2.540x10 ⁻⁵	m	meter
in	inch	2.540x10 ⁻²	m	meter
ft	foot	3.048x10 ⁻¹	m	meter
yd	yard	9.144x10 ⁻¹	m	meter
<u>MASS</u>				
oz(mass)	ounce(mass)	2.835x10 ⁻²	kg	kilogram
lb(mass)	pound	4.536x10 ⁻¹	kg	kilogram
T	ton(short)	9.072x10 ²	kg	kilogram
<u>PERMEABILITY-GAS</u>				
cc/m ² • 24 hrs • atm	centimeter ³ / meter ² • 24 hrs • atmosphere	9.87	ml/m ² • Mpa • d	milliliter/meter ² mega pascal day
cc/100in ² • 24 hrs	cc/100in ² • 24 hrs	15.50	cc/m ² • d	cc/meter ² • day
<u>PERMEABILITY-WVTR</u>				
g/100in ² • 24 hrs	gram/100 inch ² • 24 hrs	15.50	g/m ² • d	gram/meter ² • day
<u>POWER</u>				
Btu/hr	Btu/hr	2.930x10 ⁻¹	W	watt
hp	horsepower	7.460x10 ²	W	watt
<u>PRESSURE</u>				
m•bar	millibar	0.760	mm Hg	millimeters mercury(Torr)
mm Hg	millimeters mercury (Torr)	1.316	m•bar	millibar
atm	atmosphere	1.013x10 ⁵	Pa	pascal
cm Hg	centimeter	1.333x10 ³	Pa	pascal
lbs/in ² (psi)	pounds per square inch	6.895x10 ³	Pa	pascal
<u>TEMPERATURE</u>				
°F	degrees Fahrenheit	.555 (°F – 32)	°C	degrees Centigrade
°C	degrees Centigrade	(1.8x°C) + 32	°F	degrees Fahrenheit
°C	degrees Centigrade	°C + 273.15	K	Kelvin
°F	degrees Fahrenheit	<u>°F + 459.67</u> 1.8	K	Kelvin
<u>THERMAL CONDUCTIVITY</u>				
k (thermal conductivity)	Btu • in/sec • ft ² • °F	5.192x10 ²	W/m • K	watt/meter Kelvin
k (thermal conductivity)	Btu • in/hr • ft ² • °F	1.442x10 ⁻¹	W/m • K	watt/meter Kelvin
<u>VOLUME</u>				
in ³	cubic inch	1.639x10 ⁻⁵	m ³	meter ³
ft ³	cubic foot	2.832x10 ⁻²	m ³	meter ³
gal	gallon (U.S. liquid)	3.785x10 ⁻³	m ³	meter ³
<u>YIELD</u>				
in ² /lb	inches ² /pound	1.442x10 ⁻³	m ² /kg	meter ² /kilogram

Note: These are basic units, i.e. meter. Make proper adjustments (kilo, milli, centi, etc.) so that the final answer is in the form you need from .01 to 1,000.

Section 5: Metal Evaporation and Heated Boat Data

5.1 Some Factors Influencing Boat Life

ELECTRICAL CONTACT
INITIAL HEATING CYCLE
INITIAL WETTING OF THE CAVITY WITH ALUMINUM
OPERATING TEMPERATURE
UTILIZATION OF THE CAVITY AREA/CLEANING
VACUUM LEVEL
NUMBER OF THERMAL CYCLES
CORRECT RESISTIVITY
COOLING OF BOATS
BOAT DEFECTS
EVAPORATION RATE
BOAT COMPOSITION

5.2 Recommendations for the Use of Evaporation Boats

To assure optimum boat performance and to maximize boat life, the following procedures should be adhered to:

1. In order to assure adequate electrical contact during thermal cycling, both ends of each boat should be covered with flexible graphite foil and firmly pressed into place in clean copper clamps. A prerequisite for uniform electrical contact is parallel surfaces between the clamps and the boats.
2. The aluminum feed wire should be positioned so that it meets the boat approximately 30-50% from the end of the cavity.
3. For the initial heat-up cycle, the temperature of the boats should be slowly raised to operating temperature during a period of 10 to 15 minutes. This procedure assures maximum boat life and performance. Subsequent heat-up cycles can be performed at whatever rate provides the best machine productivity.
4. When the desired operating temperature has been reached (approximately 1450 ° C) and the boat cavity is uniformly wetted with molten aluminum, the coating process can begin. For optimum performance, care should be taken to operate each boat at the temperature at which a thin layer of molten aluminum covers the entire cavity during operation
5. Upon completion of the coating cycle and after the wire feeder has been stopped, the boats should be kept under power until all residual aluminum on the cavity surface has evaporated.

Adherence to these five key procedures can significantly extend boat life and improve the quality of the metallized film.

5.3 Boat Temperature vs. Approximate Boat Life

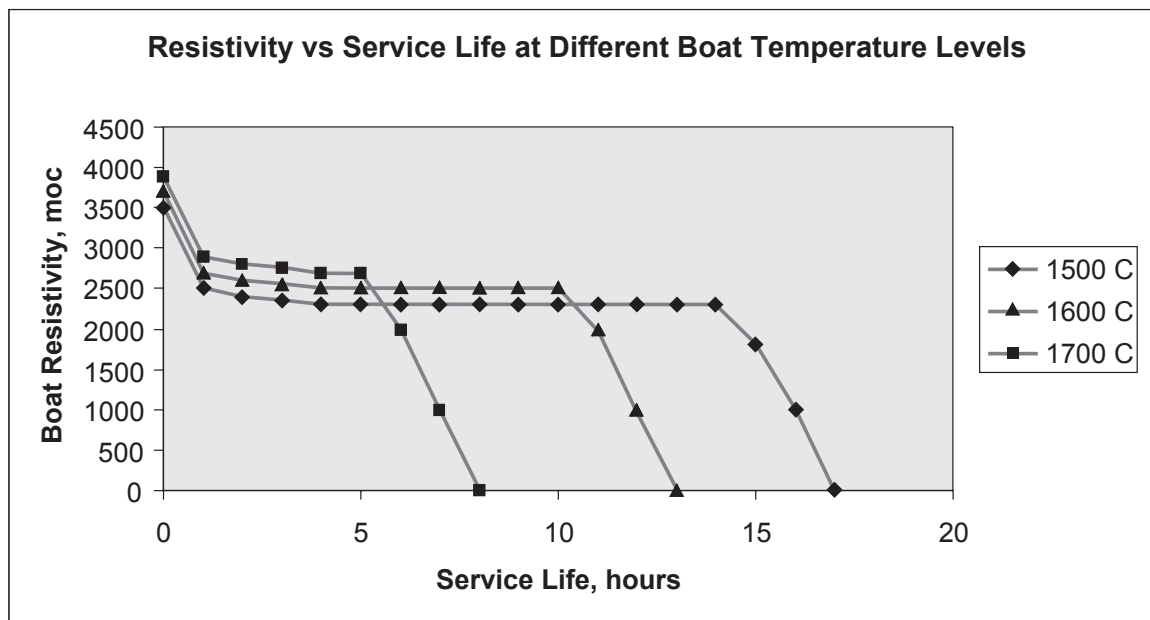
Operating Temperature (° C)	Boat Life (Hours)
1500	X
1600	3/4 X
1700	1/2 X

5.4 Boat Resistivity (HT) and Approximate Service Life at Different Temperatures For Aluminum Evaporation

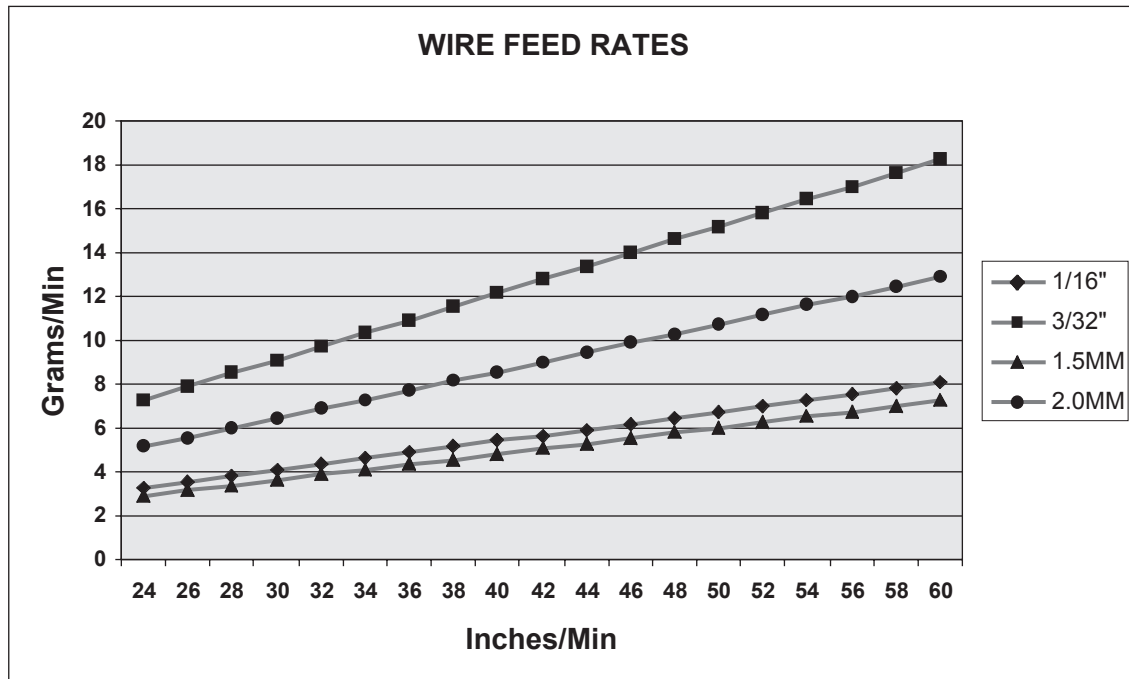
Service Life	Resistivity (moc)		
Hours	1500° C	1600° C	1700° C
0	3500	3700	3900
1	2500	2700	2900
2	2400	2600	2800
3	2350	2550	2750
4	2300	2500	2700
5	2300	2500	2700
6	2300	2500	2000
7	2300	2500	1000
8	2300	2500	0
9	2300	2500	
10	2300	2500	
11	2300	2000	
12	2300	1000	
13	2300	0	
14	2300		
15	1800		
16	1000		
17	0		

Boat size : 10 x 30 x 150 mm
 Cavity: 1.5 x 26 x 90 mm
 Resistivity at room temperature: 1000 moc
 Wire- feed rate: 8 g/min of aluminum

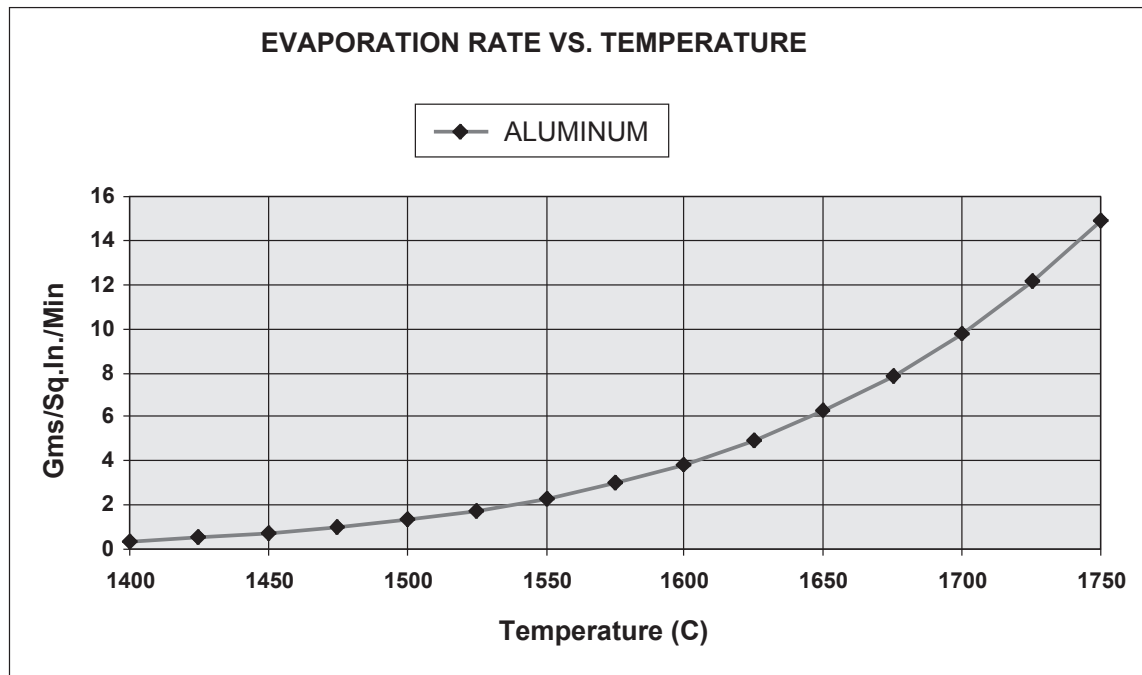
Plot of data in Table 5.4



5.5 Aluminum Wire Feed Rates (Courtesy of Advanced Ceramics Corporation)



5.6 Evaporation Rate vs. Temperature for Aluminum Metal (Courtesy of Advanced Ceramics Corporation)



$$\text{Rate of Evaporation, gm/in}^2/\text{min} = 22.586 \cdot \text{Vapor Pressure} \cdot [\text{Molecular Weight/Temperature, } ^\circ\text{K}]^{1/2}$$

Note: Aluminum vapor pressure from RCA Review, June, 1969, p 295, and molecular weight from: Principles Of Engineering Materials, Bartlett, Nix, Tetleman, pp 540-541.

5.7 Calculation of Evaporation Rates

A discussion of the evaporation rate of various materials can be found in texts dealing with the ideal gas laws^{1,2}. These discussions show that evaporation rates can be estimated from a consideration of molecular dynamics of saturated vapors. Several analyses of this sort exists and there are several sources of evaporation rate equations^{2,3} and the vapor pressure data needed for the calculation^{2,4}.

Theoretically, evaporation rates may be estimated by using the fact that for a saturated vapor, evaporation and condensation are occurring at the same rate. While it is difficult to calculate the evaporation rate, the rate of condensation may be calculated from the kinetic gas theory as the rate of molecules striking the surface of a solid or liquid and is given as Equation 5.1:

Equation 5.1

$$\alpha v = \frac{1}{4} n v_a = p / [2\pi RT]^{1/2}$$

where:

α = the condensation coefficient

v = rate of gas molecule impacts per unit area per unit time

n = number density of gas molecules

v_a = average molecular velocity

p = pressure

R = ideal gas constant

T = absolute temperature

For a vapor at equilibrium with a liquid, this is just equal to μ , the rate at which molecules evaporate from the surface, giving Equation 5.2

Equation 5.2

$$\mu = \alpha v = \frac{1}{4} n v_a = p / [2\pi RT]^{1/2}$$

The ideal gas considerations allow the calculation of the maximum possible rate of evaporation, which would occur, if all of the vapor is removed as fast as it was formed such that there was no condensation¹. Dushman² makes use of Langmuir's⁵ observation that the rate of evaporation of Tungsten in high vacuum is the same as the rate of evaporation in the presence of a saturated vapor. Dushman² also makes use of Langmuir⁵, who showed that for metal atoms condensing on a metal surface the value of α is equal to 1, to calculate the mass of gas or vapor incident on a unit area per unit time as:

Equation 5.3

$$G = mv = \frac{1}{4} \rho v_a$$

Where:

ρ = mn , the gas density

m = mass of molecules

Which can be written in terms of vapor pressure and temperature as:

Equation 5.4

$$G = 5.833 \times 10^{-2} P_{\text{mm}} [M/T]^{1/2}$$

Where:

G = gm/cm² sec

P_{mm} = vapor pressure in mm Hg or Torr

T = absolute temperature in degrees Kelvin

M = molecular weight

Equation 5.4 may be modified through its constant to allow use of different units for the pressure or to obtain different units for the rate of evaporation as used in the calculations of Figure 5.6 in terms of gm/in²/min.

For much of the tabulated vapor pressure data of metals, at pressures below 1 mm Hg, it has been observed that the vapor pressure may be represented as a function of temperature by Equation 5.5²:

Equation 5.5

$$\text{Log } P = A - B/T$$

Where, A and B are constants.

The equation for evaporation rate data G, may be rewritten and expressed in terms of the constants A and B through the vapor pressure as follows:

Equation 5.6

$$\text{Log } G = 5.7660 + 0.5 \log M + \log P_{\mu} - 0.5 \log T$$

Where:

P_{μ} is measured in microns (1 micron = 10^{-3} Torr)

Substituting in Equation 5.6 for $\log P_{\mu}$ and rearranging yields Equation 5.7:

Equation 5.7

$$\text{Log } G = C - 0.5 \log T - B/T$$

Noting that:

Equation 5.8

$$C = A + 0.5 \log M + 5.7660$$

Where:

P_{μ} measured in microns, (1 micron = 10^{-3} Torr)

Dushman² tabulates values of the constants for a wide range of metals for the pressure in microns (1 micron = 10^{-3} Torr). Table 5.8 lists the constants for metals important in commercial metallization of films and papers. Much of the data in Dushman² is based upon the data compilation of Honig⁶, in the RCA Review, which has been widely cited but is difficult to locate. An additional source of vapor pressure data is that of Nesmeyanov⁴. Nesmeyanov reviews the vapor pressure literature data for a wide range of vapor pressure determinations for the elements. Nesmeyanov⁴ fits and tabulates the constants for vapor pressure data according to Equation 5.9:

Equation 5.9

$$\text{Log } P_{\text{mm}} = A - B/T + C \cdot T + D \cdot \log T$$

as opposed to Equation 5.5 of Dushman². The constants A, B, C and D are listed for solid and liquid condensed vapors. Nesmeyanov⁴ also tabulates boiling point as a function of pressure, heats of fusion and heats of vaporization for the elements.

5.8 Evaporation Rate G, gm/cm² sec, Constants for Various Metals for Use in Equation 5.7 Taken From Dushman²

<i>Metal</i>	<i>A, for P_μ in microns 1 micron = 10⁻³ Torr</i>	<i>B</i>	<i>C</i>
Aluminum	11.79	15940	8.27
Copper	11.96	16980	8.63
Zinc	11.63	6560	8.30
Indium	11.23	12480	8.03
Tin	10.88	14870	7.69
Chromium	12.94	20000	9.56
Nickel	12.75	20960	9.4
Silver	11.85	14270	8.63
Gold	11.89	17580	8.80

5.9 Vapor Pressure Constants of Nesmeyanov⁴ For Use in Equation 5.9 for Various Elements Where the Condensed Phase of the Element is a Solid, Pressure in mm Hg

<i>Metal</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Aluminum	15.41687	16710.43	0.00198518	-1.81636
Copper	-0.70729	16869.44	-0.00091963	3.50147
Zinc	13.50846	7011.863	0.00040185	-1.58311
Indium	470.88627	26891.71	0.10315698	-179.48306
Tin	-150.16773	10870.18	-0.03682439	62.12498
Chromium	4.36192	19723.49	-0.00055955	2.03941
Nickel	-10.94190	16193.89	-0.00176813	7.29936
Silver	21.68194	15662.85	0.00069459	-4.01891
Gold	-24.40936	16439.69	-0.00239291	11.25366

5.10 Vapor Pressure Constants of Nesmeyanov⁴ for Use in Equation 5.9 for Various Elements Where the Condensed Phase of the Element is a Liquid, Pressure in mm Hg

<i>Metal</i>	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Aluminum	12.40857	15999.58	-0.00000352	-0.99926
Copper	-285.06133	-14786	-0.01263166	91.90151
Zinc	-7.91900	5330.735	-0.00161644	5.59972
Indium	10.82068	12298.11	0.00007147	-0.82159
Tin	-9.32188	14023.92	-0.00088122	5.62012
Chromium	165.22012	44507.12	0.00366345	-45.49926
Nickel	30.68264	19678.72	0.00038550	-6.08393
Silver	10.80805	14323.13	-0.00009019	-0.53927
Gold	-6.59202	16370.99	-0.00064886	4.79166

5.11 Section 5 References Cited

¹ Kennard, E. H., *Kinetic Theory Of Gases : With An Introduction to Statistical Mechanics*, McGraw-Hill Book Company, Inc. New York, First edition, sixth printing, 1938, pp 68-69

² Dushman, S., *Scientific Foundations Of Vacuum Technology*, Vol 2, Lafferty, J. M. editor, John Wiley&Sons, New York (1962), chapter 1

³ Diels, K., Jaeckel, R., *Leybold Vacuum Handbook*, Translation of *Leybold Vakuum-Taschenbuch*, 2nd edition, Springer-Verlag, Berlin, by Adam, H., Edward, J., Pergamon Press, New York, (1966), p 14.

⁴ Nesmeyanov, A. N., *Vapor Pressure Of The Chemical Elements*, Gary, R., editor, Elsevier Publishing Company, New York, 1963

⁵ Langmuir, *Phys. Rev.*, **2**, 329 (1913)

⁶ Honig, R. E., "Vapor Pressure Data of the More Common Elements", *RCA Review*, **18**, 195 (1957)

Section 6: Aluminum Wire Chemistry and Temper

6.1 Aluminum Wire Alloys

The primary concern for aluminum wire is the purity of aluminum versus the other alloying elements. All aluminum alloys with a first digit of 1 possess a minimum aluminum purity of 99%. The second digit of the series indicates any specific control used over the individual impurities in the particular alloy. The third and fourth digits are the same as the two digits to the right of the decimal point indicating minimum aluminum content required.

Alloy/Impurity	1100	1350	1188	1199
Silicon (Si)	0.95 Si+Fe	0.10	0.06	0.006
Iron (Fe)		0.4	0.06	0.006
Copper (Cu)	0.05-0.20	0.05	0.005	0.006
Manganese (Mn)	0.05	0.01	0.01	0.002
Magnesium (Mg)			0.01	0.006
Chromium (Cr)		0.01		
Zinc (Zn)	0.10	0.05	0.02	0.006
Titanium (Ti)			0.01	0.002
Others	0.15	0.10	0.01	0.003
Aluminum (AL)	99.00	99.50	99.88	99.99

All element values for aluminum wire are maximum values except the aluminum, which is minimum. The above analysis is in conformance as specified by the Aluminum Association Standards and Data.

Aluminum wire used in metallizing can be produced by cold rolling or drawing and, thus, supplied at different strengths and tempers. A soft temper or “O” temper is produced by annealing the wire after cold working it to its final dimensions. A hard temper of H18 is a result from strain hardening of about 75% (full hard). Intermediate tempers are produced by strain hardening annealed wire by an amount that increases tensile strength by one-quarter, one-half, or three-quarters the difference in strength between O and H18. These tempers are called H12, H14, and H16, respectively.

6.2 Chemical and Physical Properties of Aluminum¹

“A hard, strong, silvery-white metal, protected by oxide film from reacting with air and water. Soluble in hot concentrated HCL and NaOH solution. Hundreds of uses as metal and alloys”

Abundance: Earths crust 83,000 p.p.m.; seawater 0.01 p.p.m.

Atomic number	13
Relative atomic mass	26.981
Atomic radii, pico meters	143.1
Melting point, °K	933.52
Boiling point, °K	2740
Heat of fusion, kj/mol	10.67
Heat of vaporization, kj/mol	290.8
Solid density, kgm/m ³	2698 [298 K]
Liquid density, kgm/m ³	2390 [liquid at m.p.]
Thermal conductivity, W/mK	237 [300 K]
Electrical resistivity, ohm-m	2.6548 x 10 ⁻⁸

Vapor pressure and maximum evaporation rate as a function of temperature²

Pressure, microns	10 ⁻²	10 ⁻¹	1	10	100	1000
Temperature °C	882	972	1082	1207	1347	1547
Evaporation rate gm/cm ² sec	8.92x10 ⁻⁸	8.59x10 ⁻⁷	8.23x10 ⁻⁶	7.88x10 ⁻⁵	7.53x10 ⁻⁴	7.10x10 ⁻³

6.3 Section 6 References Cited

¹ Emsley, J., *The Elements*, Clarendon Press, Oxford, (1989), pp12-13

² Dushman, S., *Scientific Foundations Of Vacuum Technology*, Vol 2, Lafferty, J. M. editor, John Wiley&Sons, New York (1962), table 10.1, p 697.

Section 7: Surface Treatment of Webs for Aluminum Vapor Deposition

7.1 Discussion of Surface Treatment Technologies

INTRODUCTION

Web based materials for aluminum vapor deposition, often require surface preparation or chemical modification to insure adequate adhesion of the aluminum layer to the web surface or perhaps a smooth surface. With today's improved understanding of the impact of film surface chemistry there may also be good reasons to chemically modify web surfaces to enhance moisture and gas barrier¹ and other properties as well.

Today there are many web materials available for metallization with a wide range of surface chemistries and topographies from paper, non-woven polymers, cellophane and synthetic polymer films. These webs cover a wide range of chemical functionalities, which will influence what surface treatments if any, will be necessary to prepare them for metallization. In preparation for metallization the web is surface modified, or functionalized, prior to placement in the vacuum chamber and at its point of manufacture. The surface modification may also take place or be enhanced in the vacuum chamber during, but just prior to, the deposition of the aluminum. The two treatment locations will be broken down into two broad categories of atmospheric pretreatments and in-chamber pretreatments for this manual.

Atmospheric pretreatment consists of coating or chemically priming, corona treatment, flame treatment, chemical treatment and atmospheric plasma treatment of the web surface. Most atmospheric pretreatments result in the direct oxidation of the web surface. In-chamber pretreatment is performed by plasma treatment using various treater designs and various gas plasmas to optimize various film properties². The choice of the treatment method can be a key decision in determining the final web properties or as simple as a choice of what method is available, a technical matter of taste, depending on the requirements of the final web properties. The greatest care must be exercised in the definition of the final properties required for the metallized product to determine which method of surface modification is best suited for the product.

In general the web surface to be modified is relatively flat and nonporous and is composed of a natural or synthetic polymer. Even with paper, the metallizing surface is generally coated with a polymer layer to smooth the surface, which will accept the metal deposit and the fiber structure of the paper is not important in the metallization. Consequently, the discussion of treatment here will be restricted to a discussion of the methods used and their impact on natural and synthetic polymer materials. The principle focus will be on the treatment of polyester (PET), polyethylene (LDPE, HDPE) and polypropylene (OPP) films with various references to other polymers such as polystyrene, polyimides and polyamides (NYLON) when appropriate.

Most surface pretreatments are designed to chemically modify the top most layers of a web and not to penetrate into the bulk material. Coating of films serves the purpose of placing a new polymer onto the carrier film and is a surface modification by changing the near surface bulk composition of the film or paper for metallization. Consequently it will not be discussed in detail, as we will focus on various polymers interaction with the treatment method and not simple changes in surface polymer. Indeed many of the films used for metallization are coextruded multilayer polymers structures with the outer most polymer layers being selected for particular purposes such as sealability, printability or for metallization. Again we will focus on the polymers reaction to the treatment methods occurring at the surface and not with the surface polymer selection. This section will focus on the methods and mechanisms of surface oxidation of polyethylene, polypropylene and polyethylene terephthalate films, paper and foil by corona, flame and plasma treatment. Means of measuring the surface treatment and surface oxidation will also be covered.

Oxidation of a material in this context will mean the addition of an oxygen atom into the structure of the polymer. Polyethylene and polypropylene have no oxygen in their chemical structure as compared to polyethylene terephthalate, which contains oxidized carbon atoms in the form of carboxylic acid (HO-C=O) and alcohol (C-OH) functions built into the molecular structure of the polymer in the ester linkage of the polyester. Carbon atoms are capable of forming four chemical bonds and oxygen atoms can form two chemical bonds. Thus the state of oxidation of a carbon atom may be determined by counting the

number of carbon-oxygen bonds. Using this we see that a carbon atom in polyethylene molecules with two C-C and two C-H bonds is unoxidized. An alcohol (-C-OH) with one carbon-oxygen single bond is the least oxidized carbon and a carbonyl carbon (-C=O) with a carbon-oxygen double bond is more oxidized than the alcohol and the carboxylic acid, (HO-C=O) with three carbon-oxygen bonds is even more oxidized and carbon dioxide (O=C=O) with two carbon-oxygen double bonds is completely oxidized. These chemical group names and oxidation levels will be used through out this section to explain the impact of the various treatment methods on the polymers.

TREATMENT MEASUREMENT

Measurement of the surface energy or treatment level can be conducted many different ways to obtain simple wettability measurements, to obtain more detailed information about the nature of the surface forces, the level of surface oxidation and the composition of the surface chemistry. Each method is more complex, time consuming, informative and expensive than the preceding method. Each will be briefly described and its use outlined. Those wishing to study this topic further are directed to the recent book of Chan³, which reviews the many methods of polymer surface modification and characterization.

For the simple determination of the existence of treatment, pre-made treatment pens may be purchased to wipe on a film surface to check for minimum surface treatment levels. However, due to contamination of the pen tip by actual testing of the film by surface wiping, the pens should not be routinely used to determine the treatment level but as a go no-go test for the presence of treatment. In order to save money or time if no treatment pens are available, it should also be possible to determine which commercial markers will show treated and non-treated surfaces. As some markers are solvent based and others are water based it should be quite possible to test some to determine which will show proper film treatment levels for your process by the formation of a uniform mark or a mark which breaks up and does not wet your substrate.

Surface Treatment Level: This is a straightforward and fast measurement to determine the surface tension or wettability of the surface. The test is detailed in the ASTM procedure D2578 for the measurement of surface tension of polyolefin films from 30 to 56 dynes/cm. While simple to do, reading the results are somewhat subjective, and is therefore prone to operator variability based upon training and experience. The test consists of wetting a cotton tipped swab with a series of solutions of formamide and ethylene glycol monoethylether (ethyl cellosolveTM) blends, Table 7.1, and spreading the solution on the surface by wiping the wet swab lightly over the surface. Alternatively, several drops of solution may be spread or drawn down by a metering rod. Approximately one square inch of surface is covered and then the breakup of the liquid film is watched for. The surface tension is defined as the solution composition where the liquid film breaks up at two seconds. This is determined by testing the surface with a series of solutions. Great care is needed at each step, which is why the operator's experience is important. A fresh swab needs to be used for each attempt, too much fluid on the cotton will impact the reading by varying the liquid film thickness, too much pressure on the swab tip can disturb the film surface changing the result, the cotton itself must be free of sizing and the solutions need to be fresh to insure proper composition. In general, the test yields a value, which can vary 1 to 2 dynes for a single tester and between testers.

Contact Angle: This is a more time consuming test to determine the treatment level of treated films. It may be determined in several ways and has been described in great detail in many texts and publications. In the equilibrium sessile drop method, a controlled volume, stationary drop of liquid is placed on a planar surface and the angle between the solid, the saturated vapor or air and the liquid is related to the height, h, and base diameter, d, of the drop which are measured.

Equation 7.1

$$\tan \theta/2 = 2 h/d$$

The results may be reported as the contact angle in degrees, θ or $\cos \theta$, the cosine of the contact angle θ . The apparent contact angle may be affected by the chemical composition variation and the roughness of the surface. These effects may be checked, by measuring the advancing and receding contact angle. These tests are performed, by making the drop edge move across the surface by slowly increasing or decreasing the volume of the liquid drop. Then the drop is held at constant volume and the contact angle is measured

after the drop edge has come to rest. The difference between the two measurements, the hysteresis, can give information about the relative surface roughness.

The Wilhelmy plate or tensiometric methods also give the equilibrium, advancing and receding contact angles by calculating the force required to lower or raise a sample into a fluid using a micro-balance. This test is gaining more use as it is less subjective in its measurement due to instrumentation used and gives several measurements with a single sample setup.

Solid Surface Free Energy: There is no direct, reliable method for directly determining the surface tension of a solid⁴. However there are several indirect empirical and semi-empirical methods based on contact angle measurements. The solid surface tension is important to know, because of its potential for determining the potential for adhesion between materials. It has been suggested that the surface free energy of a solid surface is composed of several intermolecular forces comprising dispersive, polar and hydrogen bonding. It is postulated that two interacting materials which share these intermolecular forces should adhere together and that the work of adhesion would also be comprised of these three intermolecular forces. Measuring the contact angle of the solid surface with various liquids, allows the estimation of the solid surface energy, γ_s , which can be shown to be composed of two parts, the polar part, γ_s^p , and the dispersive part, γ_s^d . These parameters are thought to be important in that they can be used to estimate the work of adhesion between a solid and another material. The polar portion is made up of the combination of the dipolar interactions and the hydrogen bonding attractions between two surfaces. Hydrogen bonding is the intermolecular attraction of an alcohol hydrogen with a molecular dipole such that a weak bond-like attraction is formed between two molecules keeping them in close association. In treated polyethylene surfaces the auto adhesion between two films has been shown to be due to hydrogen bonds between –C–OH groups on one surface and C=O functions on the other surface⁵. For polyester the auto adhesion is between terminal phenols in one surface and carbonyls in the other⁶. Equation 7.2 is used to determine the dispersive and polar solid surface energies by measuring the contact angle with two different fluids and solving the set of two simultaneous equations for γ_s^p and γ_s^d :

Equation 7.2

$$\gamma_{LV} (1 + \cos\theta) = 2(\gamma_s^d \gamma_{LV}^d)^{1/2} + 2(\gamma_s^p \gamma_{LV}^p)^{1/2}$$

Values for γ_{LV} are tabulated in Table 7.3, for well-characterized liquids and a combination of any two sets can be used for the determination⁷. Often times water / methylene iodide or water / formamide are the liquid pairs used.

Surface Oxygen Concentration: Total surface oxygen concentration may be determined by X-ray photoelectron spectroscopy, (XPS) or as it is often termed, electron spectroscopy for chemical analysis (ESCA). In this method a sample is placed in a high vacuum chamber and bombarded with monochromatic X-rays. The X-rays interact with the uppermost atoms of the surface and eject electrons from the innermost electron shells surrounding the nucleus. The chemical environment of the atoms, i.e. the nature of the chemical bonds comprising the molecules, determine the binding energy required to remove the electron from the surface atom. By carefully measuring the kinetic energy of the ejected electrons the atoms of the surface may be determined from the binding energy and the relative concentrations of the atoms may also be determined. Electrons can only escape from the uppermost layer of the sample making this a very surface sensitive method, (upper 50 nm of sample). In its simplest form the spectra show peaks representing the binding energy of the atoms but without further detailed analysis no chemical functionality. For instance the various carbon-oxygen binding energies are very close together and in low resolution spectra appear as a single peak. For polymers, the C 1s electron spectrum is used and for treated surfaces is ratioed with the O 1s electron to determine the molecular concentrations of oxygen in the treated surface.

Surface Chemical Composition: As described above, the XPS spectrum contains a single peak for carbon-oxygen species on the surface, which is not resolved in terms of the contributions of individual surface chemical species. However, it is possible to obtain a calculated value of the surface composition by estimating the relative contributions of each chemical group and performing a curve fit to reproduce the

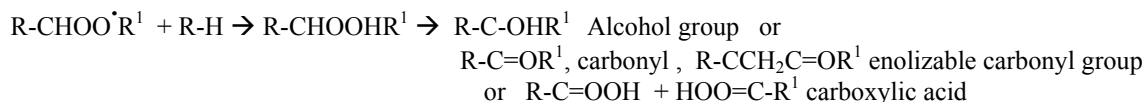
actual peak shape. This approach while not exact represents a practical approach to characterizing the treated surface and makes use of model compound studies. It has been widely used in literature studies of polymer-metal surface interactions. It is not possible to differentiate between alcohol and ether groups with this method, which is why the results can be considered speculative⁸.

The chemical functionality of a treated surface may be determined from direct chemical derivitization and analysis of the surface. This is a complex but classical chemical analysis technique where specific reagents are reacted with a surface to form unique chemical groups, which may then be analyzed by several methods such as XPS and other spectroscopic methods. While making its own set of assumptions about relative reactivity etc. in combination with XPS it allows the determination of alcohol, acid, ketone and aldehyde groups directly. By varying the reactant chemicals it is also possible to determine the amount of enolizable carbonyl groups and by a difference the amount of ether linkages.

TREATMENT MECHANISMS

A general observation of the effect of all treatment mechanisms is that they all generate oxidized polymer surfaces. Detailed studies of the various treatment methods do indicate that there are variations of the mechanism or variations in the ratio of functional groups produced, but these differences seem larger between polymers than between the different treatment methods. All the methods can be applied aggressively resulting in highly oxidized, low molecular weight materials on the substrate surface, which I will term over treatment. While it can be shown that the low molecular weight material might not be harmful for some inks or for aged or washed PET surfaces^{6, 9}, in general its impact on aluminum adhesion is the same as for low molecular weight additives which can form on the film surface and destroy the aluminum adhesion¹⁰. The low molecular weight molecules on the surface lowers adhesion of metallized layers by producing a weak interface which is not desired and hence the film surface should have been treated less for optimum metal adhesion.

Mechanism of free radical oxidation of polyethylene



The basic mechanism of treatment starts with the generation of a free radical on the polymer chain. A free radical is a very reactive chemical species consisting of an unpaired electron. The polymer radical is produced by a high-energy impact of a free electron, ultra-violet radiation or a disassociated molecule produced by the high energy or high temperature in the corona, flame or plasma treater gap. All of the treatment methods generate free electrons and/or free radicals, which interact with the substrate to create a radical on the surface. What part of the polymer molecule the radical is produced on controls the various differences in the oxidation between various polymers. Thus on polyethylene, radicals are produced on the carbon backbone of the polymer molecule, on polypropylene, the radical is produced on the backbone carbons or on the pendant methyl group and for PET the radical is produced on the aromatic ring or by chain scission. The radical formed then rapidly reacts with an oxygen molecule, O₂, which forms a peroxide. The peroxide then decomposes by various routes to form ethers, alcohols, aldehydes, ketones, or carboxylic acid groups on polyethylene and polypropylene and phenols in PET. Almost all of the studies involving self adhesion, ink adhesion and aluminum interactions show that the carbonyl groups play a key role in adhesion to the treated surface through formation of Al-O-C bonds or hydrogen bonding to alcohol groups in the inks or opposing film surface. Enolizable carbonyl groups play an important roll in adhesion because they may rearrange from the carbonyl into an unsaturated alcohol by tautomerism and thus may participate in adhesion improvement groups, as both a carbonyl or as an alcohol.

The carbonyl and alcohol groups are formed first and can be further oxidized to carboxylic acid or carbonyl groups respectively, which would be the impact of high or aggressive over treatment. Thus a definition of over treatment could be described as the treatment intensity, which forms carboxylic groups at the expense of carbonyl and alcohol groups.

TREATMENT AGING

Surface treatment values are found to decrease with time from the moment treatment is complete. This has been found in all of the treatment methods employed to date. The amount and cause of the aging is dependent on the level of treatment and the formulation of the film. In general the aging results from an migration of highly oxidized low molecular weight material on the film surface, rotation of the highly polar components attached to the polymer backbone chain into the film surface to minimize surface free energy when stored in air or the bloom* of film additives to the surface covering the surface with a low free energy layer and the transfer of film additives across a film-to-film contact plane in wound film rolls. Typical additives, which show this behavior are fatty acid amides (slip agents) and silicone oil both of which have low wetting tensions. One notable exception to the aging has been reported and that is the atmospheric plasma treatment with a helium/acetylene blend¹¹. These samples have shown high and stable surface treatment values, indicating that a different oxidized surface structure is being produced.

*The term “bloom” has its roots in the rubber vulcanization process where excess sulfur was added to cross-link rubber latex. The extra sulfur would come to the surface of the rubber¹² creating a surface pattern, which was called bloom (reported by Dr. Billmeyer as due to the pattern looking like a flower).

EFFECT OF FILM ADDITIVES

Additives are added to most film products to modify the surface properties of the film. In general there are two classes of film additives, blooming and non-blooming additives. Blooming additives are designed to migrate to the surface of a film and change it's surface properties while non-blooming, or non-migratory, additives are designed to stay in a fixed location in the film, generally on the film surface to modify properties. In general the blooming additives are called “slip” additives because they lower the films coefficient of friction (COF) while the non-blooming additives are called “anti-block” because they hold the film surfaces apart and prevent the auto adhesion or “blocking” of the film which leads to film tearing during unwinding. Migrating additives are more likely to be added to polyolefin films such as polyethylene, poly (vinyl chloride) and polypropylene than to polyester based films. Poly (vinyl chloride) films are also highly modified, or plasticized, to allow processing of the material into flexible PVC films.

What happens to a film that has been surface treated is that the surface, is covered by the slip additives and the treated surface is hidden under the new surface layer. At first the treatment seems to have gone away, while in fact it is under the new surface layer. Washing the surface will remove the slip and the treatment should reappear. Depending upon the chemistry of the additive it might be possible to raise the apparent treatment of the film. However, most migratory slip agents are relatively non-polar which would give rise to a lowered treatment measurement. The principle detriment of the slip agents for metallization is that this surface coating lowers the COF because it is a weakly adhering surface layer. If metal is deposited over this slip layer the adhesion of the aluminum to the underlying film surface will not develop as described in the patent to All and Duncan¹⁰. Typical slip additives are fatty acid amides such as, erucamide or stearamide, and high viscosity silicone oils. Also, polymer stabilizers may migrate to the film surface. These blooming additives can also interfere with adhesion of ink or polymer to aluminized surfaces, as they will form a weak layer by transfer from one film surface to another surface while the film is wound in roll form. Storage time and temperature will affect the amount of film additive, which migrates and transfers to the surface of the film.

Anti block particles are often added to reduce the amount of slip additive required to lower the COF of the film. In addition for vacuum winding of film they are used to fix the separation distance between film layers (replace the air layer) and improve winding and tracking over the rolls in the metallizer. Because they are in the surface of the film and stick above the surface they might cause film scratching if film layers move across each other. Some particles are imbedded in the film surface while others are more loosely bound and will fall out of the film and cause build up on equipment.

The additives are required in film because the final products would not work in the chain of use required for the film. Consequently, film suppliers balance the relative amounts and types of additives to balance the many different properties needed by the film for successful winding, metallization, lamination and machinability at the end use of the film and film structures. The proper choice of film type and treatment

method and treatment location will be important in determining how well the film performs in the metallizer and how well the metallizer can handle a wide range of products such as by surface cleaning by treatment.

ATMOSPHERIC TREATMENT TECHNOLOGIES

Surface treatment of polymers have been studied extensively in terms of the chemical modifications to PET, OPP and HDPE films by corona, flame and ultraviolet (UV) methods. While the newer atmospheric plasma treaters have not been studied extensively, the mechanism of treatment should be essentially the same and similar results are to be expected for the straightforward treatment of the films. Each method will be discussed to show the equipment and methods used.

Corona Treatment

In corona treatment a high frequency electric circuit is established which contains an air gap capacitor. The electric voltage is raised to the point where electric breakdown of the air occurs and a current begins to flow across the gap. The electrons in the gap are accelerated by the high potential across the gap and impact gas molecules in the air creating ions and additional electrons, which are accelerated toward the grounded surface. This causes an avalanche of high-energy electrons, which impact the grounded surface in the circuit. During the breakdown of the air, ozone is formed and an intense blue light is emitted from the air gap and on close inspection, many small sparks can be seen in the air gap. Thus the electrons, ozone and the UV light can all participate in the chemical modification of the polymer surface. The high frequency AC circuit constantly raises and lowers the voltage in the gap to keep the ionization process going. The high frequency is used to prevent treatment skips as the voltage cycles.

In commercial practice the air gap capacitor consists of an electrode held at high voltage and a grounded metal roll covered by a dielectric material to increase the voltage drop in the air and prevent arcing to the grounded roll. This dielectric surface may be formed from an epoxy, a hapolon or silicone rubber sleeve, a glass or a ceramic coating. All of this equipment is placed inside a box, which is vented to remove ozone by high airflow.

In order to treat the film it is passed through the air gap during breakdown and the high-energy electrons impact the polymer creating free radicals which rapidly react with oxygen or water molecules in the air or dissolved in the film, forming hyper-peroxides in the upper layer of the film. These hyper-peroxides then decompose into a range of chemical species such as alcohol, epoxy, ether, ketones, aldehydes and carboxylic acids. The level of treatment obtained will be determined by the energy density, KW/m^2 dissipated in the treater gap.

While the corona treatment process may be used successfully for many products, due to the high frequency and high voltage used in the process it is possible to create oxidized sites on the film surface against the roll (the back side of the film), as in all atmospheric installations there is always a thin layer of air between the film and the roll surface. While the backside treatment can be managed it can never be eliminated. For metallizable film this is problem, which must be controlled and is a principle source of metallized product defects. Many manufacturing process variables and equipment condition factors are interrelated in determining the base line level and the distribution of backside treatment. Factors effecting the level of backside treatment are: treater roll surface finish, nip roll condition and alignment, film line speed, film web tension, film gauge uniformity, and treater power level.

Flame Treatment

In the flame treatment process air and combustible gas are mixed and sent to a burner where a high velocity flame is started on a burner grid in close proximity to the film surface. There are two basic flame configurations, which may be used, a short compact flame or a longer "softer" flame. In the past only the long flame was used and the process was primarily used for treating complex 3 dimensional articles. Currently, the largest commercial applications for film treatment use the short flame with a well-defined flame front. The treatment process is dependent upon the air/gas ratio, the air/gas flow rate, position of the film surface relative to the flame (distance from the burner) and the gas composition all of which combine to determine the flame temperature and the oxidizing or reducing nature of the flame. Treatment level as determined by surface oxygen concentration is found to pass through an optimum with air/gas ratio, to

increase with air/gas flow rate and to decrease with increasing distance from the burner. Air/gas ratio controls the flame temperature and the oxidizing condition of the flame. The air/gas flow rate is total thermal power while the position in the flame front affects both flame temperature and oxidizing environment at the flame/film interface.

Process control has been greatly improved over the years and the focus is on the air/gas composition control, either thru oxygen composition of the combustion blend or by measuring the flame temperature of a side stream, which is burned and heats a thermocouple. Control is through feed back to control the gas input at a venturi mixing valve in the process air supply to the burner. The safety and reliability of these new systems have been greatly improved during the last twenty years making this a reliable film treatment method. Due to the high temperature of the flame, the film must be cooled on a water-cooled roll to protect it from the thermal load of the flame. Generally the burner is placed below the water-cooled roll and the film is passed above the burner against the roll surface. This is to allow the formation of a stable flame front and a stable treatment zone but results in the highest thermal load on the film. A key advantage of flame treatment, for metallizable base films is that the flame treatment mechanism can not yield back side treatment as there is no mechanism for having the flame front behind the film.

The chemical species produced and treatment mechanisms for polyethylene surfaces are similar to those found with corona treatment. However, due to the complex chemistry of the flame, additional mechanisms may be present causing the balance in chemical species to be somewhat different. Also, the treater may be setup to yield potentially different film product performances for products treated by both methods. Also, it has been shown that the air/gas ratio controls the surface chemistry developed^{13,14}.

In the case of polypropylene, the mechanism of oxidation does appear to be different than that of corona treatment. In flame treatment, the XPS evidence of the oxidation groups formed shows that the pendent methyl group undergoes a sequential oxidation from alcohol to carbonyl to carboxylic acid group and the polymer chain is not attacked¹⁵.

Atmospheric Plasma

This is a new treatment technology just recently described and patented¹⁶. It is being commercialized and has the potential to permit more flexibility for chemical surface modification of polymer films by control of treatment gas composition. It is a low voltage (5000 volts) discharge from a hollow cathode electrode using helium as the discharge medium. The use of helium allows a plasma discharge to form at much lower voltages than in generating a corona in air. The use of helium for generating the discharge and due to the control of helium in the treater bar, the concentration of helium behind the film and between the roll surface should be close to zero, which should eliminate the potential for backside treatment as the voltage is too low to create a plasma with air. In construction and conception this new process is conceptually between a corona and a flame treater. As in flame treatment the helium gas is delivered to the treatment gap by passing it through an electrode faced with a sintered metal surface. But like a corona treater the plasma in the treatment gap is excited by a high voltage high frequency alternating current as the pores of the treater function as hollow cathodes. In many ways it may be characterized as a "cold flame". Treatment of the film surface is effected with 100% helium or with blends of helium/oxygen or helium/nitrogen blends. Initial reports of the treatment result show an oxidized surface just as with flame and corona treatment. The treatment level is found to age just as with other forms of surface oxidation and the extent of treatment is in the same range. It is not yet known how well the relative concentrations of carbon-oxygen species may be controlled at the surface. Another exciting aspect of this technology is the use of hydrocarbon gas molecules blended with the helium carrier gas. This has the potential to dramatically change the potential treatment chemistry and hence treatment results relative to flame and corona treaters. The use of acetylene has shown some dramatic surface treatment results with both Teflon and HDPE surfaces in that high treatment level of 60 dynes may be developed and the dyne level does not age with time¹⁵. This would indicate the formation of a new surface structure than for flame, corona or for simple helium plasma treatment.

Metallization

Another method of surface treatment, which is not widely discussed, as such is metallization itself. Harold Schonhorn et al. at Bell Labs generated an extensive body of knowledge on the effects of metal evaporation

with various metals on polymer surface properties. For instance, using aluminum deposition on fluoropolymers as a means of improving adhesive bond strength of gold metallized Teflon™ after stripping the aluminum from the surface¹⁷. It was shown by XPS studies that the Teflon™ surface had reduced fluorine abundance and increased oxygen attached to the carbon of the polymer. Thus the aluminization process of the Teflon™ affected a surface oxidation of the polymers. This effect can explain the development of excellent adhesion of aluminum layers to untreated polymers sometimes observed when the wrong, i.e. untreated, surface is metallized. There are several studies of these effects in the literature today^{18,19}.

In-Chamber Plasma Treatment

There are several principle plasma generation technologies in use: the original DC and AC magnetron sources with reactive gas blends and a new hollow cathode AC plasma generator²⁰ with reactive gas blends. The essential features of all the plasma generators is that they supply high energy electrons in sufficient quantities to generate heavy ions and additional electrons from the gas blend, for bombardment of a film surface. In traditional magnetrons the ions and electrons are constrained in a ring shaped magnetic field above the generator and the film is passed thru or near the upper regions of the plasma (Figure 7.10). These systems may also be magnetically enhanced to open the field lines towards the film. In the hollow cathode design the magnetic field above the hollow cathodes are modified by permanent magnets placed behind the film as it passes through the treater. This has the effect of inserting the ions and electrons in the open magnetic field and directing them towards the film surface (Figure 7.11) as the polarity of the generator cycles from positive to negative. These various electrical systems in conjunction with the gases fed into the various plasma generators clean and activate the surface for chemical modification just prior to the deposition of the aluminum. The results published by the various plasma treater manufacturers all show improvements in the metallized film properties of barrier, metal adhesion and metal quality with various gases, gas blends and flow rates. The future of the various processes lies in determining which one allows the most flexibility and operating reliability in generating polymer surface modifications for property enhancement in addition to improved adhesion.

While the treatment occurs just prior to the deposition of the aluminum it is likely that only oxygen species are generated on the film surface under the metal layer even for plasmas generated without added oxygen. This can be inferred from the results described above where the deposition of aluminum results in oxidized surfaces. In general, the plasma systems used commercially today are based upon argon or blends of argon and oxygen. Where does the oxygen come from for treatment using argon plasma in the vacuum? One source would be dissolved oxygen in the film itself followed by oxidation of any intermediate species formed after venting of the film to air and on rewinding for conversion. This is likely due to the high oxygen transport of polymer films. Also, there are very few reports of any other form of stable chemical species other than oxygen groups on treated films.

It is apparent from discussing the plasma pretreatment process with suppliers of metallization equipment and plasma treaters, that the process is not new, especially for those suppliers that market sputtering equipment or machines for capacitor applications. Indeed, according to leading metallizing equipment manufacturers, technologies for the pretreatment of films in the metallizing chamber have been in use for many years. However, the principle uses were to pre-clean the web and for chemical modification of the film surface to improve bonding. Generally this was accomplished with a planar magnetron based technology with or without magnetic enhancement, and modified to use argon gas to create plasma to clean or etch expensive films to accept sputtered layers. In these cases the process speed was not a limit, and the cost of extra equipment and increased process complexity were warranted by the high price and quality demands of the products. However, today the use of in chamber pretreatments is expanding from the low speed sputtering applications into the high-speed process typical of aluminum metallization of packaging films. This change has been sparked by the realization, with existing film designs, of improved barrier properties²¹, metal layer improvements^{22, 23, 16} and metal adhesion²¹ in the aluminized films produced. These improvements are brought about by the chemical modification of the film surfaces with the evolving pretreatment technologies.

Summary of Surface Treatment for Aluminum Deposition

All of the treatment processes practiced by the film manufacturers suffer in some regard relative to treatment at the time of metallization because of the passage of time from the point of film manufacture to the metallization chamber. As the rolls are stored in wound form, the surface to be metallized may be contaminated from chemical additives in the film and contact with the opposite film surface. In addition, most surface oxidation techniques give surfaces, which decrease in surface energy with time due to the diffusion of chemical groups into the surface of the film. Consequently, plasma treatment at the point of metallization would seem to hold a great deal of promise because it can supply higher treatment surfaces at deposition due to no ageing, provide better metal layers due to the surface cleaning and may supply better surface chemistry due to new gas blends for the surface functionalization which is sought. There is however a potential disadvantage to the technology especially when used with existing films which, are treated by the film manufacturer. As is known in the film industry, it is possible to over-treat the film and cause a reduction in metal bonding. Thus when installing and starting up a new pretreatment process it is worthwhile working with your film supplier to obtain film with reduced surface treatment to permit the plasma treatment process to be optimized for film properties.

Table 7.2 Dyne Level Testing Solution Compositions²⁴

% Formamide By volume	% Cellosolve By volume	Wetting Tension Dyne/cm
0	100	30
2.5	97.5	31
10.5	89.5	32
19.0	81.0	33
26.5	73.5	34
35.0	65.0	35
42.5	57.5	36
48.5	51.5	37
59.0	41.0	39
63.5	36.5	40
67.5	32.5	41
71.5	28.5	42
74.7	25.3	43
78.0	22.0	44
80.3	19.7	45
83.0	17.0	46
87.0	13.0	48
90.7	9.3	50
93.7	6.3	52
96.5	3.5	54
99.0	1.0	56

Table 7.3 High Surface Energy Test Mixture²⁵

% Formamide By volume	% Water By volume	Wetting Tension Dyne/cm
100.0	0	56
81.2	18.81	58
65.0	35.0	60
47.0	53.0	62
30.6	69.4	64
18.2	81.8	66
8.6	91.4	68
3.6	96.5	70

**Table 7.4 Surface Free Energy of Various Liquids at 20 degrees C
(From Table 3.1 of Rance²⁶)**

Liquid	γ_{LV} mJ m ⁻²	γ_L^d mJ m ⁻²	$\gamma_{LV} - \gamma_L^d$ mJ m ⁻²
Water	72.8	21.8	51.0
Glycerol	63.4	37.0	26.4
Formamide	58.2	39.5	18.7
Methylene Iodide	50.8 50.76	49.5 (50.76)	1.3 -
Ethane-1, 2-diol	48.3	29.3	19.0
Dimethyl sulphoxide	43.54	34.86	8.68
Tri-cresyl phosphate	40.70	36.24	4.46
Pyridine	38.00	37.16	0.84
Dimethyl formamide	37.30	32.42	4.88
2-ethoxyethanol	28.6	23.6	5.0
n-hexadecane	27.6	(27.6)	-

**Table 7.5 Components of Surface Energies for Various Polymers
(Taken from table 4 of Brewis and Briggs²⁷)**

1 dyne/cm = 1 mNm⁻¹

Polymer	γ_s mNm ⁻¹	γ_s^d mNm ⁻¹	γ_s^p mNm ⁻¹
LDPE	33.1	32.0	1.1
Poly (vinyl fluoride)	36.7	31.3	5.4
Polypropylene	30.2	30.2	0
Polypropylene Flame treated	37.6	33.5	4.1
Poly (methylene oxide)	33.3	21.8	11.5
Poly (ethylene terephthalate)	41.3	37.8	3.5
Polytetrafluoroethylene	18.0	16.9	1.1
Polytetrafluoroethylene (etched)	50.6	36.1	14.5

Table 7.6 Inherent Surface Energy Levels of Polymer Substrates (Data from Markgraf²⁸)

Polymer, Un-treated	Dyne/cm Level
Nylon	46
Polyester, Mylar	43
Cellulosics	42
PVDC, Saran	40
Vinyl, acrylic	39
Ionomer (high)	37
Ionomer (low), polystyrene	33
Polyethylene, (LD & HD)	31
Polypropylene	29

Table 7.7 Critical Surface Tensions of Polymers²⁹ (From table 3 of Brewis and Briggs³⁰)

1 dyne/cm = 1 mNm⁻¹

Polymer	Critical Surface Tension mNm ⁻¹
Polyethylene, PE	31
Polypropylene, PP	31
Poly (vinyl chloride), PVC	39
Polystyrene, PS	43
Poly (ethylene terephthalate), PET	43
Poly (hexamethylene adipamide), Nylon 6,6	46

Figure 7.8 Schematic of Corona Treatment

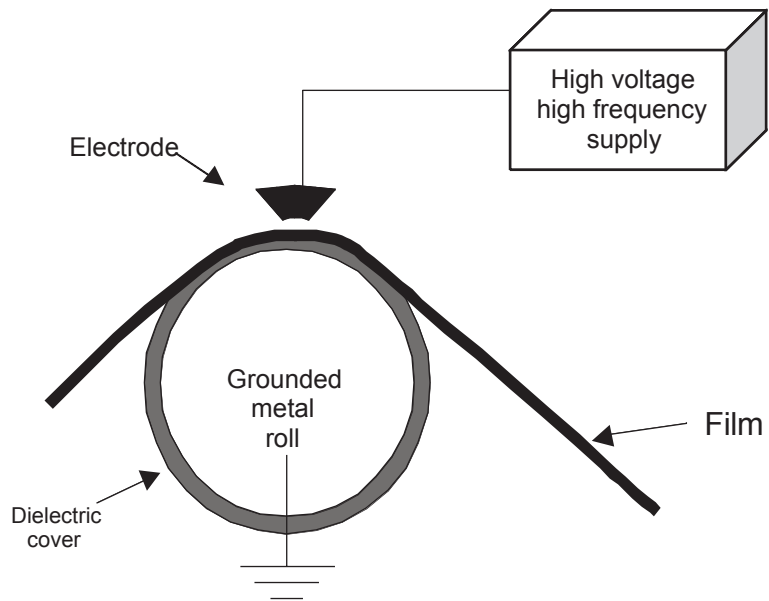


Figure 7.9 Schematic of Flame Treatment System

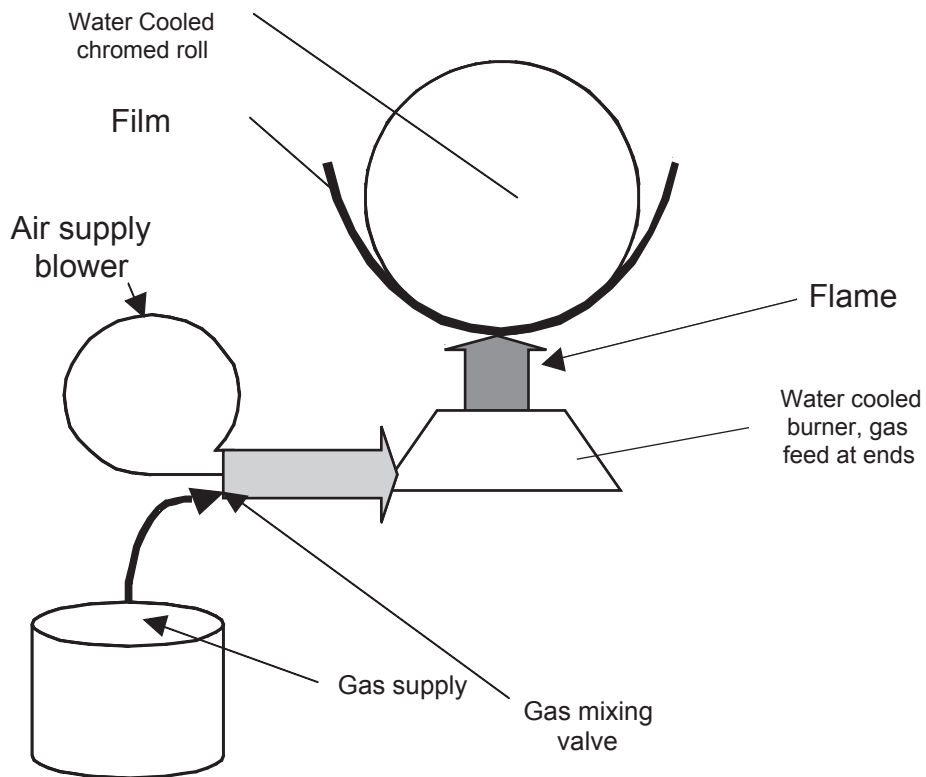


Figure 7.10 Schematic of Atmospheric Plasma System

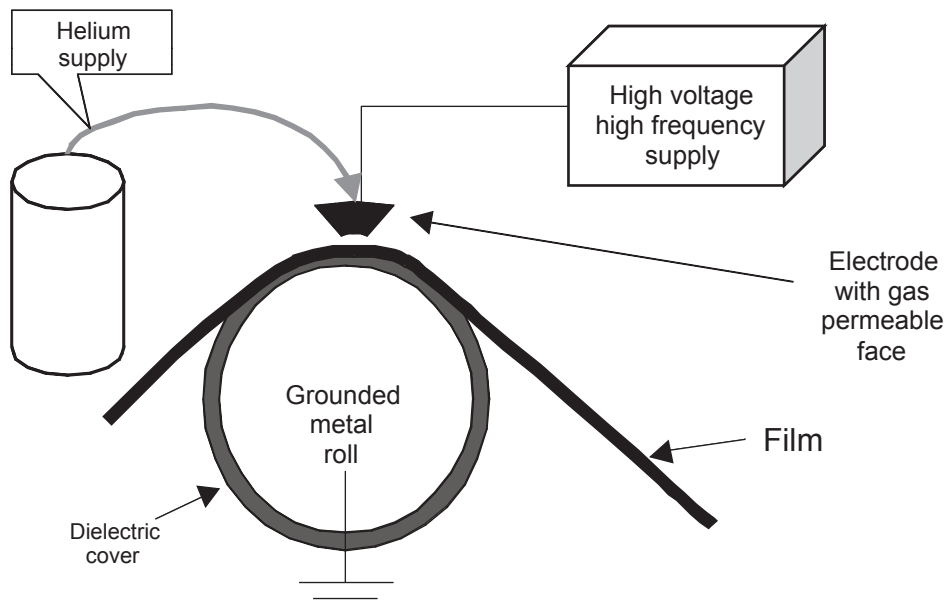


Figure 7.11 Schematic Diagram of Magnatron Plasma Treater

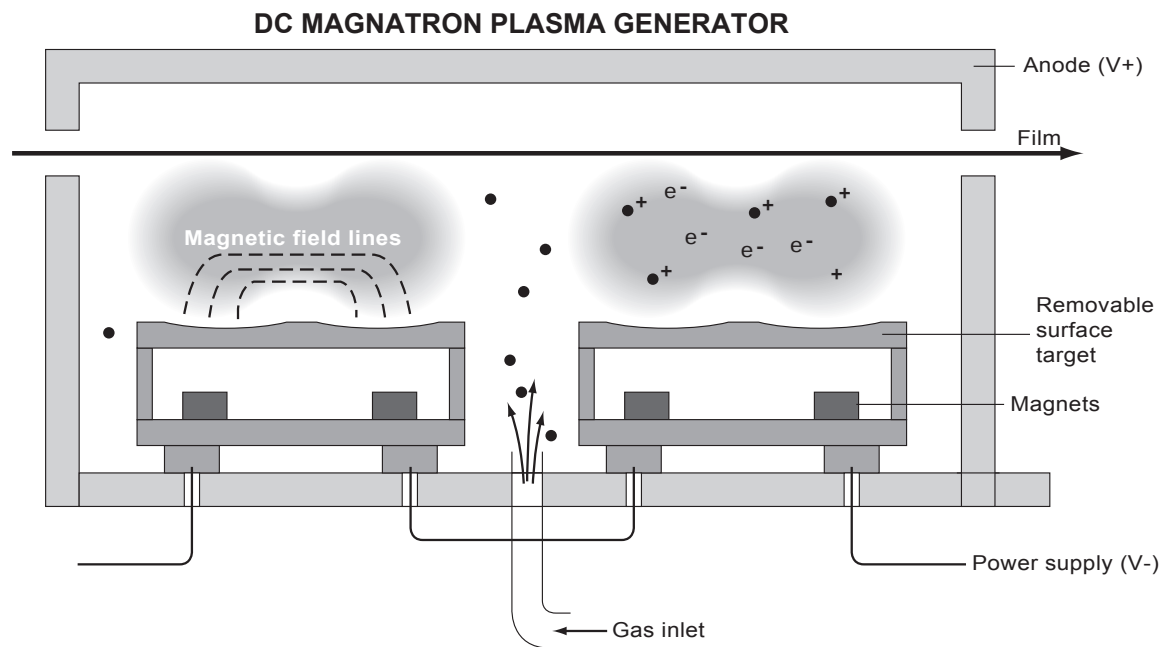


Figure 7.12 Schematic Diagram of Hollow-Cathode Plasma Treater

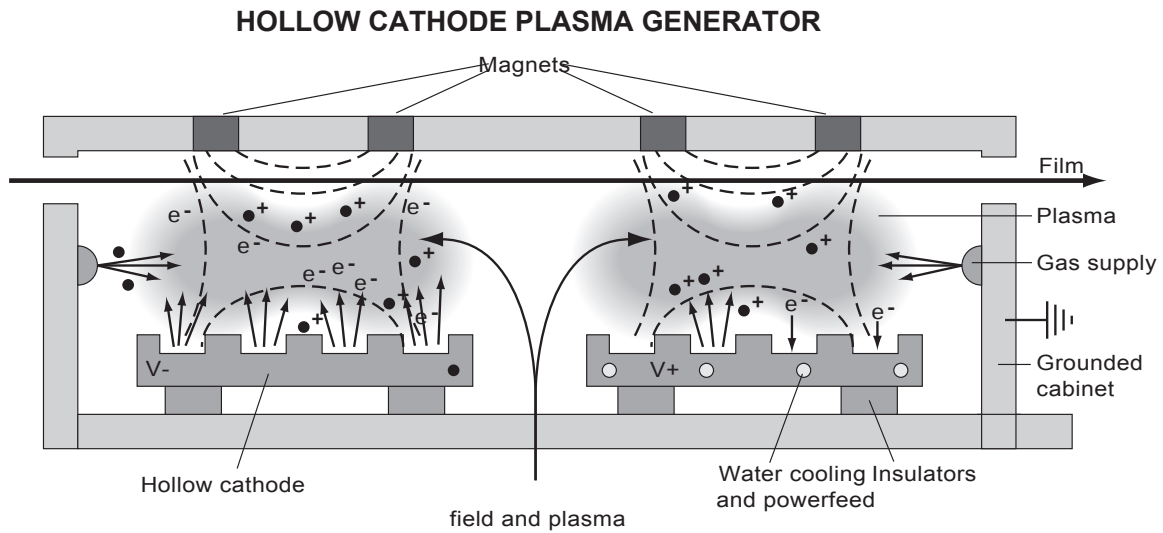
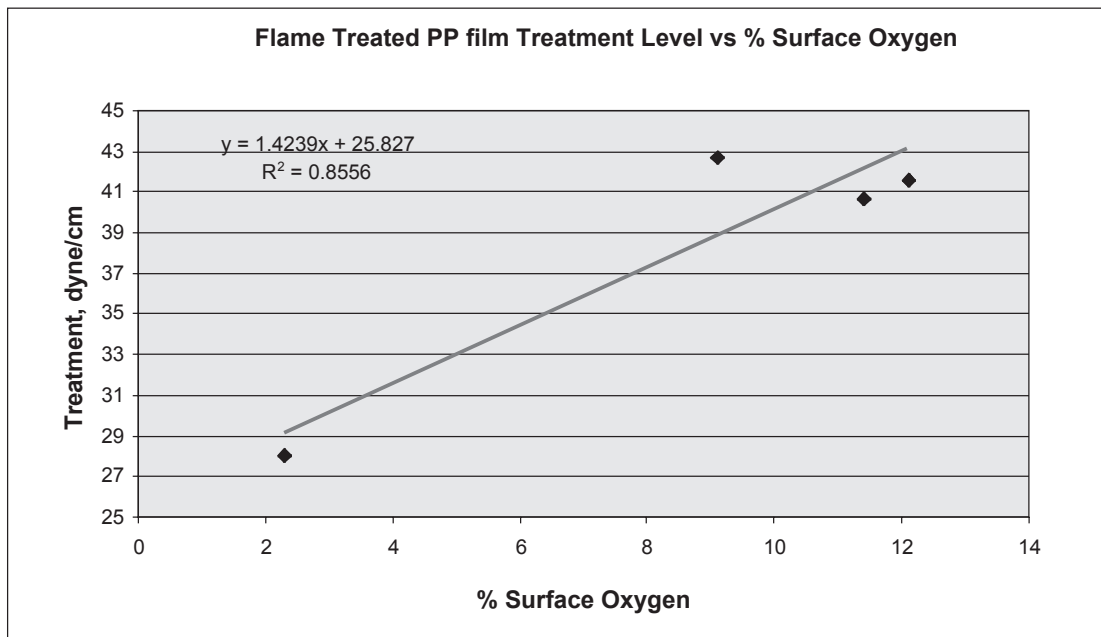


Figure 7.13 Flame Treated PP Film Surface Oxygen Level as Function of Treatment Level for One Pass Treatment (Data from, Garbassi et al.³¹)



7.14 Section 7 References Cited

- ¹ Mount III, E. M., Wagner Jr., J.R., *TAPPI Technical Proceedings, Polymers, Laminations & Coatings Conference*, 1998, p875
- ² Mount III, E. M., "Plasma Pretreatment for Metallizing Packaging Film", *Converting Magazine*, **19(3)**, (2001), pp124-131.
- ³ Chan, Chi-Ming, *Polymer Surface Modification and Characterization*, Hanser Publishers, New York, 1993.
- ⁴ Chan, Chi-Ming, *Polymer Surface Modification and Characterization*, Hanser Publishers, New York, (1993), p 36.
- ⁵ Owens, D. K., "Mechanism of Corona-Induced Self-Adhesion of Polyethylene Film", *J. Appl. Polym. Sci.*, **19**, (1975), pp. 265-271.
- ⁶ Briggs, D., Rance, D. G., Kendall, C. R., Blythe, A. R., "Surface Modification of Polyethylene Terephthalate by Electric Discharge Treatment", *Polymer*, **21**, (1980), pp. 895-900
- ⁷ Rance, D. G., "Thermodynamic Approach to Adhesion Problems", Chapter 3, *Industrial Adhesion Problems*, Brewis, D. M., Briggs, D., editors, Wiley-Interscience: John Wiley & Sons, New York, 1985, 48-86.
- ⁸ Reilley, C., Everhart, D., Ho, F., in *Applied Electron Spectroscopy for Chemical Analysis*, Windawi, H., and Ho, F., eds. Wiley New York, 1982, Chap. 6.
- ⁹ Brewis, D.M., Briggs, D., "Adhesion to Polyethylene and Polypropylene", *Polymer*, **22**, pp. 7-16, (1981)
- ¹⁰ All, F., Duncan, G. US patent 4,345,005
- ¹¹ Decker, W., Yializis, A., "Atmospheric Plasma Treatment for Enhanced Surface Functionalization of Moving Webs", *XIV International Web Coating Conference*, Reno, NV, October 2000.
- ¹² Billmeyer, F.W., *Textbook of Polymer Science*, Second Ed., Wiley-Interscience, New York, (1971), p543
- ¹³ Briggs, D., Brewis, D.M., Konieczko, M. B., "X-Ray Photoelectron Spectroscopy Studies of Polymer Surfaces", *J. Material Sci.*, **14**, (1979), 1344-1348
- ¹⁴ Brewis, D.M., Briggs, D., "Adhesion to Polyethylene and Polypropylene", *Polymer*, **22**, pp7-16, (1981)
- ¹⁵ Garbassi, F., Occhiello, E., Polato, F., Brown, A., "Surface Effect of Flame Treatments on Polypropylene", *J. Material Science*, **22**, (1987), pp1450-1456
- ¹⁶ Yializis, A., Pirzada, S. A., Decker, W., US Patent 6,118,218, "Steady State Glow-Discharge Plasma At Atmospheric Pressure", Issued Sep. 12, 2000.
- ¹⁷ Roberts, R. F., Ryan, F. W., Schonhorn, H., Sessler, G. M., and West, E., "Increase of Gold-Teflon FEB Joint Strength by Removal of Deposited Aluminum Prior to Gold Deposition for Electret Applications", *Jour. Appl. Polym. Sci.*, **20**, 255-265, (1976)
- ¹⁸ Vogel, S.S.L., Schonhorn, H., *J. Appl. Polym. Sci.*, **23**, 495, (1979)
- ¹⁹ Bodo, P., Sundgren, J.-E., "The Adhesion of Evaporated Metal Films to Polyethylene Substrates", *Surface And Interface Analysis*, **9**, 437-440, (1986)
- ²⁰ Yializis, A., "Apparatus for Plasma Treatment of Moving Webs", US Patent 6,066,826, Issued May 23, 2000.
- ²¹ Moosheimer, U., "Vacuum Web Coating Of Polymer Films: Some Requirements of the Food Packaging Industry", *Fraunhofer Institute Fur Lebensmitteltechnologie und Verpackung, Giggenghauser Strasse 35*, D-85354 Friesing, 15 pages,
- ²² Yializis, A., Ellwanger, R., Boufelfel, A., "Superior Polymer Webs via in situ Surface Functionalization", *Proceedings of the 39th Annual Technical Conference of the Society Of Vacuum Coaters*, (1996), pp 384-391.

²³ Yializis, A., Ellwanger, R. E., Harvey, J., "Barrier Degradation in Aluminum Metallized Polypropylene Films", *Proceedings of the 40th Annual Technical Conference of the Society Of Vacuum Coaters*, (1997), pp 371-375.

²⁴ *Annual Book of ASTM Standards*, Part 31, "Tension of Polyethylene and Polypropylene Films", D2578-67(reapproved 1972)

²⁵ Markgraf, D. A., "Practical Aspects of Determining the Intensity of Corona Treatment", *Tappi Journal*, (1985), February, pp 74-77

²⁶ Rance, D. G., "Thermodynamic Approach to Adhesion Problems", Chapter 3, *Industrial Adhesion Problems*, Brewis, D. M., Briggs, D., editors, Wiley-Interscience: John Wiley & Sons, New York, 1985, p 60.

²⁷ Brewis, D.M., Briggs, D., "Adhesion to Polyethylene and Polypropylene", *Polymer*, **22**, (1981), p 9.

²⁸ Ibid., Markgraf, Table II, p. 76.

²⁹ Ibid., Markgraf.

³⁰ Ibid., Brewis.

³¹ Garbassi, F., Occhiello, E., Polato, F., Brown, A., "Surface Effect of Flame Treatments on Polypropylene", *J. Material Science*, **22**, (1987), pp1450-1456

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